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Djuranovic et al.

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(54) **INCORPORATION OF INTERNAL
POLYA-ENCODED POLY-LYSINE
SEQUENCE TAGS AND THEIR VARIATIONS
FOR THE TUNABLE CONTROL OF
PROTEIN SYNTHESIS IN BACTERIAL AND
EUKARYOTIC CELLS**

FOREIGN PATENT DOCUMENTS

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patent is extended or adjusted under 35
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claimer.

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2310/20 (2017.05); **C12N 2800/80** (2013.01)

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CPC **C12N 15/64**
See application file for complete search history.

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ABSTRACT

The present disclosure relates to modulation of protein
expression.

8 Claims, 74 Drawing Sheets
(33 of 74 Drawing Sheet(s) Filed in Color)
Specification includes a Sequence Listing.

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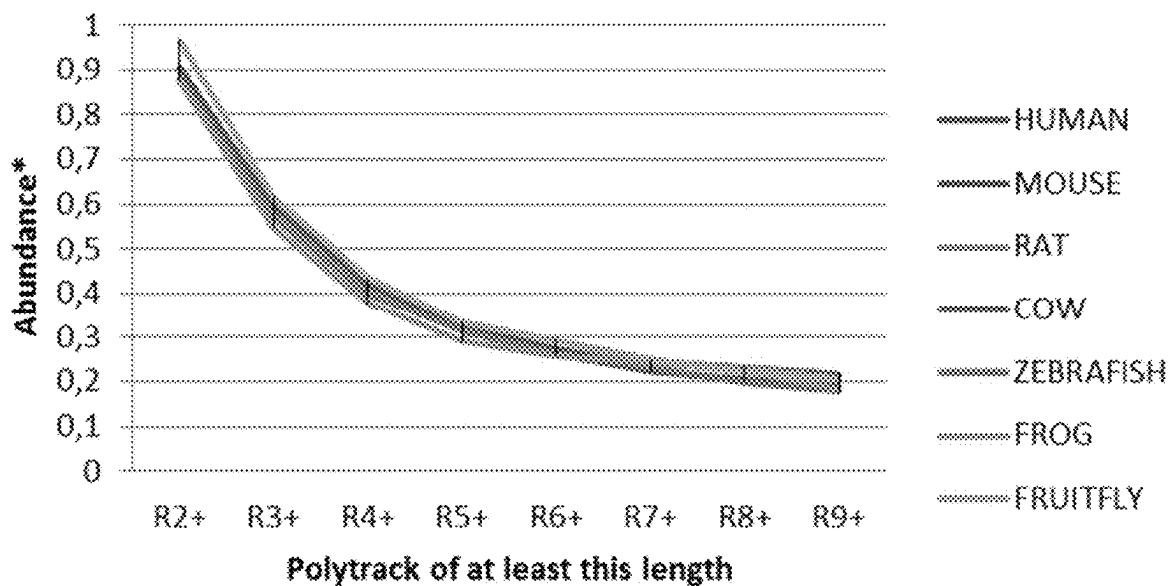
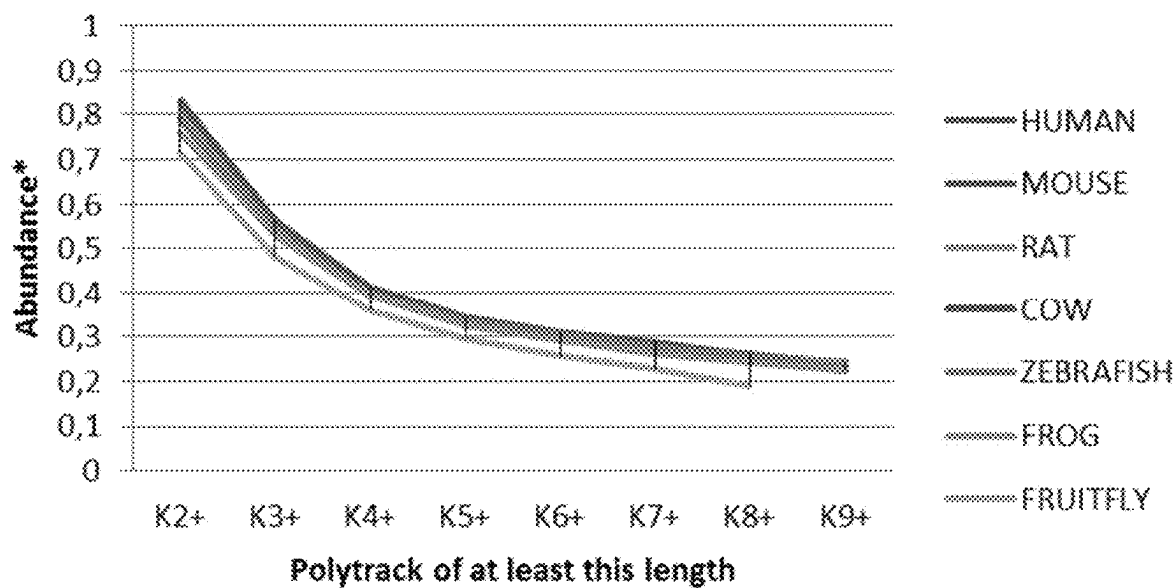
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**FIG. 1A****FIG. 1B**



$X = (AAA)_{6-12}, (AAG)_{6-12}, (GAA)_{12}, (CGA)_{12}, (AGG)_{12}, TAA$

FIG. 2A

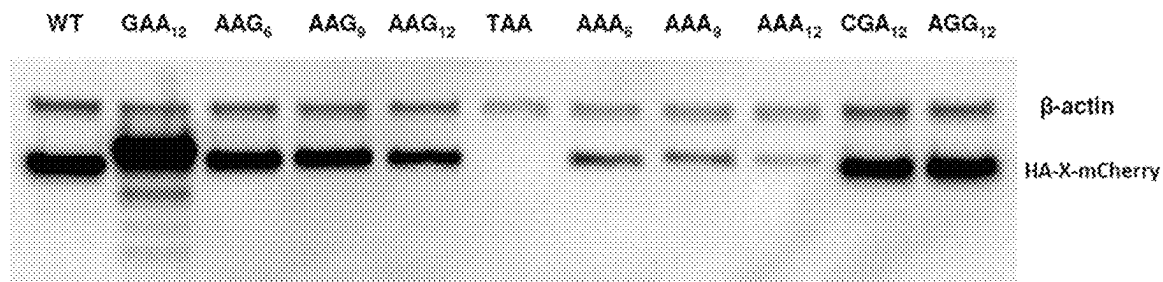
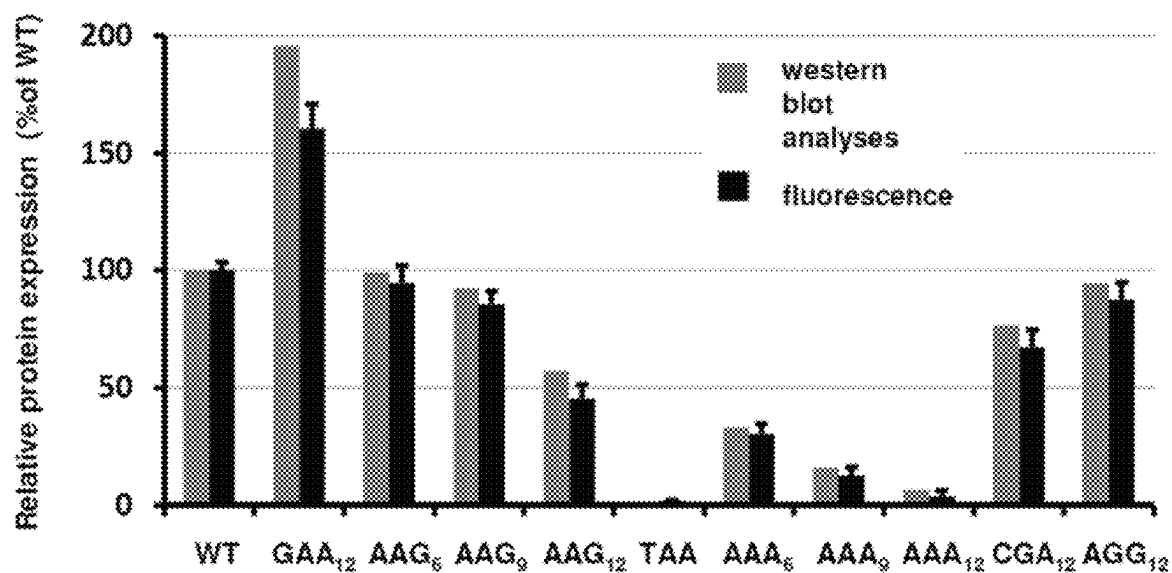
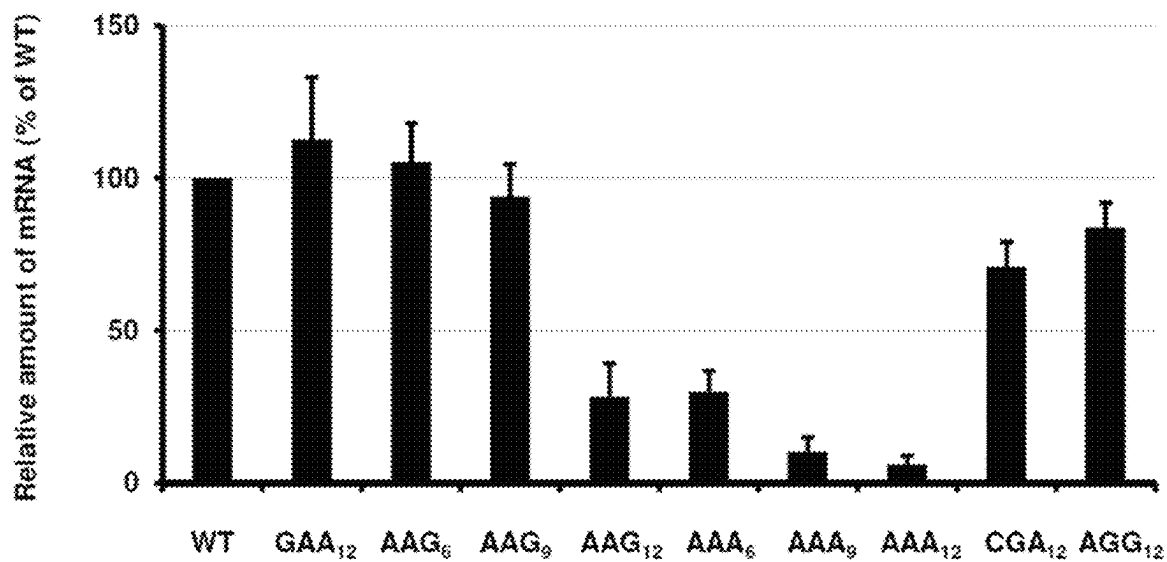
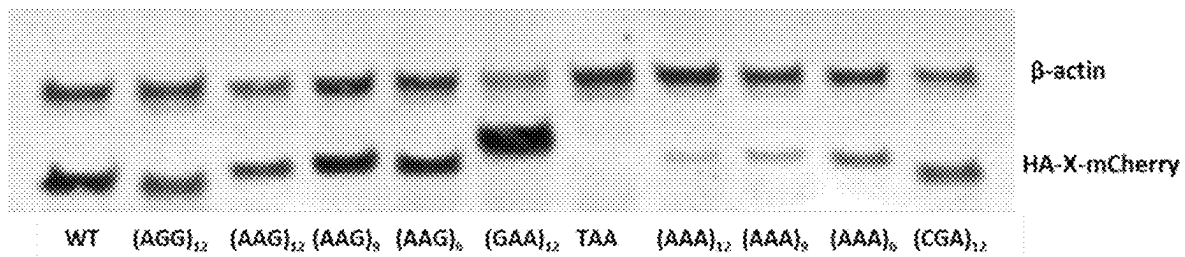
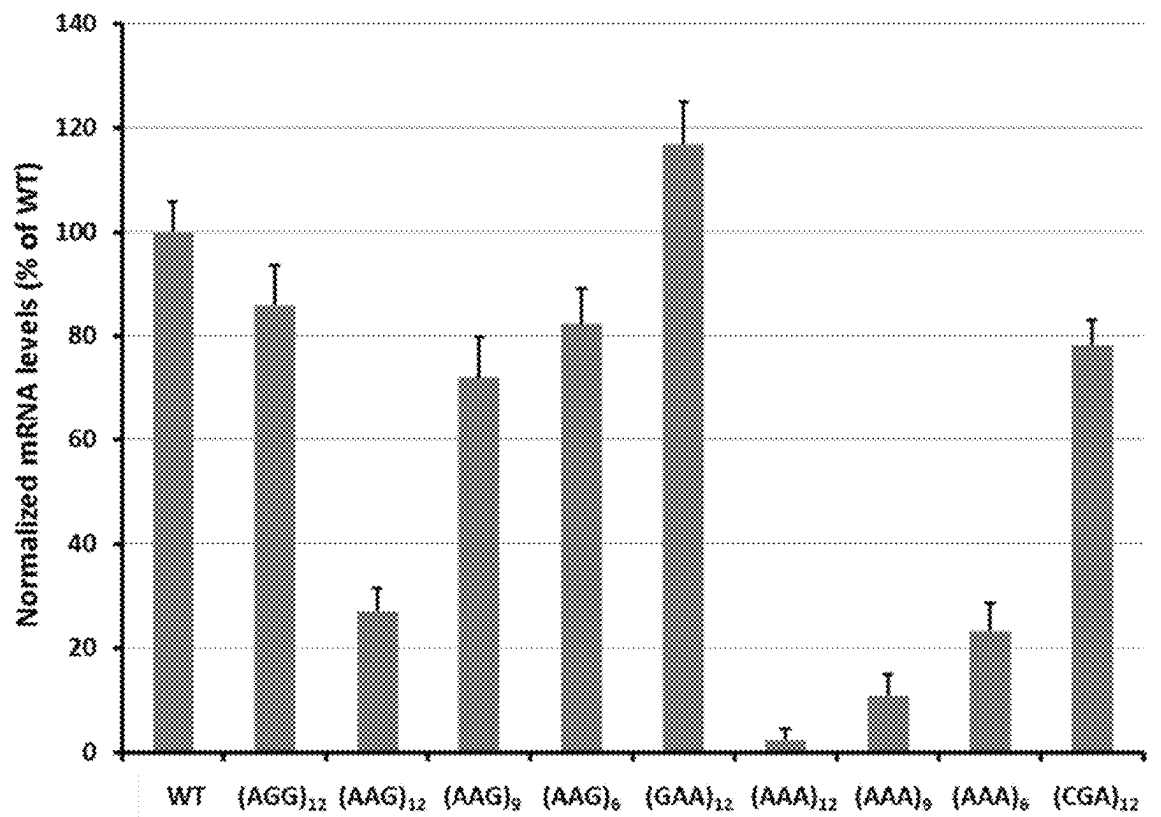
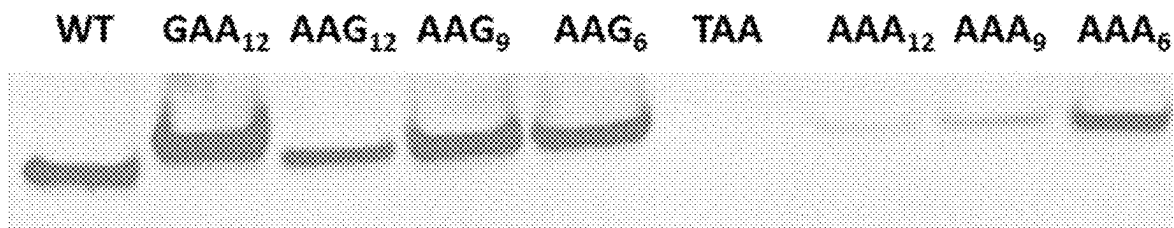
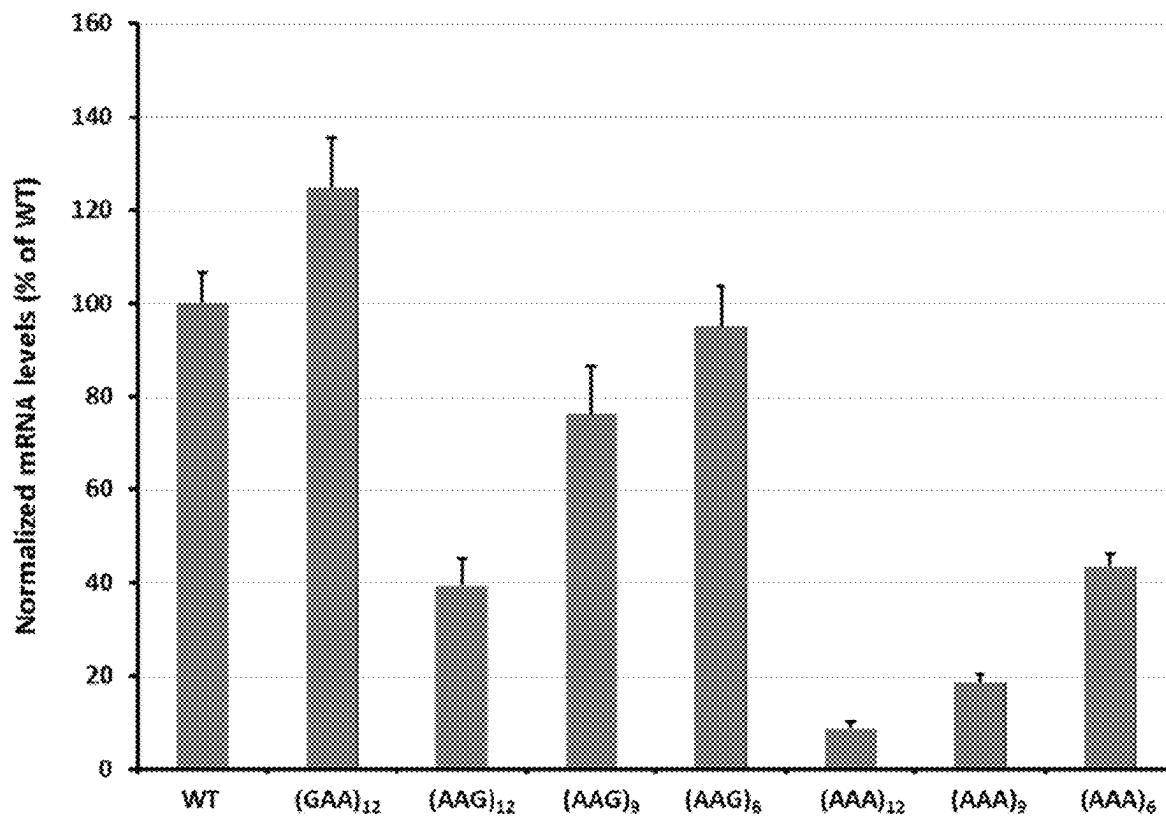
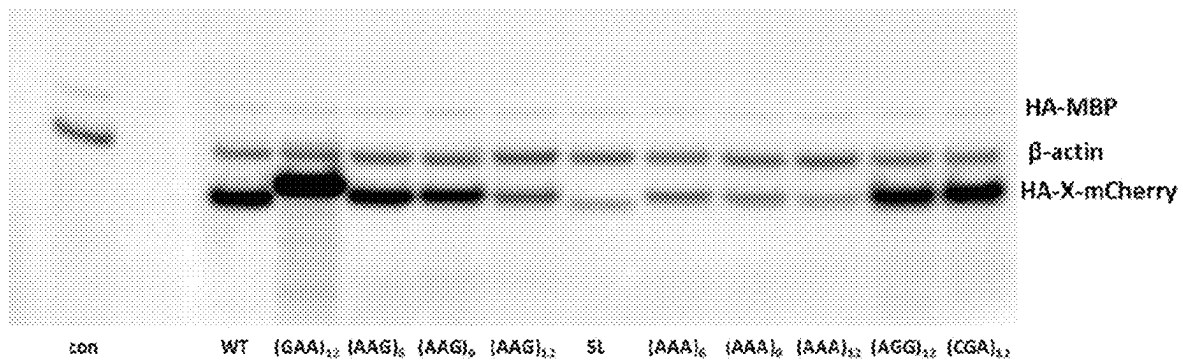
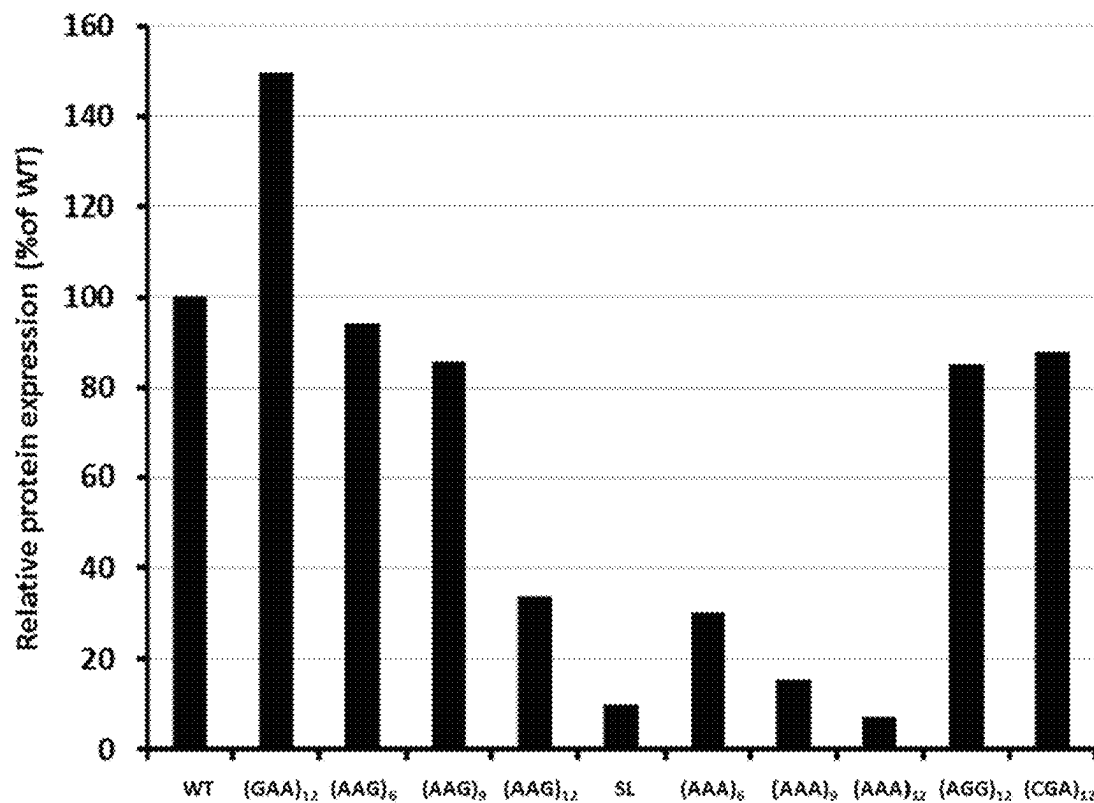


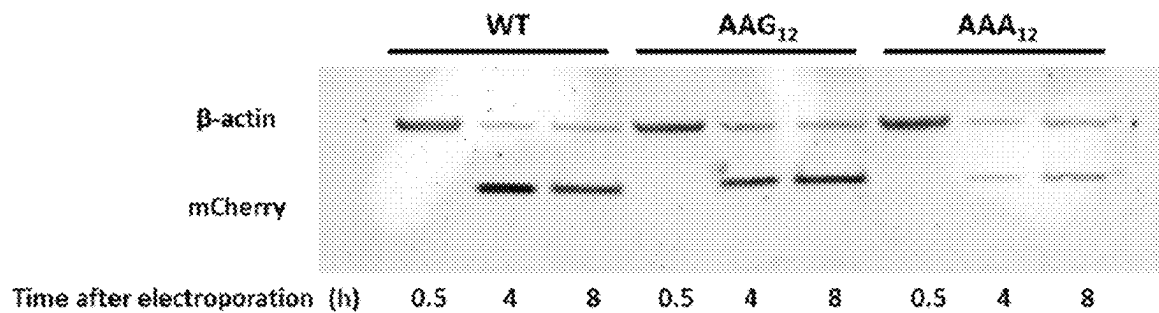
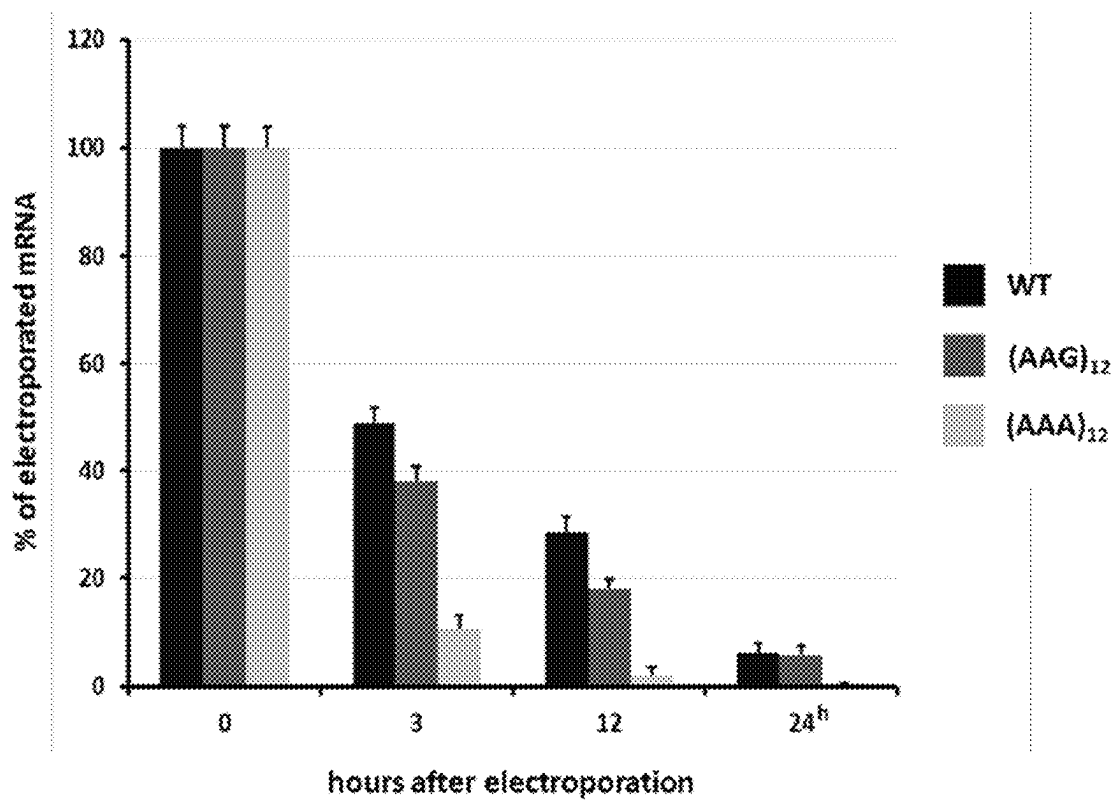
FIG. 2B

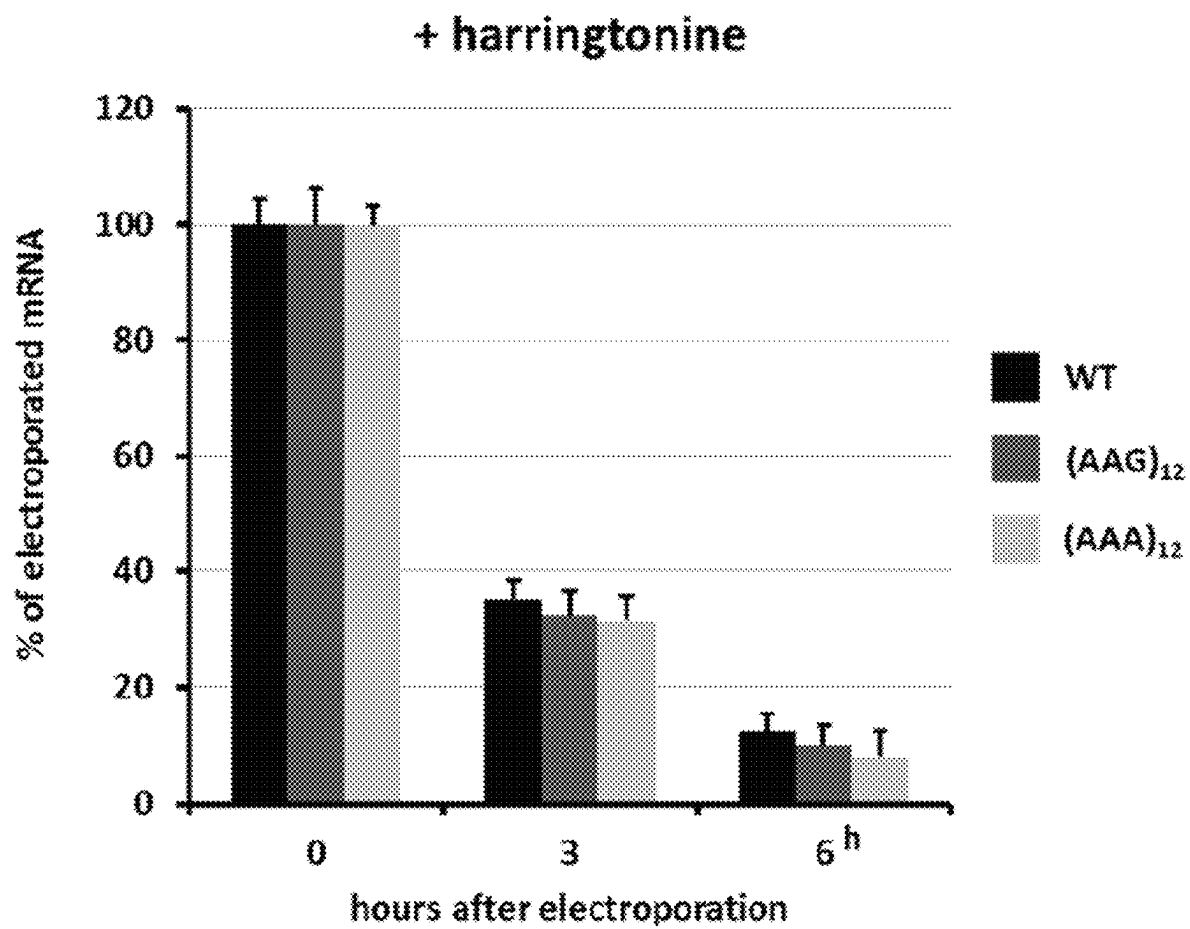
**FIG. 2C****FIG. 2D**

**FIG. 3A****FIG. 3B**

**FIG. 4A****FIG. 4B**

**FIG. 5A****FIG. 5B**

**FIG. 6A****FIG. 6B**

**FIG. 6C**

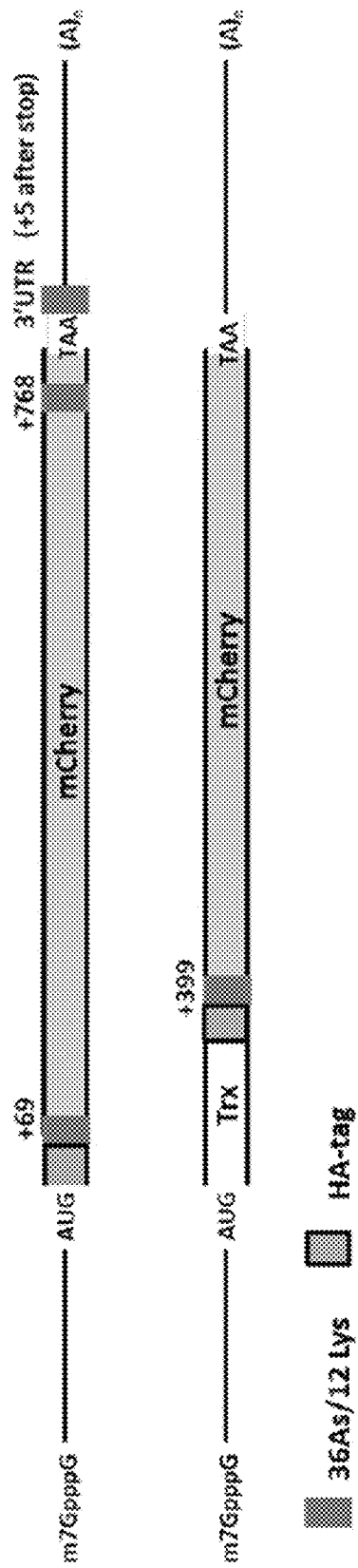


FIG. 7A

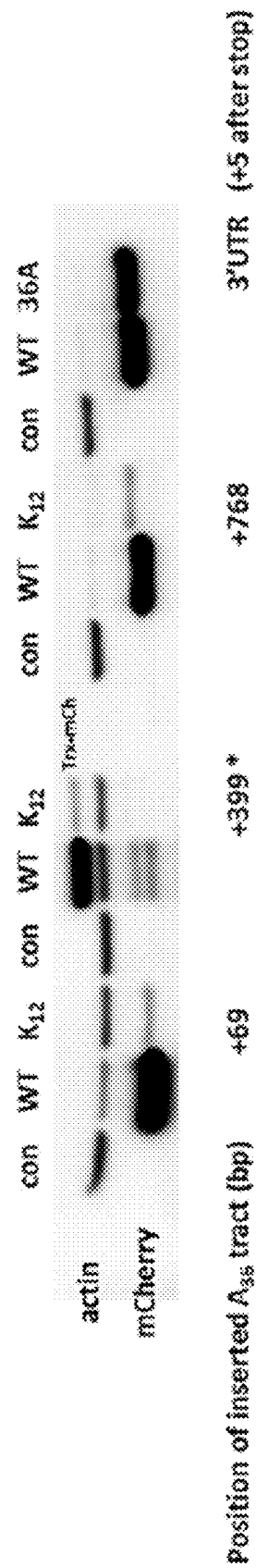
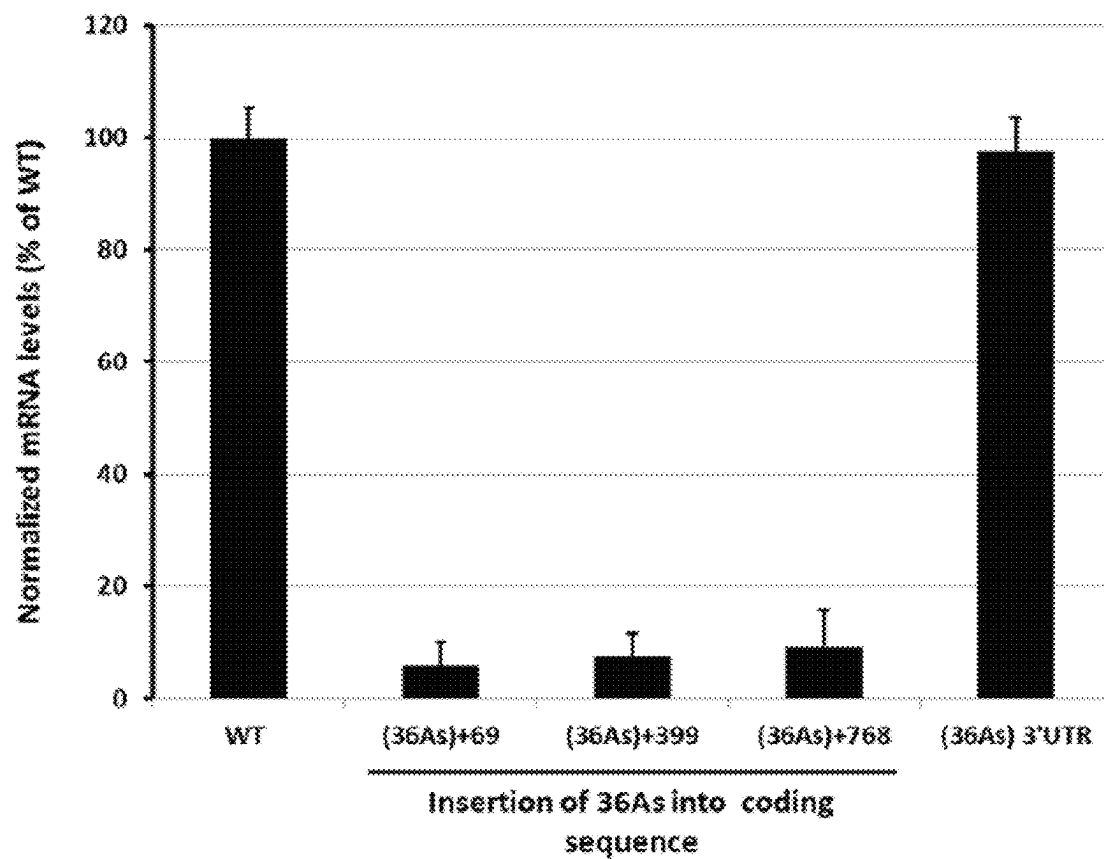
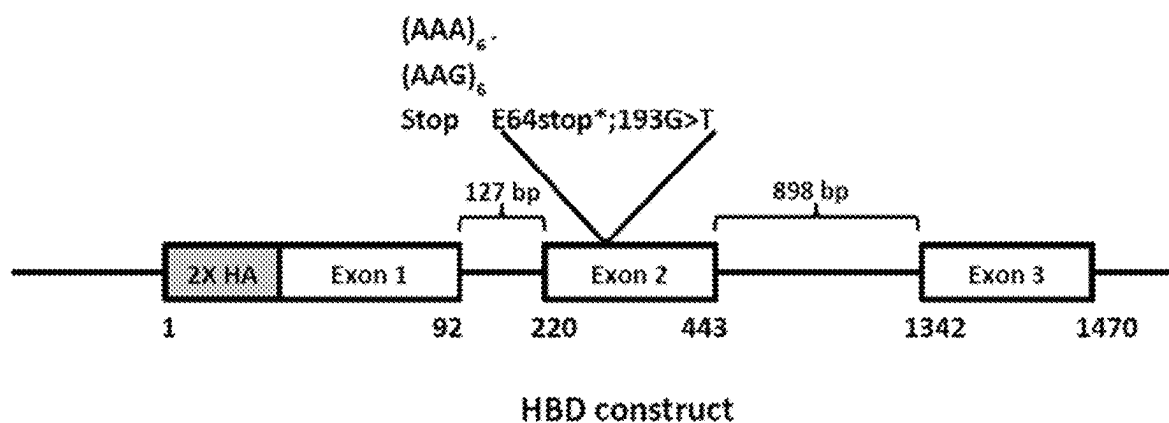
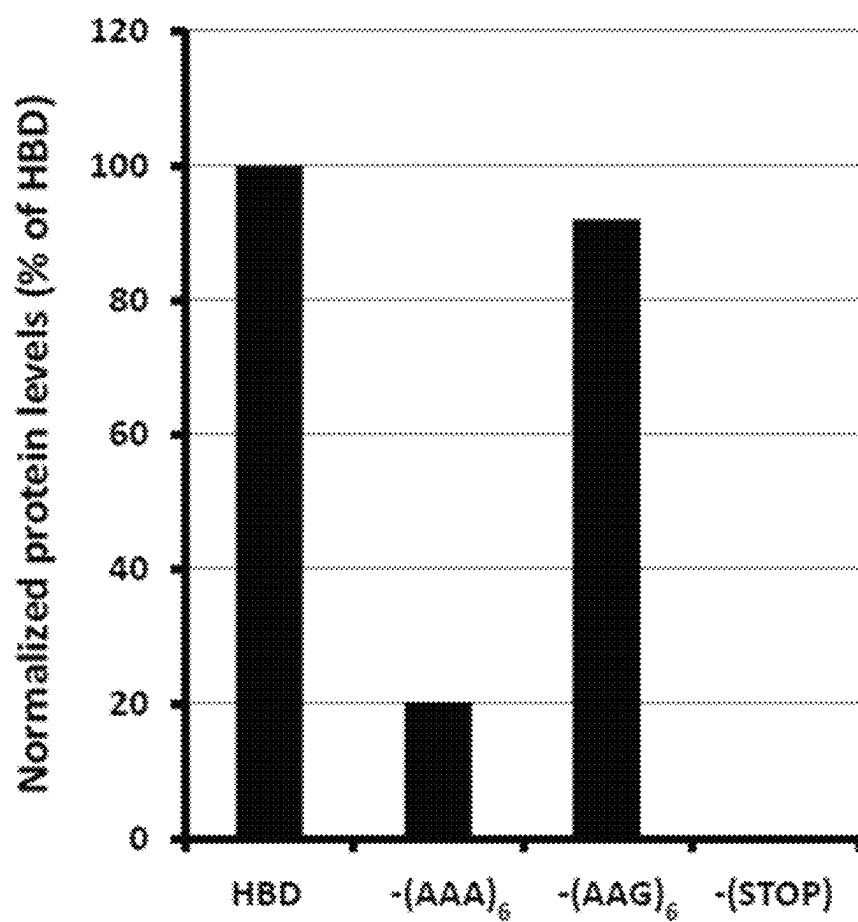
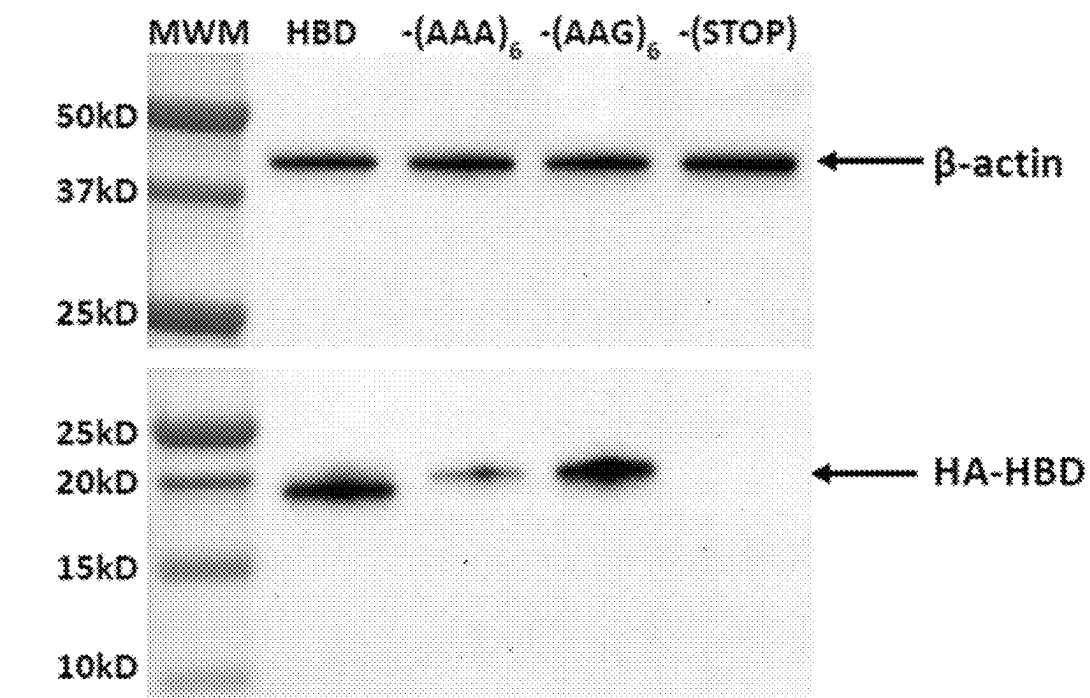
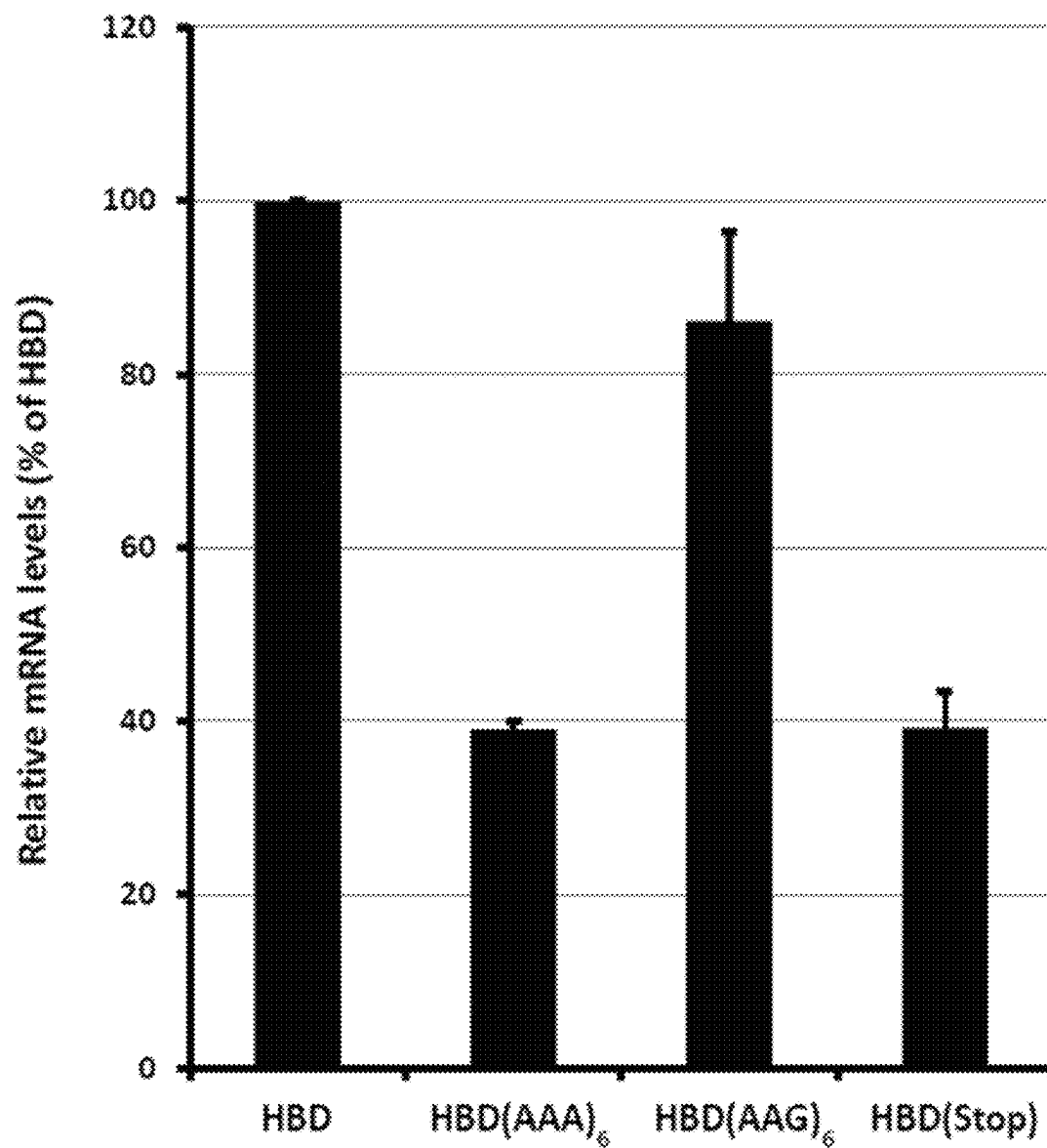


FIG. 7B

**FIG. 7C****FIG. 8A**

**FIG. 8B**

**FIG. 8C**

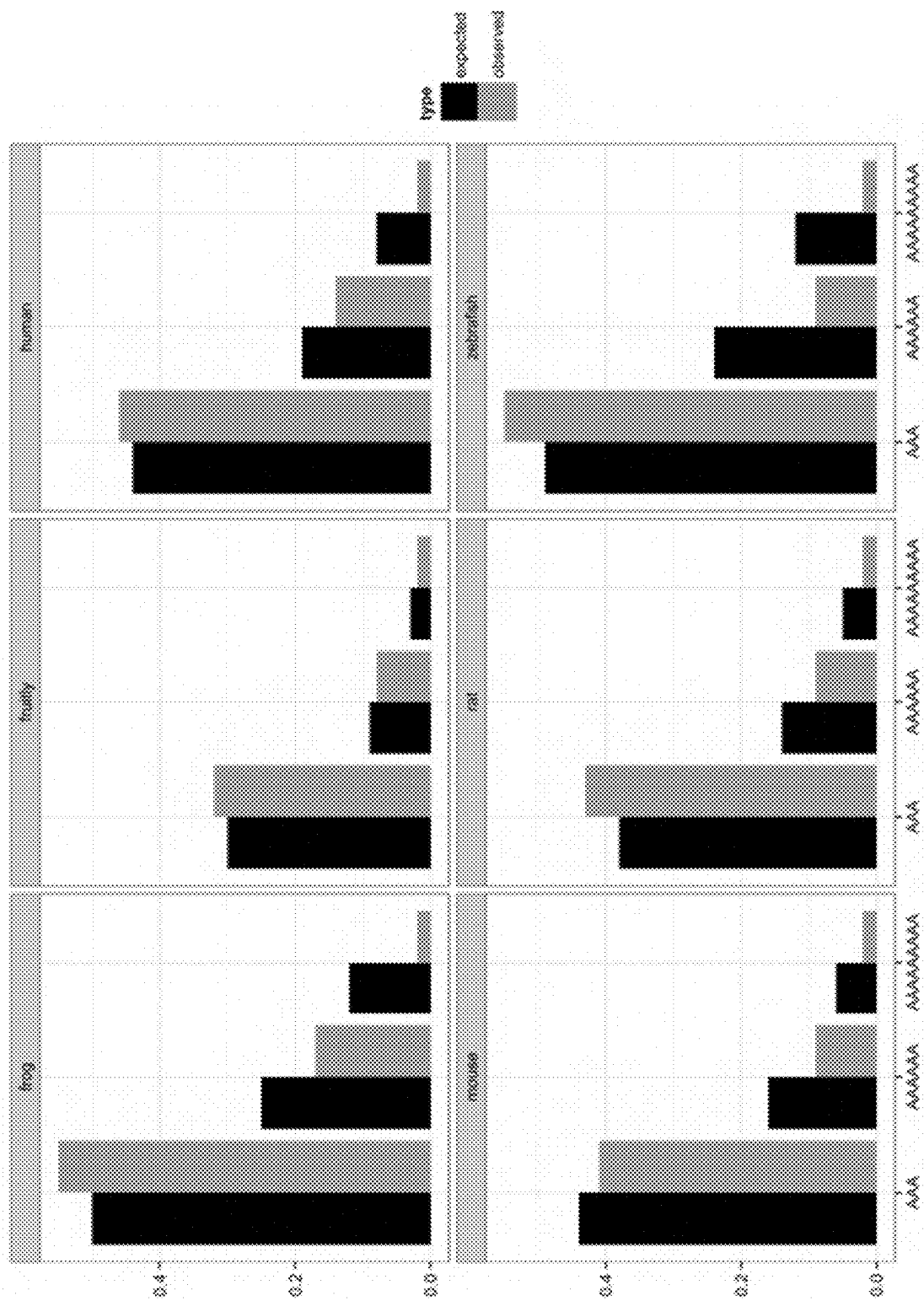
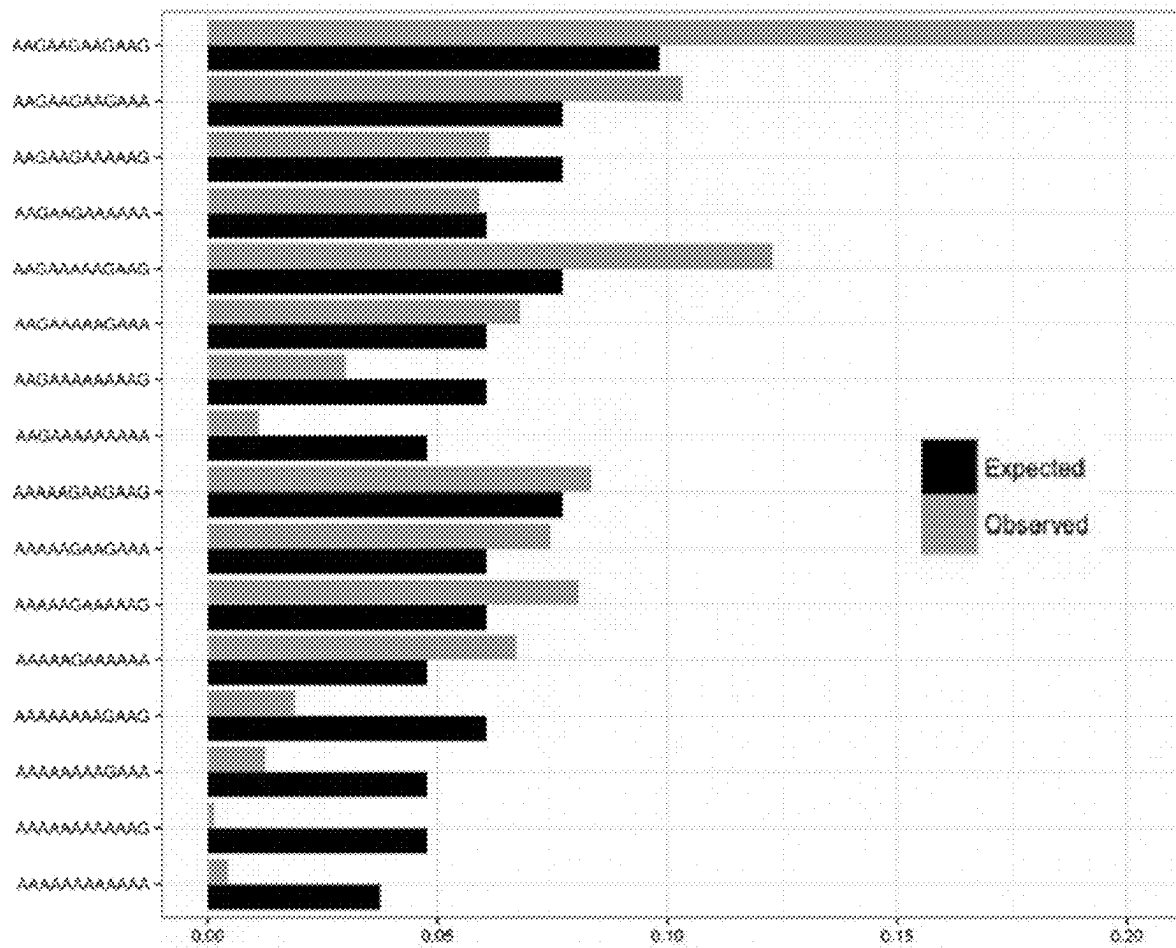


FIG. 9

**FIG. 10**

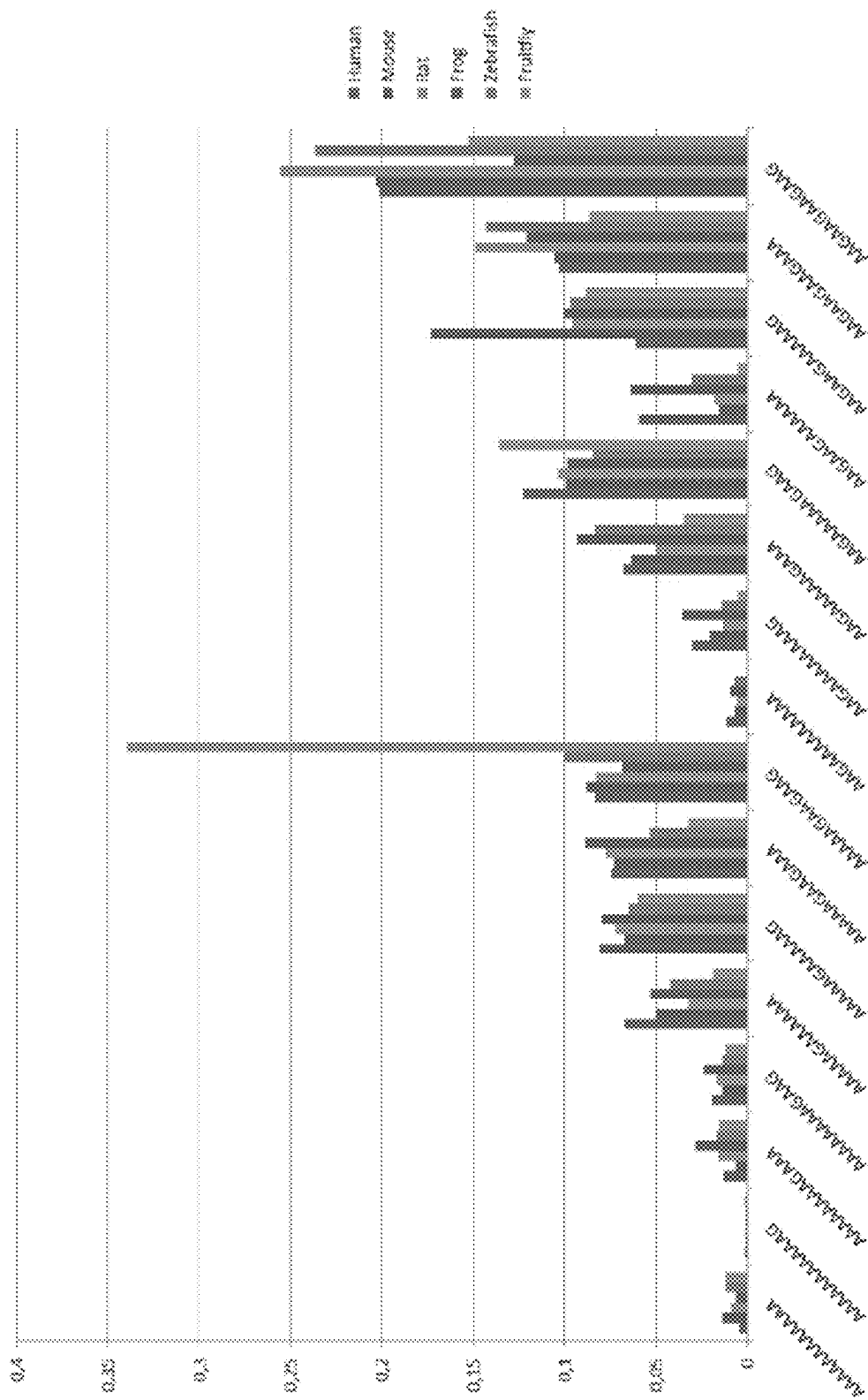


FIG. 11

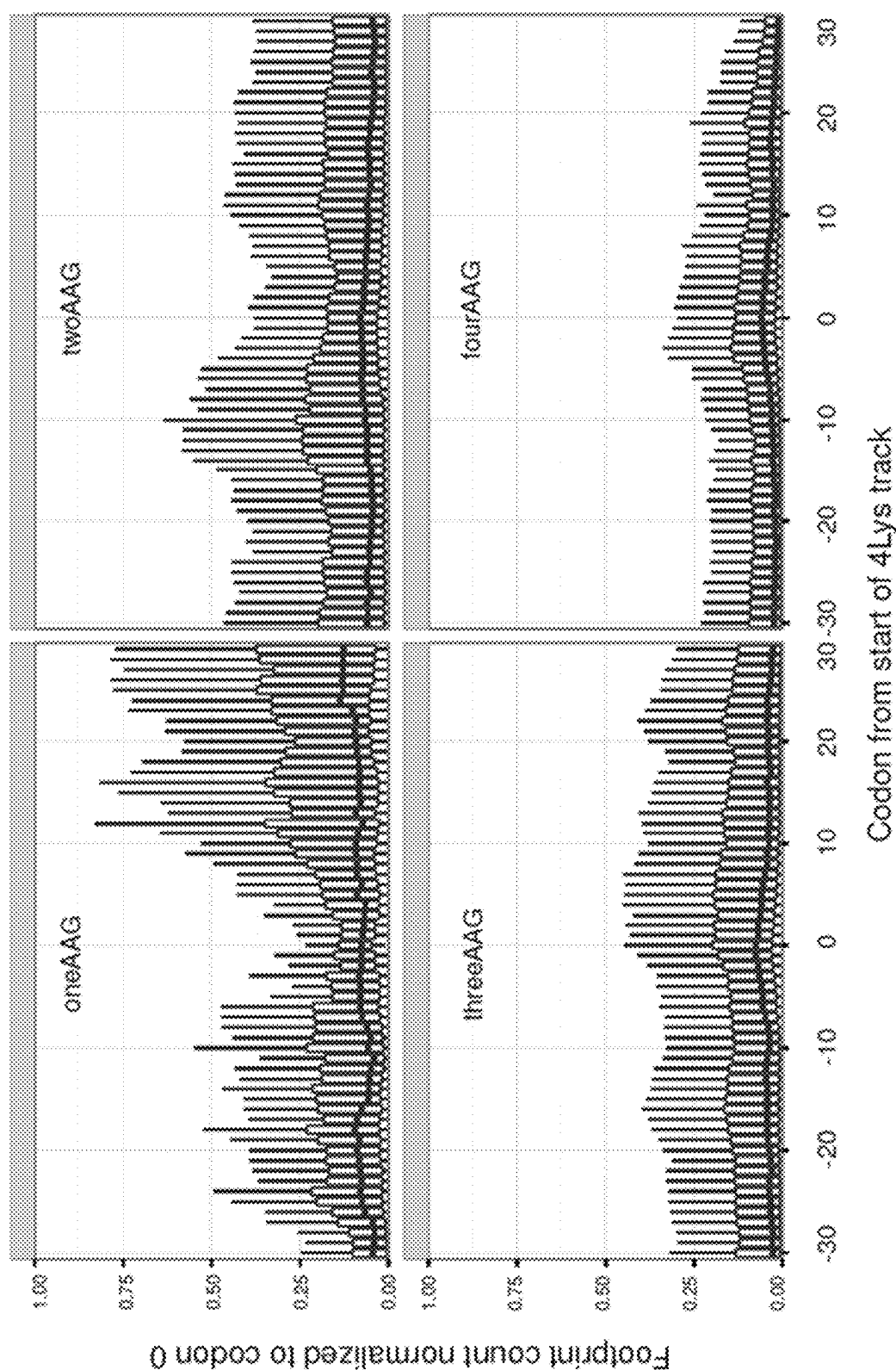
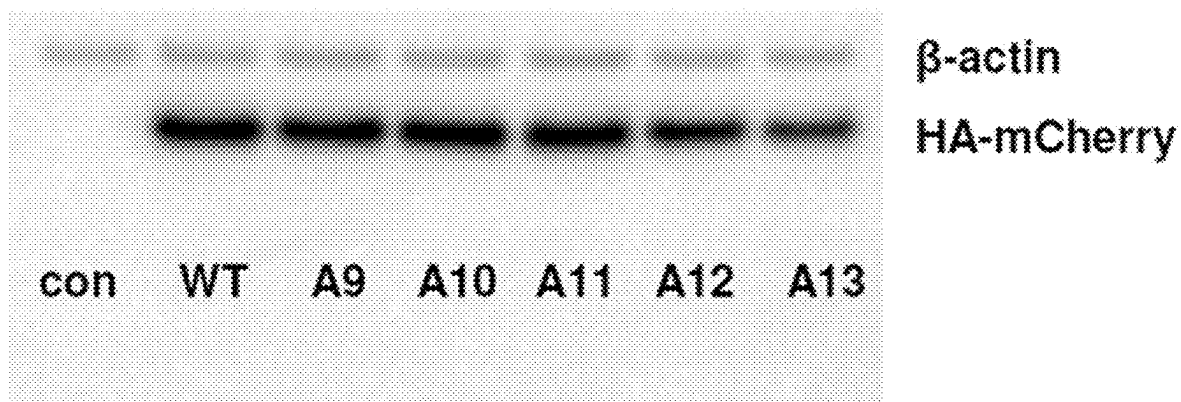
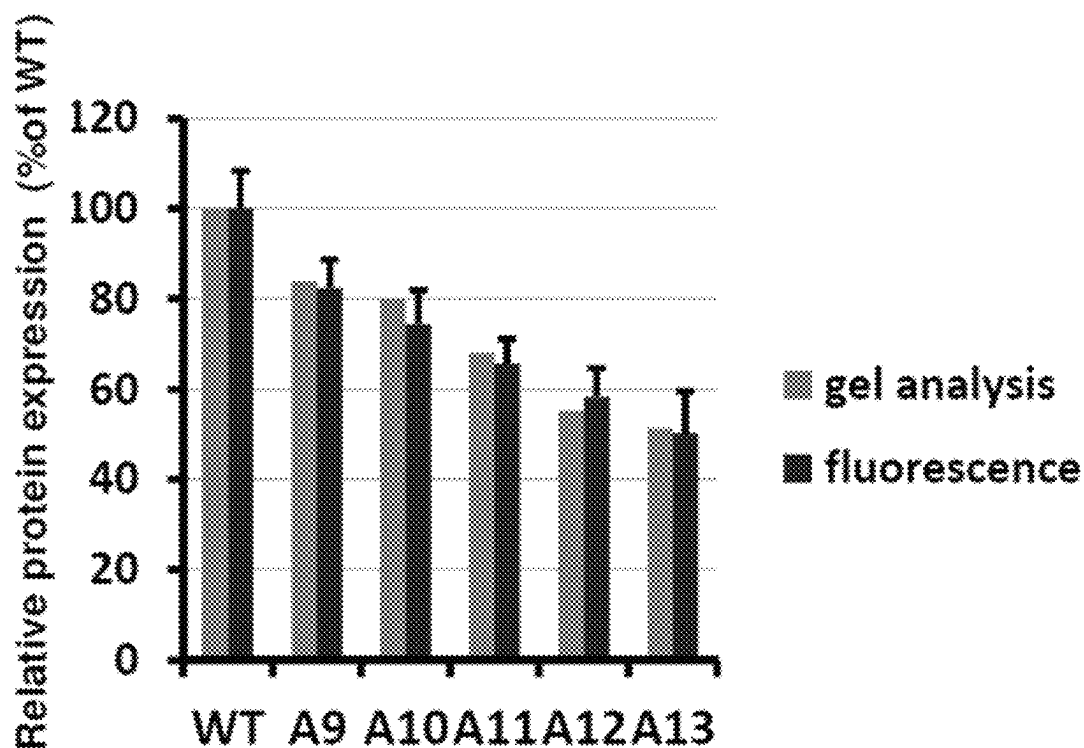
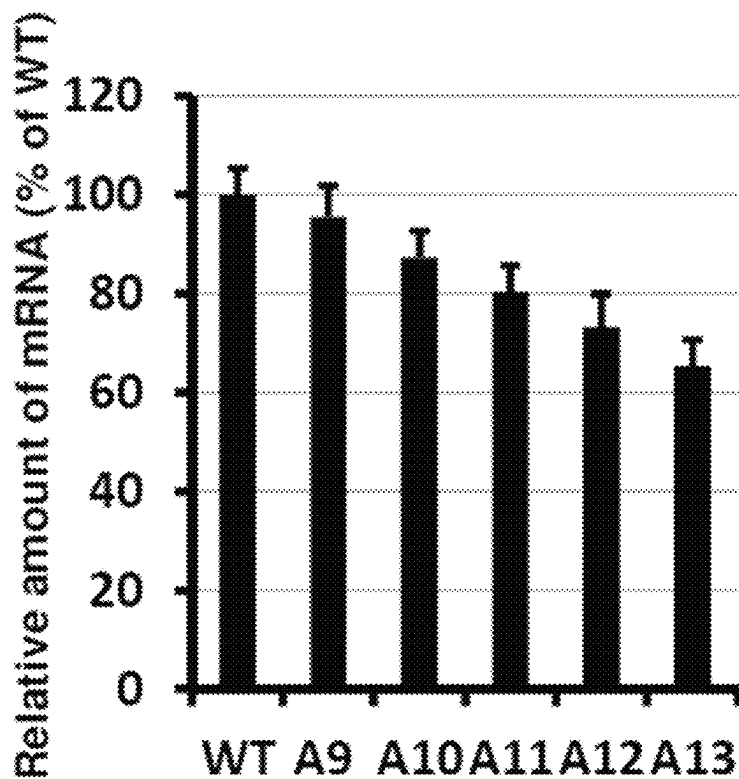


FIG. 12A

WT	GCAGCGGTGAGC A A V S
A9	GCAGCGAAAAAAAAAGTGAGC A A K K K V S
A10	GCAGCAAAAAAAAAAGTGAGC A A K K K V S
A11	GCAGCAAAAAAAAAACCGTG A A K K K T V
A12	GCAGCAAAAAAAAAACCGTG A A K K K N V
A13	GCAGAAAAAAAAACCGTG A E K K K N V

FIG. 12B**FIG. 12C**

**FIG. 12D****FIG. 12E**

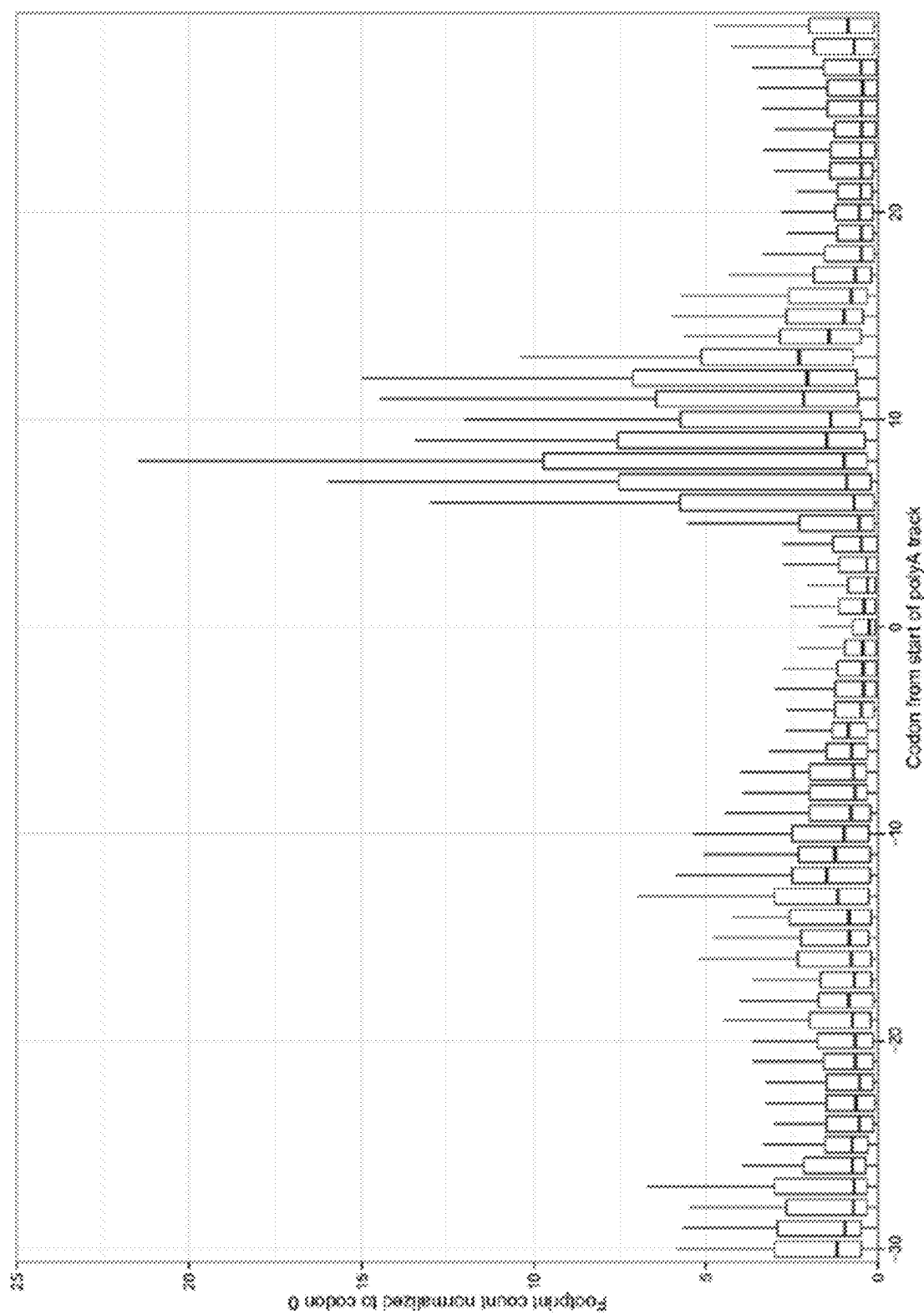


FIG. 13A

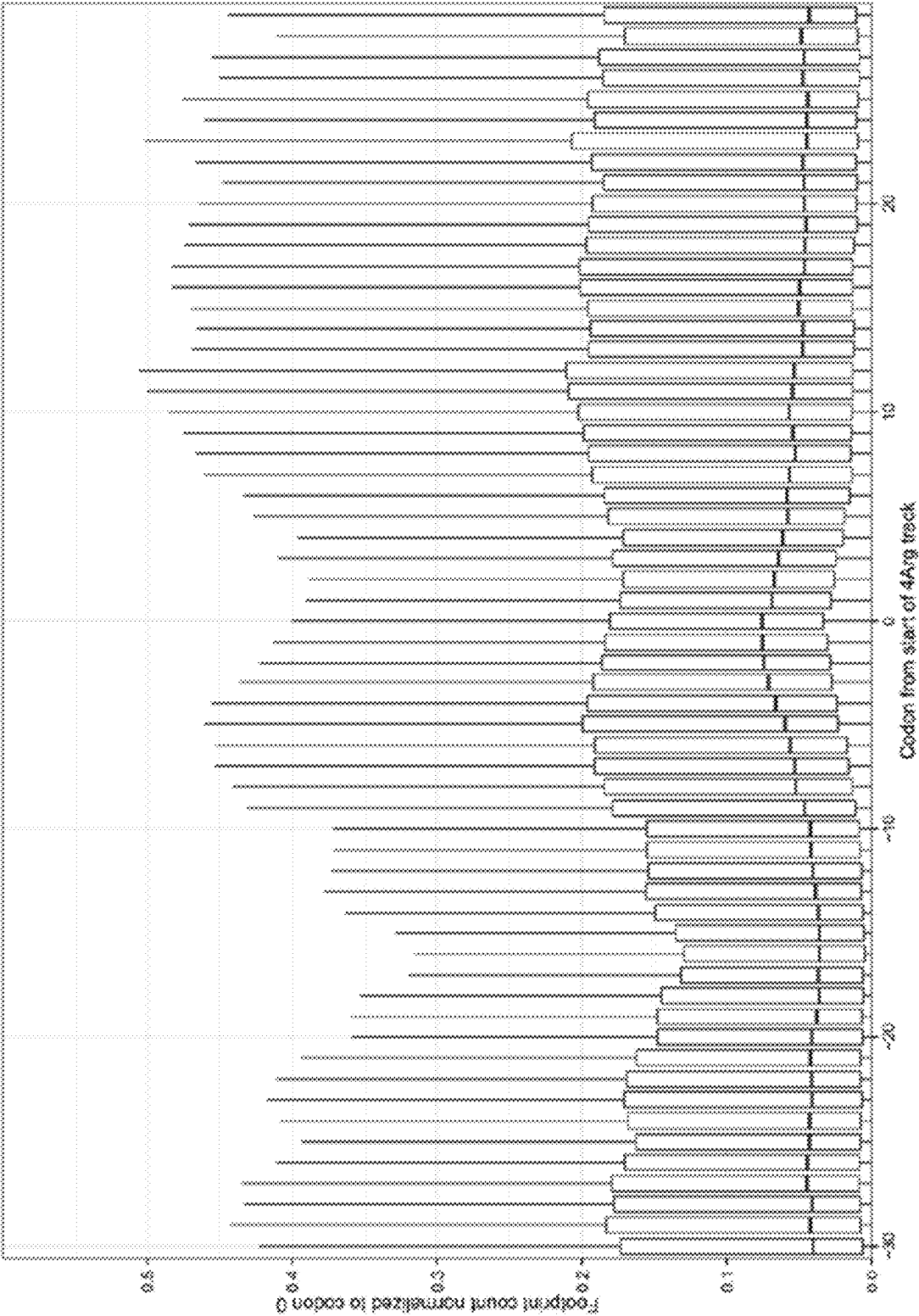
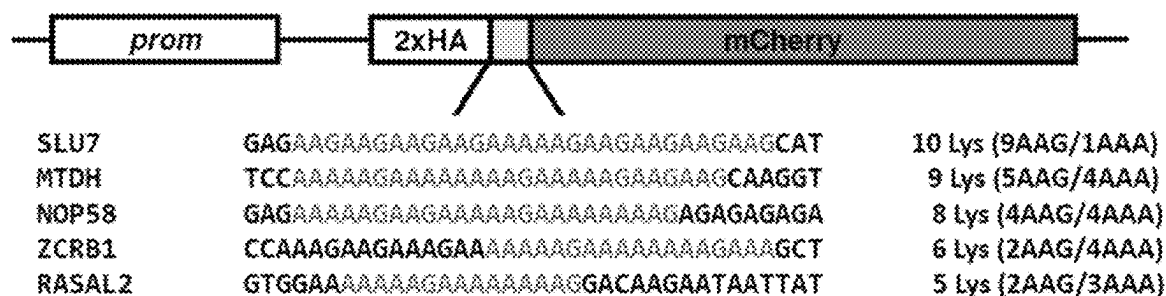
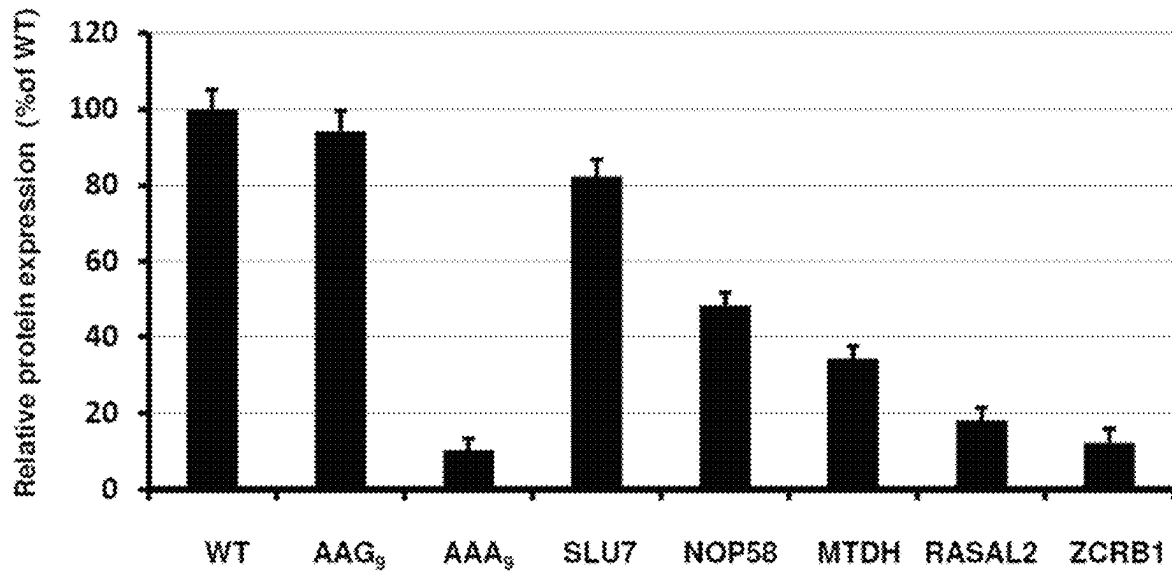
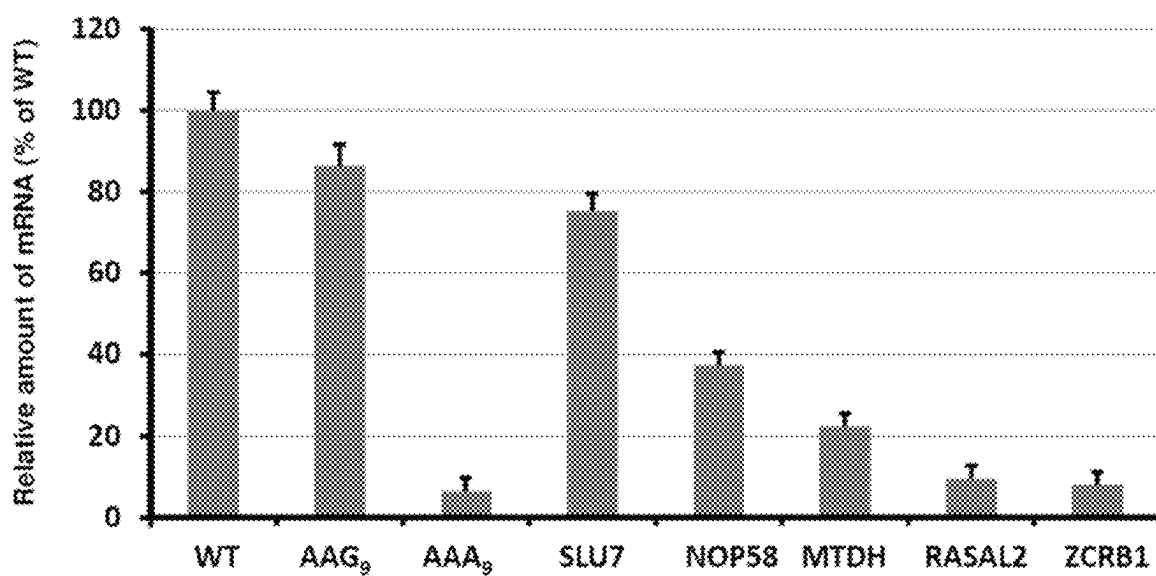
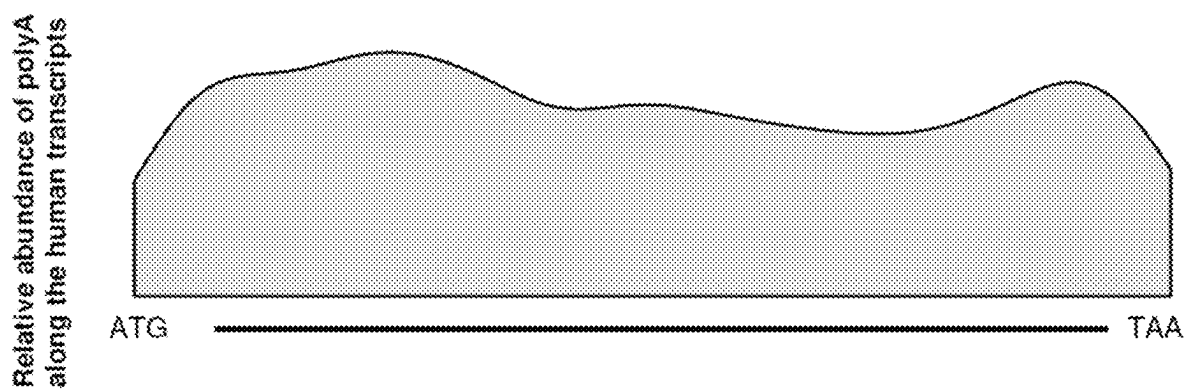


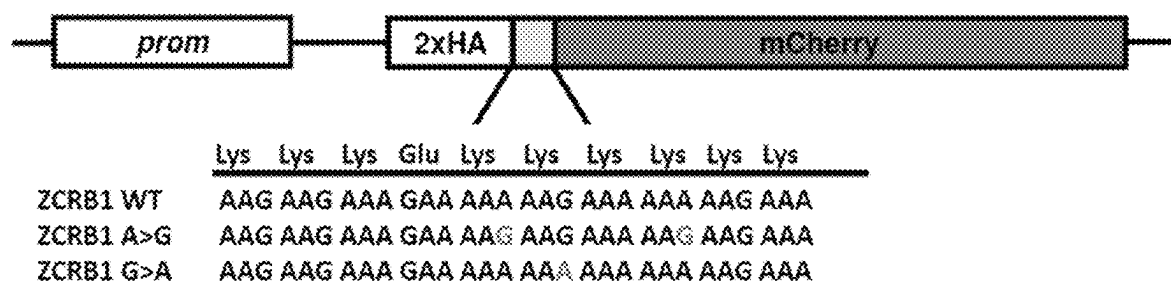
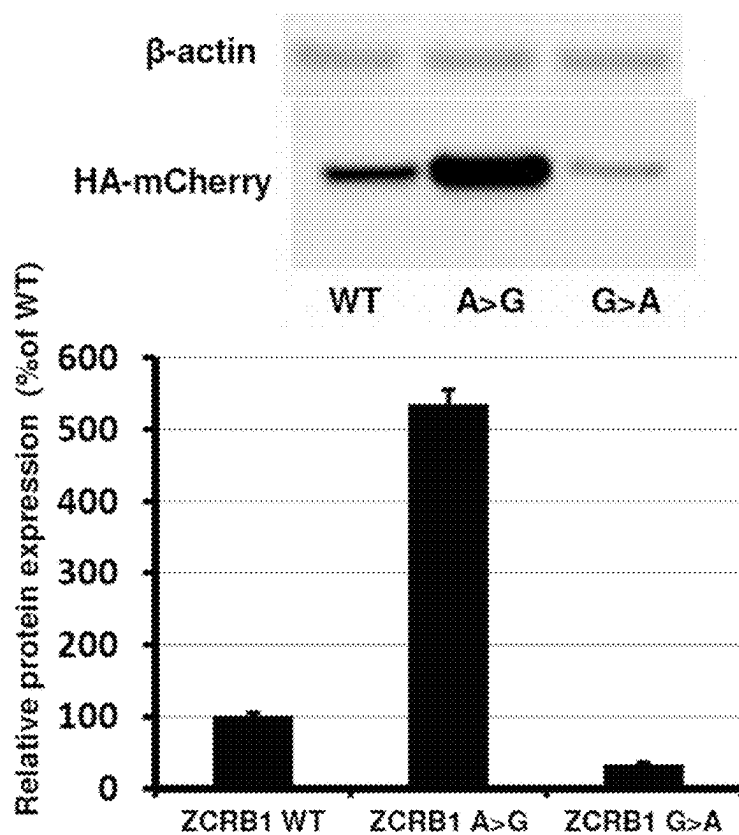
FIG. 13B

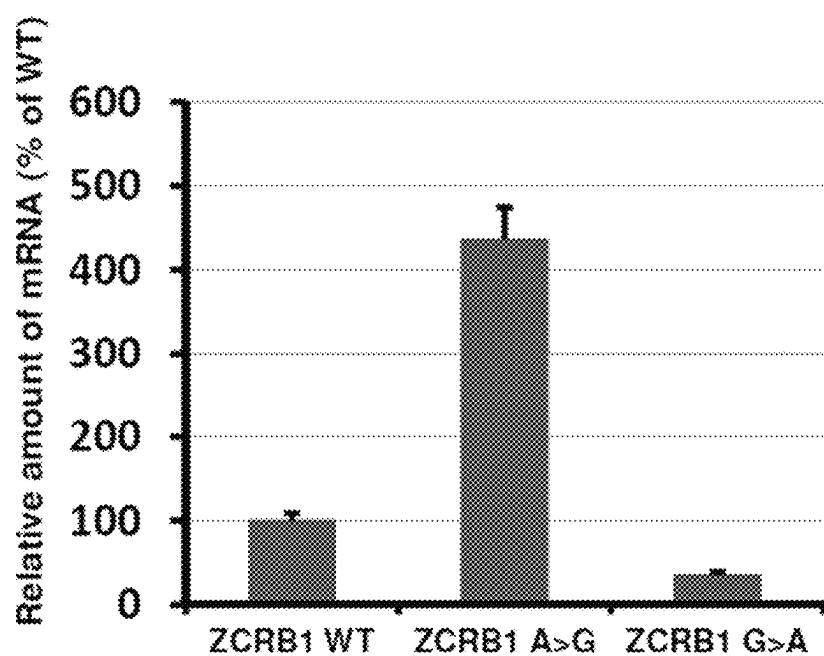
**FIG. 14A****FIG. 14B**

**FIG. 14C****FIG. 14D**

AA seq	...AspValGluLysLysLysLysLysAspLysAsnAsn...
Human	...gatgtggaaaaaaagaaaaaaaaaggacaagaataat...
Pig	...gacgtggaaaaaaagaaaaaaaaaggacaagaataat...
Mouse	...gatgtggaaaaaaagaaaaaaaaaggacaagaataat...
Hamster	...gatgtggaaaaaaagaaaaaaaaaggacaagaataat...
Chicken	...gatgtggaaaagaagaaaaaaaaaggacaaaaataat...
Zebrafish	...gatgtggaaaagaagaaaaaaaaaggacaaaaacaac...
Frog	...gatgttgaaaagaagaaaaaaaaagataaaaaacaac...

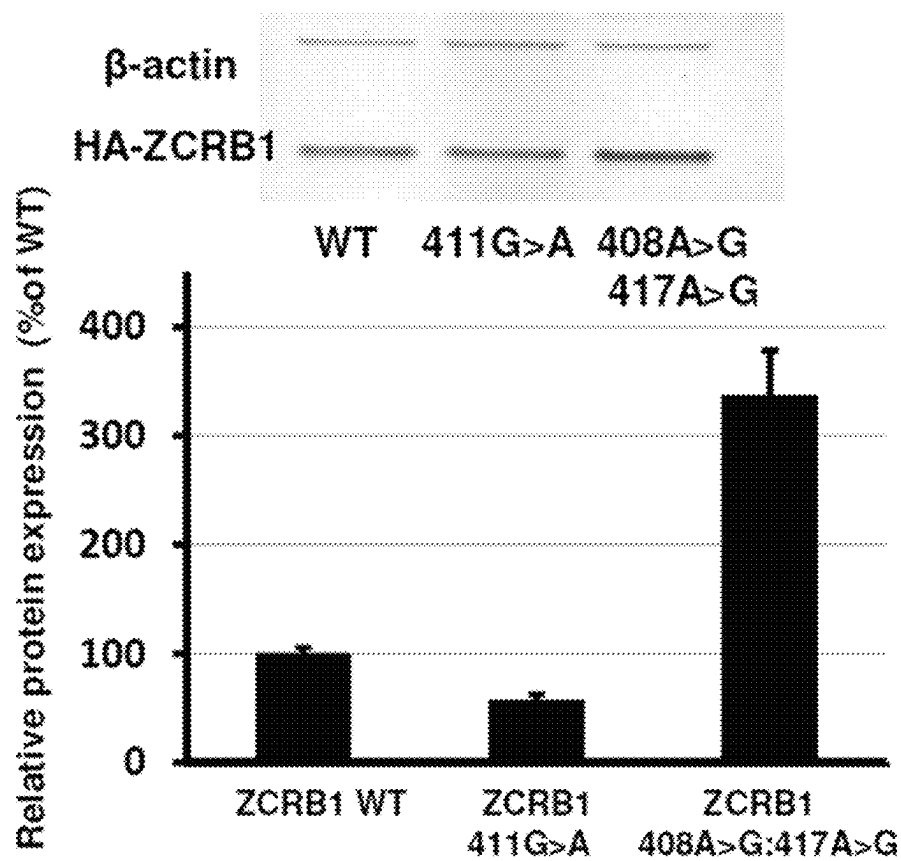
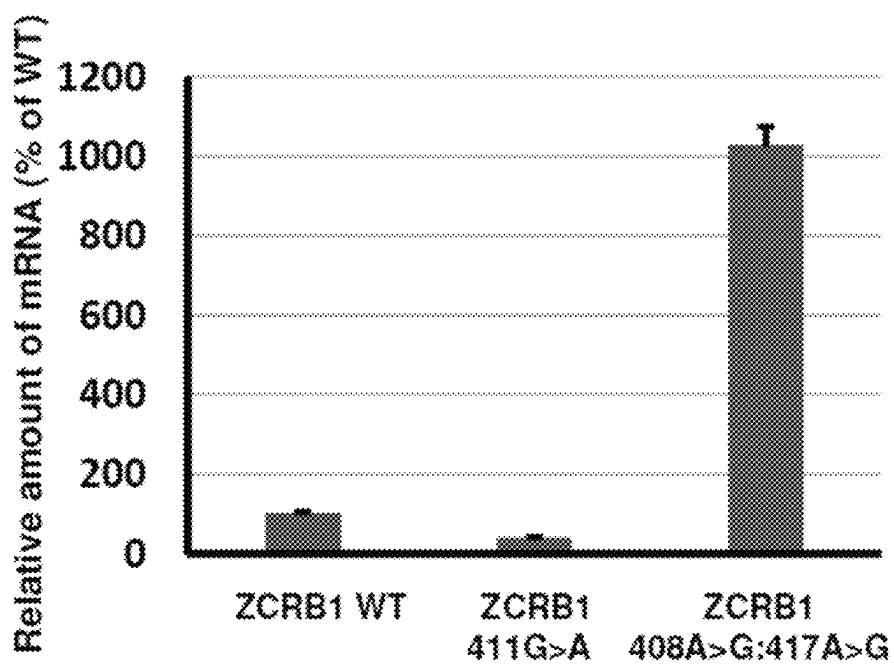
FIG. 15

**FIG. 16A****FIG. 16B**

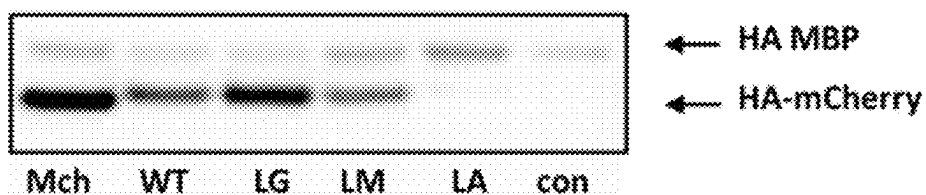
**FIG. 16C**

	Pro	Lys	Lys	Lys	Glu	Lys	Lys	Lys	Lys	Lys	Lys	Ala
ZCRB1 WT	CCA	AAG	AAG	AAA	GAA	AAA	AAG	AAA	AAA	AAG	AAA	GCT
ZCRB1 411G>A	CCA	AAG	AAG	AAA	GAA	AAA	AAA	AAA	AAA	AAG	AAA	GCT
ZCRB1 408A>G;417A>G	CCA	AAG	AAG	AAA	GAA	AAG	AAG	AAA	AAG	AAG	AAA	GCT

FIG. 16D

**FIG. 16E****FIG. 16F**

	Ser	Lys	Lys	Lys	Lys	Lys	Lys	Lys	Lys	Lys	Gln	
Lyr (WT)	TCC	AAA	AAG	AAA	AAA	AAG	AAA	AAG	AAG	AAG	CAA-	(8As/9K)
Lyr (LG)	TCC	AAG	AAG	AAG	AAG	AAG	AAG	AAG	AAG	AAG	CAA-	(2As/9K)
Lyr (LM)	TCC	AAA	AAA	AAA	AAA	AAG	AAA	AAG	AAG	AAG	CAA-	(14As/9K)
Lyr (LA)	TCC	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	AAA	CAA-	(27As/9K)

FIG. 17A**FIG. 17B**

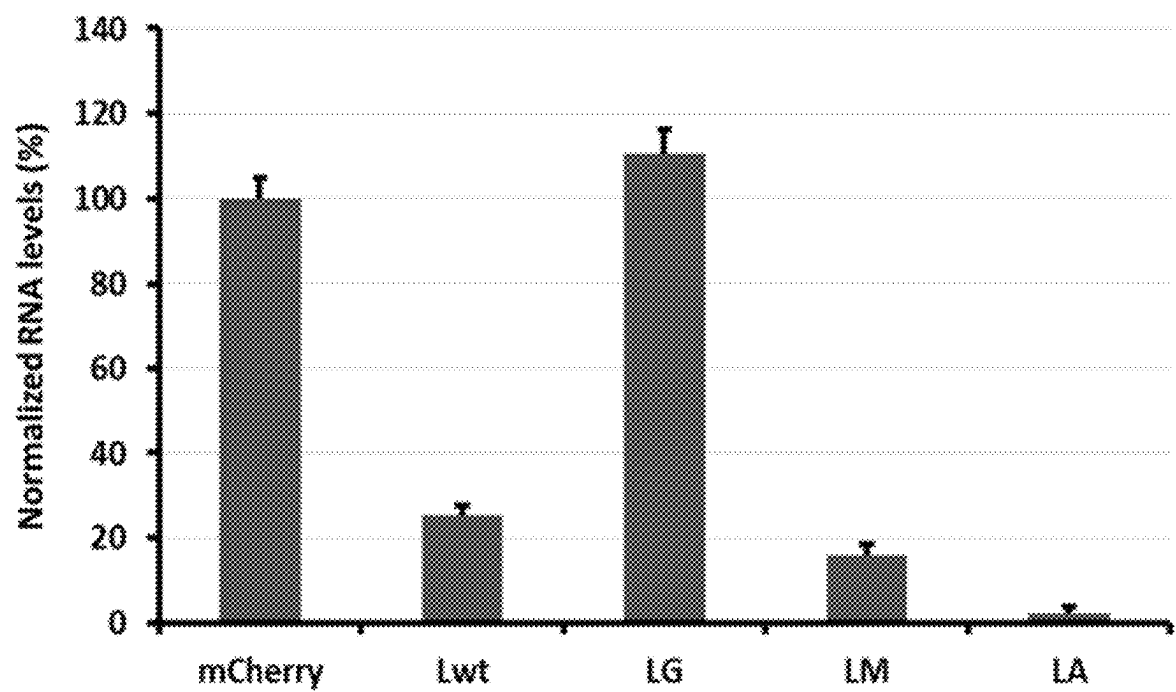
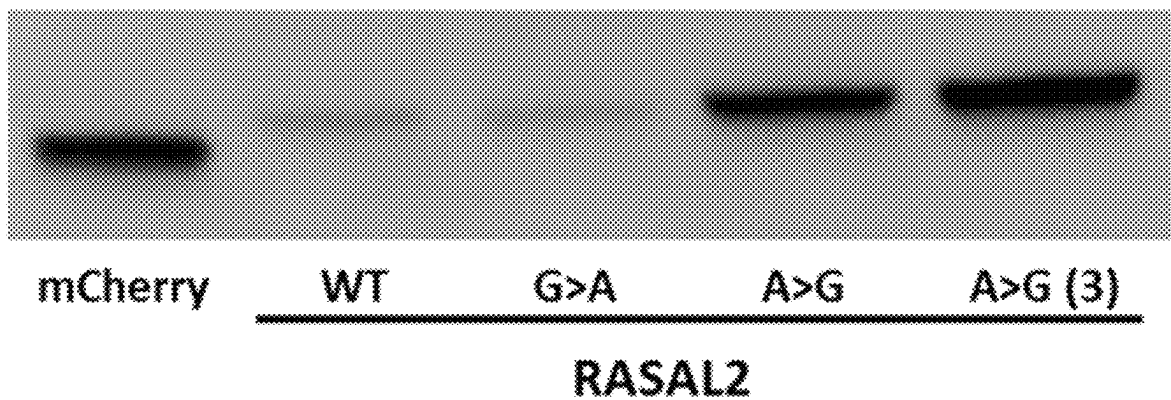
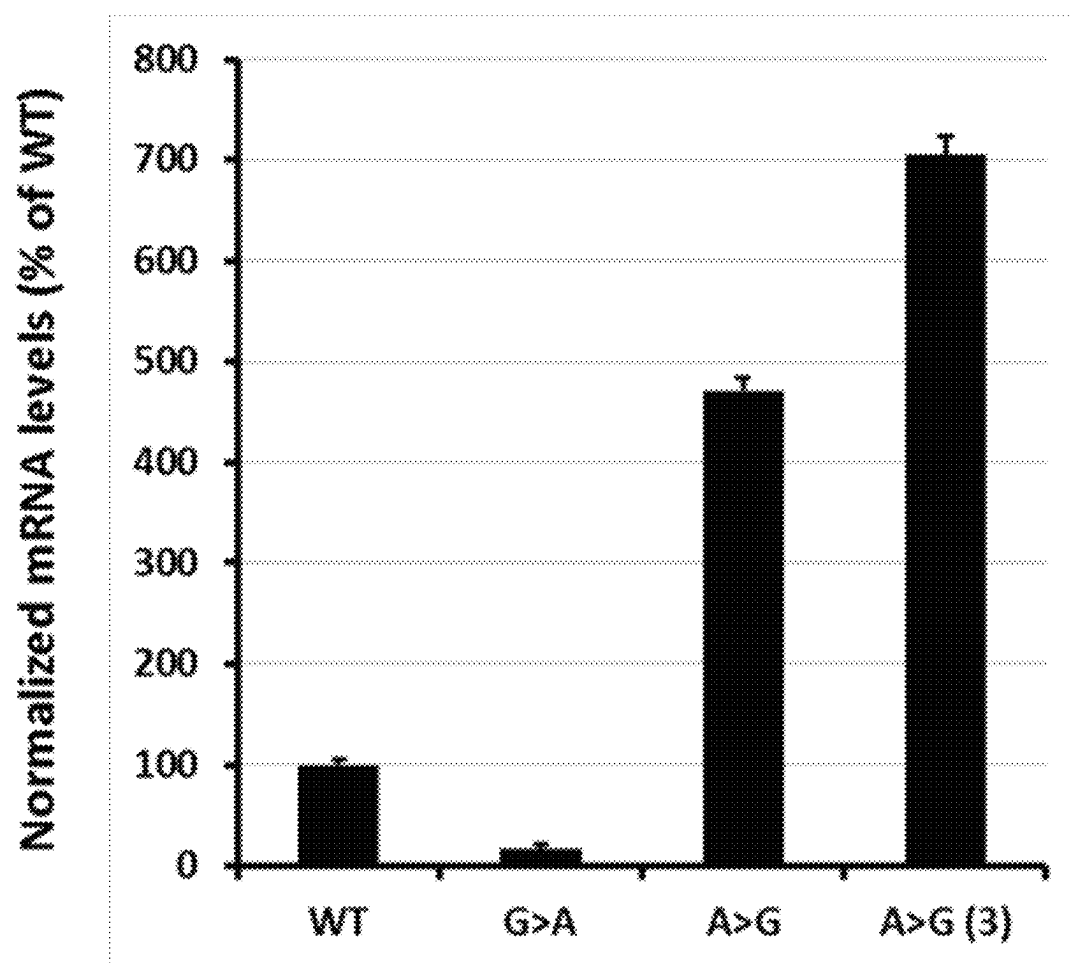


FIG. 17C

	<u>Glu</u>	<u>Lys</u>	<u>Lys</u>	<u>Lys</u>	<u>Lys</u>	<u>Lys</u>	<u>Asp</u>	<u>Lys</u>	<u>Asn</u>	<u>Asn</u>
RASAL2 WT	GAA	AAA	AAG	AAA	AAA	AAG	GAC	AAG	AAT	AAT
RASAL2 G>A	GAA	AAA	AAA	AAA	AAA	AAG	GAC	AAG	AAT	AAT
RASAL2 A>G	GAA	AAA	AAG	AAA	AAG	AAG	GAC	AAG	AAT	AAT
RASAL2 A>G (3)	GAA	AAG	AAG	AAG	AAG	AAG	GAC	AAG	AAT	AAT

FIG. 18A

**FIG. 18B****FIG. 18C**

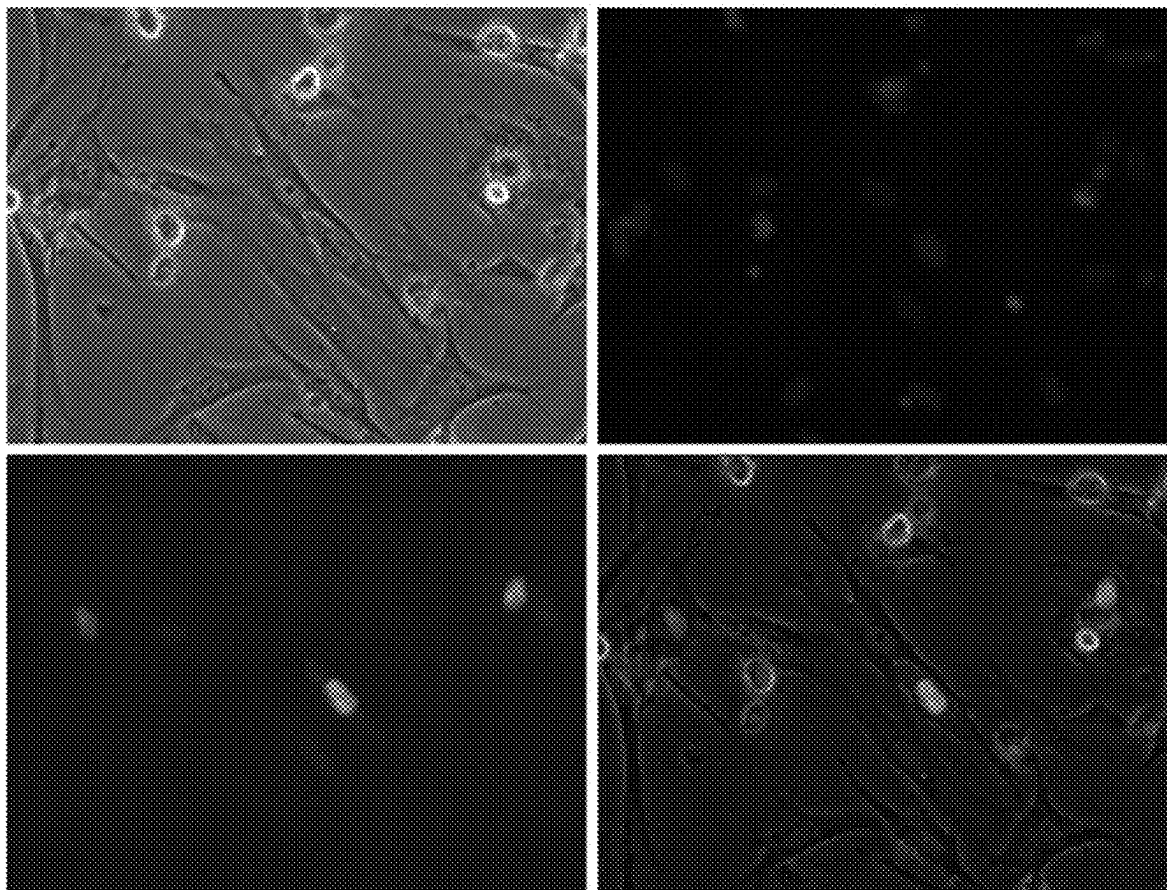


FIG. 19A

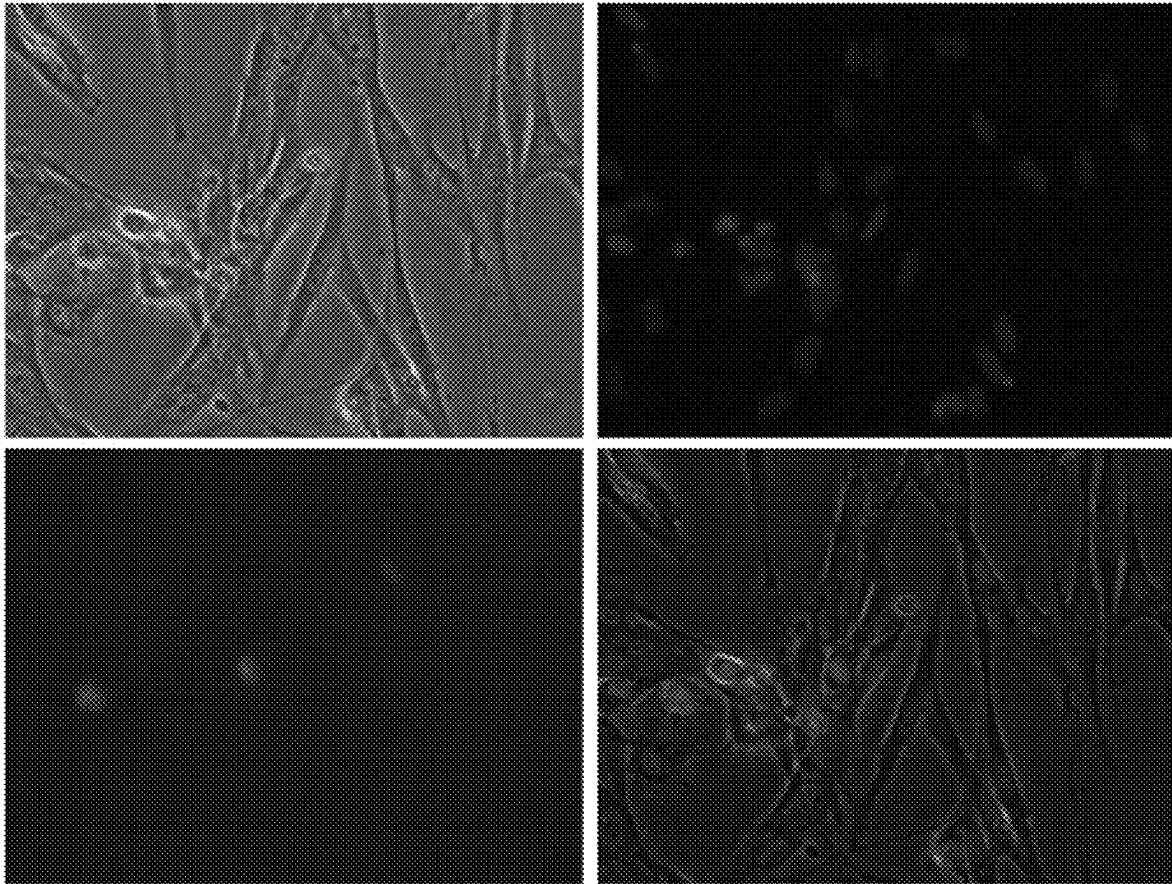


FIG. 19B

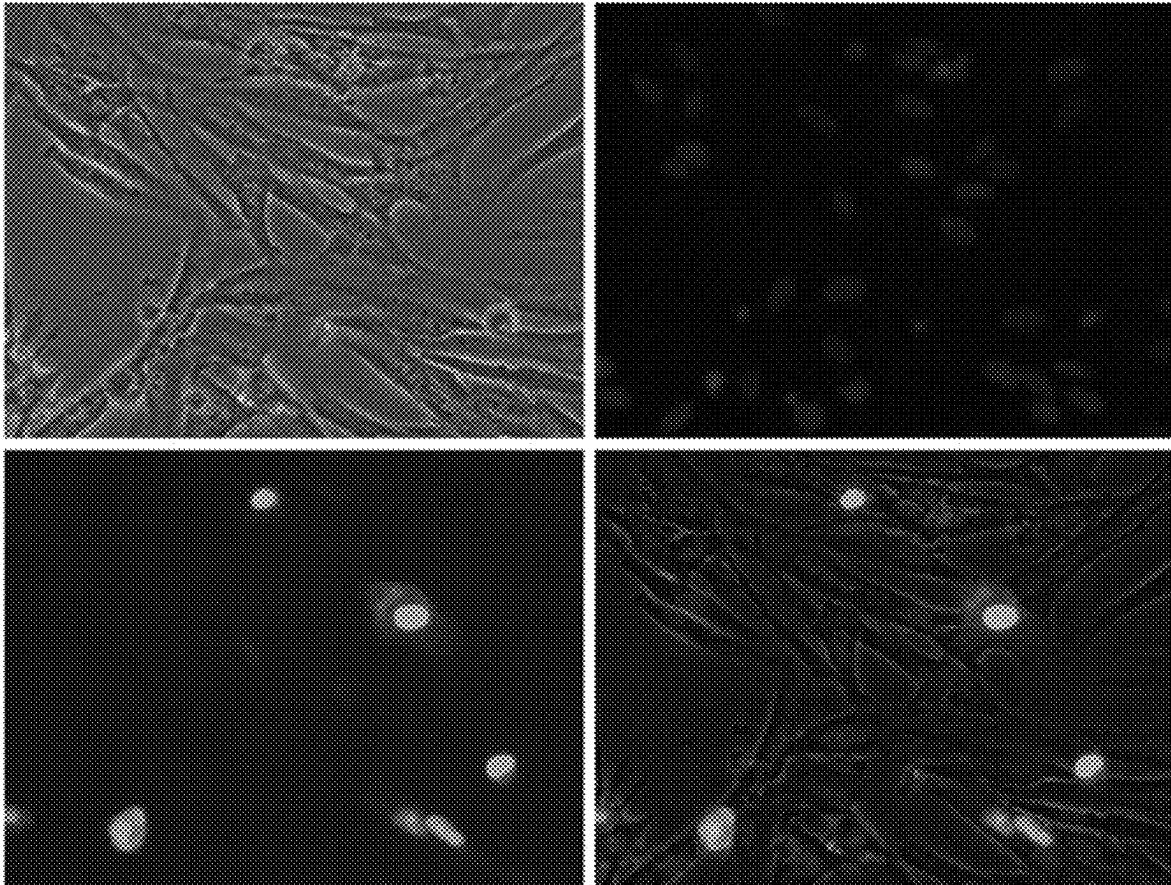
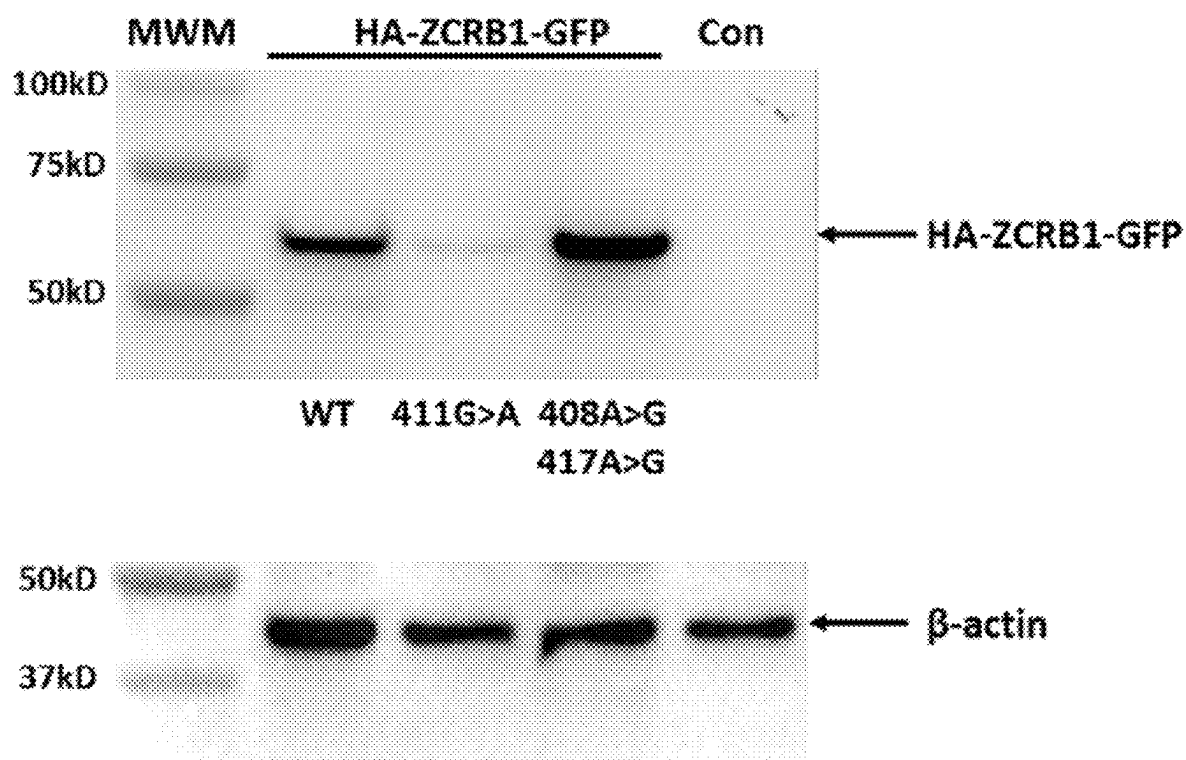
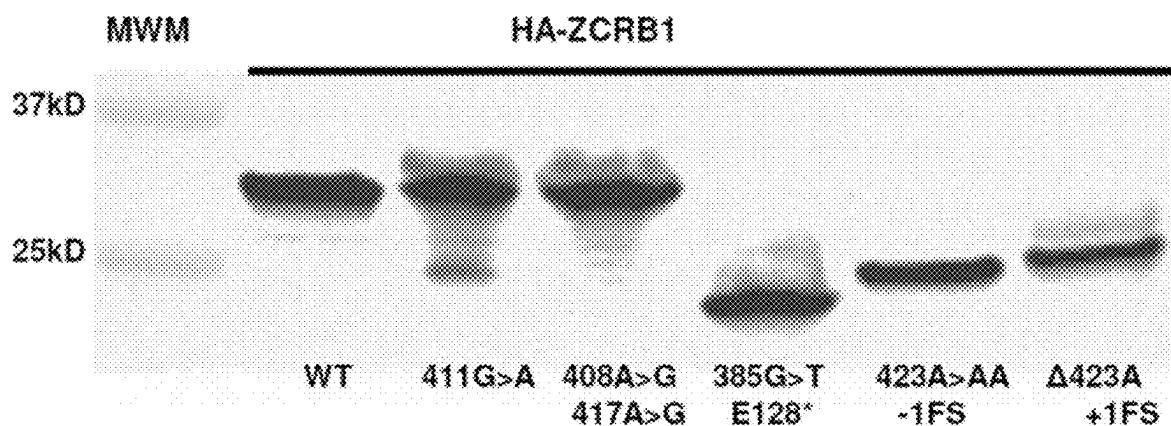


FIG. 19C

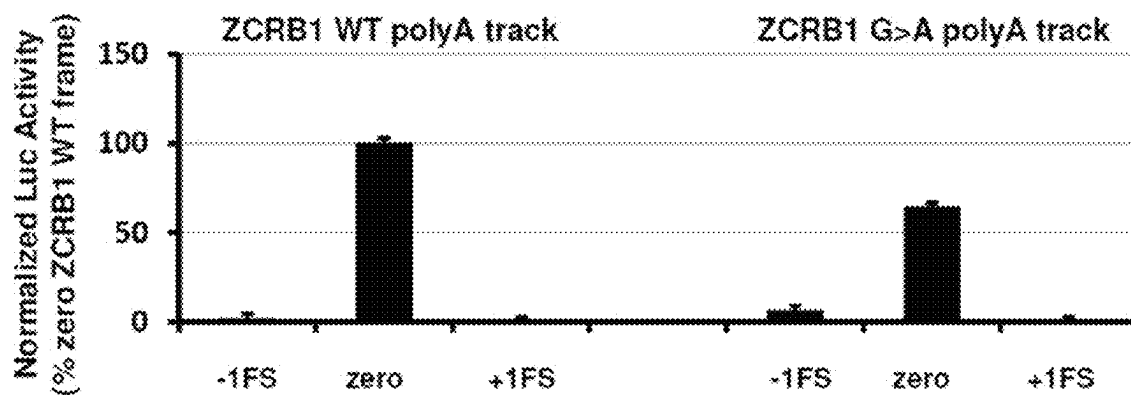
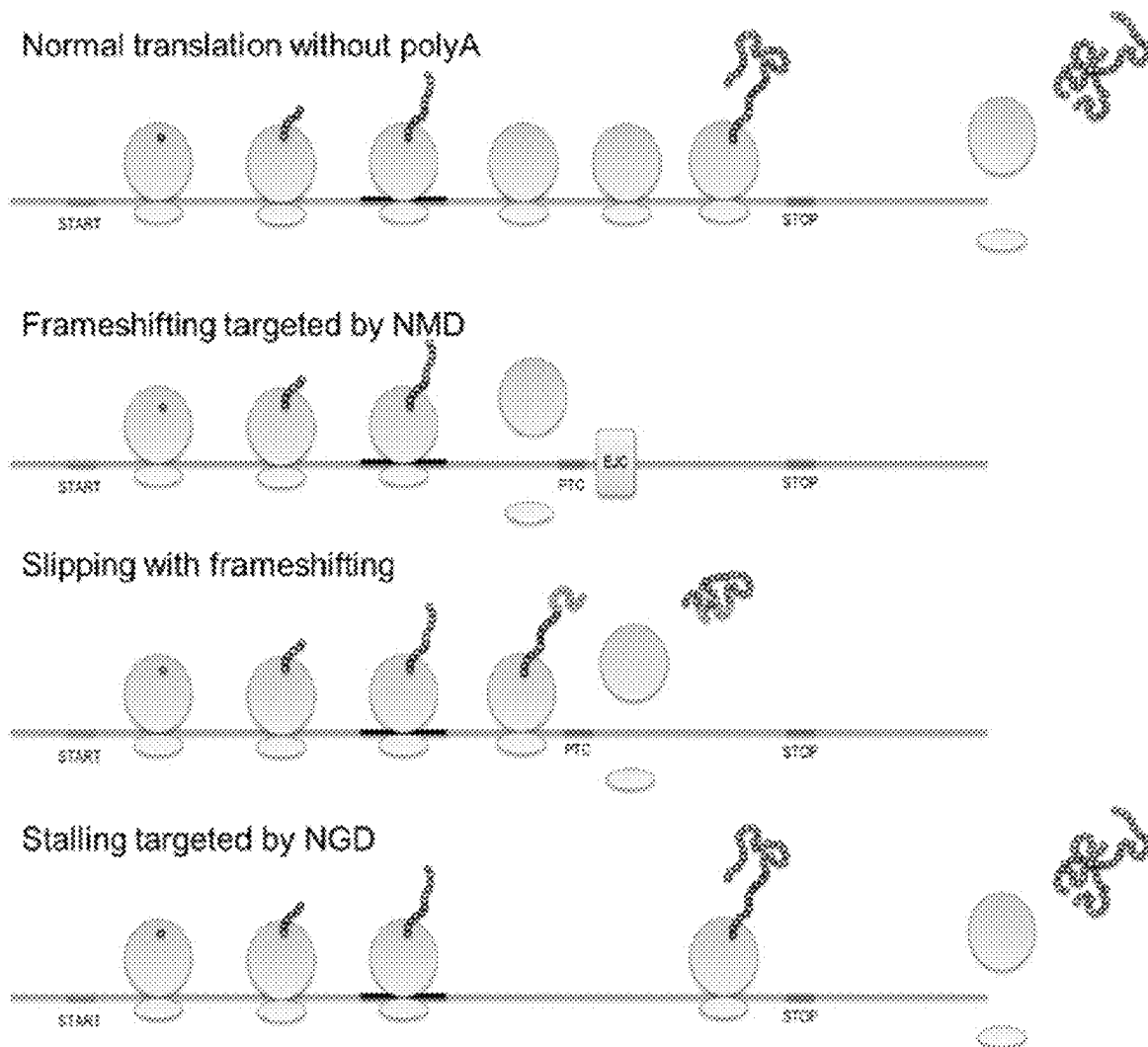
**FIG. 19D**

**FIG. 20A**

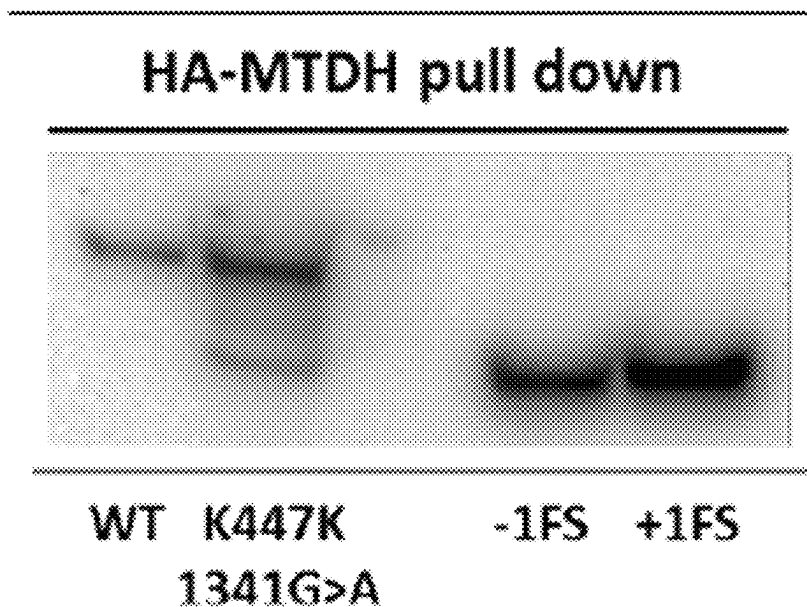
LUC con

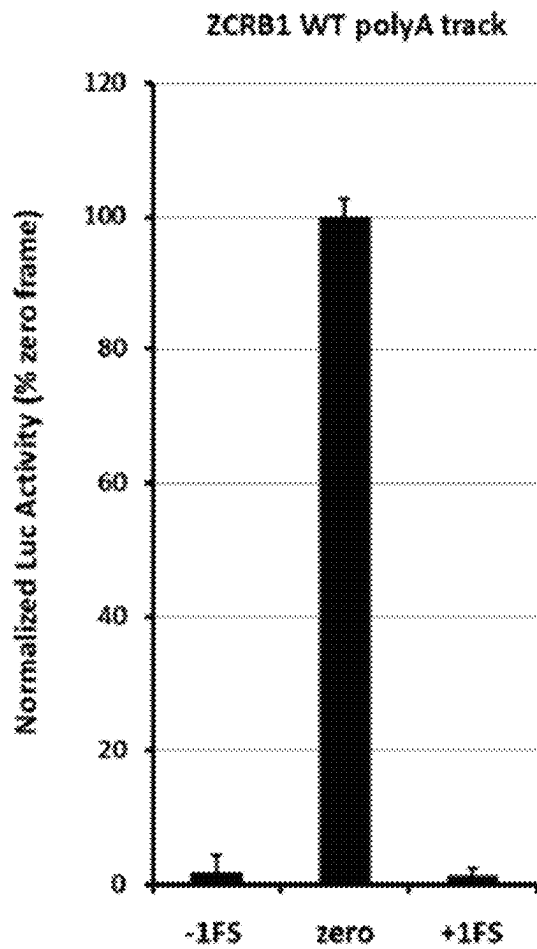
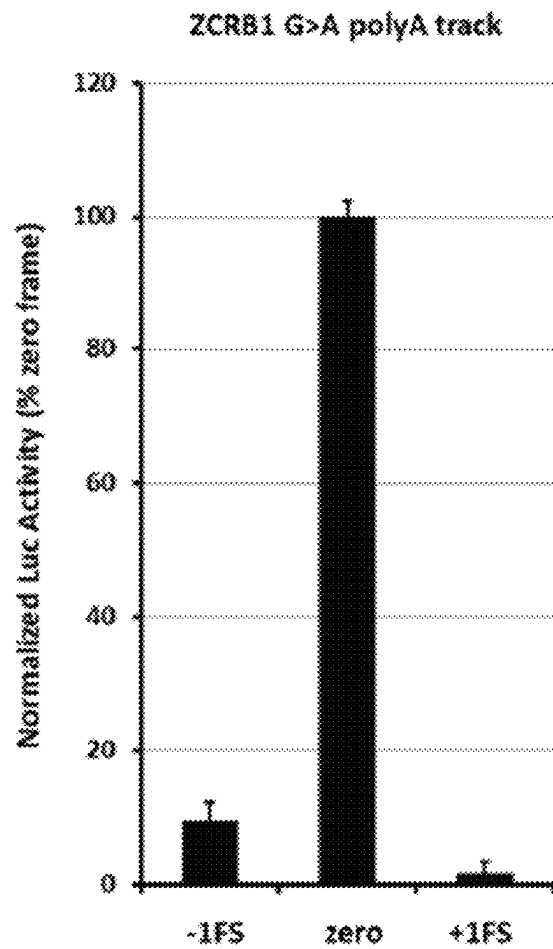
ZCRB WT - LUC	ATG-2xHA-SEQ ID NO: 84-LUC
ZCRB -1 - LUC	ATG-2xHA-SEQ ID NO: 85-LUC
ZCRB +1 - LUC	ATG-2xHA-SEQ ID NO: 86-LUC
ZCRB 417G>A - LUC	ATG-2xHA-SEQ ID NO: 87-LUC
ZCRB 417G>A-1 - LUC	ATG-2xHA-SEQ ID NO: 88-LUC
ZCRB 417G>A+1 - LUC	ATG-2xHA-SEQ ID NO: 89-LUC

FIG. 20B

**FIG. 20C****FIG. 20D**

	442	443	444	445	446	447	448	449	450	451	452
	...	Ser	Lys	Lys	Lys	Lys	Lys	Lys	Lys	Lys	Gln...
MTDH WT	...	TCC	AAA	AAG	AAA	AAA	AAG	AAA	AAG	AAG	CAA...
MTDH K447K	...	TCC	AAA	AAG	AAA	AAA	AAA	AAG	AAG	AAG	CAA...
											1341G>A

FIG. 21A**FIG. 21B**

**FIG. 22A****FIG. 22B**

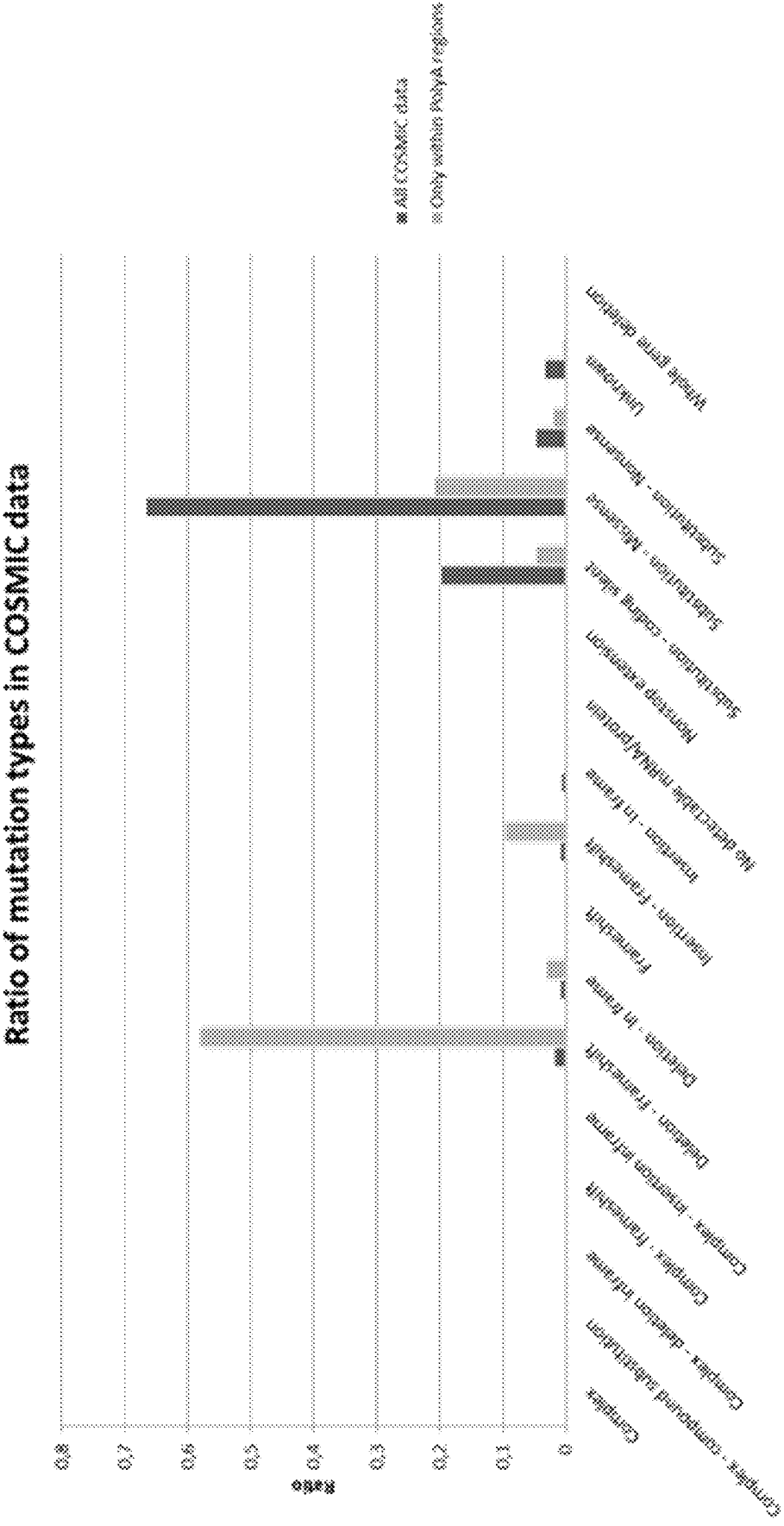


FIG. 23

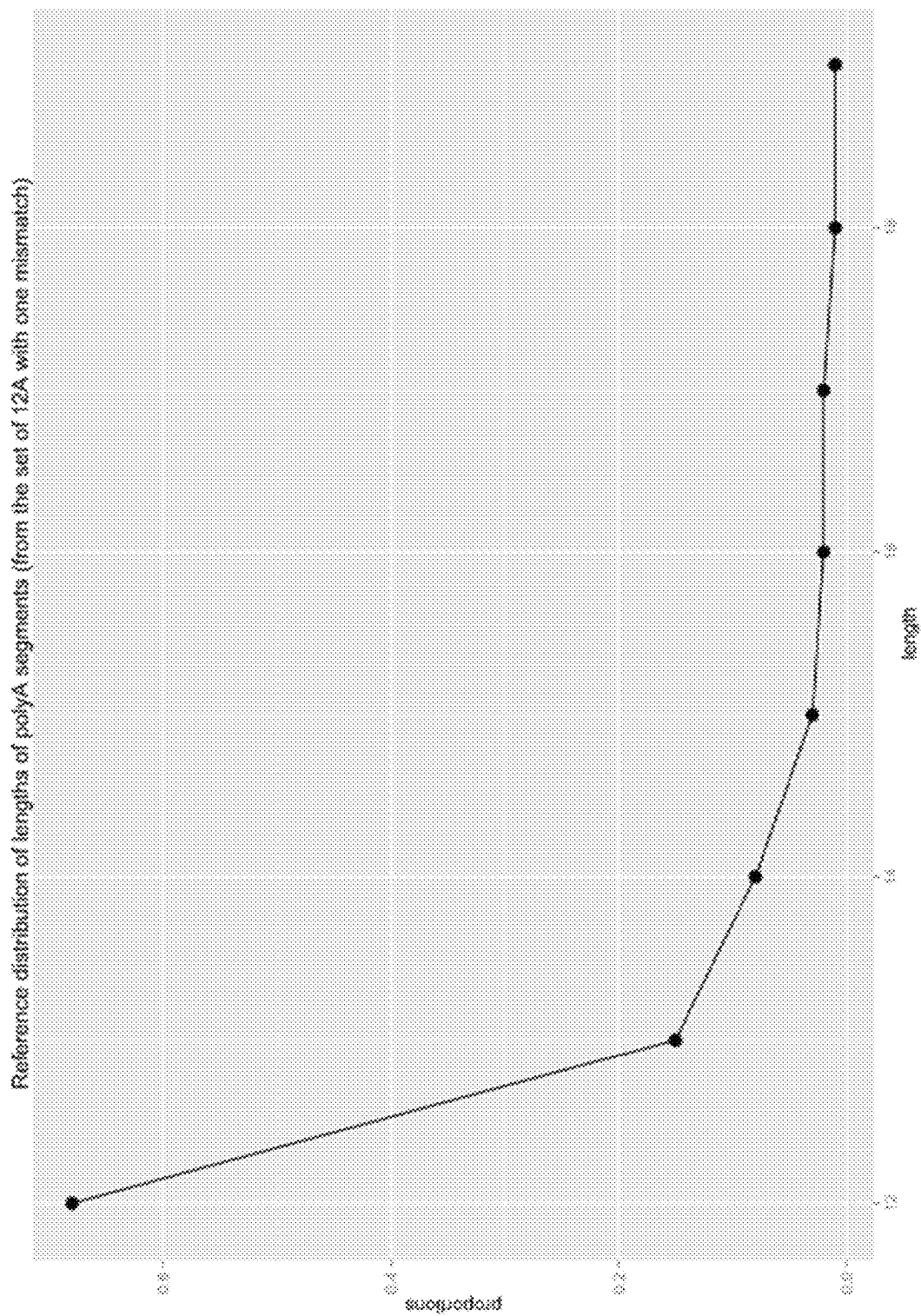
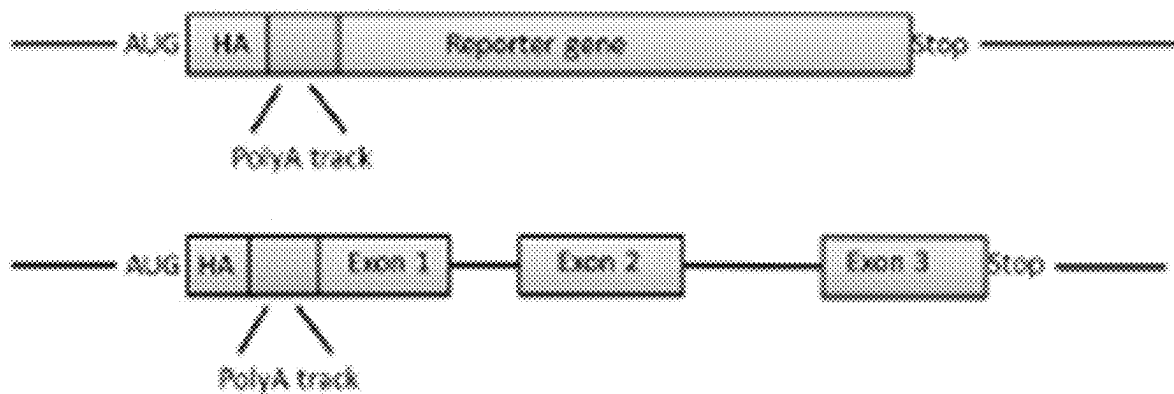
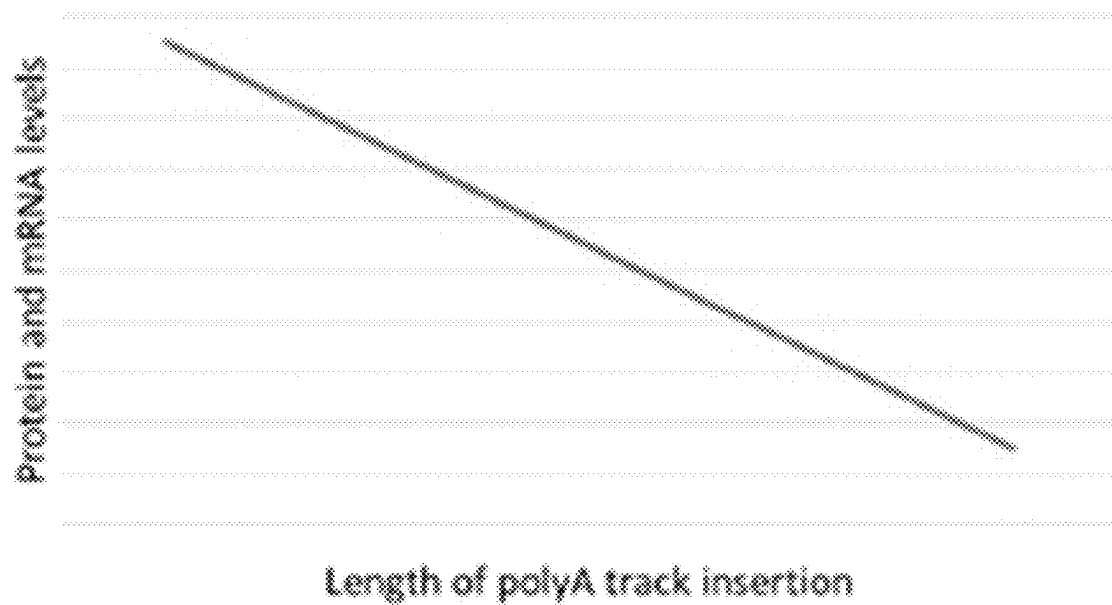
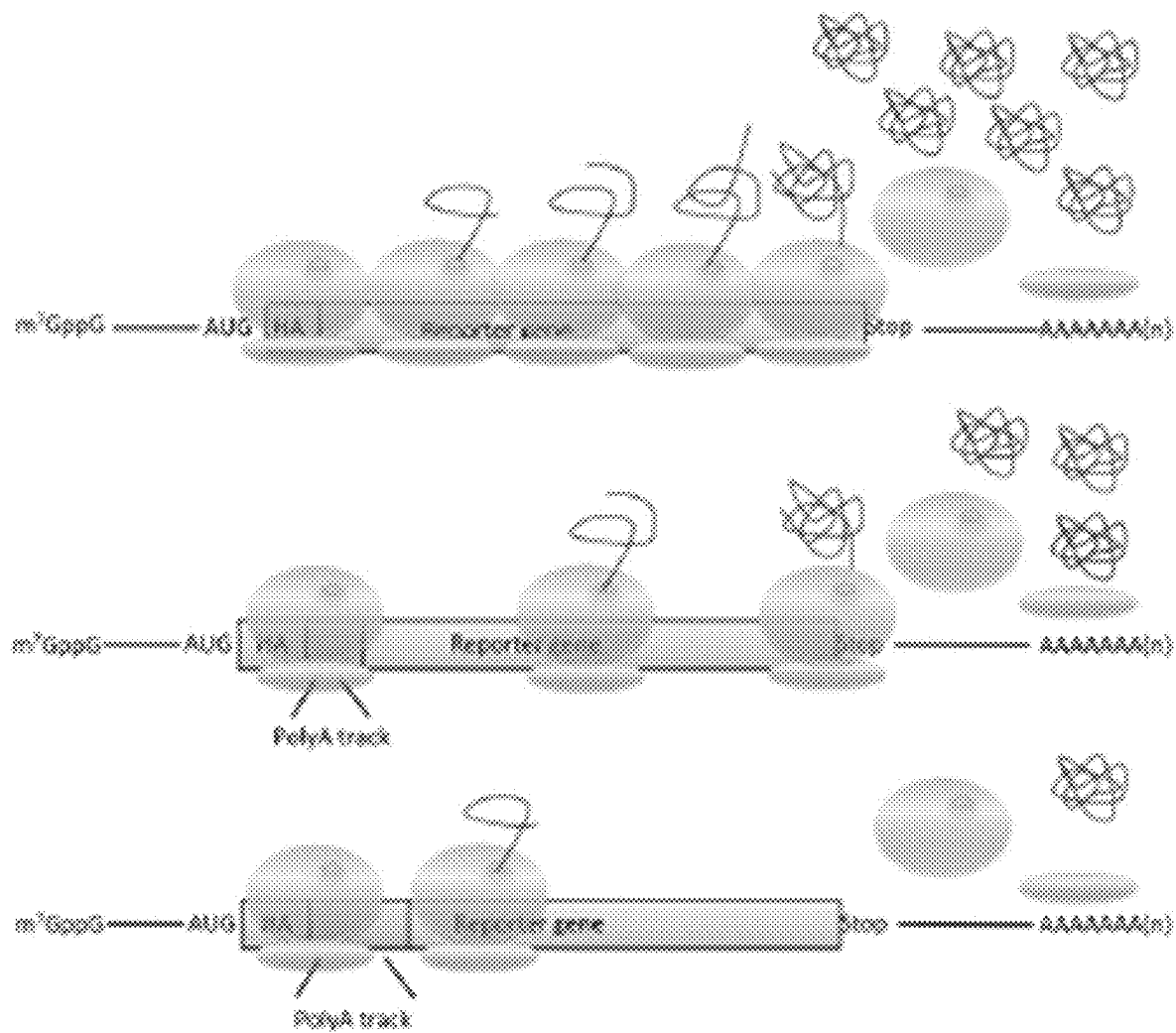
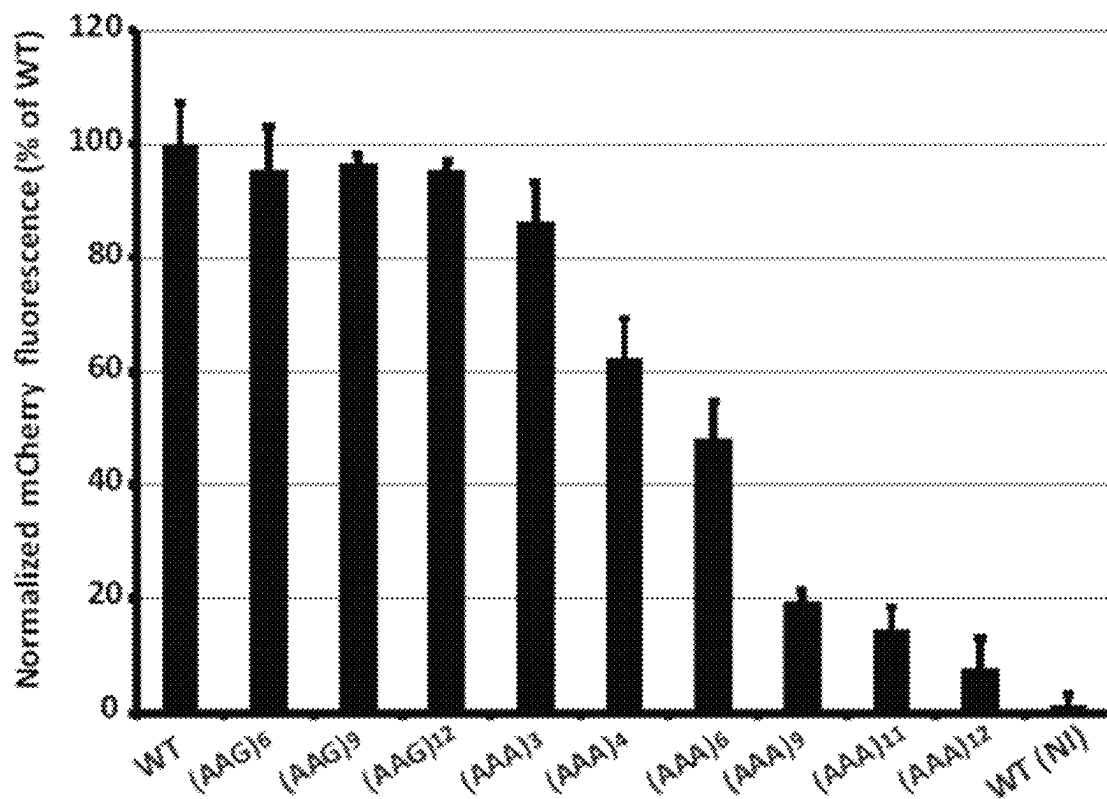
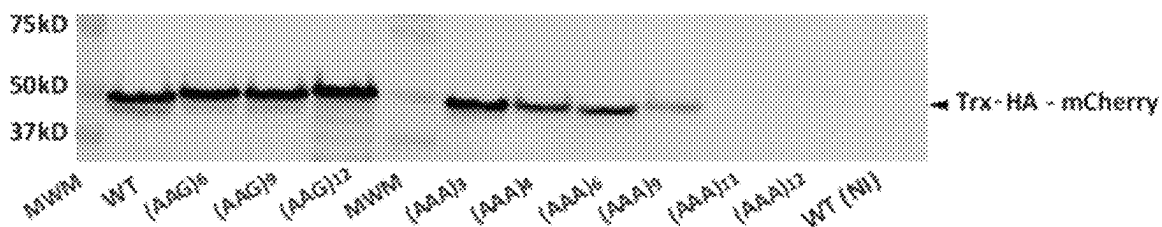
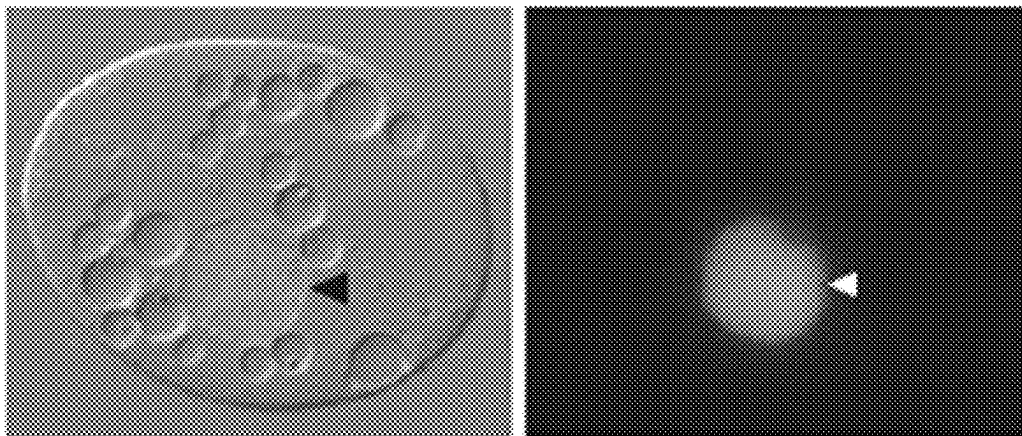
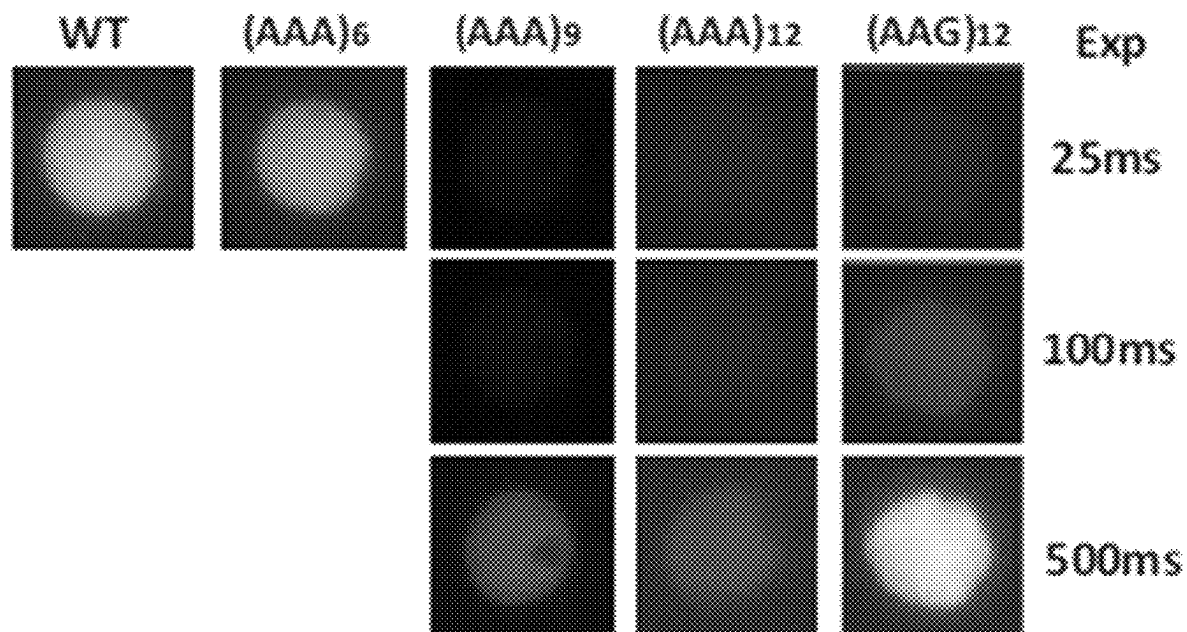


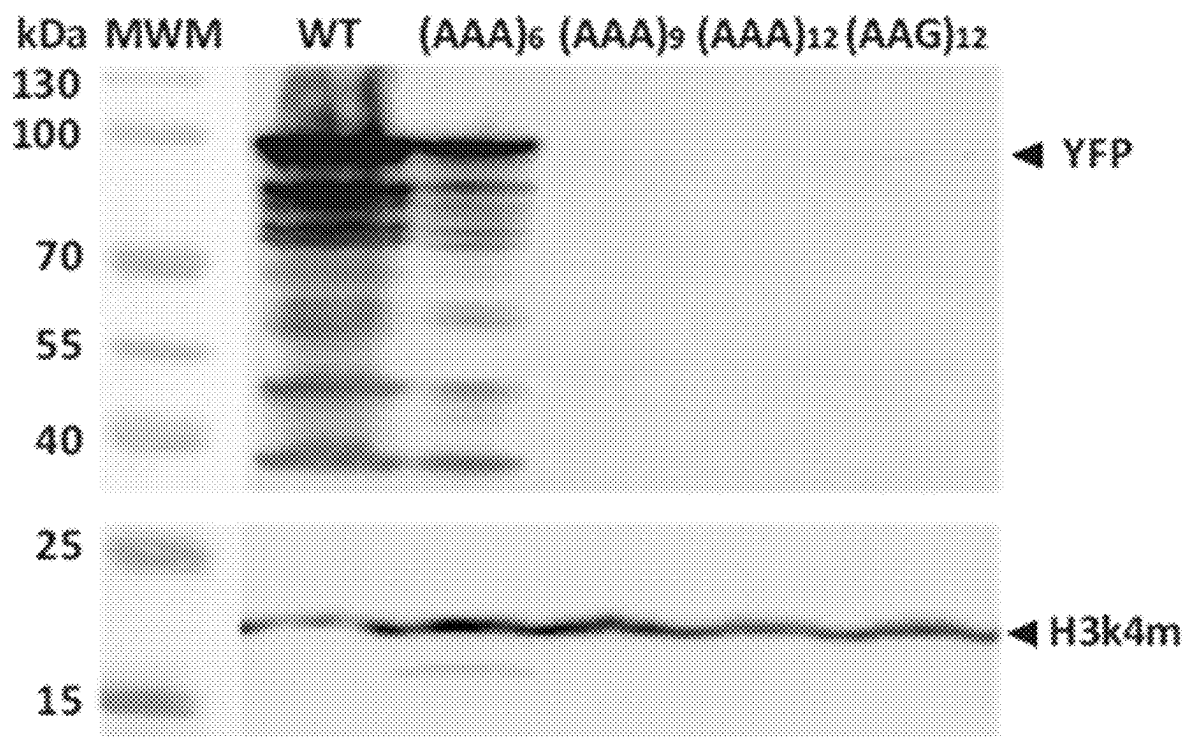
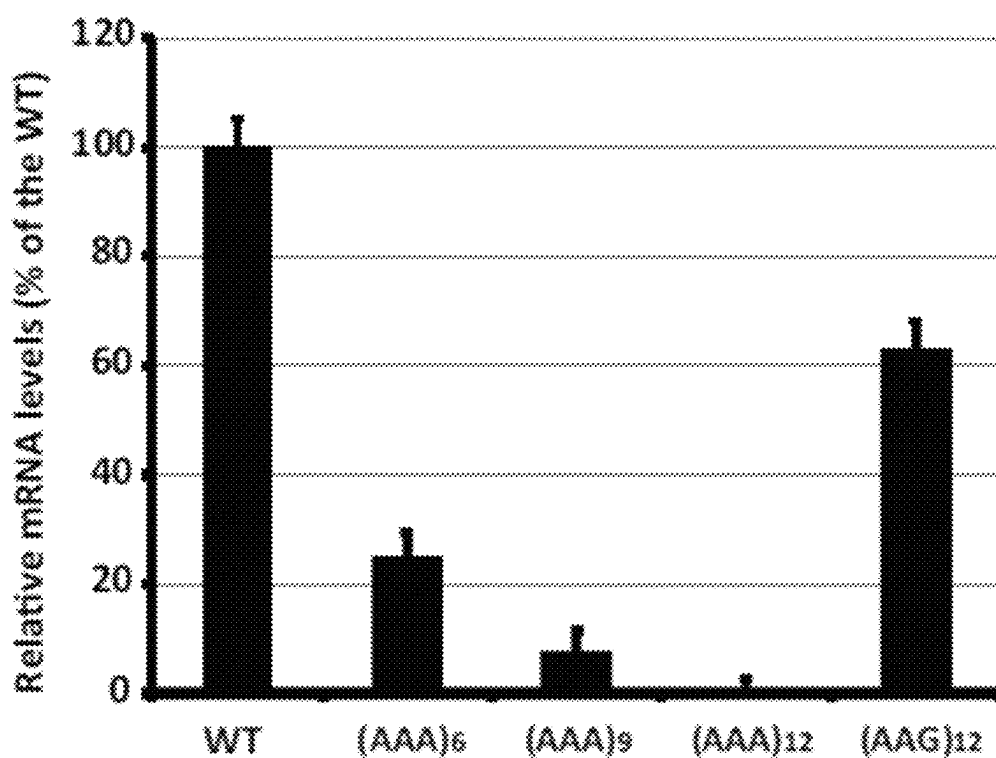
FIG. 24

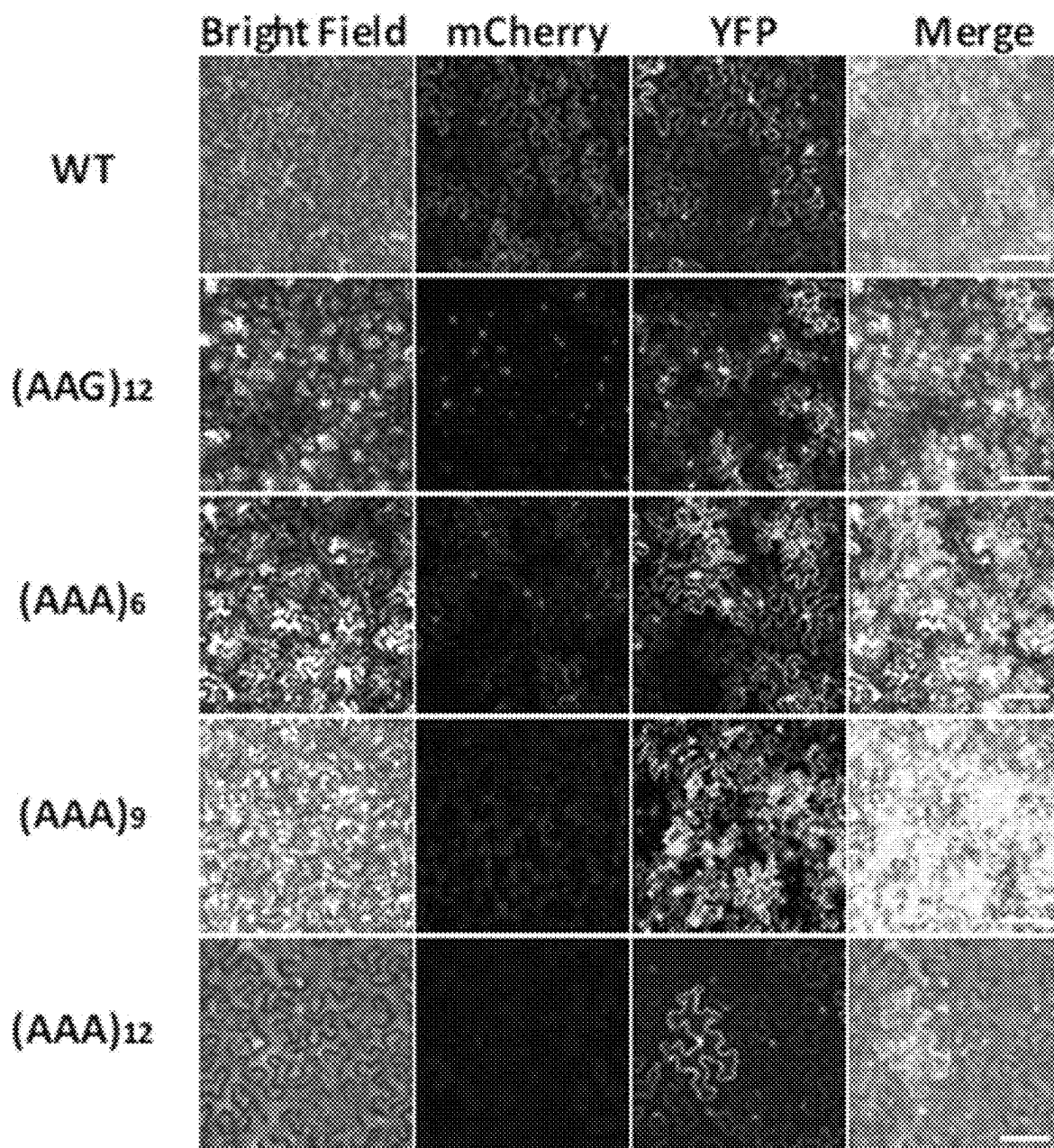
**FIG. 25A****FIG. 25B**

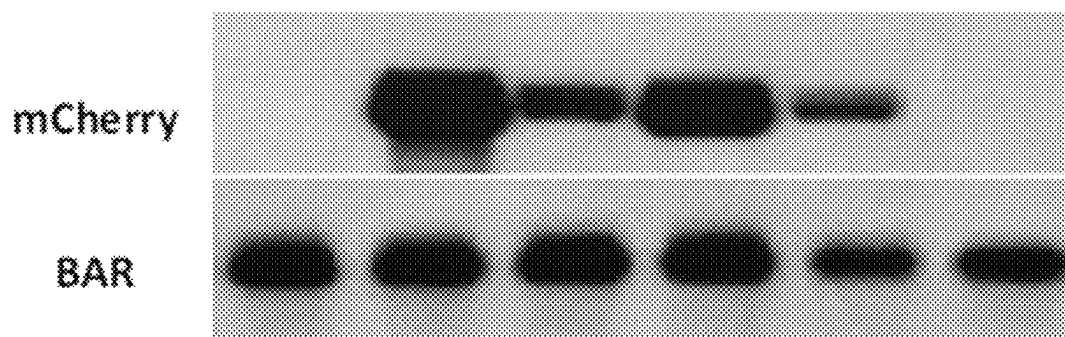
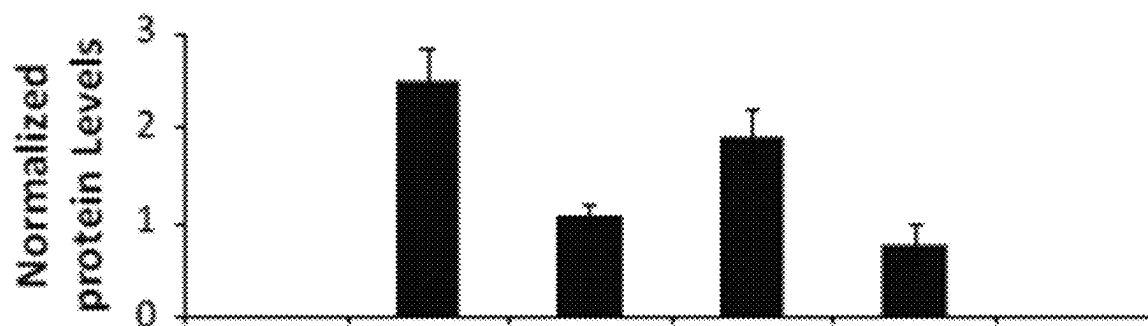
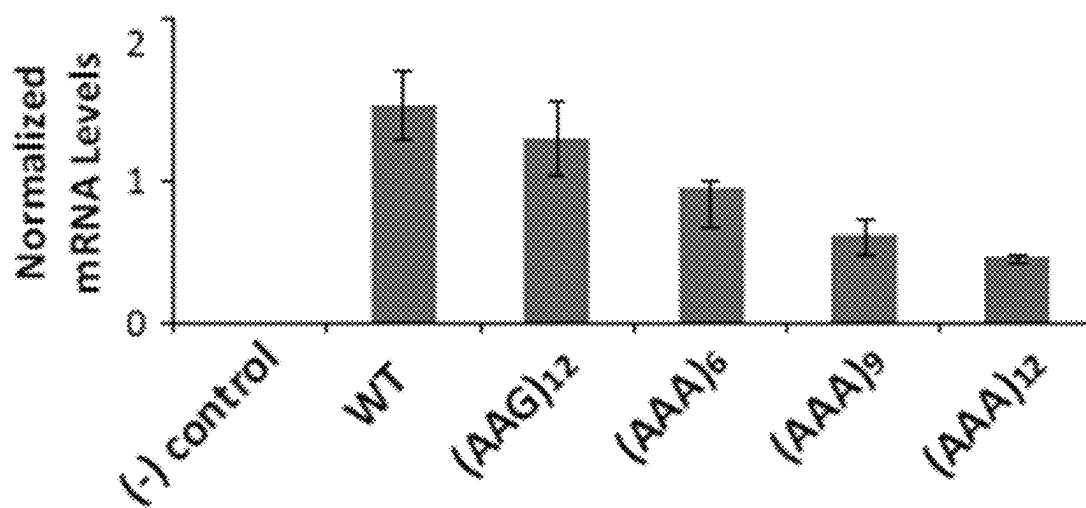
**FIG. 25C**

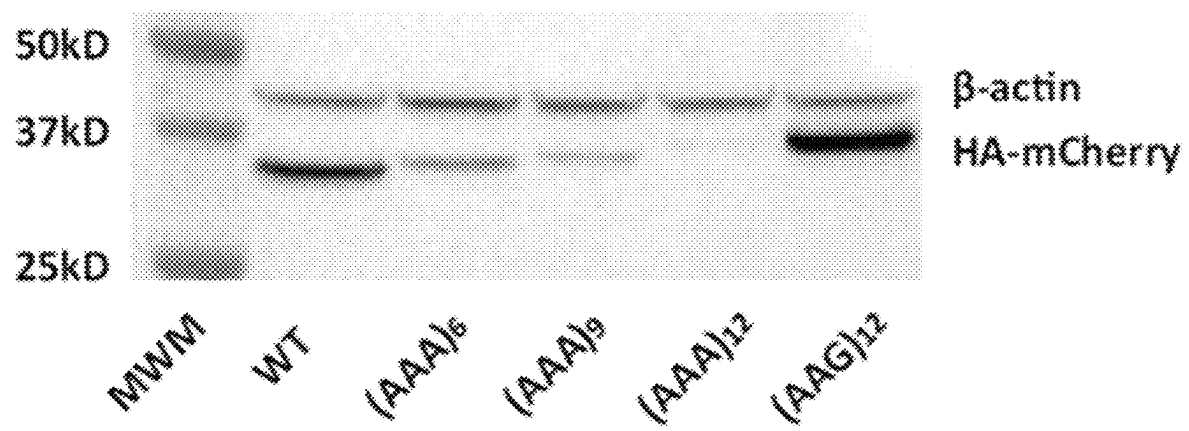
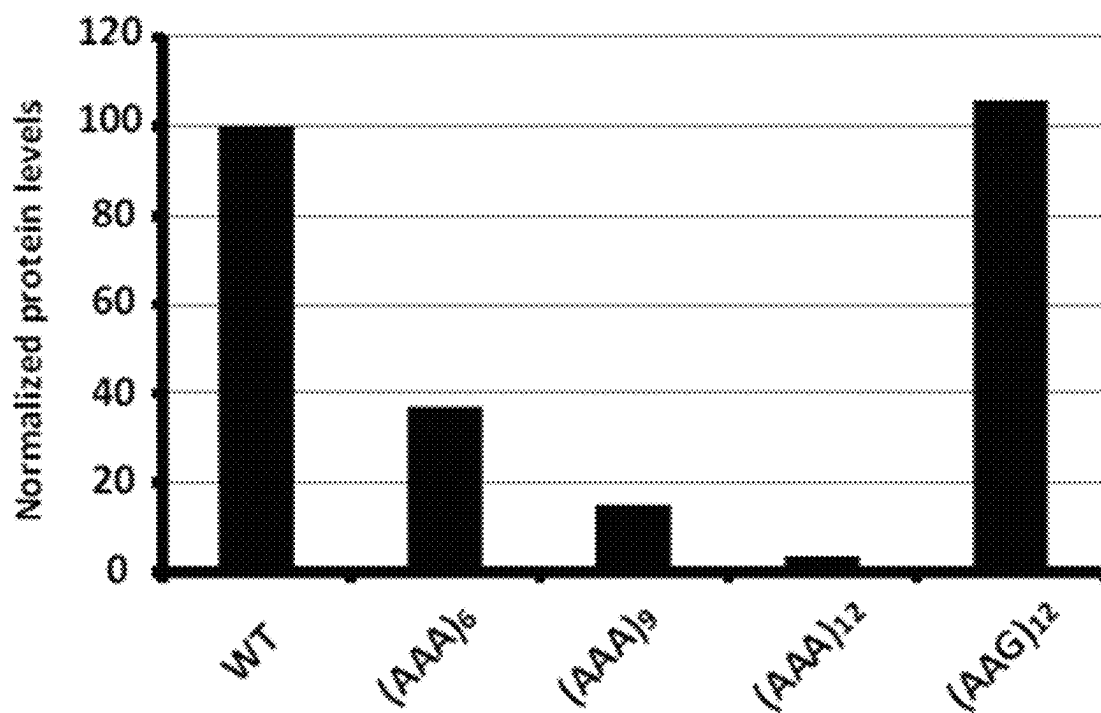
**FIG. 26A****FIG. 26B**

**FIG. 26C****FIG. 26D**

**FIG. 26E****FIG. 26F**

**FIG. 27A**

**FIG. 27B****FIG. 27C****FIG. 27D**

**FIG. 27E****FIG. 27F**

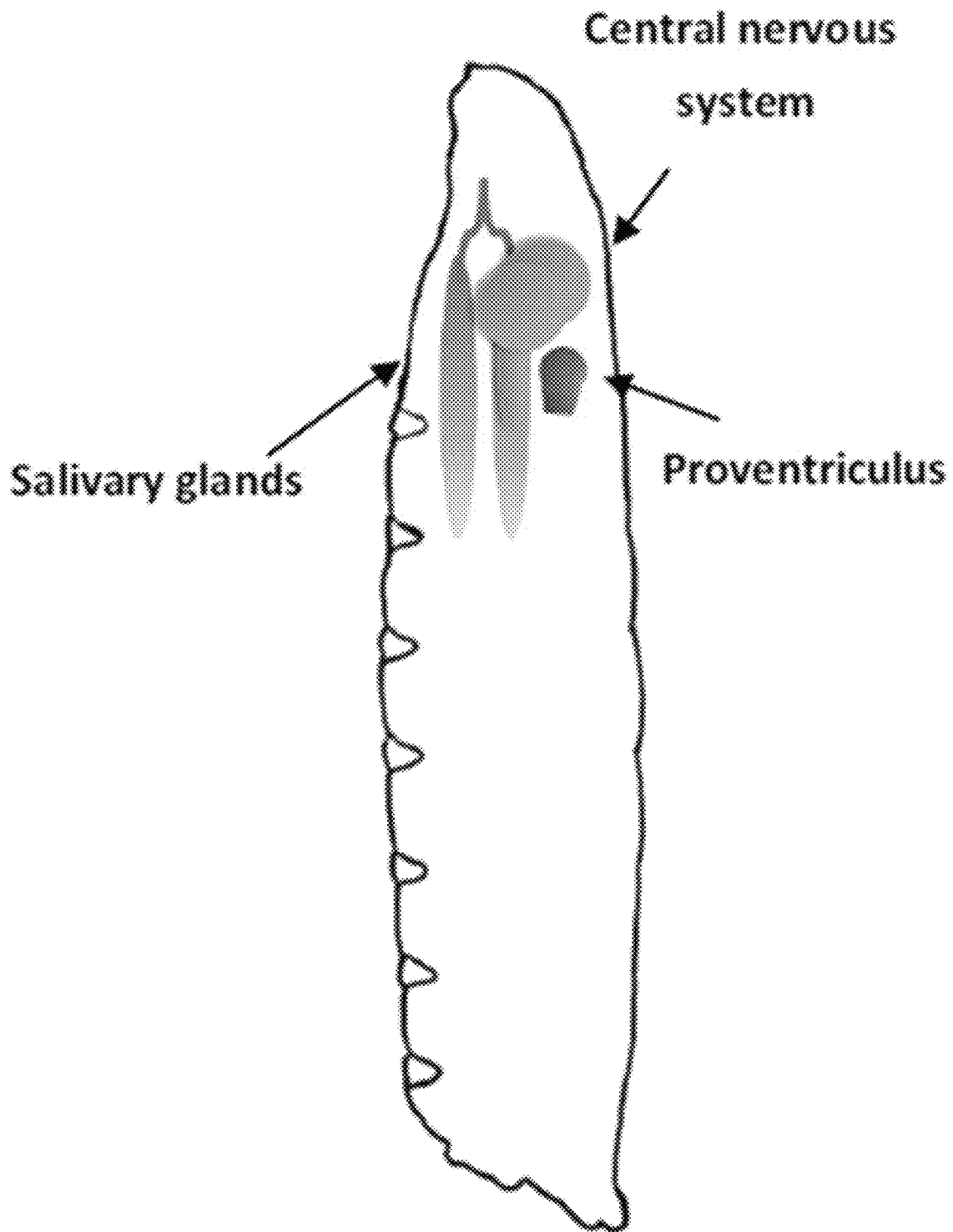
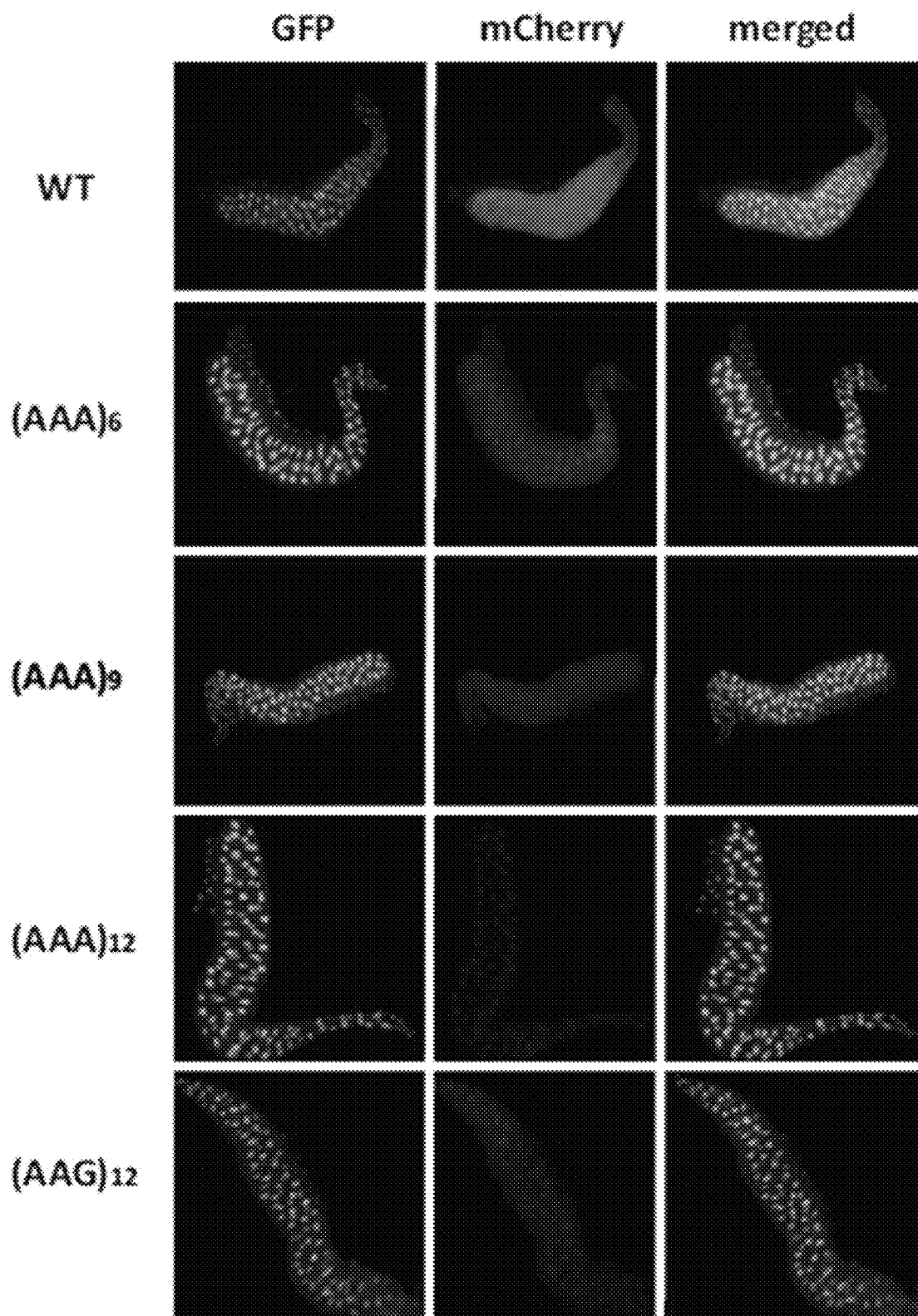
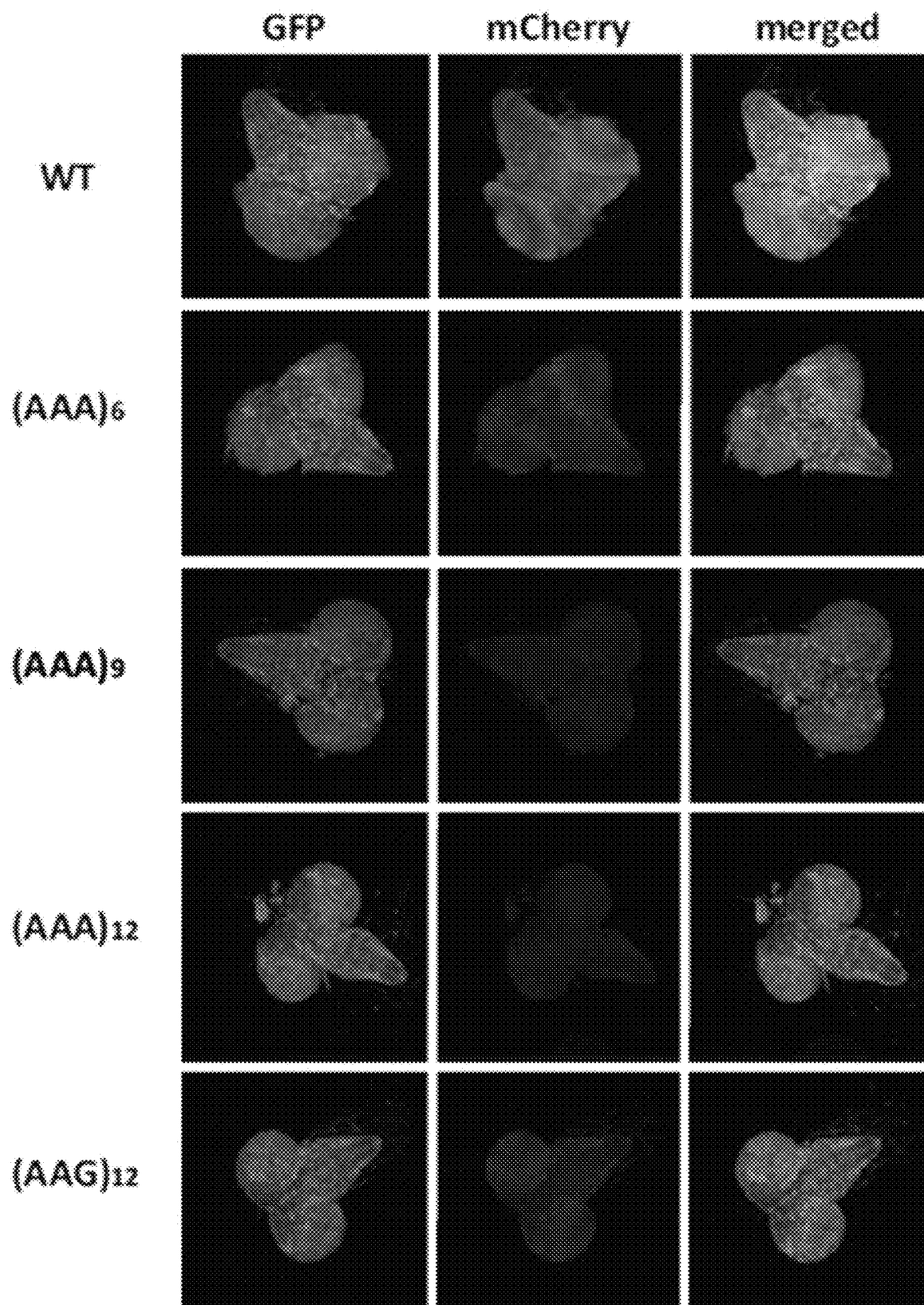
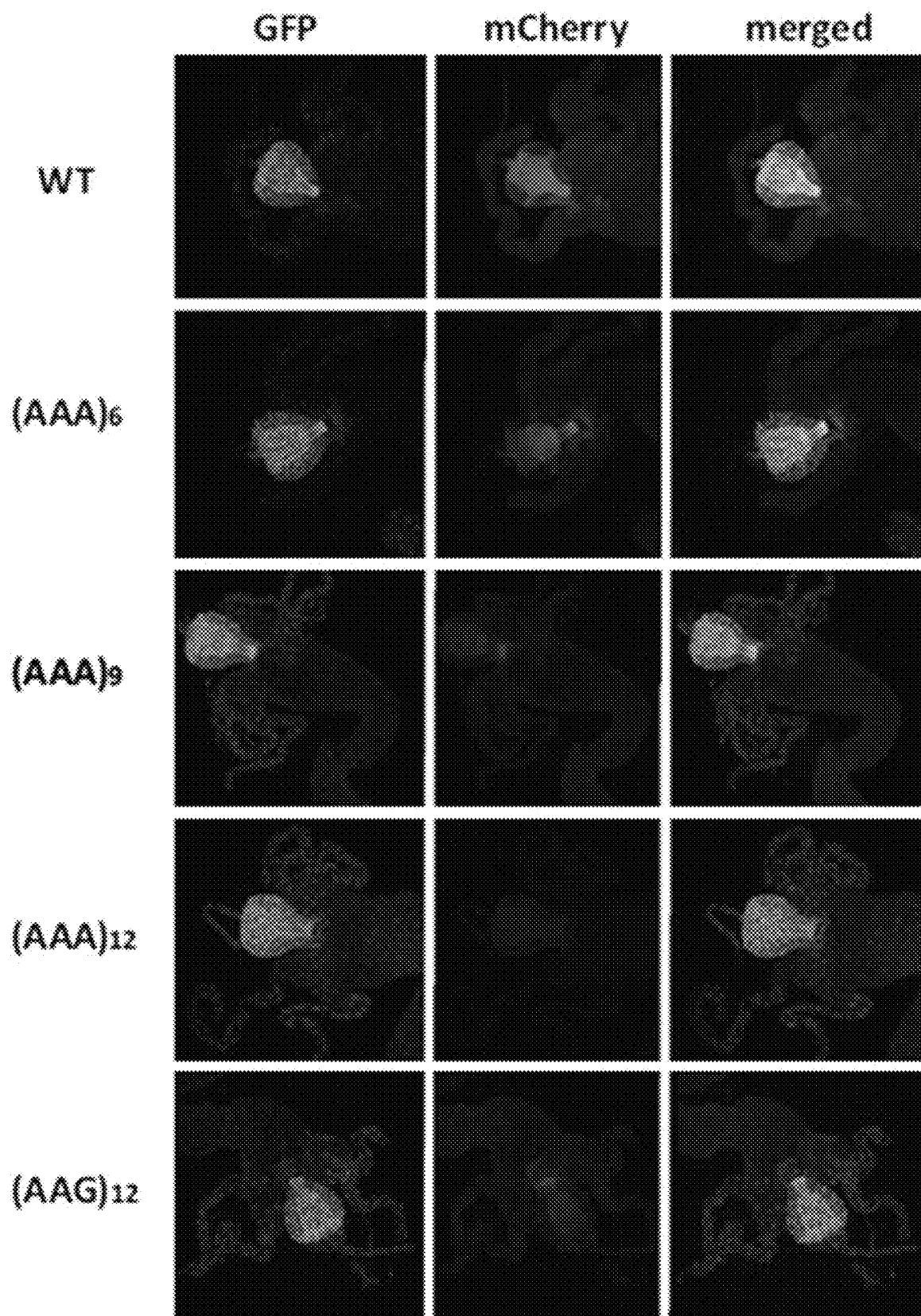
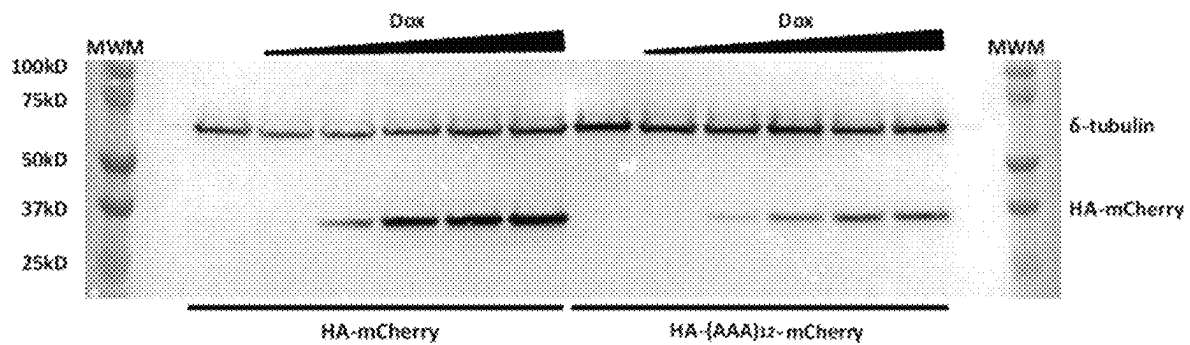
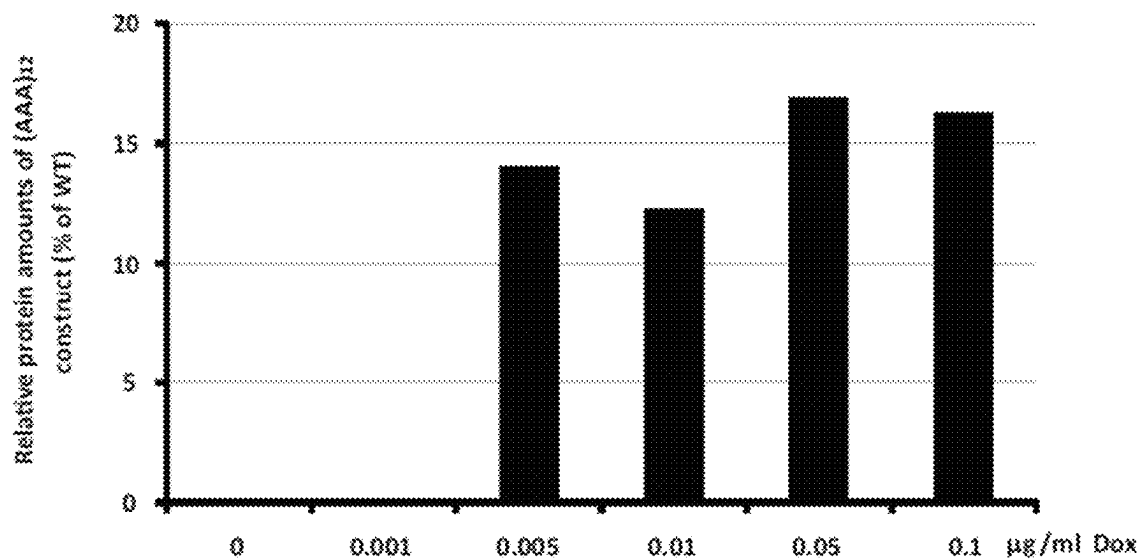


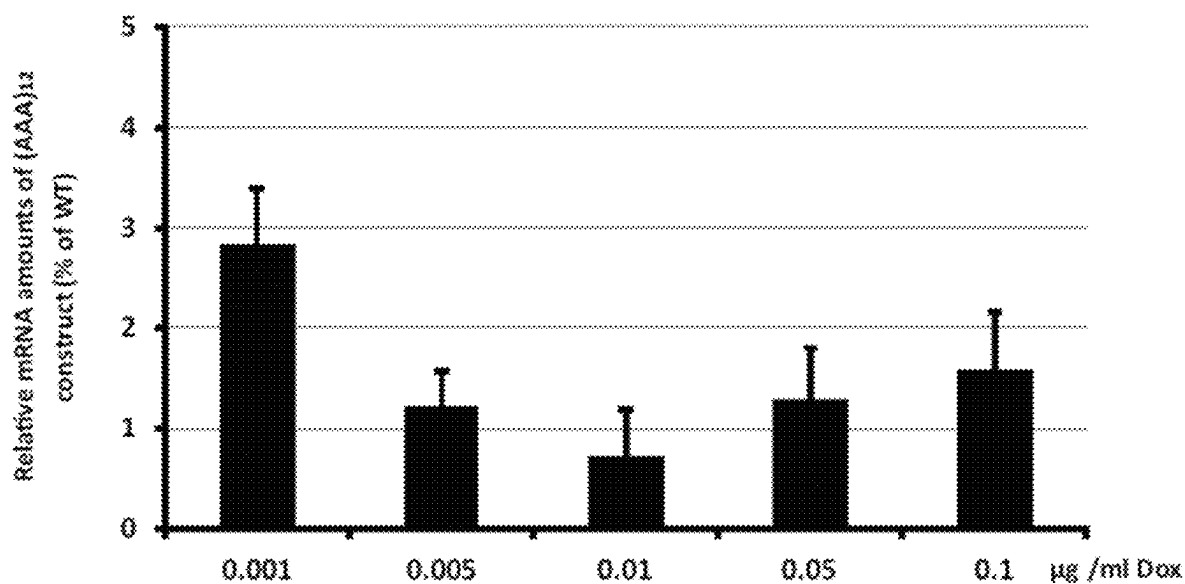
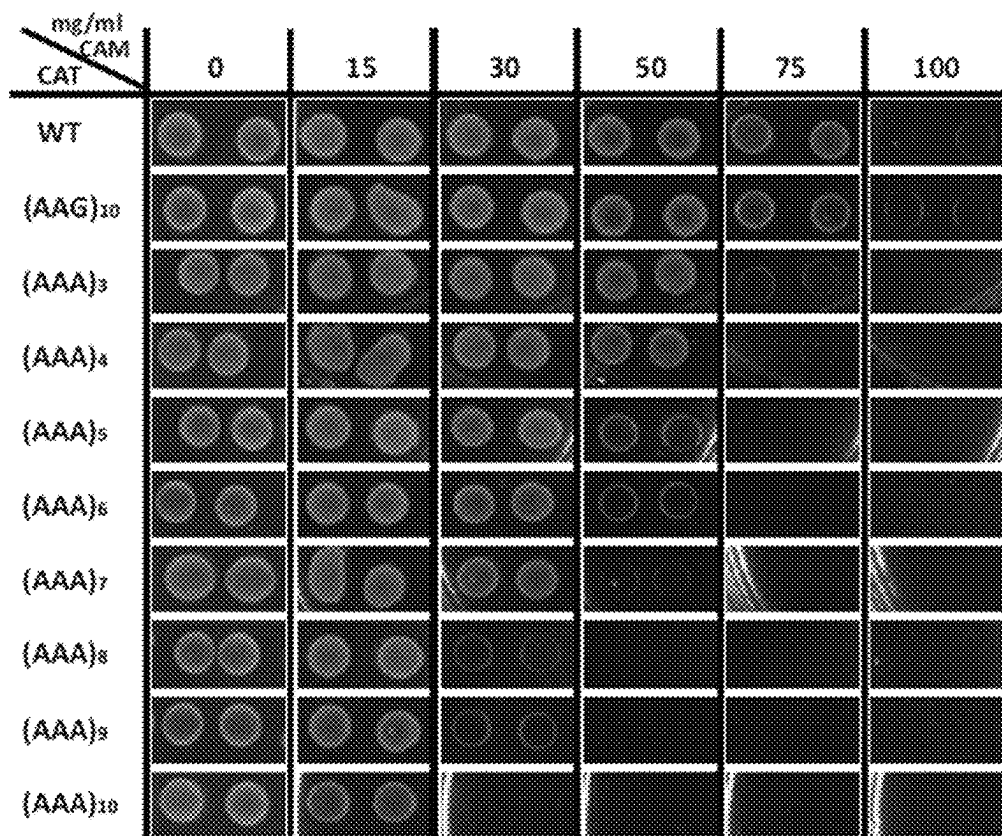
FIG. 28A

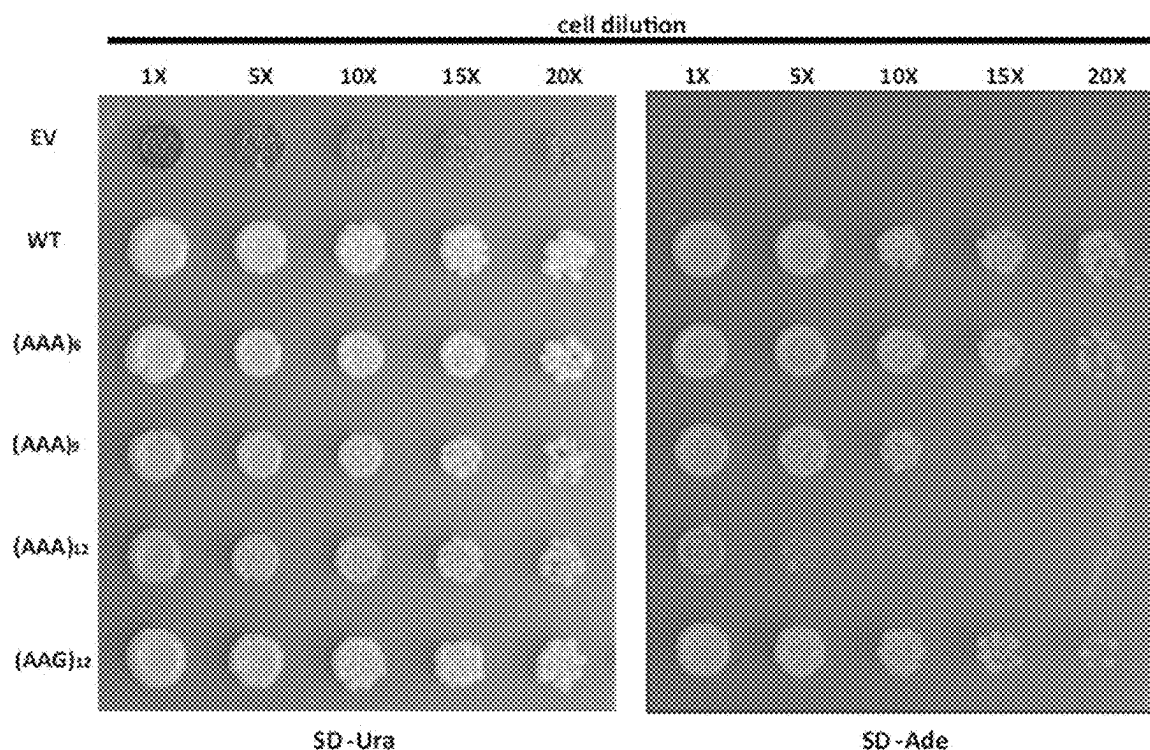
**FIG. 28B**

**FIG. 28C**

**FIG. 28D**

**FIG. 29A****FIG. 29B**

**FIG. 29C****FIG. 30A**

**FIG. 30B**

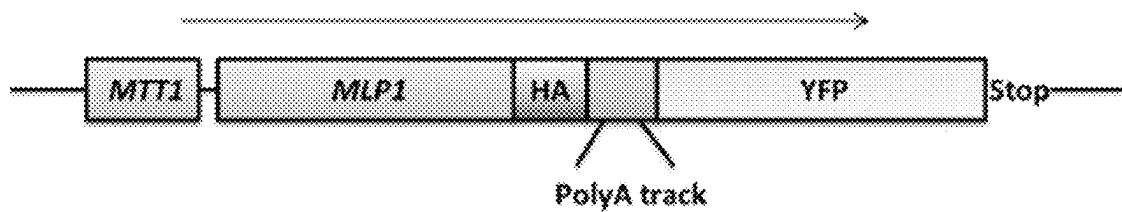
X = 6-12xLys_{AAG} or 3-12Lys_{AAA}

WT (TRX-HA) – Ala – Val – (mCherry)
- GCA - GTG -

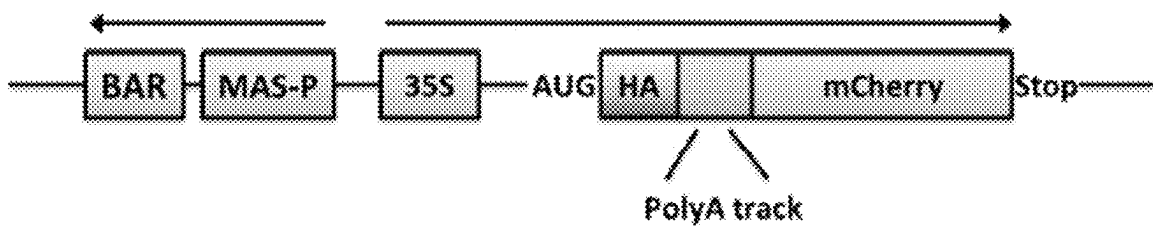
(AAG)_n (TRX-HA) – Ala – (Lys)₆₋₁₂ – Val – (mCherry)
- GCA - (AAG)₆₋₁₂ - GTG -

(AAA)_n (TRX-HA) – Ala – (Lys)₆₋₁₂ – Val – (mCherry)
- GCA - (AAA)₆₋₁₂ - GTG -

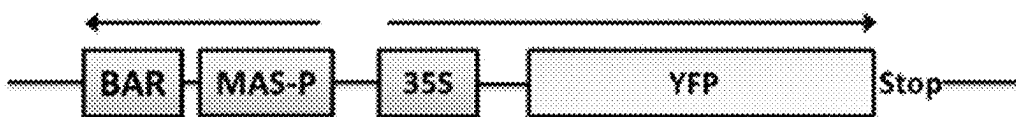
FIG. 31

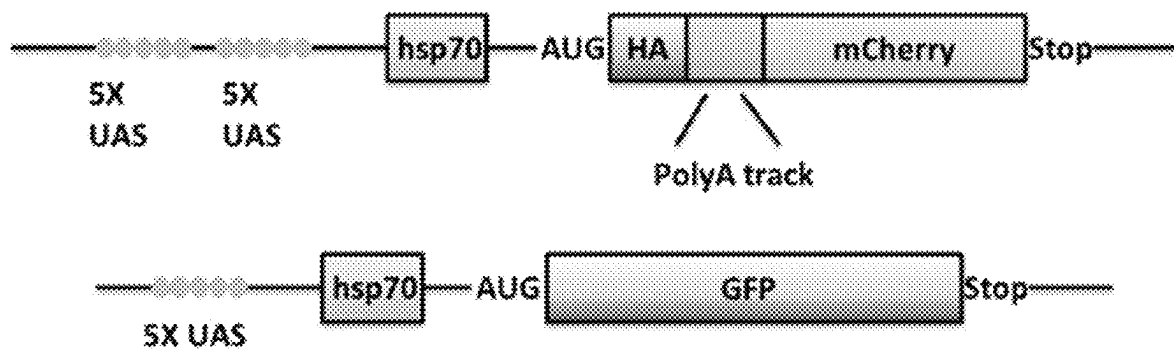
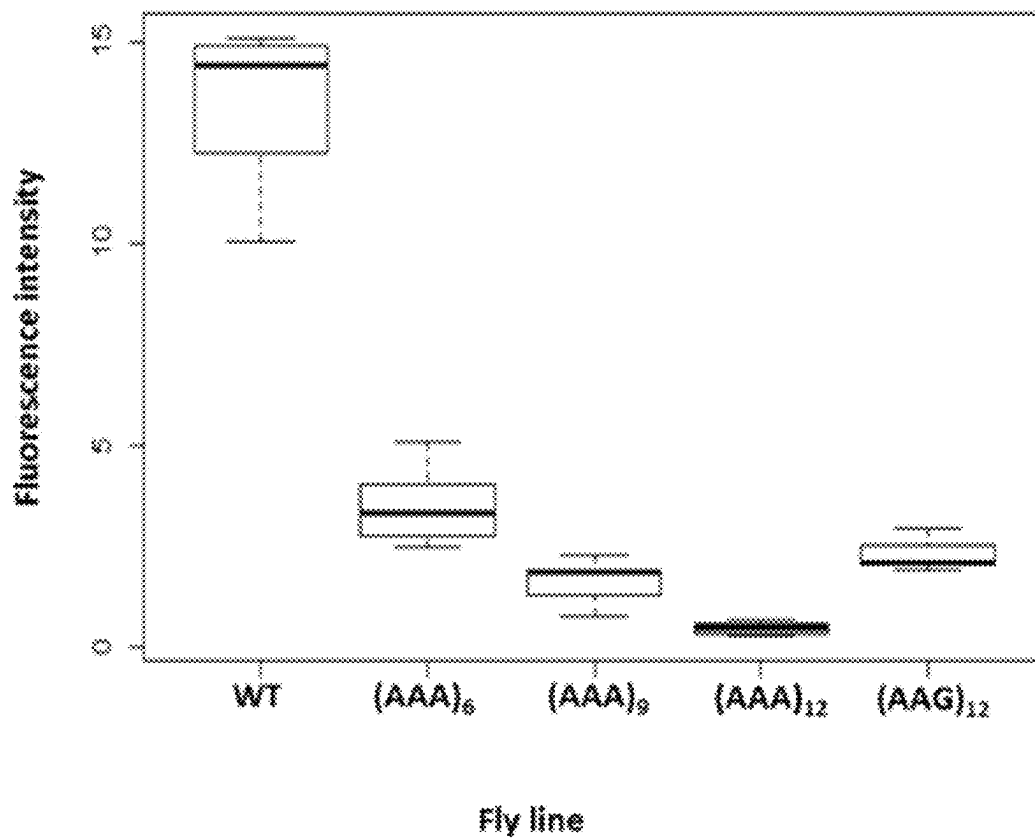
Tetrahymena thermophila reporter**FIG. 32***N. benthamiana* reporter

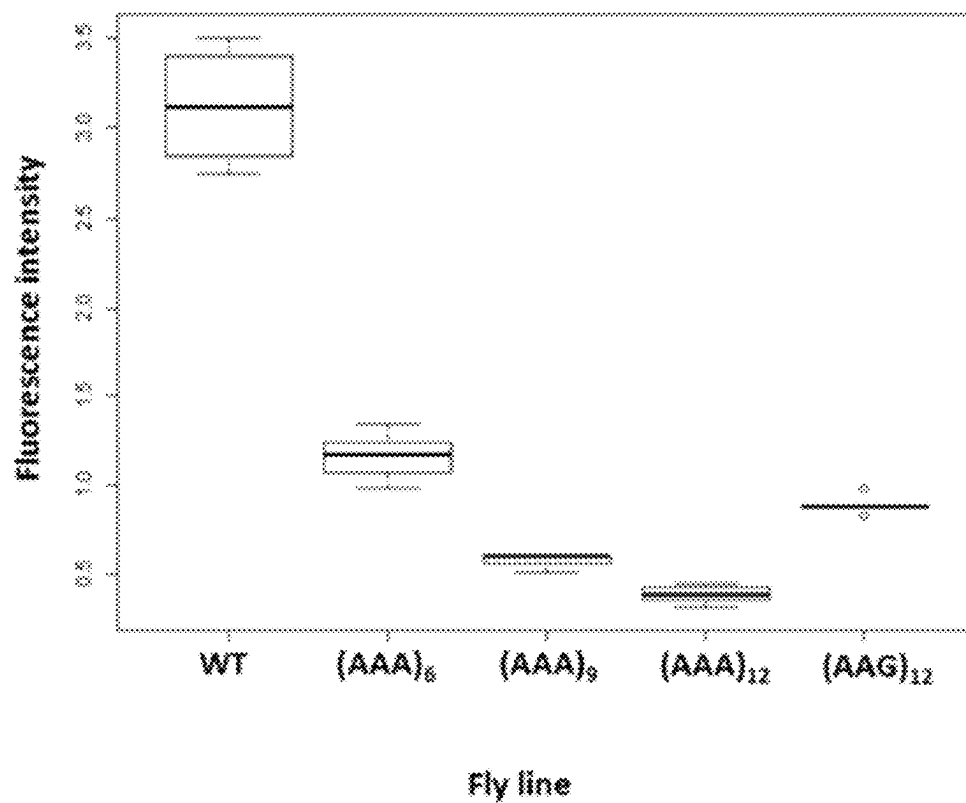
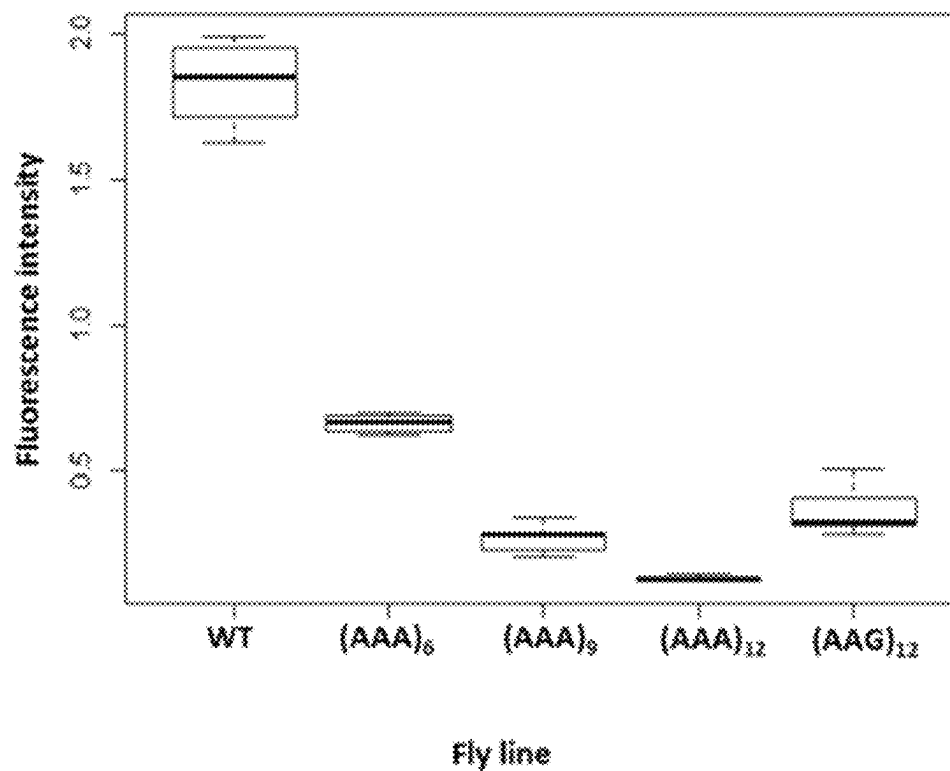
pEarleyGate 100

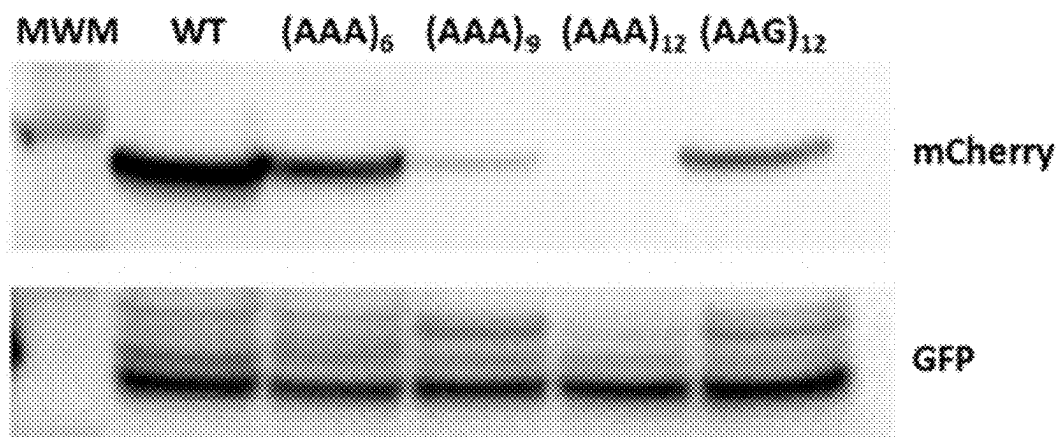
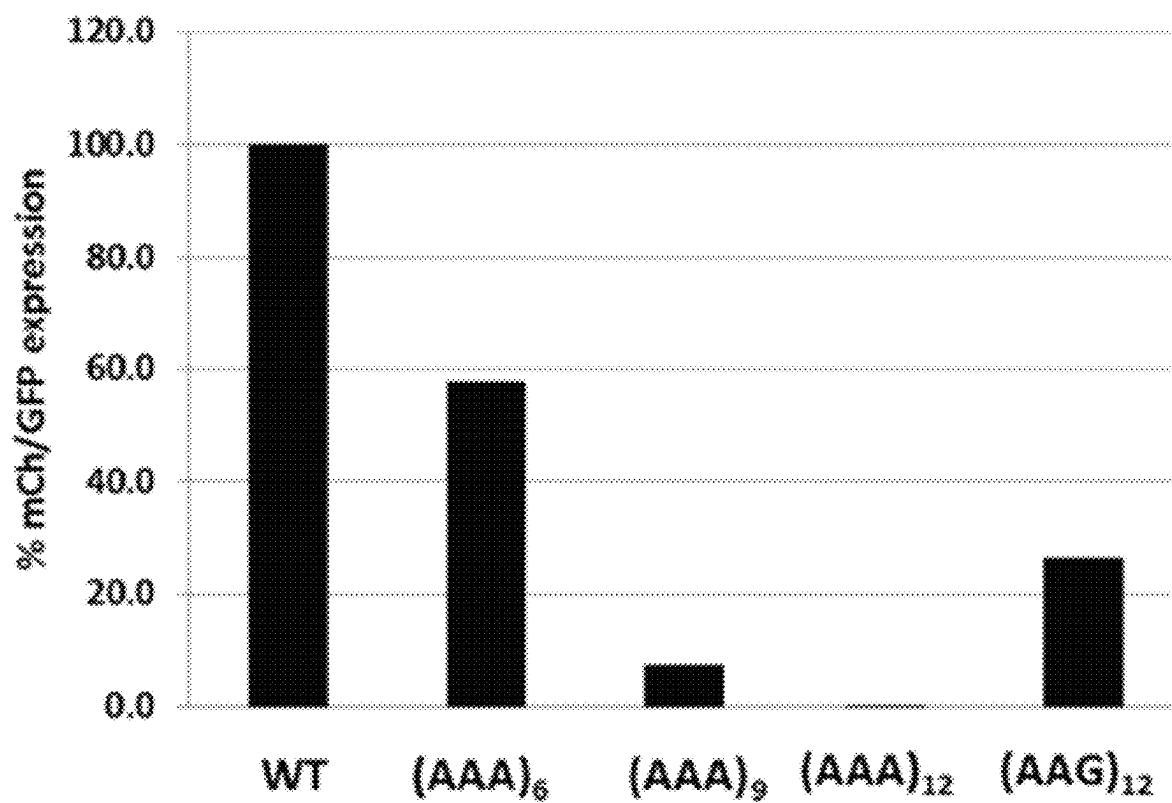


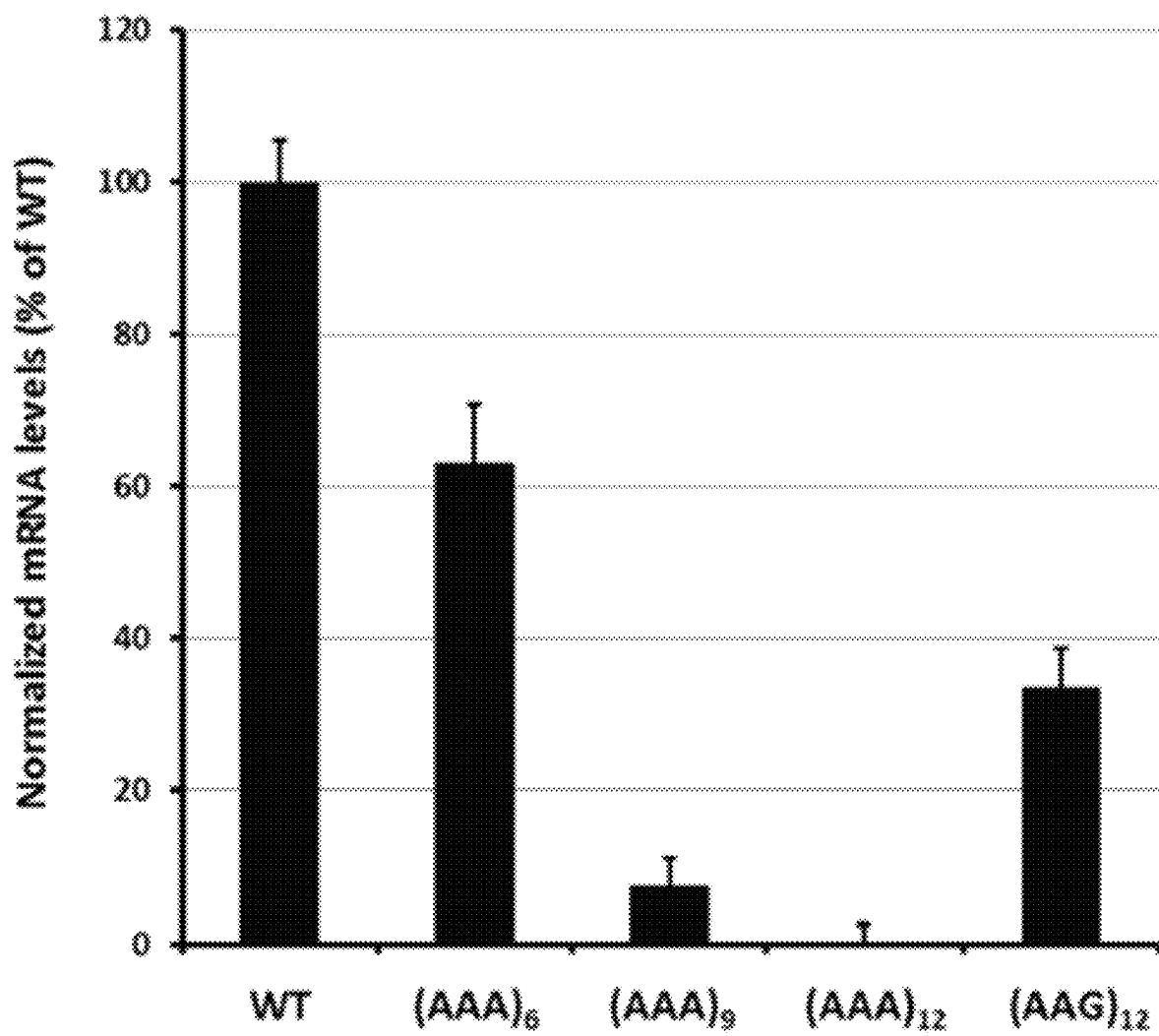
pEarleyGate 104

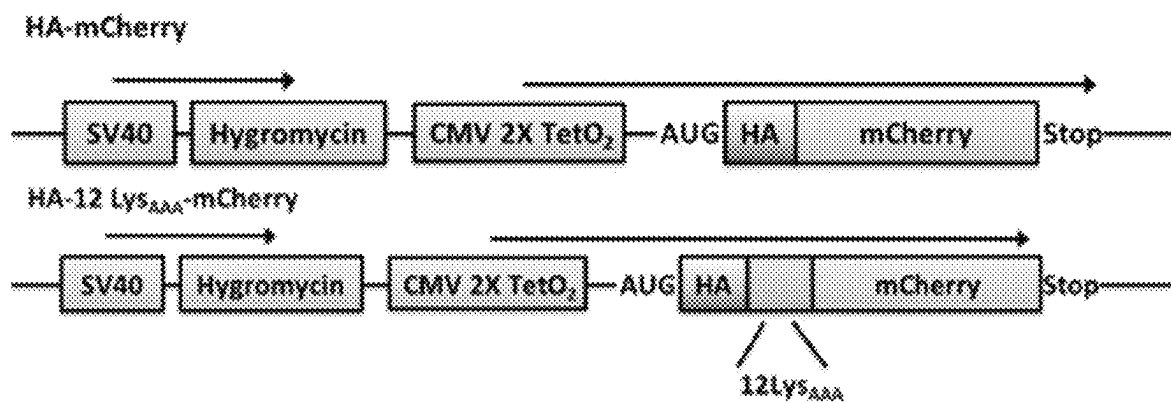
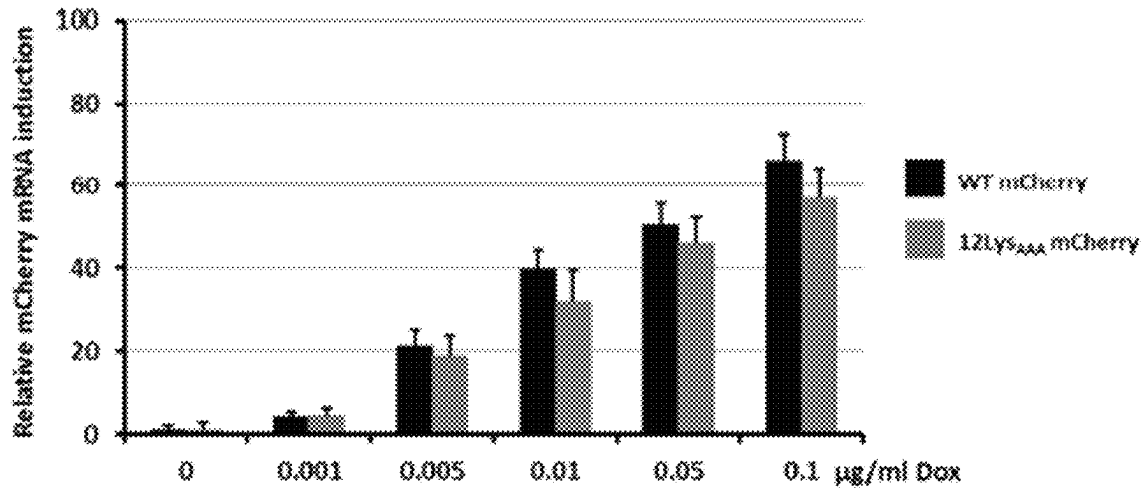
**FIG. 33**

**FIG. 34****FIG. 35A**

**FIG. 35B****FIG. 35C**

**FIG. 36A****FIG. 36B**

**FIG. 37**

**FIG. 38A****FIG. 38B**

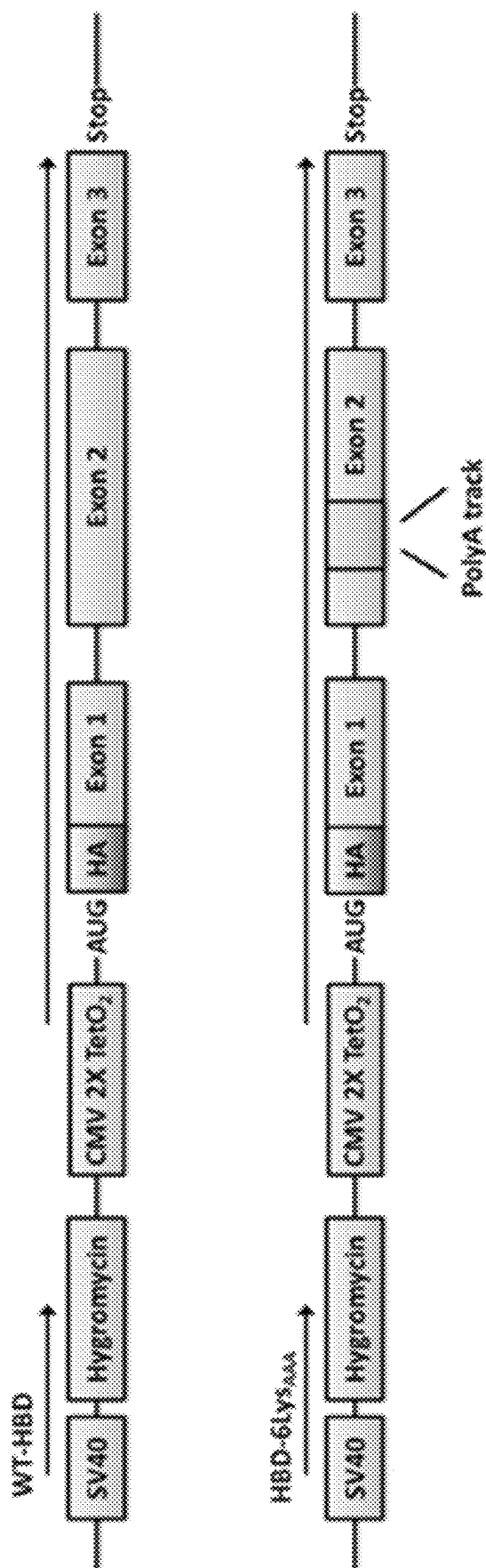
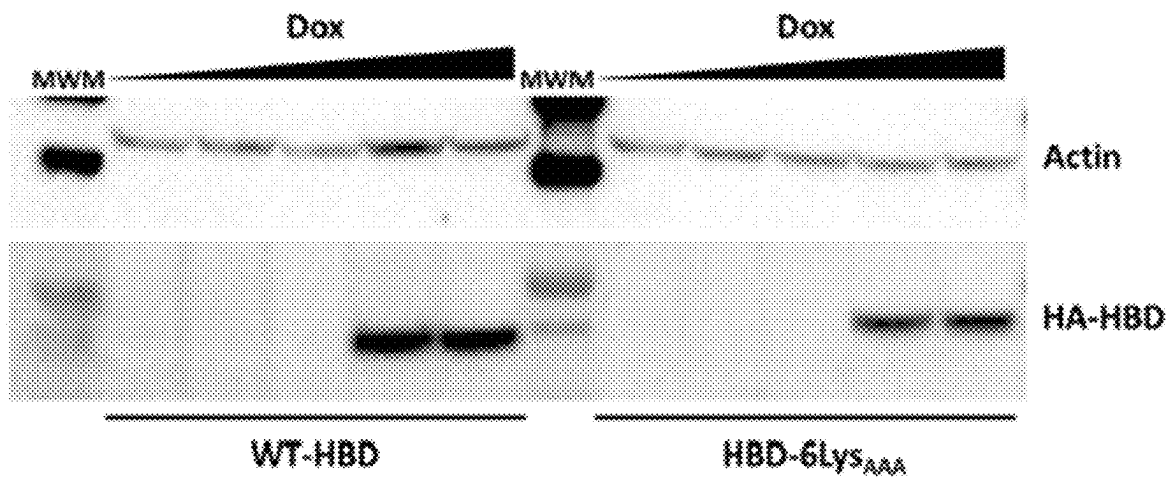
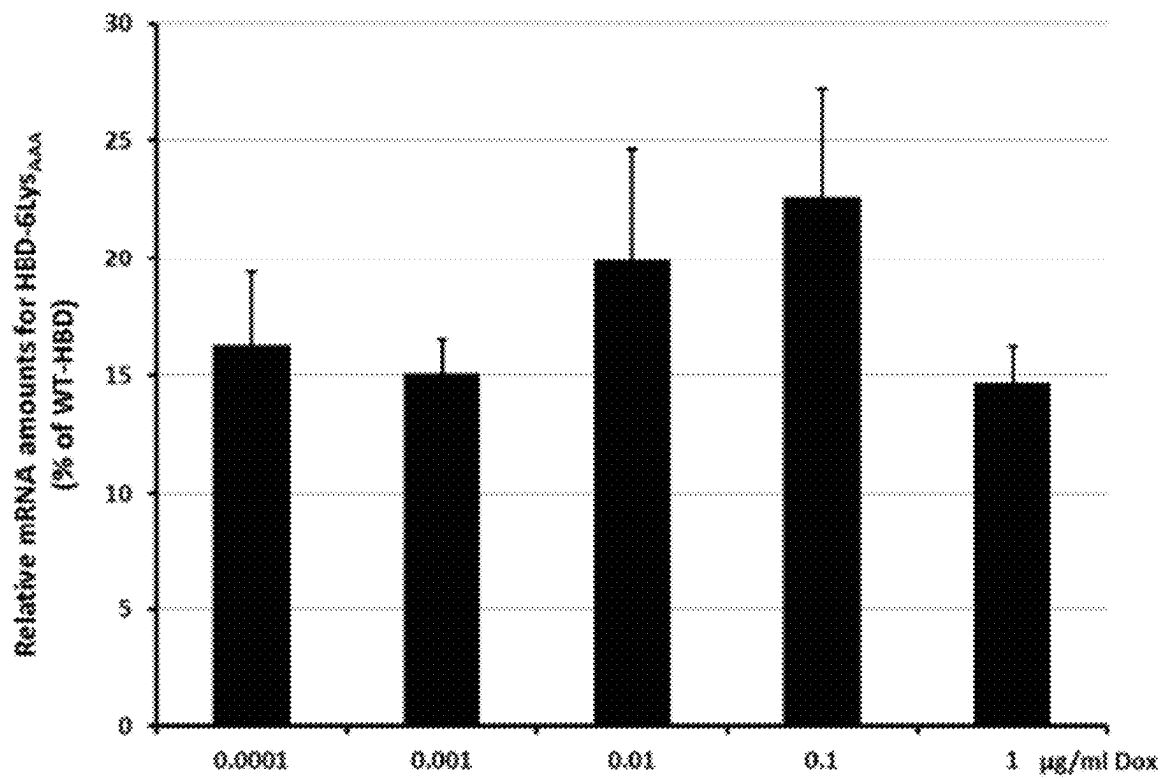
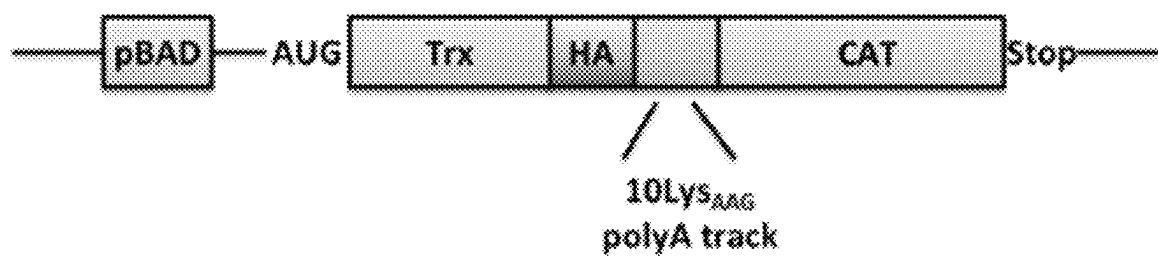
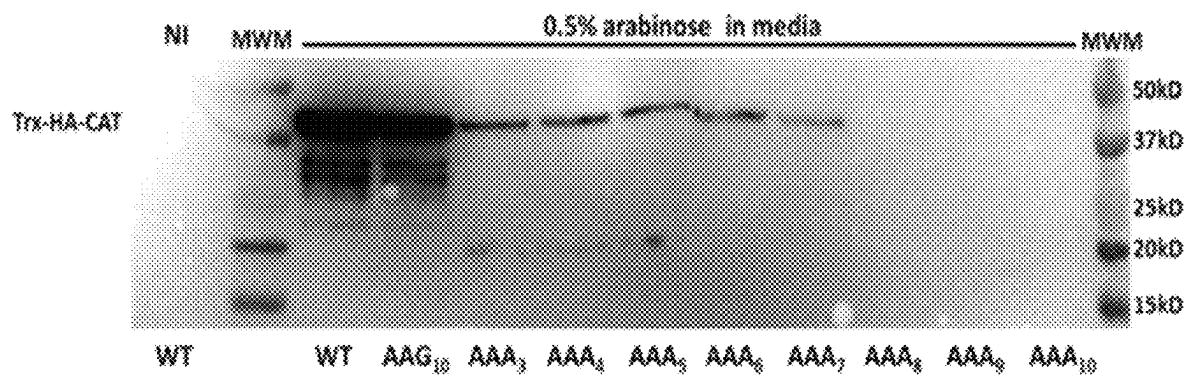
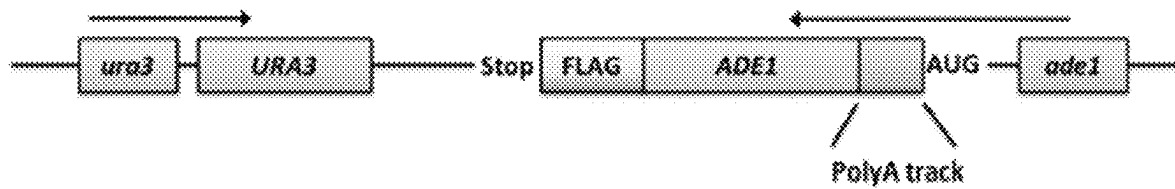
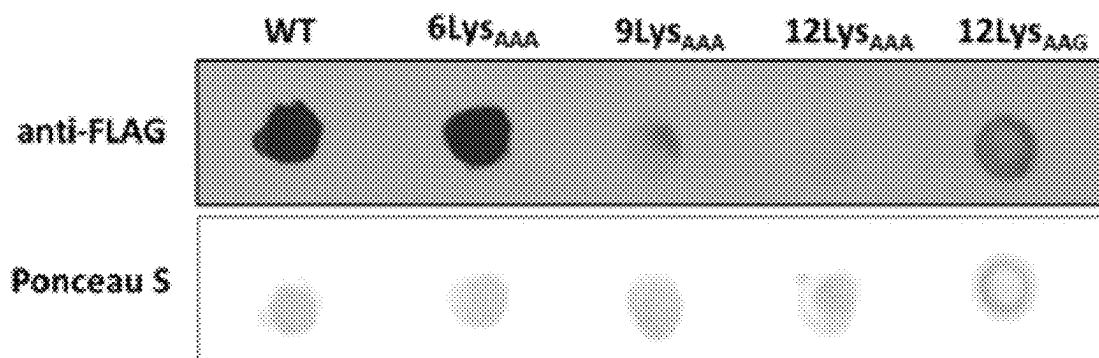


FIG. 39

**FIG. 40****FIG. 41**

**FIG. 42****FIG. 43**

**FIG. 44A****FIG. 44B**

WT		GCAGTG
		A V
K2/A9	(AN)	GCAAAAAAAAAAATGTG
		A K K N V
K3/A9	(AS)	GCGAAAAAAAAAATCC
		A K K K S
K2/A10	(QN)	CAAAAAAAAAAATGTG
		Q K K N V
K3/A10	(EV)	GAAAAAAAAAAGGTG
		E K K K V
K3/A10	(QV)	CAAAAAAAAAAAGGTG
		Q K K K V
K3/A11	(AN)	GCGAAAAAAAAAATGTG
		A K K K N V
K4/A11	(AV)	GCGAAAAAAAAAAGGTG
		A K K K K V

FIG. 45A

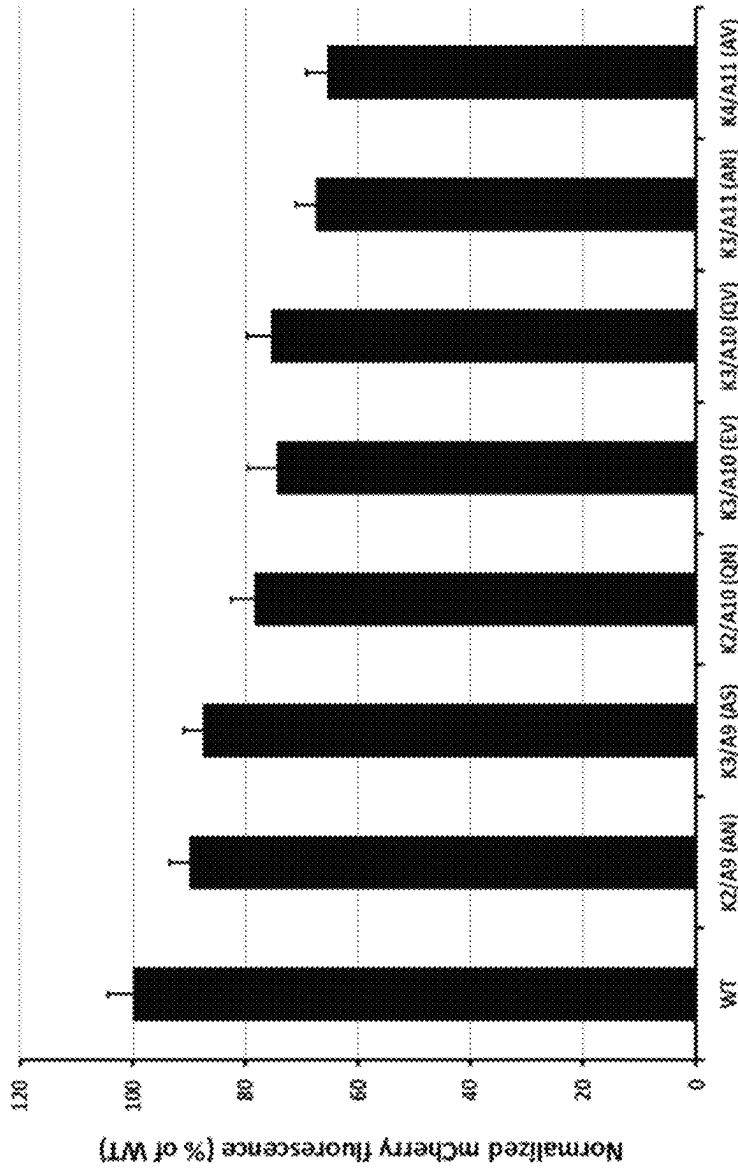
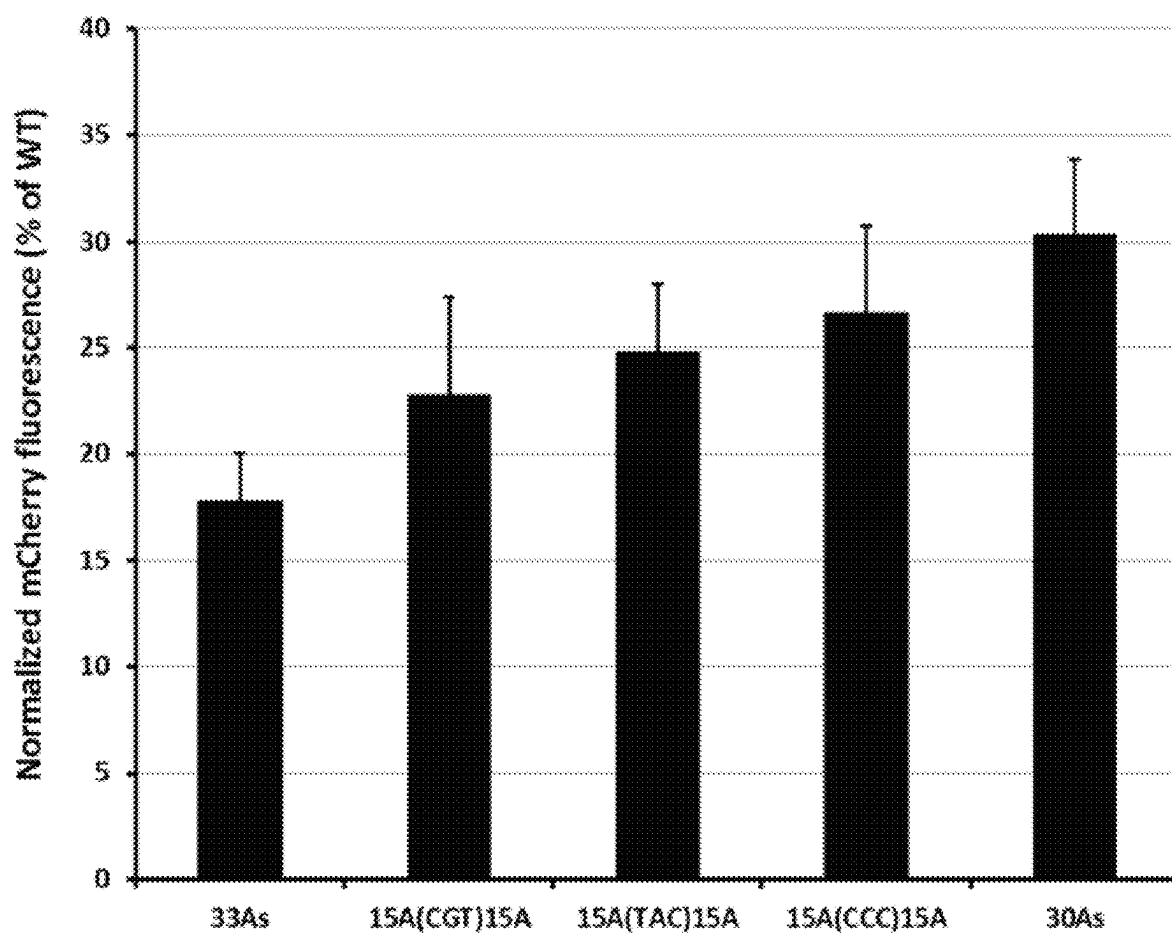
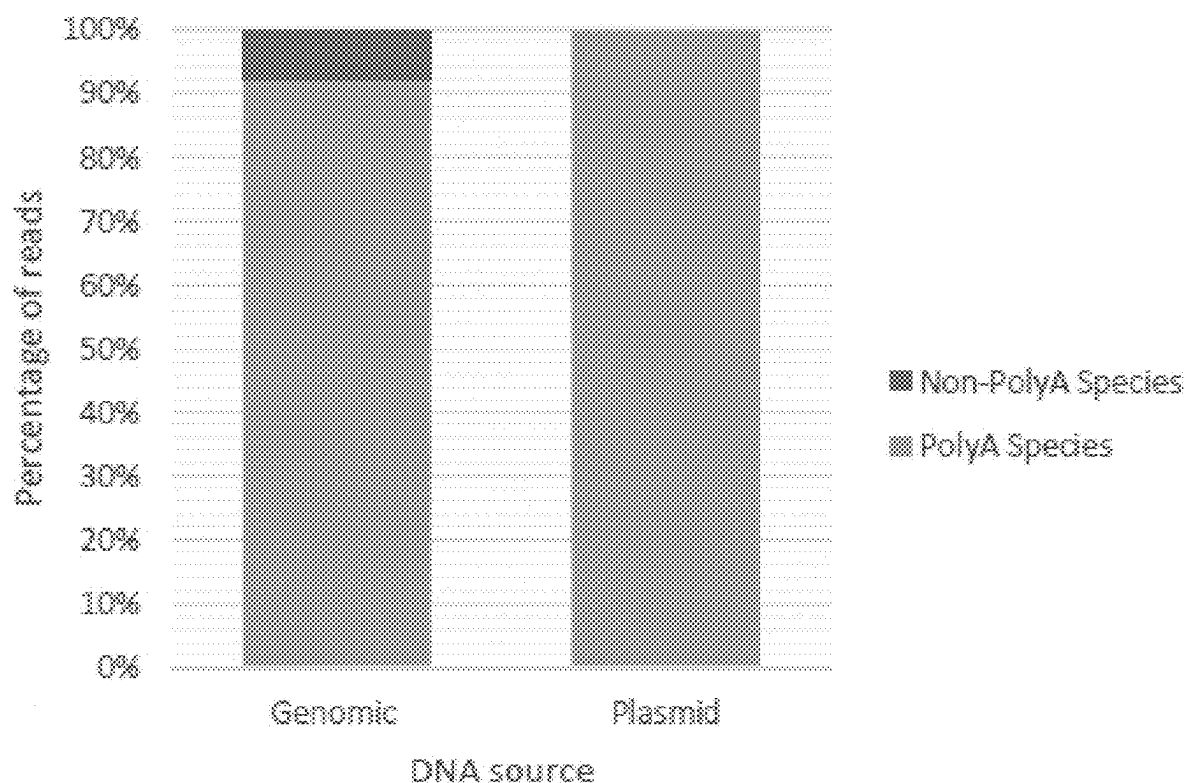


FIG. 45B

33As	-GTACCGGATTATCGGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAGTGAGCAAGG- (Lys) ₁₁
15A (CTG) 15A	-GTACCGGATTATCGGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAGTGAGCAAGG- (Lys) ₅ (Arg) (Lys) ₅
15A (TAC) 15A	-GTACCGGATTATCGGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAGTGAGCAAGG- (Lys) ₅ (Tyr) (Lys) ₅
15A (CCC) 15A	-GTACCGGATTATCGGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAGTGAGCAAGG- (Lys) ₅ (Pro) (Lys) ₅
30As	-GTACCGGATTATCGGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAGTGAGCAAGG- (Lys) ₁₀

FIG. 45C

**FIG. 45D**

**FIG. 46A**

Non-PolyA species of genomic reads (7.73%):

TGCACCCAAAAAATTTACAAAAAACCGTGAGCAAGGGCGA

Bold portion maps to chromosome X

Underline portion maps to chromosome 3L

FIG. 46B

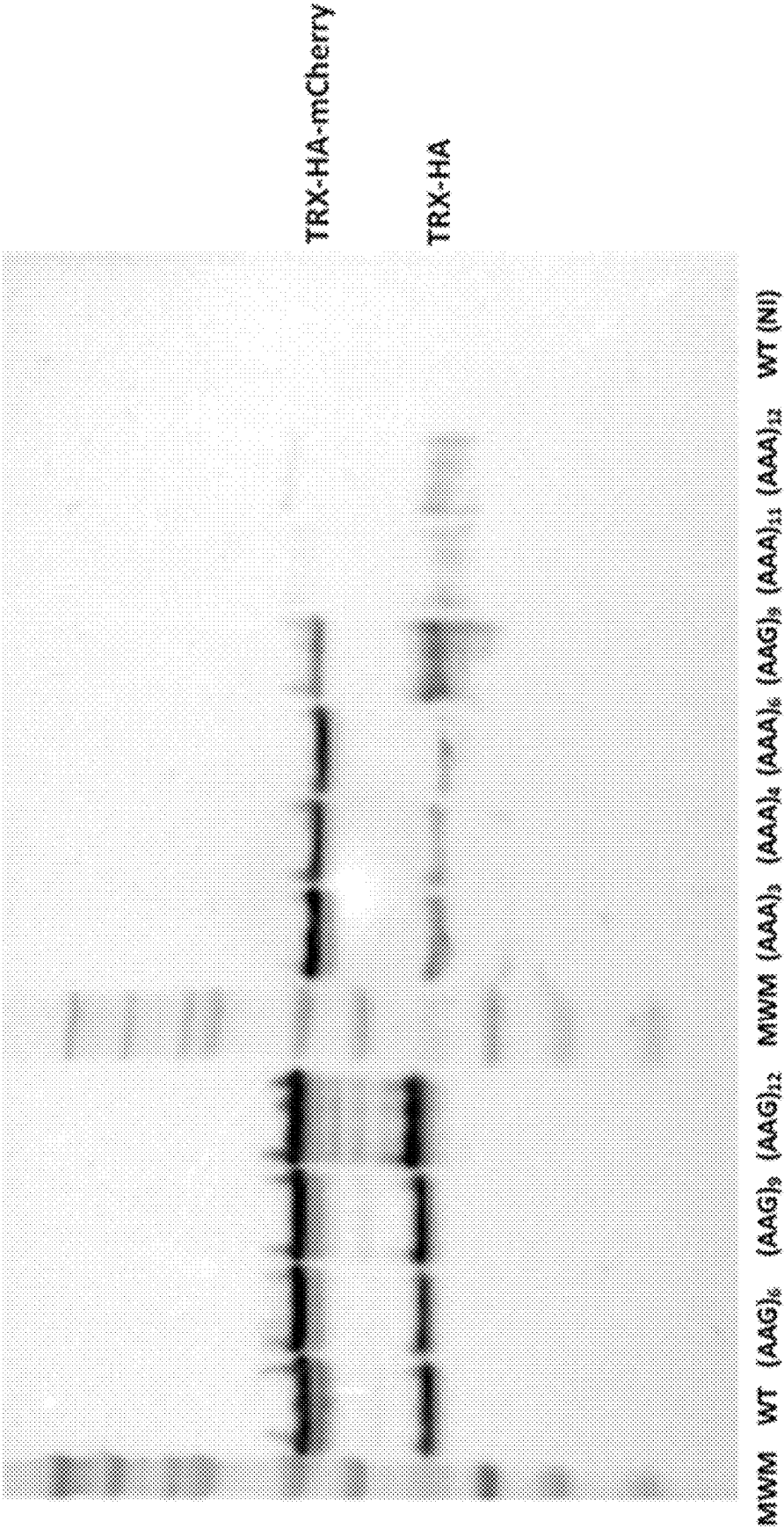
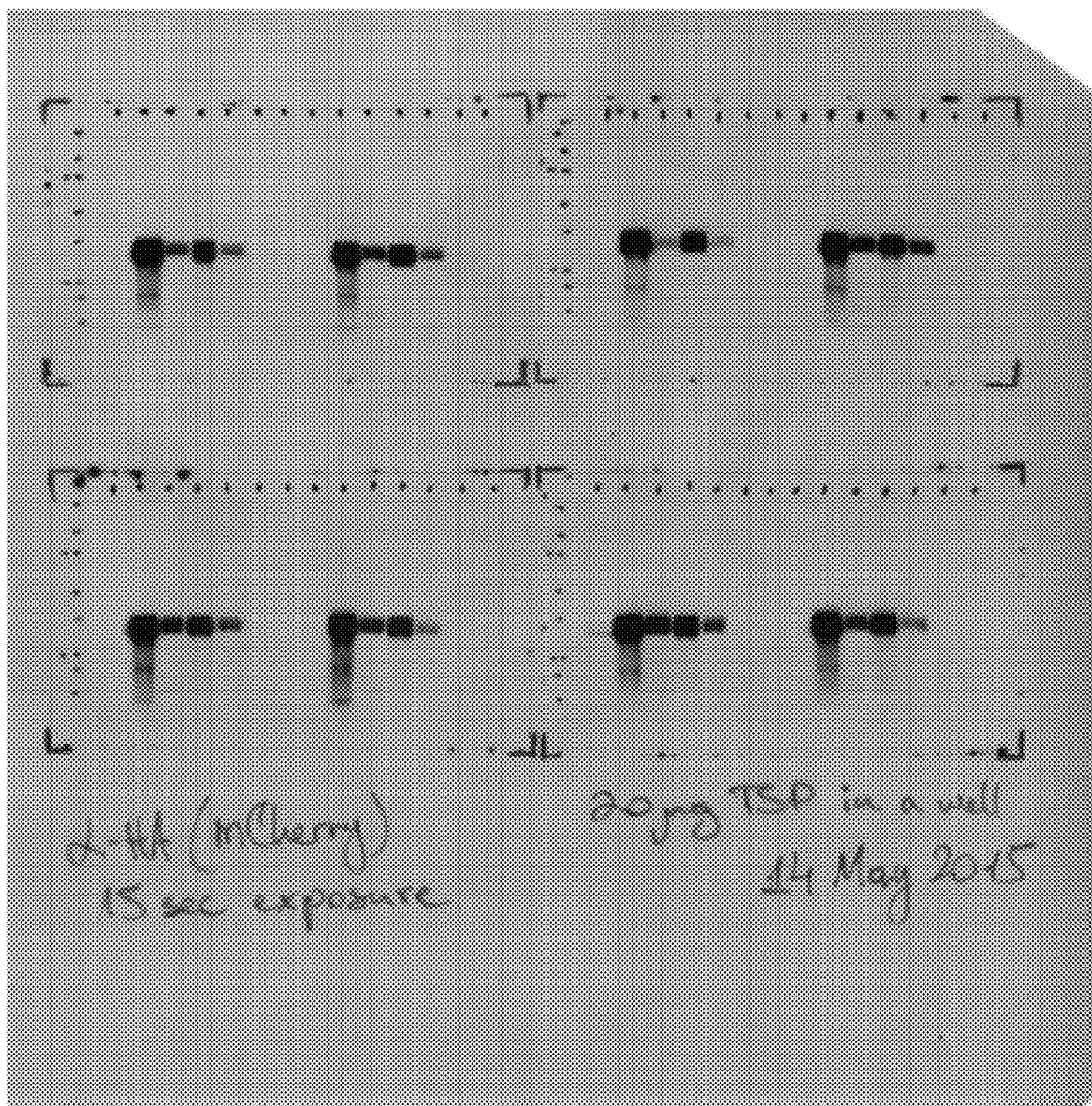
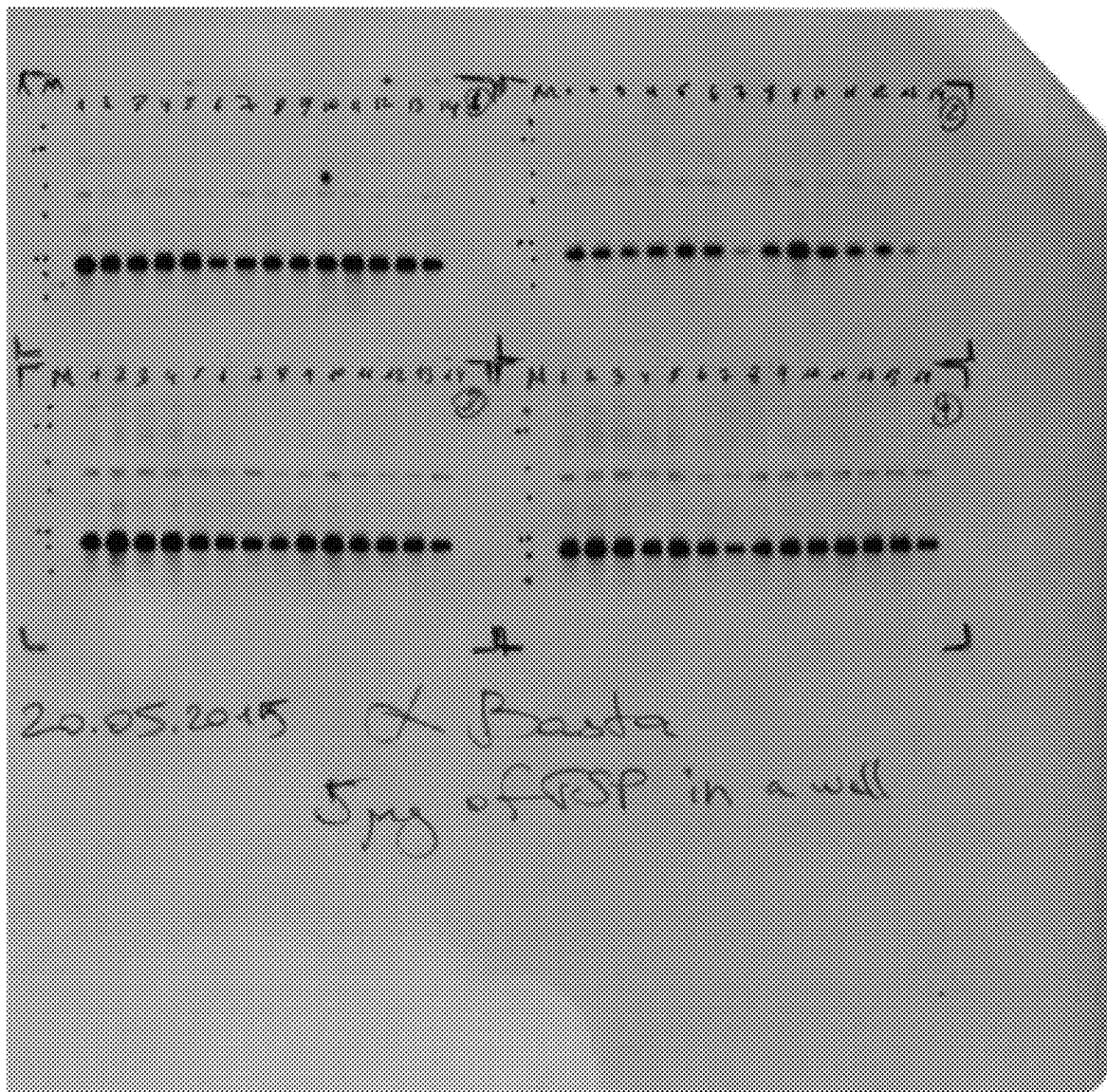
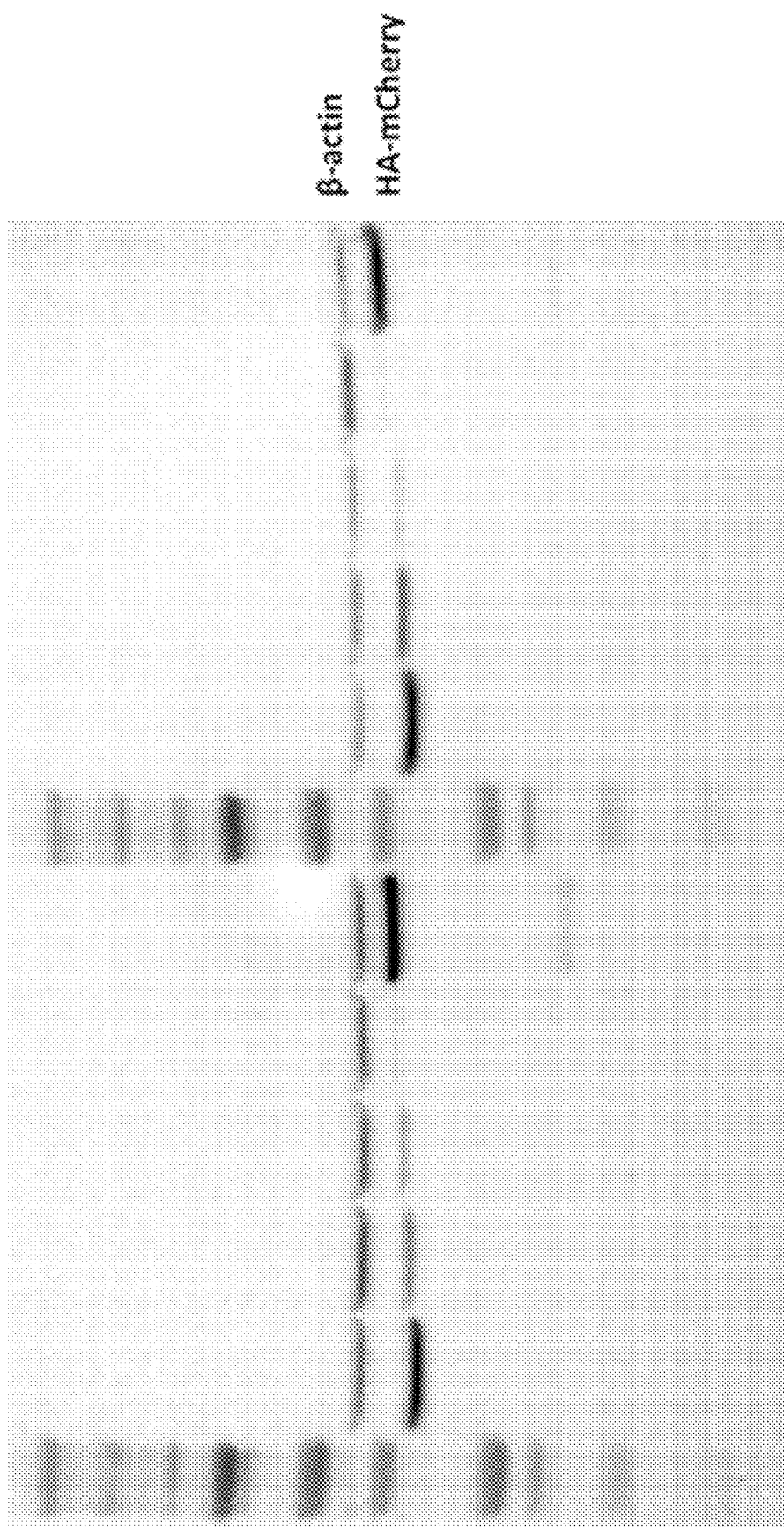
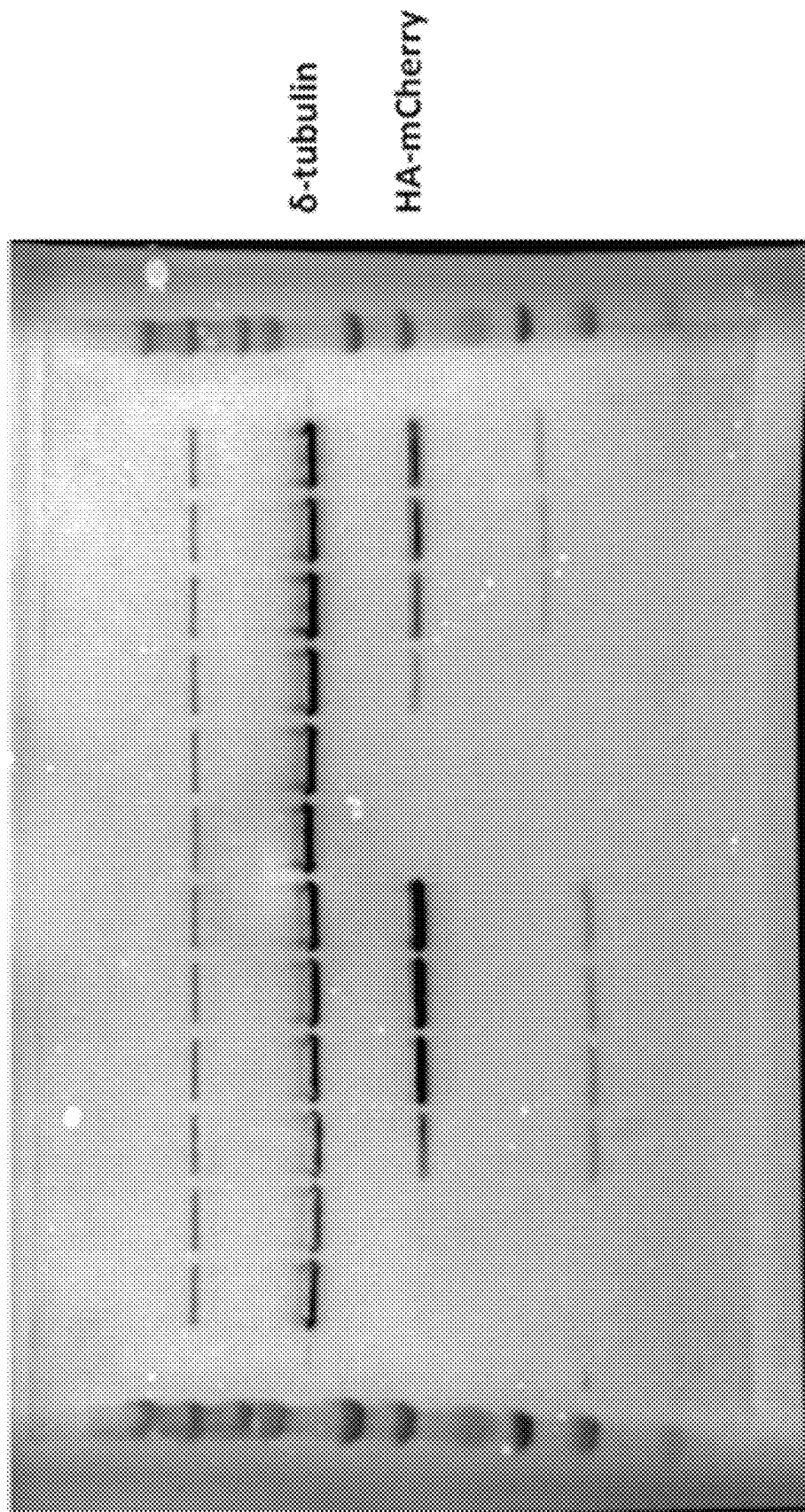


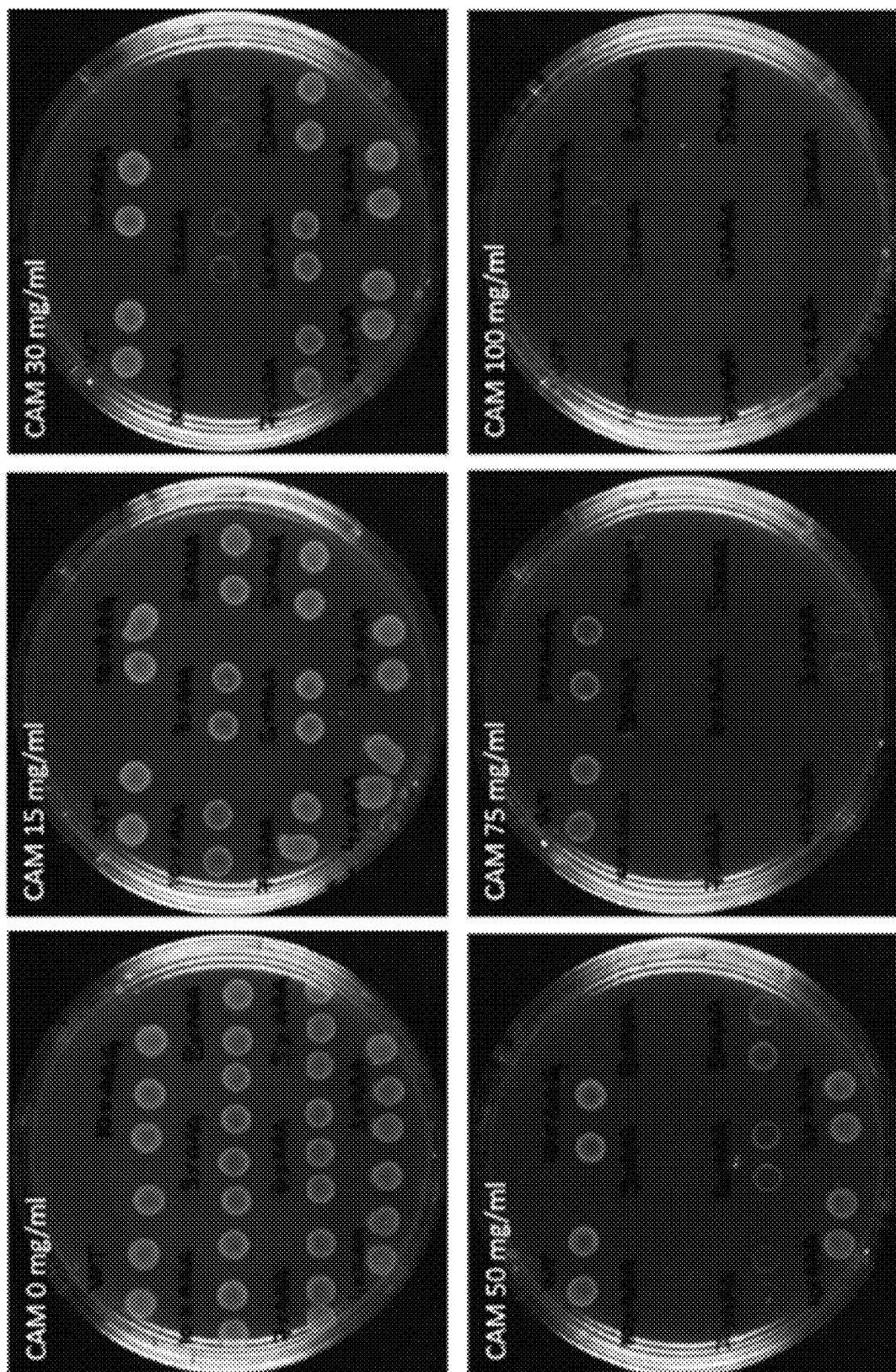
FIG. 47

**FIG. 48A**

**FIG. 48B**

**FIG. 49**

**FIG. 50**

**FIG. 51**

1

**INCORPORATION OF INTERNAL
POLYA-ENCODED POLY-LYSINE
SEQUENCE TAGS AND THEIR VARIATIONS
FOR THE TUNABLE CONTROL OF
PROTEIN SYNTHESIS IN BACTERIAL AND
EUKARYOTIC CELLS**

CROSS REFERENCE TO RELATED
APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/317,791, filed Jan. 14, 2019 which claims the benefit of PCT Application number PCT/US17/41766, filed Jul. 12, 2017, which claims the benefit of U.S. Provisional Application 62/361,307, filed Jul. 12, 2016, U.S. Provisional Application No. 62/427,518, filed Nov. 29, 2016, U.S. Provisional Application No. 62/437,464, filed Dec. 21, 2016, and U.S. Provisional Application No. 62/438,017, filed Dec. 22, 2016, each of the disclosures of which is hereby incorporated by reference in its entirety.

GOVERNMENTAL RIGHTS

This invention was made with government support under T32 GM007067 and RO1 GM112824 awarded by the National Institutes of Health (NIH). The government has certain rights in the invention.

REFERENCE TO A SEQUENCE LISTING

This application contains a Sequence Listing that has been submitted in Extensible Markup Language (.xml) and is hereby incorporated by reference in its entirety. The XML copy, created on Mar. 13, 2023, is named 047563-755867.xml, and is 148.025 bytes in size.

FIELD OF THE INVENTION

The present disclosure relates to modulation of protein expression.

BACKGROUND OF THE INVENTION

Gene expression in cells is a multistep process that involves transcription of genetic material from DNA to RNA and ultimately translation of mRNA into protein. These processes are subject to stringent control at all levels. Translational regulation generally controls the amount of protein generated from a given mRNA. While a majority of translational regulation mechanisms target the recruitment of ribosomes to the initiation codon, the protein synthesis machinery can also modulate translation, elongation, and termination (Dinman and Berry (2007) Cold Spring Harb. Monogr. Arch.; Hershey et al. (2012) Cold Spring Harb. Perspect. Biol. 4).

Pausing during the translational cycle—so-called ribosome stalling—is one mechanism by which the level of translation elongation can be regulated. Ribosome stalling is recognized by components of mRNA surveillance pathways, no-go decay (NGD) and non-stop decay (NSD), resulting in endonucleolytic cleavage of the stalled mRNA, ribosome rescue and proteolytic degradation of incomplete protein products (Shoemaker and Green (2012) Nat. Struct. Mol. Biol. 19, 594-601). NGD and NSD act on aberrant mRNAs that trigger translational arrest, as observed with damaged bases, stable stem-loop structures (Doma and Parker (2006) Nature 440, 561-564), rare codons (Letzring et al. (2010)

2

RNA N. Y. N. 16, 2516-2528) or mRNAs lacking stop codons (non-stop mRNAs) (Dimitrova et al. (2009) J. Biol. Chem. 284, 10343-10352). However, these mechanisms also act on more specific types of translational pauses, such as runs of codons that encode consecutive basic amino acids (Kuroha et al. (2010) EMBO Rep. 11, 956-961; Brandman et al. (2012) Cell 151, 1042-1054). It is thought that polybasic runs, as well as translation of the poly(A) tail in the case of non-stop mRNAs, cause ribosome stalling through interaction of the positively charged peptide with the negatively charged ribosome exit channel (Lu and Deutsch (2008) J. Mol. Biol. 384, 73-86). Presumably, the strength of the stall is dependent on the length and composition of the polybasic stretch, and thus the impact on overall protein expression might vary (Shoemaker and Green (2012) Nat. Struct. Mol. Biol. 19, 594-601). Given this logic, it seems plausible that such an amino acid motif may act as a gene regulatory element that would define the amount of protein translated and the stability of the mRNA. For example, structural and biophysical differences between lysine and arginine residues as well as potential mRNA sequence involvement could act to further modulate this process.

Most studies investigating the effects of polybasic sequences during translation have used reporter sequences in *E. coli* (Koutmou et al. (2015) eLIFE 10.7554/eLife.05534), yeast (Brandman et al. (2012) Cell 151, 1042-1054; Tsuboi et al. (2012) Mol. Cell. 46, 518-529) or in vitro rabbit reticulocyte lysate (Lu and Deutsch (2008) J. Mol. Biol. 384, 73-86). However, detailed mechanistic information about the nature of the stall in endogenous targets through genome-wide analyses has not yet been conducted.

SUMMARY OF THE INVENTION

In an aspect, the disclosure provides a method for modulating the level of expression of a polypeptide in a cell, the method comprising modulating the amount of consecutive adenine (A) nucleotides in at least one lysine codon in an open reading frame of a polynucleotide sequence encoding the polypeptide in the cell, thereby modulating the level of expression of the polypeptide in the cell.

In another aspect, the disclosure provides an expression vector comprising: a) a cloning site for inserting at least one polynucleotide sequence encoding a polypeptide to be expressed, and at least one polynucleotide tag sequence comprising at least one AAG lysine codon that increases expression of the at least one polynucleotide sequence when the expression vector is introduced into a cell; or b) a cloning site for inserting at least one polynucleotide sequence encoding a polypeptide to be expressed, and at least one polynucleotide tag sequence comprising at least one AAA lysine codon that decreases expression of the at least one polynucleotide sequence when the expression vector is introduced into a cell.

In another aspect, the disclosure provides an expression vector comprising: a) at least one engineered polynucleotide sequence encoding a polypeptide to be expressed, the at least one engineered polynucleotide sequence comprising at least one engineered synonymous mutation of at least one AAA lysine codon to at least one AAG lysine codon in a coding sequence of the at least one polynucleotide sequence, wherein the synonymous mutation increases expression of the polypeptide to be expressed when the expression vector is introduced into a cell; or b) at least one engineered polynucleotide sequence encoding a polypeptide to be expressed, the at least one engineered polynucleotide sequence comprising at least one engineered synonymous

3

mutation of at least one AAG lysine codon to at least one AAA lysine codon in a coding sequence of the at least one polynucleotide sequence, wherein the synonymous mutation decreases expression of the polypeptide to be expressed when the expression vector is introduced into a cell.

In yet another aspect, the disclosure provides a method of decreasing translation of a protein in a cell, by increasing the quantity of consecutive adenine nucleotides in an open reading frame (ORF) or an untranslated region (UTR) adjacent to the ORF in genomic DNA (gDNA). The gDNA may be modified by using a clustered regularly interspaced short palindromic repeats (CRISPR) enzyme system, zinc-finger nuclease (ZFN), or transcription activator-like effector nuclease (TALEN).

BRIEF DESCRIPTION OF THE FIGURES

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

FIG. 1A and FIG. 1B show the distribution of polyarginine (FIG. 1A) and polylysine (FIG. 1B) runs of different length in several organisms. Abundance is normalized to the number of residues of certain kind in all protein isoforms (see Materials and Methods, Example 1).

FIG. 2A, FIG. 2B, FIG. 2C, and FIG. 2D show a cartoon of reporter constructs used in electroporation experiments (FIG. 2A), western blot analyses of HA-X-mCherry constructs 48 hours after electroporation (FIG. 2B; HA and β -actin antibodies), normalized protein expression using Licor western blot analyses or in vivo mCherry fluorescence measurement (FIG. 2C; β -actin or fluorescence of co-expressed GFP construct were used for normalization of the data, respectively; each bar represents percentage of wild type mCherry (WT) expression/fluorescence), and normalized RNA levels of HA-X-mCherry constructs (FIG. 2D; neomycin-resistance gene was used for normalization of qRT-PCR data; each bar represents percentage of wild type mCherry (WT) mRNA levels).

FIG. 3A and FIG. 3B show the expression of HA-X-mCherry reporters in Chinese Hamster Ovary cells. Western blot analysis of reporter expression was normalized to β -actin levels (FIG. 3A). qRT-PCR analyses of mRNA abundance was normalized to neomycin resistance gene and presented as fraction of mRNA levels for mCherry construct without insert (FIG. 3B).

FIG. 4A and FIG. 4B show the expression of HA-X-mCherry reporters in *Drosophila* S2 cells. Western blot analysis of reporter expression was normalized to total protein amount (FIG. 4A). qRT-PCR analyses of mRNA abundance was normalized to levels of endogenous GAPDH mRNA and presented as fraction of mRNA levels for mCherry construct without insert (FIG. 4B).

FIG. 5A and FIG. 5B show the expression of HA-X-mCherry reporters from T7-RNA polymerase in vitro transcribed mCherry mRNAs in neonatal human fibroblasts (HDFs). HA-MBP mRNAs were in vitro transcribed and co-electroporated into HDFs as a control for electroporation efficiency and western blot normalization. β -actin was used as a control for the total protein amounts (FIG. 5A). Each lane was subjected to Bio-Rad quantification analyses to determine the levels of expression shown on the graph below (FIG. 5B).

FIG. 6A, FIG. 6B, and FIG. 6C show the differential stability of electroporated mRNAs from HA-X-mCherry reporters is translation dependent. In vitro transcribed

4

mCherry mRNAs were electroporated in HDFs. Protein (FIG. 6A) and mRNA (FIG. 6B) levels were assessed by western blot analyses or qRT-PCR. HA-(AAA)12-mCherry construct shows significant reduction in protein levels as well as in mRNA stability. Addition of translation initiation inhibitor, harringtonine, completely abolishes effect on mRNA stability (FIG. 6C).

FIG. 7A, FIG. 7B, and FIG. 7C show that insertion of polylysine mCherry constructs in the coding sequence results in the same protein reduction and decreased mRNA stability. FIG. 7A shows the scheme of assayed mCherry constructs. Thioredoxin (Trx) fusion protein was used instead of insertion of polylysine run in the middle of mCherry gene. Positions of 12 lysine (36As) insertions and HA-tag in the constructs are labeled with blue and gray boxes, respectively. Numbers above reporter indicate distance of 36As insertion from the first nucleotide in the coding sequence. FIG. 7B shows that protein expression was monitored by western blot analyses using HA and beta-actin antibody. FIG. 7C shows that qRT-PCR analyses of mRNA abundance was normalized to neomycin resistance gene and presented as fraction of mRNA levels for WT mCherry construct without insert.

FIG. 8A, FIG. 8B, and FIG. 8C show the expression of HA-tagged hemoglobin (delta chain; HBD) constructs with natural introns in HDF cells. FIG. 8A shows a scheme of HBD gene with position of poly lysine and stop codon insertions. Position and length of introns as well as exons in HBD constructs are indicated. FIG. 8B shows western blot analysis of HA-HBD construct expression normalized to β -actin levels. FIG. 8C shows qRT-PCR analyses of mRNA abundance normalized to neomycin resistance gene and presented as fraction of mRNA levels for WT HBD construct without insert.

FIG. 9 shows a comparison of usage of AAA in single, double and triple lysine runs across several organisms. Expected values (black bars) are based on Kazusa database, while observed (yellow bars) are calculated from all isoforms of proteins available in NCBI RefSeq database.

FIG. 10 shows observed codon usage in all isoforms of human proteins vs. expected (based on the proportions 0.44 to 0.56, AAA to AAG for all lysines) in the tracks of four consecutive lysines. From top to bottom along the y-axis, the sequences correspond to SEQ ID NOs 71-86 respectively.

FIG. 11 shows codon distribution in four-lysine tracks in different organisms. All protein isoforms sequences and sequences of corresponding mRNAs were taken into account. The script checks all tracks of four consecutive lysines, even when overlapping (if there is a track of five lysines, it will report two nucleotide strings of length 12). From left to right along the x-axis, the sequences correspond to SEQ ID Nos 87-102 respectively.

FIG. 12A, FIG. 12B, FIG. 12C, FIG. 12D, and FIG. 12E shows HA-(A9-A13)-mCherry construct sequences and protein expression. FIG. 12A shows the occupancy of ribosomal footprints for regions around different codon combinations for four lysine tracks. All combinations of one, two, three and four AAG codons per group are shown. Data for four AAA codons is not shown because only a single gene has such a sequence. The upper and lower "hinges" correspond to the first and third quartiles (the 25th and 75th percentiles). The upper and lower whiskers extend from hinges up or down at maximum of 1.5*IQR of the respective hinge. FIG. 12B shows the sequences of HA-(A9-A13)-mCherry constructs used in electroporation experiments. The WT nucleotide and amino acid sequences shown are SEQ ID NO: 71 and 72, respectively. The A9 nucleotide and amino acid

5

sequences shown are SEQ ID NO: 10 and 73, respectively. The A10 nucleotide and amino acid sequences shown are SEQ ID NO: 11 and 73, respectively. The A11 nucleotide and amino acid sequences shown are SEQ ID NO: 12 and 74, respectively. The A12 nucleotide and amino acid sequences shown are SEQ ID NO: 13 and 75, respectively. The A14 nucleotide and amino acid sequences shown are SEQ ID NO: 15 and 76, respectively. FIG. 12C shows western blot analyses of HA-(A9-A13)-mCherry constructs 48 hours after electroporation (HA and β -actin antibodies). FIG. 12D shows normalized protein expression using Licor western blot analyses or in vivo mCherry fluorescence measurement. β -actin or fluorescence of co-expressed GFP construct were used for normalization of the data, respectively. Each bar represents percentage of wild type mCherry (WT) expression/fluorescence. FIG. 12E shows normalized RNA levels of HA-X-mCherry constructs. Neomycin resistance gene was used for normalization of qRT-PCR data. Each bar represents percentage of wild type mCherry (WT) mRNA levels.

FIG. 13A and FIG. 13B show the occupancy of ribosomal footprints from three different data sets: FIG. 13A shows the region around polyA tracks and FIG. 13B shows the region around four arginine tracks, all codons combinations together. The upper and lower "hinges" correspond to the first and third quartiles (the 25th and 75th percentiles). The upper and lower whiskers extend from hinges at 1.5*IQR of the respective hinge.

FIG. 14A, FIG. 14B, FIG. 14C, and FIG. 14D show the sequences of polylysine runs from human genes incorporated into HA-X-mCherry constructs (FIG. 14A; continuous runs of lysine residues are labeled; number of lysine residues and ratio of AAG and AAA codons for each construct are indicated), normalized protein expression using in vivo mCherry reporter fluorescence (FIG. 14B; fluorescence of co-transfected GFP was used to normalize the data; each bar represents percentage of wild type mCherry (WT) expression/fluorescence), normalized RNA levels of HA-X-mCherry constructs (FIG. 14C; neomycin resistance gene was used for normalization of qRT-PCR data; each bar represents percentage of wild type mCherry (WT) mRNA levels), and smoothed Gaussian kernel density estimate of positions of polyA tracks along the gene (FIG. 14D; position of polyA segment is expressed as a ratio between number of first residue of polyA track and length of a gene). In FIG. 14A, SLU7 is SEQ ID NO: 15; MTDH is SEQ ID NO: 16; NOP58 is SEQ ID NO: 17; ZCRB1 is SEQ ID NO: 18; and RASAL2 is SEQ ID NO: 19.

FIG. 15 shows the sequence conservation of RAS Activating-Like protein 2 gene (RASAL2) at DNA and protein sequence. Polylysine sequence and nucleotides forming polyA track are indicated in red and bold letters, respectively. The amino acid sequence shown is SEQ ID NO: 111. The human, mouse, and hamster sequence shown is SEQ ID NO: 112. The pig sequence shown is SEQ ID NO: 113. The chicken sequence shown is SEQ ID NO: 114. The zebrafish sequence shown is SEQ ID NO: 114. The frog sequence shown is SEQ ID NO: 116.

FIG. 16A, FIG. 16B, FIG. 16C, FIG. 16D, FIG. 16E, and FIG. 16F show the scheme of constructs with ZCRB1 gene polyA tracks used for analyses of synonymous mutations (FIG. 16A), western blot analyses and normalized protein expression of ZCRB1 reporter constructs with synonymous mutations (FIG. 16B; HA and β -actin antibodies; each bar represents percentage of wild type ZCRB1-mCherry (WT) expression), normalized RNA levels of ZCRB1 reporter constructs with synonymous mutations (FIG. 16C; neomy-

6

cin resistance gene was used for normalization of qRT-PCR data; each bar represents percentage of wild type ZCRB1-mCherry construct (WT) mRNA levels), the scheme of full-length HA-tagged ZCRB1 gene constructs (FIG. 16D; position and mutations in polyA tracks are indicated), western blot analysis and normalized protein expression of ZCRB1 gene constructs with synonymous mutations (FIG. 16E; each bar represents percentage of wild type HA-ZCRB1 (WT) expression, and normalized RNA levels of ZCRB1 gene constructs (FIG. 16F; neomycin resistance gene was used for normalization of qRT-PCR data). In FIG. 16A, the nucleotide sequence shown for ZCRB1 WT corresponds to nucleotides 3-30 of SEQ ID NO: 20; the nucleotide sequence shown for ZCRB1 411G>A corresponds to nucleotides 3-30 of SEQ ID NO: 21; the nucleotide sequence shown for ZCRB1 408A>G;417A>G corresponds to nucleotides 3-30 of SEQ ID NO: 22; and the amino acid sequence corresponds to residues 2-11 of SEQ ID NO: 77. In FIG. 16D, ZCRB1 WT is SEQ ID NO: 20; ZCRB1 411G>A is SEQ ID NO: 21; ZCRB1 408A>G; 417A>G is SEQ ID NO: 22; and the amino acid sequence is SEQ ID NO: 77.

FIG. 17A, FIG. 17B, and FIG. 17C show synonymous mutations in mCherry reporter with metadherin (MTDH, Lyric(Lyr)) polyA track. FIG. 17A shows a scheme of reporter sequences with G>A and A>G synonymous mutations. The amino acid sequence shown is SEQ ID NO: 82, and the nucleotide sequences shown are SEQ ID NO: 78-81, respectively. FIG. 17B shows western blot analyses of reporter constructs with synonymous mutations. FIG. 17C shows normalized mRNA levels for reporter sequences with wild type MTDH polyA-track (Lwt) and corresponding mutants. mRNA levels are represented as fractions of wild type mCherry levels.

FIG. 18A, FIG. 18B, and FIG. 18C show synonymous mutations in mCherry reporter with RASAL2 polyA track. FIG. 18A shows a scheme of reporter sequences with G>A and A>G synonymous mutations. The amino acid sequence is SEQ ID NO: 83. The RASAL2 WT nucleic acid sequence corresponds to nucleotide 3032 of SEQ ID NO: 23. The RASAL2 G>A nucleic acid sequence corresponds to nucleotides 3-32 of SEQ ID NO: 24. The RASAL2 A>G nucleic acid sequence corresponds to nucleotides 3-32 of SEQ ID NO: 25. The RASAL2 A>G (3) nucleic acid sequence corresponds to nucleotides 3-32 of SEQ ID NO: 26. FIG. 18B shows western blot analyses of reporter constructs with synonymous mutations. FIG. 18C shows normalized mRNA levels for reporter sequences with wild type RASAL2 polyA-track (Lwt) and corresponding mutants. mRNA levels are represented as fractions of wild type mCherry levels.

FIG. 19A, FIG. 19B, FIG. 19C, and FIG. 19D show expression analysis of N-terminally HA- and C-terminally GFP-tagged ZCRB1 gene and its synonymous mutants in HDF cells using Evos-FL microscopy. Cell images were taken 24 hours post electroporation using same optical settings. Cell nuclei were made visible using Hoechst 33342 dye. Images of HDF cells expressing double-tagged ZCRB1 wild type (WT) protein (FIG. 19A) ZCRB1 K137K:411 G>A (FIG. 19B) and ZCRB1 K136K:408 A>G; K139K:417 A>G (FIG. 19C) mutants. Images for each channel (trans, DAPI and GFP) were taken separately and overlay image was composed using EVOS FL digital software. FIG. 19D shows western blot analyses of HA-ZCRB1-GFP proteins from HDF cells using HA-antibody. Western blot analyses were normalized using beta-actin levels as loading controls.

FIG. 20A, FIG. 20B, FIG. 20C, and FIG. 20D show a scheme using luciferase constructs and luciferase expres-

sion. FIG. 20A shows immunoprecipitation of HA-ZCRB gene constructs using anti-HA magnetic beads. ZCRB1 WT, synonymous (single 411 G>A or double 408 A>G; 417 A>G), non-sense (385 G>T, insertion of stop codon prior poly(A) track), deletion (423ΔA, equivalent to +1 frame-shift) or insertion (423 A>AA, equivalent to -1 frame-shift) mutant constructs are labelled respectively. FIG. 20B shows a scheme of luciferase constructs used to estimate frame-shifting potential for ZCRB1 WT and 411 G>A mutant polyA tracks. FIG. 20C shows luciferase levels (activity) from -1, "zero" and +1 frame constructs of wild type and G>A mutant ZCRB1 polyA track are compared. Bars represent normalized ratio of ZCRB1 G>A and ZCRB1 WT poly(A) tracks elucidates changes in the levels of luciferase expression in all three frames. FIG. 20D shows a model for function of poly(A) tracks in human genes. Poly(A)-tracks lead to three possible scenarios: Frameshifting consolidated with NMD which results in reduced output of wild type protein; Frameshifting with synthesis of both out of frame and wild type protein; and non-resolved stalling consolidated by endonucleolytic cleavage of mRNA and reduction in wild type protein levels, as in NGD pathway. Scheme for translation of mRNAs without poly(A) tracks is shown for comparison.

FIG. 21A and FIG. 21B show that the introduction of COSMIC database reported synonymous mutation K447K (1341 G>A) in full length recombinant MTDH gene. FIG. 21A shows the sequence of the wild type and K447K (G>A) mutant of MTDH gene. The amino acid sequence shown is SEQ ID NO: 82, and the nucleotide sequences shown are SEQ ID NO: 78 and 90, respectively. FIG. 21B shows western blot analyses of HA-tagged WT and K474K mutant MTDH proteins. Major additional protein product corresponds to frame-shifted MTDH protein products (-1 and +1 FS) created by insertion or deletion of one nucleotide following the last nucleotide in polyA-track.

FIG. 22A and FIG. 22B show the frame-shifting efficiency of polyA tracks from ZCRB1 WT (FIG. 22A) and ZCRB G>A mutant (FIG. 22B) measured by luciferase activity. Values for -1 and +1 frame-shifts (FS) for WT and mutant polyA track are presented as fractions of luciferase activity coming from expression from zero frame construct of WT (FIG. 22A) or mutant G>A sequence (FIG. 22B).

FIG. 23 shows the proportion of mutation types in polyA segments vs all mutation types. Data has been generated from COSMIC database. There is a dramatic shift in the distribution of mutations in polyA segments from substitutions (all COSMIC data) to frameshifts (polyA segments).

FIG. 24 shows the normalized distribution of lengths for polyA regions identified as 12 As allowing for one mismatch up to length 19 in human transcripts.

FIG. 25A, FIG. 25B, and FIG. 25C show design and mechanism of polyA track tag regulated gene expression. FIG. 25A shows scheme of inserted polyA tracks in the reporter genes used in this study. Hemagglutinin (HA) tag (gray) and polyA tracks (red) were introduced in the coding region of the reporter genes next to the start AUG codon. Exon boundaries as well as termination codon (Stop) are indicated. FIG. 25B shows proposed correlation between gene products levels, mRNA and protein, and the length of inserted polyA track tags. The reduction in levels of both reporter protein and mRNA is dependent on increasing length of consecutive adenine nucleotides in the coding sequence. FIG. 25C shows scheme of translation of eukaryotic reporter mRNA with or without inserted polyA tracks. The length of inserted polyA track tag determines the protein output of the regulated reporter gene as indicated by the

number of globular protein structures. Features of the eukaryotic mRNAs (m7GpppG—cap, AUG—start codon, Stop—termination codon and polyA tail), as well as HA-tag, position of the polyA track tag, ribosome and nascent polypeptide chain are illustrated in the scheme.

FIG. 26A, FIG. 26B, FIG. 26C, FIG. 26D, FIG. 26E, and FIG. 26F show regulation of reporter gene by polyA tracks in the single cell prokaryotic and eukaryotic organisms. FIG. 26A shows percentage of mCherry fluorescence of tested LysAAG ((AAG)6-12) and LysAAA((AAA)3-12) insertion constructs compared to wild type fluorescence (WT, no insertion construct) 2 hours after promoter induction with 0.1% arabinose (w/v) in the media. mCherry fluorescence was assayed at excitation wavelength of 475±9 nm and emission was detected at 620±9 nm. Error bars indicate mean mCherry fluorescence values±standard deviation for three individual *E. coli* colonies for each construct. Background levels of mCherry expression can be estimated from the fluorescence of the wild type construct that was not induced with the addition of arabinose in the media (WT (NI)). FIG. 26B shows set of constructs analyzed for mCherry fluorescence was additionally assessed for protein expression levels by Western blot analysis. Equal amounts of *E. coli* cell lysates with Thioredoxin(Trx) fusion proteins were used for analysis. Fusion proteins were detected using HA-tag specific antibody. Positions of the fusion protein (Trx-HA-mCherry) and sizes of molecular weight markers (MWM) are indicated. FIG. 26C shows representative differential interference contrast microscopy (left panel) and the corresponding fluorescence image (right panel -25 msec exposure) of a *T. thermophila* cell expressing the wild type (WT) MLP1-HA-YFP fusion. Arrowheads denote the position of the macronucleus. FIG. 26D shows MLP1-HA-YFP accumulation within macronuclei of live *T. thermophila* cells expressing an allelic series of fusion proteins WT, (AAA)6-12, and (AAG)12, was visualized by epifluorescence microscopy. Different exposures times are indicated on the right to demonstrate the relative accumulation of each variant. FIG. 26E shows western blot analysis was performed with whole cell lysates made from *T. thermophila* cells expressing the MLP-HA-YFP fusion proteins. Protein from equivalent cell numbers was loaded in each lane and detected using YFP specific antibody (top panel) and normalized to the nuclear histone species, histone H3 trimethyl-lysine 4 (H3K4m) (bottom panel). Positions of the full-length fusion protein (YFP), normalization control (H3K4m), and sizes of molecular weight markers (MWM) are indicated. Degradation of excess fusion protein is readily apparent as faster migrating species below the full-length MLP1-HA-YFP. FIG. 30F shows steady state levels of fusion gene constructs measured by qRT-PCR. Relative levels of the mRNA for (AAG)12 and (AAA)6-12 are presented as percentage of the wild type (WT) construct mRNA levels. Error bars represent mean±standard deviation values (n=3).

FIG. 27A, FIG. 27B, FIG. 27C, FIG. 27D, FIG. 27E, and FIG. 27F show regulation of reporter gene by polyA tracks in the eukaryotic tissue cultures. FIG. 27A shows fluorescence images of *N. benthamiana* epidermal cells transiently expressing wild type (WT), (AAG)12 and (AAA)6-12 mCherry constructs. YFP expression was used as a transfection control. (FIG. 27B) Western blot analysis, (FIG. 27C) protein level estimate and (FIG. 27D) mRNA levels for transfected (-) insert control and WT, (AAG)12 and (AAA)6-12 mCherry constructs expressed transiently in *N. benthamiana* epidermal cells. FIG. 27B shows primary HA-tag antibody was used for detection of HA-mCherry constructs.

Phosphinotricin acetyl transferase (BAR) specific antibody was used as a loading and normalization control (FIG. 27C). Levels of mCherry protein from different constructs were derived from detected band intensities normalized for BAR accumulation detected in the same sample. Error bars represent mean values $\pm$ standard error from biological replicates (n=8). FIG. 27D shows mRNA levels for different mCherry constructs were calculated as cycle threshold (Ct) values and normalized to BAR gene mRNA values. Error bars represent mean values $\pm$ standard error from biological replicates (n=3). FIG. 27E shows western blot analysis of transient mCherry constructs expression in HeLa cells. WT, 12LysAAG ((AAG)12) and 6-12LysAAA ((AAA)6-12) mCherry proteins were detected using HA-tag specific primary antibody. β -actin was used as a loading control and was detected using specific antibody. Positions of the fusion protein (HA-mCherry), normalization control (β -actin) and sizes of molecular weight markers (MWM) are indicated. FIG. 27F shows quantification of the mCherry protein levels from detected western blot intensities. Levels of mCherry were normalized to β -actin band intensities and represented as a percentage of the wild type construct values.

FIG. 28A, FIG. 28B, FIG. 28C, and FIG. 28D show PolyA tracks regulate mCherry reporter gene expression in different organs of *D. melanogaster*. FIG. 28A shows diagram of third instar fruit fly larva showing approximate location of salivary glands (SG, blue), central nervous system (CNS, green) and proventriculus (PV, red). Fluorescence imaging of formaldehyde fixed SG (FIG. 28B), CNS (FIG. 28C) and PV (FIG. 28D), dissected from larvae expressing wild type (WT), (AAG)12 and (AAA)6-12 mCherry constructs. mCherry and GFP indicate images acquired by selective fluorescence filter setting. Overlay of mCherry and GFP fluorescence is shown in the merged panel.

FIG. 29A, FIG. 29B, and FIG. 29C show PolyA tracks regulate mCherry reporter expression independently of the promoter strength. FIG. 29A shows western blot analysis of the cell lysates from T-Rex HEK293 stable cell lines expressing doxycycline (Dox) inducible wild type (HA-mCherry) and 12LysAAA insertion construct (HA-(AAA)12-mCherry) from a single locus. Dox concentration in the media was varied from 0 to 0.1 μ g/ml. Constitutively expressed δ -tubulin was used as a loading control and was detected using specific antibody. Positions of the fusion protein (HA-mCherry), normalization control (δ -tubulin) and sizes of molecular weight markers (MWM) are indicated. FIG. 29B shows quantification of the mCherry protein levels from detected western blot intensities. Levels of mCherry were normalized to δ -tubulin band intensities and represented as a percentage of the wild type construct values. Numbers indicate concentration of Dox in the media. FIG. 29C shows steady state mRNA levels of the 12LysAAA insertion construct ((AAA)12) measured by qRT-PCR. Relative levels of the mRNA for (AAA)12 are presented as percentage of the wild type (WT) construct mRNA levels. Error bars represent mean $\pm$ standard deviation values (n=3). Numbers indicate final concentration of Dox in the media.

FIG. 30A and FIG. 30B show regulation of drug resistance and metabolic survival by insertion of polyA track tags in genes from *E. coli* and *S. cerevisiae*. FIG. 30A shows survival of *E. coli* cells expressing wildtype (WT), 10LysAAG ((AAG)10) and 3-10LysAAA ((AAA)3-10 chloramphenicol acetyltransferase (CAT) constructs on chloramphenicol (CAM) selective media. Pulse induced *E. coli* cells, expressing different CAT constructs, were plated

on selective antibiotic plates with varying amounts of CAM in the media (0-100 mg/ml). Two independent clones were assessed for each construct. *E. coli* colonies were imaged 16 hours after plating. FIG. 30B shows assays for ADE1 gene regulation by polyA tracks ((AAA)6-12). Ability of *S. cerevisiae* ade1 Δ cells to produce sufficient levels of functional Ade1 protein were assayed by reintroduction of single copy vector with wild type (WT), 12LysAAG ((AAG)12) and 6-12LysAAA ((AAA)6-12) Ade1 construct. Empty vector (EV) served as a negative control. Yeast colonies show differential red coloration, on the selective SD-Ura media, which is proportional to the activity of Ade1 protein. Adenine dropout media (SD-Ade) selects for yeast cells expressing sufficient amounts of active Ade1 protein. Dilutions of the yeast cultures showing relative survival and growth are indicated.

FIG. 31 shows a diagram of a mCherry expression construct used in *E. coli*. Position of the inducible arabinose promoter (pBAD), Thioredoxin (Trx), double HA-tag (HA), insertion sequence (X) and fluorescent reporter (mCherry) are indicated. Examples of WT, LysAAG and LysAAA insertions and the resulting protein and DNA sequences of reporter constructs are shown.

FIG. 32 shows a diagram of a mCherry expression construct used in *T. thermophila*. Position of the inducible metallothionein promoter (MTT1), Macronucleus-Localized Protein 1 (MLP1), double HA-tag (HA) and fluorescent reporter (eYFP) are indicated. Red box designates the position of in frame inserted polyA tracks and 12 LysAAG sequences. WT construct contains no insertions at this position.

FIG. 33 shows diagram of mCherry and YFP expression constructs used in *N. benthamiana*. Position of the mannopine synthase promoter (MAS-P), the cauliflower mosaic virus 35S promoter and its upstream enhancer (35S), phosphinotricin acetyl transferase (BAR, herbicide resistance gene for selection of transgenic plants), double HA-tag (HA) and fluorescent reporters, mCherry and YFP, are indicated. Red box designates the position of in frame inserted polyA tracks and 12 LysAAG sequences. WT construct contains no insertions at this position.

FIG. 34 shows diagram of mCherry and GFP expression constructs used in *D. melanogaster*. Position of the heat shock protein 70 promoter (hsp70), upstream activating sequences (UAS, GAL4 DNA binding sequence), double HA-tag (HA) and fluorescent reporters, mCherry and GFP, are indicated. Red box designates the position of in frame inserted polyA tracks and 12 LysAAG sequences. WT construct contains no insertions at this position. Tub-GAL4 driver line used for the expression of mCherry and GFP was derived from BSC42734.

FIG. 35A, FIG. 35B, and FIG. 35C show quantification of mCherry fluorescence in *D. melanogaster* Salivary Glands (SG), Central Nervous System (CNS), and Proventriculus (PV). Normalized mCherry fluorescence intensity of WT, 12 LysAAG and 6-12 LysAAA in *D. melanogaster* SG (FIG. 35A), CNS (FIG. 35B) and PV (FIG. 35C). GFP fluorescence was excited by a 488 nm laser and mCherry by a 561 nm laser. All microscopy parameters were constant between tissues, except master gain which was set as follows: SG—488 nm laser master gain was 509, 561 nm laser was 560, CNS—488 nm laser master gain was 625, 561 nm laser was 720, and PV—488 nm laser master gain was 618, 561 nm laser was 616. Fluorescence intensity was measured as an average intensity from each tissue image (Zen 9 software) and plotted as a ratio of mCherry to GFP intensity. Box plots indicate median intensity ratio per construct (n $\geq$ 5).

11

FIG. 36 shows western blot analysis and quantification of mCherry protein from third instar *D. melanogaster* larvae. Five third instar fruit fly larvae expressing either WT, 12 LysAAG or 6-12LysAAA constructs were frozen, homogenized, sonicated and lysed in SDS sample buffer. Equal amounts of lysate were analyzed by SDS-PAGE followed by western blot transfer. mCherry protein was detected using HA-tag antibody (Santa Cruz Biotechnology Inc) and relative amounts were calculated based on the GFP expression control. GFP protein was detected using GFP specific antibody (Clontech). Relative amounts of mCherry expression are shown as percentage of WT-mCherry expression.

FIG. 37 shows mCherry mRNA abundance in whole third instar *D. melanogaster* larvae measured by RT-qPCR. Five third instar fruit fly larvae expressing either WT, 12 LysAAG or 6-12LysAAA constructs were frozen, homogenized, and lysed in RiboZol (Ambion). mCherry RNA abundance was measured by RT-qPCR. Relative amounts of mCherry mRNA were normalized to levels of Elongation Factor 1 alpha-100 (EF1) and shown as percentage of WT-mCherry levels. Error bars indicate mean±standard.

FIG. 38A and FIG. 38B show a diagram of mCherry expression constructs and their transcriptional activation in Flp-InTM T-RexTM 293 stable cell lines. FIG. 38A shows scheme of genetic loci expressing WT (HA-mCherry) and 12LysAAA insertion construct (HA-12LysAAA-mCherry) in stable Flp-InTM T-RexTM 293 cell lines. Position of the SV40 promoter (SV40), hygromycin B phosphotransferase (Hygromycin), antibiotic resistance gene for selection of single insertion constructs, doxycycline-inducible CMV promoter (CMV 2× TetO2), double HA-tag (HA) and fluorescent reporter (mCherry) are indicated. Red box designates the position of in frame inserted 12 LysAAA sequence. WT construct contains no insertions at this position. FIG. 38B shows relative folds of transcriptional activation for WT and 12LysAAA mCherry loci were calculated from mRNA levels for each construct at different levels of induction by doxycycline (Dox, 0.001-0.1 µg/ml). RT-qPCR data for each construct was normalized to the mRNA levels of constitutively expressed hygromycin B phosphotransferase gene. Fold induction was calculated over the non-induced samples for each construct separately. Error bars indicate mean±standard deviation.

FIG. 39 shows diagram of human beta globin delta chain (HBD) expression constructs in Flp-InTM T-RexTM 293 stable cell lines. Scheme of genetic loci expressing WT-HBD and HBD-6LysAAA constructs in stable Flp-InTM T-RexTM 293 cell lines. Position of the SV40 promoter (SV40), hygromycin B phosphotransferase (Hygromycin), antibiotic resistance gene for selection of single insertion constructs, doxycycline-inducible CMV promoter (CMV 2× TetO2), double HA-tag (HA) and HBD reporter (mCherry) are indicated. Red box designates the position of in frame inserted 6 LysAAA sequence. WT construct contains no insertions at this position.

FIG. 40 shows western blot analysis of HBD protein abundance during Dox induction. Western blot analysis of the cell lysates from Flp-InTM T-RexTM 293 stable cell lines expressing doxycycline (Dox) inducible wild type (WT-HBD) and 6 LysAAA insertion construct (HBD-6LysAAA) from a single locus. Dox concentration in the media was varied from 0.0 to 1 µg/ml. Constitutively expressed β-actin (Actin) was used as a loading control and was detected using specific antibody. Positions of the HA-tagged HBD protein (HA-HBD), normalization control (β-actin) and molecular weight marker (MWM) are indicated.

12

FIG. 41 shows ratio of WT-HBD and HBD-6LysAAA mRNA abundance from Flp-InTM T-RexTM 293 stable cell lines. Steady state mRNA levels of the 6LysAAA insertion construct (HBD-6LysAAA) measured by qRT-PCR. Relative levels of the mRNA for HBD-6LysAAA are presented as percentage of the wild type HBD (WT-HBD) construct mRNA levels. Error bars represent mean±standard deviation values (n=3). Numbers indicate final concentration of Dox in the media.

FIG. 42 shows diagram of chloramphenicol acetyltransferase (CAT) expression construct used in *E. coli*. Position of the inducible arabinose promoter (pBAD), Thioredoxin (Trx), double HA-tag (HA), insertion sequences (10 LysAAG and polyA track (3-10LysAAA) and reporter gene (CAT) are indicated.

FIG. 43 shows expression of arabinose-inducible fusion Thioredoxin-HA-CAT constructs in *E. coli*. Western blot analysis of the lysates from *E. coli* cells expressing arabinose-inducible wild type Trx-HA-CAT (WT), 10 LysAAG (AAG10) and 3-10 LysAAA insertion constructs (AAA3-10). Cells were induced with 0.5% (w/v) of arabinose in the media for 30 minutes. Equal number of cells were harvested, lysed in SDS sample buffer and loaded on SDS-PAGE gel. Trx-HA-CAT fusion proteins were detected using HA-tag specific antibody. Positions of the Trx-HA-CAT proteins and molecular weight marker (MWM) are indicated. NI represents negative control; WT construct without induction.

FIG. 44A and FIG. 44B shows a diagram of N-succinyl-5-aminoimidazole-4-carboxamide ribotide synthetase (ADE1) construct and expression of ADE1 constructs in *S. cerevisiae*. FIG. 44A shows position of the orotidine 5'-phosphate decarboxylase promoter and gene (ura3 and URA3, respectively), ADE1 promoter and gene (ade1 and ADE1, respectively) and FLAG-tag (FLAG) are indicated. Red box (insert) designates the position of in frame inserted 12 LysAAG and 6-12 LysAAA sequences. WT construct contains no insertions at this position. FIG. 44B shows dot blot of yeast cell lysates expressing FLAG-tagged WT, 12 LysAAG and 6-12 LysAAA ADE1 protein from endogenous ade1 promoter. ADE1 protein was detected using anti-Flag (Sigma) antibody. 20 µg of total protein was spotted onto a nitrocellulose membrane for each construct. Ponceau S staining is used as loading control.

FIG. 45A, FIG. 45B, FIG. 45C, and FIG. 45D. Quantification of mCherry fluorescence with modified polyA track sequences. PolyA tracks designed with flanking XAA and AAY codons, where X and Y denote C/G or T/C/G nucleotides respectively, were inserted into the mCherry reporter. The number of lysine residues (K) and adenine residues (A) are noted as well as the two amino acids flanking lysine. The nucleotide and amino acid sequence for K2/A9 (AN) are SEQ ID NO: 91 and 92, respectively. The nucleotide and amino acid sequence for K3/A9 (AS) are SEQ ID NO: 93 and 94, respectively. The nucleotide and amino acid sequence for K2/A10 (QN) are SEQ ID NO: 95 and 96, respectively. The nucleotide and amino acid sequence for K3/A10 (EV) are SEQ ID NO: 97 and 98, respectively. The nucleotide and amino acid sequence for K3/A10 (QV) are SEQ ID NO: 99 and 100, respectively. The nucleotide and amino acid sequence for K3/A11 (AN) are SEQ ID NO: 101 and 102, respectively. The nucleotide and amino acid sequence for K4/A11 (AV) are SEQ ID NO: 103 and 104, respectively. (FIG. 45A). Normalized mCherry fluorescence intensity. (FIG. 45B). PolyA tracks with non-lysine codons interrupting the consecutive AAA codons were inserted into the mCherry reporter. The number of adenosine residues and interrupting codon and the protein sequences are indicated.

13

The nucleic acid sequences for 33As, 15A(CTG)15A, 15A (TAC)15A, 15A(CCC)15A, and 30As are SEQ ID NO: 105, 106, 107, 108, and 109, respectively. (FIG. 45C). Normalized mCherry fluorescence intensity. Error bars represent standard deviation from three different colonies (FIG. 45B and FIG. 45D).

FIG. 46A and FIG. 46B. Illumina sequencing of polyA track genomic insertion. Approximately 30 generations after insertion of HA-(AAA)12-mCherry, genomic DNA was sequenced to examine mutation rate of long PolyA tracks. The fraction of sequencing reads which contain a polyA track vs reads that do not are shown for both genomic DNA and plasmid DNA (FIG. 46A). The non-polyA species of reads from genomic DNA are shown. The nucleic acid sequence shown is SEQ ID NO: 110 (FIG. 46B).

FIG. 47. Image of the full western blot used in FIG. 26B. Degradation of frameshifted product TRX-HA is indicated. Orientation of the gel is as in the original figure.

FIG. 48A and FIG. 48B Eight biological replicas for experiment represented in FIG. 27B. Loading of samples is in the same order as in the original figure for both mCherry and BAR (BASTA) western blots.

FIG. 49. Complete image of the western blot used in FIG. 27E. Loading and orientation of the western blot is the same as in the original figure. Samples from two biological replicas (left side and right side) are shown on this image.

FIG. 50. Image of the full western blot represented FIG. 29A. Low molecular weight band for degradation product (reacting with HA-antibody) and unspecific high molecular band (reacting with δ -tubulin antibody) are visible on the image. Orientation of the gel is the same as in the original figure.

FIG. 51. Original plates for the experiments described in FIG. 30A. Two colonies for each construct are excised from the plates to make the final figure.

DETAILED DESCRIPTION OF THE INVENTION

The disclosure provides a method for the translational control of gene expression in cells (e.g., eukaryotic and bacterial cells). In broad terms, the disclosure is based on polyA-triggered ribosome stalling and frameshifting leading to mRNA degradation and an alteration of protein output. The disclosure involves inserting a series of protein sequence tags that differ by several codons to allow tunable amounts of translation, and thus protein output in cells. Control of protein production can be further modulated by differential localization of such sequence tags (e.g. N- or C-terminal) as well as through proteasome and non-sense mediated decay (NMD) inhibition.

Aspects of the disclosure allow for control of protein production at the level of translation based on the insertion into genes of interest of predefined lengths of tagging sequences encoding poly-lysine with iterated AAA and/or AAG codons. These strings of codons induce site-specific cleavage of the mRNA, likely through stalling and frameshifting of the ribosome, as well as inhibition of endogenous stalling and frameshifting of the ribosome, respectively. As such, this system allows differential control of protein expression based on single or multiple base changes within a polylysine track situated within a coding sequence. A wide range of protein output can be achieved by inserting a variety of polyA (e.g., synonymous G-A lysine mutations) and disrupted polyA sequences (e.g., synonymous A-G lysine mutations) within an ORF of a gene.

14

The presently disclosed subject matter can be used for a variety of research, diagnostic, and/or therapeutic applications for which tunable regulation of protein expression in cells is desired. In one example, the presently disclosed subject matter can be used to achieve site specific mRNA cleavage triggered by a ribosome translating a polynucleotide comprising a polylysine track comprising at least one AAA lysine codon in its coding sequence. In another example, the presently disclosed subject matter involves peptidyl-tRNA drop off on polyA sequences in eukaryotes, archaea, and bacteria. In yet another example, the presently disclosed subject matter can be used to achieve differential expression of recombinant proteins based on single or multiple base changes inside of a polynucleotide tag sequence comprising at least one AAA lysine codon and/or at least one AAG lysine codon, or at least one polylysine track comprising such lysine codons. In a further example, the presently disclosed subject matter provides different sequence tags that specify differential effects on protein output in bacterial, archaeal, and eukaryotic cells.

In sum, the presently disclosed subject matter can be used for the regulation of protein translation in eukaryotic, archaeal, and bacterial systems, the tunable down regulation of essential cellular genes, the controlled expression of proteins and analysis of these effects on cell homeostasis, assays for translational control at various steps of the translation cycle, and the estimation of mRNA translation and turnover, among other uses.

I. Methods for Modulating the Level of Expression of Polypeptides in Cells

In an aspect, the presently disclosed subject matter provides a method for modulating the level of expression of a polypeptide in a cell, the method comprising modulating the amount of consecutive adenine (A) nucleotides in at least one lysine codon in an open reading frame of a polynucleotide sequence encoding the polypeptide in the cell, thereby modulating the level of expression of the polypeptide in the cell.

As used herein, “modulating” broadly means to cause or facilitate a qualitative or quantitative change, alteration, or modification in a molecule, a process, pathway, or phenomenon of interest. As used herein, “expression” refers to the process by which a polynucleotide is transcribed from a DNA template (such as into an mRNA or other RNA transcript) and/or the process by which a transcribed mRNA is subsequently translated into peptides, polypeptides, or proteins. Transcripts and encoded polypeptides may be collectively referred to as “gene product.” If the polynucleotide is derived from genomic DNA, expression may include splicing of the mRNA in a cell.

A “gene,” as used herein, refers to a polynucleotide containing at least one open reading frame that is capable of encoding a particular protein after being transcribed and translated. As used herein, a “gene product” is the biochemical material, either RNA or protein, resulting from expression of a gene. A measurement of the amount of gene product is sometimes used to infer how active a gene is. As used herein, “gene expression” is the process by which information from a gene is used in the synthesis of a functional gene product. As used herein, a “reporter gene” refers to a gene that produces a gene product that is easily detected. Examples of reporter genes include, but are not limited to, bioluminescent, fluorescent, computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), single-photon emission computed tomography (SPECT) reporter genes, and the like. In some aspects, the reporter gene is a bioluminescent reporter

15

gene (e.g., firefly luciferase). In some aspects, the reporter gene is a fluorescent reporter gene (e.g., a fluorescent protein (GFP, mCherry, etc.).

The terms “polynucleotide”, “polynucleotide sequence”, “nucleotide sequence”, “nucleic acid” and “oligonucleotide” are used interchangeably. They refer to a polymeric form of nucleotides of any length, either deoxyribonucleotides or ribonucleotides, or analogs thereof. Polynucleotides may have any three dimensional structure, and may perform any function, known or unknown. The following are non-limiting examples of polynucleotides: coding or non-coding regions of a gene or gene fragment, loci (locus) defined from linkage analysis, exons, introns, messenger RNA (mRNA), transfer RNA, ribosomal RNA, short interfering RNA (siRNA), short-hairpin RNA (shRNA), micro-RNA (miRNA), ribozymes, cDNA, recombinant polynucleotides, branched polynucleotides, plasmids, vectors, isolated DNA of any sequence, isolated RNA of any sequence, nucleic acid probes, and primers. A polynucleotide may comprise one or more modified nucleotides, such as methylated nucleotides and nucleotide analogs. If present, modifications to the nucleotide structure may be imparted before or after assembly of the polymer. The sequence of nucleotides may be interrupted by non-nucleotide components. A polynucleotide may be further modified after polymerization, such as by conjugation with a labeling component.

The terms “polypeptide,” “peptide” and “protein” are used interchangeably herein to refer to polymers of amino acids of any length. The polymer may be linear or branched, it may comprise modified amino acids, and it may be interrupted by non amino acids. The terms also encompass an amino acid polymer that has been modified; non-limiting examples of such modifications include disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation, such as conjugation with a labeling component. As used herein the term “amino acid” includes natural and/or unnatural or synthetic amino acids, including glycine and both the D or L optical isomers, and amino acid analogs and peptidomimetics.

As used herein, “modulating the level of expression” refers to causing or facilitating a qualitative or quantitative change, alteration, or modification in the amount of at least one polypeptide produced in a cell as a result of translation of at least one polynucleotide sequence (e.g., mRNA) encoding such at least one polypeptide. The phrase “modulating the amount of consecutive A nucleotides” means changing the linear sequence of nucleotides that are linked together by phosphodiester bonds in at least one polynucleotide sequence encoding at least one polypeptide of interest to be expressed in a cell a way that increases or decreases the number of contiguous A nucleotides in a targeted region of such polynucleotide sequence.

The presently disclosed subject matter demonstrates that a synonymous lysine mutation consisting of a single AAG-to-AAA codon in a polyA or polylysine track of a nucleic acid sequence (e.g., gene or mRNA) encoding a protein of interest decreases expression of the protein and mRNA stability. Conversely, the presently disclosed subject matter demonstrates that a synonymous lysine mutation consisting of a single AAA-to-AAG codon in a polyA or polylysine track of a nucleic acid sequence (e.g., gene or mRNA) encoding a protein of interest increases expression of the protein and mRNA stability. Put differently, increasing the length of consecutive A nucleotides, for example, by changing selected AAG lysine codons to AAA lysine codons reduces protein expression and mRNA stability, whereas decreasing the length of consecutive A nucleotides, for

16

example, by changing selected AAA lysine codons to AAG lysine codons increases protein expression and mRNA stability.

Accordingly, some aspects of the presently disclosed subject matter contemplate methods for decreasing the level of expression of a polypeptide in a cell, for example, by increasing the amount of consecutive A nucleotides in at least one lysine codon in an open reading frame of a polynucleotide sequence encoding at least one polypeptide in at least one cell.

The terms “decrease”, “reduced”, “reduction”, “decrease” or “inhibit” are all used herein generally to mean a decrease by a statistically significant amount. However, for avoidance of doubt, “reduced”, “reduction”, “decrease” or “inhibit” means a decrease by at least 10% as compared to a reference level, for example a decrease by at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90%, where the decrease is less than 100%. In one embodiment, the decrease includes a 100% decrease (e.g. absent level as compared to a reference sample), or any decrease between 10-100% as compared to a reference level.

In some embodiments, modulating the amount of consecutive A nucleotides in at least one lysine codon comprises increasing the amount of consecutive A nucleotides in at least one AAG lysine codon in the open reading frame of the polynucleotide sequence encoding the polypeptide in a cell, thereby decreasing the level of expression of the polypeptide in the cell. In the contexts of decreasing the level of expression of a polypeptide in a cell, the methods contemplated herein can decrease protein translation and mRNA stability by at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80, 90%, or as much as 100% as compared to a reference level (e.g., an objective measure of the level of expression of at least one polypeptide in a cell employing the method compared to the level of expression of at least one polypeptide in the cell in the absence of employing the method).

The presently disclosed subject matter contemplates using any technique available to the skilled artisan for increasing the amount of consecutive A nucleotides in at least one AAG lysine codon in an open reading frame of a polynucleotide sequence encoding at least one polypeptide in at least one cell.

In some embodiments, increasing the amount of consecutive A nucleotides in the at least one lysine codon comprises introducing at least one synonymous G to A nucleotide mutation into at least one AAG lysine codon. As used herein, “synonymous” in the context of a “nucleotide mutation” refers to a change in the nucleotide of a codon which does not alter the amino acid encoded by such codon. To introduce at least one synonymous G to A nucleotide mutation into at least one AAG lysine codon in an open reading frame of at least one polynucleotide sequence, at least one AAG lysine codon must be identified in the open reading frame. Methods of identifying at least one AAG lysine codon in an open reading frame of at least one polynucleotide encoding at least one polypeptide of interest are well known to the skilled artisan. For example, the position of at least one AAG lysine codon in an open reading frame of at least one polynucleotide encoding at least one polypeptide can be determined using publicly available sequence databases and bioinformatics tools (e.g., BLAST searching, mRNA mapping, etc.). When the position of at least one AAG lysine codon in an open reading frame of at least one polynucleotide encoding at least one polypeptide is determined, at

least one synonymous G to A nucleotide mutation can be introduced into such at least one AAG lysine codon, for example, using site directed mutagenesis. It should be appreciated, however, it is the number of consecutive A nucleotides that controls the levels of protein expression and mRNA stability. As such, it may be advantageous to identify at least one AAG lysine codon that is flanked by an upstream codon that ends with an A nucleotide (e.g., UUA (Leu), AUA (Ile), GUA (Val), UCA (Ser), CCA (Pro), ACA (Thr), GCA (Ala), UAU (Tyr), CAA (Gln), AAA (Lys), GAA (Glu), CGA (Arg), AGA (Arg), and GGA (Gly)) and/or is flanked by a downstream codon that begins with an A nucleotide (e.g., AUU (Ile), AUC (Ile), AUA (Ile), AUG (Met), ACU (Thr), ACC (Thr), ACA (Thr), ACG (Thr), AAU (Asn), AAC (Asn), AAA (Lys), AAG (Lys), AGU (Ser), AGC (Ser), AGA (Arg), and AGG (Arg)) for introduction of at least one synonymous G to A nucleotide mutation, for example, to increase the number of consecutive A nucleotides in the polynucleotide sequence with the resultant decrease in the level of expression of at least one polypeptide encoded by such polynucleotide sequence. Accordingly, in some embodiments, the method comprises introducing at least one synonymous G to A nucleotide mutation into at least one AAG lysine codon that is flanked by an upstream codon that ends with an A nucleotide and/or is flanked by a downstream codon that begins with an A nucleotide.

Generally, it is believed that the greater the increase in the number of consecutive A nucleotides in at least one polynucleotide sequence encoding at least one polypeptide, the greater the decrease will be in the level of expression of the at least one polypeptide. The skilled artisan will appreciate that the increase in the number of consecutive A nucleotides introduced into at least one polynucleotide sequence in this manner will be limited by the number of AAG lysine codons in the open reading frame of such polynucleotide sequence, as well as those AAG lysine codons that are flanked by upstream codons ending with A nucleotides and/or are flanked by downstream codons beginning with A nucleotides.

In some embodiments, at least one synonymous G to A nucleotide mutation is introduced into at least one AAG lysine codon in an open reading frame of at least one polynucleotide sequence. In some embodiments, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least 10, at least 11, at least 12, or at least n synonymous G to A nucleotide mutations (where n is a positive integer greater than or equal to 13 and less than or equal to the number of AAG lysine codons in a particular polynucleotide sequence) are introduced into at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least 10, at least 11, at least 12, or at least n AAG lysine codons (where n is a positive integer greater than or equal to 13 and less than or equal to the number of AAG lysine codons in a particular polynucleotide sequence) in an open reading frame of at least one polynucleotide sequence.

In other embodiments, increasing the amount of consecutive A nucleotides in the at least one lysine codon comprises inserting at least one AAA lysine codon into the open reading frame of the polynucleotide sequence encoding the polypeptide in the cell. It should be appreciated that any amount of at least one AAA lysine codons can be inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide in a cell. In general, the greater the amount of consecutive A nucleotides inserted into an open reading frame of at least one polynucleotide sequence, the greater the decrease in the level of

expression of at least one polypeptide encoded by the at least one polynucleotide sequence. In this way, levels of expression of at least one polypeptide can be controlled in a cell. In some embodiments, at least one (AAA) lysine codon is inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest in a cell. In some embodiments, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least 10, at least eleven, at least 12, or at least n AAA lysine codons ((AAA)_n) (where n is a positive integer greater than or equal to 13) are inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest in a cell. In some embodiments, at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 21, at least 22, at least 23, at least 24, at least 25, at least 26, at least 27, at least 28, at least 29, at least 30, at least 31, at least 32, at least 33, at least 34, at least 35, at least 36, or at least n A nucleotides ((A)_n) (where n is a positive integer greater than or equal to 37) are inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest in a cell.

The AAA lysine codons and/or consecutive A nucleotides can be inserted in the form of one or more polynucleotide sequence tags designed for tunable expression of at least one polypeptide in a cell. In some embodiments, at least one polyA polynucleotide sequence tag can be inserted in an open reading frame, for example, in between an upstream codon ending with an A nucleotide and a downstream codon beginning with an A nucleotide. In some embodiments, two or more polyA polynucleotide sequence tags can be inserted adjacent to each other or spaced apart by intervening polynucleotide sequences in the open reading frame.

The presently disclosed subject matter contemplates insertion of at least one AAA lysine codon, a consecutive number of A nucleotides, and/or a polyA polynucleotide sequence tag into any portion of an open reading frame in at least one polynucleotide encoding at least one polypeptide in a cell. In some embodiments, at least one AAA lysine codon is inserted into a coding sequence of the open reading frame. In some embodiments, two or more polyA polynucleotide sequence tags are inserted into a coding sequence of at least one polypeptide. In some embodiments, at least one AAA lysine codon is inserted into a 5' untranslated region (UTR) of at least one polynucleotide encoding a polypeptide in a cell. In some embodiments, a polyA polynucleotide sequence tag is inserted into a 5' UTR of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAA lysine codon is inserted into an exon of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, a polyA polynucleotide sequence tag is inserted into an exon of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAA lysine codon is inserted into an exon/intron boundary of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, a polyA polynucleotide sequence tag is inserted into an exon/intron boundary of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAA lysine codon is inserted into an intron of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, a polyA polynucleotide sequence tag is inserted into an intron of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least

one AAA lysine codon is inserted into a 3' UTR of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, a polyA polynucleotide sequence tag is inserted into a 3' UTR of at least one polynucleotide encoding a polypeptide of interest.

In some embodiments, at least one polyA polynucleotide sequence tag comprises one or more AAG lysine codons, for example, for tunable expression of at least one polypeptide of interest encoded by at least one polynucleotide into which the at least one polyA polynucleotide sequence tag is inserted. Examples of such polyA nucleotide sequence tags include at least one AAA lysine codon preceded or followed by at least one AAG lysine codon, a first AAA lysine codon and a second AAA lysine codon flanking an AAG lysine codon, alternating AAG and AAA lysine codons (e.g., (AAG-AAA)_n (where *n* is a positive integer greater than or equal to 1), triple repeats comprising combinations of AAG and AAA lysine codons (e.g., (AAA-AAG-AAA)_n, (AAG-AAA-AAA)_n, (AAA-AAA-AAG)_n, where each *n* is a positive integer greater than or equal to 1), quadruple repeats, etc. In some embodiments, *n* is 1. In some embodiments, *n* is 2. In some embodiments, *n* is 3. In some embodiments, *n* is 4. In some embodiments, *n* is 5. In some embodiments, *n* is 6. In some embodiments, *n* is 6. In some embodiments, *n* is 8. In some embodiments, *n* is 9. In some embodiments, *n* is 10. In some embodiments, *n* is 11. In some embodiments, *n* is 12.

Some aspects of the presently disclosed subject matter contemplate methods for increasing the level of expression of a polypeptides in a cell, for example, by decreasing the amount of consecutive A nucleotides in at least one lysine codon in an open reading frame of a polynucleotide sequence encoding at least one polypeptide in at least one cell.

The terms "increased", "increase", "enhance" or "activate" are all used herein to generally mean an increase by a statically significant amount; for the avoidance of any doubt, the terms "increased", "increase", "enhance" or "activate" means an increase of at least 10% as compared to a reference level, for example an increase of at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% increase or any increase between 10-100% as compared to a reference level, or at least about a 2-fold, or at least about a 3-fold, or at least about a 4-fold, or at least about a 5-fold or at least about a 10-fold increase, or any increase between 2-fold and 10-fold or greater as compared to a reference level.

In some embodiments, modulating the amount of consecutive A nucleotides in the at least one lysine codon comprises decreasing the amount of consecutive A nucleotides in the at least one AAA lysine codon in an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest in a cell, thereby increasing the level of expression of the polypeptide in the cell. In the contexts of increasing the level of expression of a polypeptide in a cell, the methods contemplated herein can increase protein translation and mRNA stability, by at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80, 90%, or as much as 100%, at least about a 2-fold, or at least about a 3-fold, or at least about a 4-fold, or at least about a 5-fold or at least about a 10-fold increase, or any increase between 2-fold and 10-fold or greater as compared to a reference level (e.g., an objective measure of the level of expression of at least one polypeptide in a cell employing the method compared to the

level of expression of at least one polypeptide in the cell in the absence of employing the method).

The presently disclosed subject matter contemplates using any technique available to the skilled artisan for decreasing the amount of consecutive A nucleotides in at least one lysine codon in an open reading frame of a polynucleotide sequence encoding at least one polypeptide of interest. In some embodiments, decreasing the amount of consecutive A nucleotides in at least one lysine codon comprises introducing at least one synonymous A to G nucleotide mutation into the at least one AAA lysine codon. To introduce at least one synonymous A to G nucleotide mutation into at least one AAA lysine codon in an open reading frame of at least one polynucleotide sequence, at least one AAA lysine codon must be identified in the open reading frame using methods which are well known to the skilled artisan. When the position of at least one AAA lysine codon in an open reading frame of at least one polynucleotide encoding at least one polypeptide is determined, at least one synonymous A to G nucleotide mutation can be introduced into such at least one AAA lysine codon, for example, using site directed mutagenesis. It should be appreciated, however, it is the number of consecutive A nucleotides that controls the levels of protein expression and mRNA stability. As such, it may be advantageous to identify at least one AAA lysine codon flanked by an upstream codon that ends with an A nucleotide (e.g., UUA (Leu), AUA (Ile), GUA (Val), UCA (Ser), CCA (Pro), ACA (Thr), GCA (Ala), UAU (Tyr), CAA (Gln), AAA (Lys), GAA (Glu), CGA (Arg), AGA (Arg), and GGA (Gly)) and/or flanked by a downstream codon that begins with an A nucleotide (e.g., AUU (Ile), AUC (Ile), AUA (Ile), AUG (Met), ACU (Thr), ACC (Thr), ACA (Thr), ACG (Thr), AAU (Asn), AAC (Asn), AAA (Lys), AAG (Lys), AGU (Ser), AGC (Ser), AGA (Arg), and AGG (Arg)) to introduce at least one synonymous A to G nucleotide mutation into, for example, to decrease the number of consecutive A nucleotides in the polynucleotide sequence with the resultant increase in the level of expression of at least one polypeptide encoded by such polynucleotide sequence. Accordingly, in some embodiments, the method comprises introducing at least one synonymous A to G nucleotide mutation into at least one AAA lysine codon flanked by an upstream codon that ends with an A nucleotide and/or flanked by a downstream codon that begins with an A nucleotide.

Generally, it is believed that the greater the decrease in the number of consecutive A nucleotides in at least one polynucleotide sequence encoding at least one polypeptide of interest, the greater the increase will be in the level of expression of the at least one polypeptide of interest. The skilled artisan will appreciate that the decrease in the number of consecutive A nucleotides in at least one polynucleotide sequence will be limited by the number of consecutive A nucleotides in such sequence, and in particular embodiments by the number of AAA lysine codons in the open reading frame of such polynucleotide sequence.

In some embodiments, at least one synonymous A to G nucleotide mutation is introduced into at least one AAA lysine codon in an open reading frame of at least one polynucleotide sequence encoding a polypeptide of interest. In some embodiments, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least 10, at least 11, at least 12, or at least *n* synonymous A to G nucleotide mutations (where *n* is a positive integer greater than or equal to 13 and less than or equal to the number of AAA lysine codons in a particular polynucleotide sequence) are introduced into at least two, at least three, at least four, at least five, at least six, at least

21

seven, at least eight, at least nine, at least 10, at least 11, at least 12, or at least n AAA lysine codons (where n is a positive integer greater than or equal to 13 and less than or equal to the number of AAA lysine codons in a particular polynucleotide sequence) in an open reading frame of at least one polynucleotide sequence.

In other embodiments, decreasing the amount of consecutive A nucleotides in at least one lysine codon comprises inserting at least one AAG lysine codon into the open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest. It should be appreciated that any amount of at least one AAG lysine codons can be inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest. In general, the greater the amount of AAG lysine codons inserted into an open reading frame of at least one polynucleotide sequence, the greater the increase in the level of expression of at least one polypeptide encoded by the at least one polynucleotide sequence. In this way, levels of expression of at least one polypeptide can be controlled in a cell. In some embodiments, at least one (AAG) lysine codon is inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest. In some embodiments, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, at least 10, at least eleven, at least 12, or at least n AAG lysine codons ((AAG) n) (where n is a positive integer greater than or equal to 13) are inserted into an open reading frame of at least one polynucleotide sequence encoding at least one polypeptide of interest.

The AAG lysine codons can be inserted in the form of one or more poly AAG lysine polynucleotide sequence tags designed for tunable expression of at least one polypeptide in a cell. In some embodiments, at least one polyAAG lysine polynucleotide sequence tag can be inserted in an open reading frame. In some embodiments, two or more polyAAG lysine polynucleotide sequence tags can be inserted in the open reading frame.

The presently disclosed subject matter contemplates insertion of at least one AAG lysine codon, and/or a polyAAG lysine polynucleotide sequence tag into any portion of an open reading frame in at least one polynucleotide encoding at least one polypeptide of interest. In some embodiments, at least one AAG lysine codon is inserted into a coding sequence of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one poly AAG lysine polynucleotide sequence tag is inserted into a coding sequence of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAG lysine codon is inserted into a 5' untranslated region (UTR) of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, a polyAAG lysine polynucleotide sequence tag is inserted into a 5' UTR of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAG lysine codon is inserted into an exon/intron boundary of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, a polyAAG lysine polynucleotide sequence tag is inserted into an exon/intron boundary of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAG lysine codon is inserted into an intron of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one polyAAG lysine polynucleotide sequence tag is inserted into an intron of at least one polynucleotide encoding a polypeptide of interest. In some embodiments, at least one AAG lysine codon is inserted into a 3' UTR of at least one

22

polynucleotide encoding a polypeptide of interest. In some embodiments, at least one polyAAG lysine polynucleotide sequence tag is inserted into a 3' UTR of at least one polynucleotide encoding a polypeptide of interest.

In some embodiments, at least one polyAAG lysine polynucleotide sequence tag comprises one or more AAA lysine codons, for example, for tunable expression of at least one polypeptide of interest encoded by at least one polynucleotide into which the at least one polyAAG lysine polynucleotide sequence tag is inserted. Examples of such polyAAG lysine polynucleotide sequence tags include at least one AAG lysine codon preceded or followed by at least one AAA lysine codon, a first AAG lysine codon and a second AAG lysine codon flanking an AAA lysine codon, alternating AAA and AAG lysine codons (e.g., (AAA-AAG) n (where n is a positive integer greater than or equal to 1), triple repeats comprising combinations of AAA and AAG lysine codons (e.g., (AAG-AAA-AAG) n , (AAA-AAG-AAG) n , (AAG-AAG-AAA) n , where each n is a positive integer greater than or equal to 1), quadruple repeats, etc. In some embodiments, n is 1. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4. In some embodiments, n is 5. In some embodiments, n is 6. In some embodiments, n is 6. In some embodiments, n is 8. In some embodiments, n is 9. In some embodiments, n is 10. In some embodiments, n is 11. In some embodiments, n is 12.

In some embodiments, at least one AAG lysine codon is inserted into at least one polynucleotide sequence for expression at an N-terminus or C-terminus of at least one polypeptide of interest to be expressed in a cell. In some embodiments, at least one polyAAG lysine polynucleotide sequence tag comprises a N-terminus tag. In some embodiments, at least one polyAAG lysine polynucleotide sequence tag comprises a C-terminus tag.

Those skilled in the art will appreciate that the manner in which at least one synonymous lysine mutation (e.g., at least one synonymous A to G nucleotide mutation or at least one synonymous G to A nucleotide mutation) is introduced into at least one lysine codon, the manner in which at least one lysine codon (e.g., at least one AAA lysine codon or at least one AAG lysine codon) is inserted into an open reading frame of at least one polynucleotide sequence encoding a polypeptide of interest, and/or the manner in which at least one polylysine sequence tag (e.g., at least one polyA track or polyA polynucleotide sequence tag or at least one polyAAG lysine track or polyAAG lysine polynucleotide sequence tag) is inserted into an open reading frame of at least one polynucleotide sequence encoding a polypeptide of interest depends on whether at least one polynucleotide encoding at least one polypeptide of interest is an endogenous polynucleotide sequence in a cell, or an exogenous polynucleotide sequence encoding a heterologous protein to be expressed in a cell.

In some embodiments, at least one polynucleotide sequence comprises an endogenous polynucleotide sequence and the step of modulating the amount of consecutive A nucleotides in the at least one lysine codon includes selecting an endogenous polynucleotide sequence in the cell that comprises at least one lysine codon, and editing the endogenous polynucleotide sequence in the cell. The presently disclosed subject matter contemplates editing endogenous polynucleotide sequences in cells that is available to the skilled artisan. In some embodiments, editing the endogenous polynucleotide sequence in the cell comprises contacting the cell with an engineered nuclease selected from the group consisting of a CRISPR-Cas system,

CRISPR-Cpf1 system, a zinc finger nuclease (ZFN), a transcription activator-like effector nuclease (TALEN), and a meganuclease.

In other embodiments, at least one polynucleotide sequence comprises an exogenous polynucleotide sequence and the step of modulating the amount of consecutive A nucleotides in the at least one lysine codon includes providing an expression vector comprising an exogenous polynucleotide sequence comprising at least one AAA lysine codon or at least one AAG lysine codon inserted thereinto operably linked to a promoter that drives expression of the exogenous polynucleotide in the cell, and contacting the cell with the expression vector.

In some embodiments, at least one lysine codon comprises at least one polylysine track selected from the group consisting of AAA lysine codons, AAG lysine codons, and combinations thereof. In some embodiments, at least one polylysine track comprises between 4 and 36 A nucleotides. In some embodiments, at least one polylysine track comprises at least 11 consecutive A nucleotides in at least three consecutive lysine codons.

In some embodiments, exogenous polynucleotide sequences comprising at least one lysine codon and/or at least one polylysine track, and/or at least one polyA polynucleotide sequence tag and/or at least one polyAAG lysine polynucleotide sequence tag can be synthesized utilizing in vitro transcription methods which are well known to the skilled artisan.

In some embodiments, at least one lysine codon, at least one polyA polynucleotide sequence tag, and/or at least one polylysine track is not a polyA tail. In some embodiments, at least one lysine codon, at least one polyA polynucleotide sequence tag, and/or at least one polylysine (polyA) track is not located in the 3' UTR or downstream of the 3' UTR.

Ii. Expression Vectors

Aspects of the presently disclosed subject matter relate to expression vectors for the tunable expression of polypeptides of interest in cells. The presently disclosed expression vectors comprise at least one polynucleotide comprising at least one lysine codon, at least one polylysine track, at least one polyA polynucleotide sequence tag, and/or at least one polyAAG lysine polynucleotide sequence tag.

In general, and throughout this specification, the term "vector" refers to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. Vectors include, but are not limited to, nucleic acid molecules that are single-stranded, double-stranded, or partially double-stranded; nucleic acid molecules that comprise one or more free ends, no free ends (e.g. circular); nucleic acid molecules that comprise DNA, RNA, or both; and other varieties of polynucleotides known in the art. One type of vector is a "plasmid," which refers to a circular double stranded DNA loop into which additional DNA segments can be inserted, such as by standard molecular cloning techniques. Another type of vector is a viral vector, wherein virally-derived DNA or RNA sequences are present in the vector for packaging into a virus (e.g. retroviruses, replication defective retroviruses, adenoviruses, replication defective adenoviruses, and adeno-associated viruses). Viral vectors also include polynucleotides carried by a virus for transfection into a host cell.

Certain vectors are capable of autonomous replication in a host cell into which they are introduced (e.g. bacterial vectors having a bacterial origin of replication and episomal mammalian vectors). Other vectors (e.g., non-episomal mammalian vectors) are integrated into the genome of a host cell upon introduction into the host cell, and thereby are

replicated along with the host genome. Moreover, certain vectors are capable of directing the expression of genes to which they are operatively-linked. Such vectors are referred to herein as "expression vectors." Common expression vectors of utility in recombinant DNA techniques are often in the form of plasmids.

Recombinant expression vectors can comprise a nucleic acid of the presently disclosed subject matter in a form suitable for expression of the nucleic acid in a host cell, which means that the recombinant expression vectors include one or more regulatory elements, which may be selected on the basis of the host cells to be used for expression, that is operatively-linked to the nucleic acid sequence to be expressed.

Within a recombinant expression vector, "operably linked" is intended to mean that the nucleotide sequence of interest is linked to the regulatory element(s) in a manner that allows for expression of the nucleotide sequence (e.g. in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell).

The term "regulatory element" is intended to include promoters, enhancers, internal ribosomal entry sites (IRES), and other expression control elements (e.g. transcription termination signals, such as polyadenylation signals and poly-U sequences). Such regulatory elements are described, for example, in Goeddel (1990) *Gene Expression Technology: Methods in Enzymology* 185, Academic Press, San Diego, Calif. Regulatory elements include those that direct constitutive expression of a nucleotide sequence in many types of host cell and those that direct expression of the nucleotide sequence only in certain host cells (e.g., tissue-specific regulatory sequences). In some embodiments, the promoter is an inducible promoter that is active in response to specific stimuli.

A cell-specific promoter may direct expression primarily in a desired cell of interest, such as muscle cell, a neuron, a skin cell, a blood cell, an immune cell, a liver cell, a pancreatic cell, a spleen cell, etc. In some embodiments, the promoter is a tissue-specific promoter that is active in specific tissues. In some embodiments, the promoter is a tumor-specific promoter that is active specifically in tumor cells. Regulatory elements may also direct expression in a temporal-dependent manner, such as in a cell-cycle dependent or developmental stage-dependent manner, which may or may not also be tissue or cell-type specific.

In some embodiments, a vector comprises one or more pol III promoters, one or more pol II promoters, one or more pol I promoters, or combinations thereof. Examples of pol III promoters include, but are not limited to, U6 and H1 promoters. Examples of pol II promoters include, but are not limited to, the retroviral Rous sarcoma virus (RSV) LTR promoter (optionally with the RSV enhancer), the cytomegalovirus (CMV) promoter (optionally with the CMV enhancer) (e.g., Boshart et al. (1985) *Cell* 41:521-530), the SV40 promoter, the dihydrofolate reductase promoter, the β -actin promoter, the phosphoglycerol kinase (PGK) promoter, and the EF1 α promoter.

Also encompassed by the term "regulatory element" are enhancer elements, such as WPRE; CMV enhancers; the R-U5' segment in LTR of HTLV-I (Takebe et al. (1988) *Mol. Cell. Biol.* 8:466-472); SV40 enhancer; and the intron sequence between exons 2 and 3 of rabbit β -globin (O'Hare et al. (1981) *Proc. Natl. Acad. Sci. USA.* 78(3):1527-31). It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed, the level of expression desired, etc. A vector can be introduced into host

cells to thereby produce transcripts, proteins, or peptides, including fusion proteins or peptides, encoded by nucleic acids as described herein (e.g., clustered regularly interspersed short palindromic repeats (CRISPR) transcripts, proteins, enzymes, mutant forms thereof, fusion proteins thereof, etc.). Advantageous vectors include lentiviruses and adeno-associated viruses, and types of such vectors can also be selected for targeting particular types of cells.

As used herein the term “wild type” is a term of the art understood by skilled persons and means the typical form of an organism, strain, gene or characteristic as it occurs in nature as distinguished from mutant or variant forms.

As used herein the term “variant” should be taken to mean the exhibition of qualities that have a pattern that deviates from what occurs in nature.

The terms “non-naturally occurring” or “engineered” are used interchangeably and indicate the involvement of the hand of man. The terms, when referring to nucleic acid molecules or polypeptides mean that the nucleic acid molecule or the polypeptide is at least substantially free from at least one other component with which they are naturally associated in nature and as found in nature.

“Complementarity” refers to the ability of a nucleic acid to form hydrogen bond(s) with another nucleic acid sequence by either traditional Watson-Crick or other non-traditional types. A percent complementarity indicates the percentage of residues in a nucleic acid molecule which can form hydrogen bonds (e.g., Watson-Crick base pairing) with a second nucleic acid sequence (e.g., 5, 6, 7, 8, 9, 10 out of 10 being 50%, 60%, 70%, 80%, 90%, and 100% complementary). “Perfectly complementary” means that all the contiguous residues of a nucleic acid sequence will hydrogen bond with the same number of contiguous residues in a second nucleic acid sequence. “Substantially complementary” as used herein refers to a degree of complementarity that is at least 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% over a region of 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 30, 35, 40, 45, 50, or more nucleotides, or refers to two nucleic acids that hybridize under stringent conditions.

As used herein, “stringent conditions” for hybridization refer to conditions under which a nucleic acid having complementarity to a target sequence predominantly hybridizes with the target sequence, and substantially does not hybridize to non-target sequences. Stringent conditions are generally sequence-dependent, and vary depending on a number of factors. In general, the longer the sequence, the higher the temperature at which the sequence specifically hybridizes to its target sequence. Non-limiting examples of stringent conditions are described in detail in Tijssen (1993), Laboratory Techniques In Biochemistry And Molecular Biology-Hybridization With Nucleic Acid Probes Part 1, Second Chapter “Overview of principles of hybridization and the strategy of nucleic acid probe assay”, Elsevier, N.Y.

“Hybridization” refers to a reaction in which one or more polynucleotides react to form a complex that is stabilized via hydrogen bonding between the bases of the nucleotide residues. The hydrogen bonding may occur by Watson Crick base pairing, Hoogsteen binding, or in any other sequence specific manner. The complex may comprise two strands forming a duplex structure, three or more strands forming a multi stranded complex, a single self hybridizing strand, or any combination of these. A hybridization reaction may constitute a step in a more extensive process, such as the initiation of PCR, or the cleavage of a polynucleotide by an

enzyme. A sequence capable of hybridizing with a given sequence is referred to as the “complement” of the given sequence.

Vectors may be introduced and propagated in a prokaryote. In some embodiments, a prokaryote is used to amplify copies of a vector to be introduced into a eukaryotic cell or as an intermediate vector in the production of a vector to be introduced into a eukaryotic cell (e.g. amplifying a plasmid as part of a viral vector packaging system). In some embodiments, a prokaryote is used to amplify copies of a vector and express one or more nucleic acids, such as to provide a source of one or more proteins for delivery to a host cell or host organism. Expression of proteins in prokaryotes is most often carried out in *Escherichia coli* with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion proteins.

Fusion vectors add a number of amino acids to a protein encoded therein, such as to the amino terminus of the recombinant protein. Such fusion vectors may serve one or more purposes, such as: (i) to increase expression of recombinant protein; (ii) to increase the solubility of the recombinant protein; and (iii) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification. Often, in fusion expression vectors, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such enzymes, and their cognate recognition sequences, include Factor Xa, thrombin and enterokinase. Example fusion expression vectors include pGEX (Pharmacia Biotech Inc; Smith and Johnson (1988) Gene 67: 31-40), pMAL (New England Biolabs, Beverly, Mass.) and pRIT5 (Pharmacia, Piscataway, N.J.) that fuse glutathione S-transferase (GST), maltose E binding protein, or protein A, respectively, to the target recombinant protein.

Examples of suitable inducible non-fusion *E. coli* expression vectors include pTrc (Amrann et al. (1988) Gene 69:301-315) and pET 11d (Studier et al. (1990) Gene Expression Technology: Methods in Enzymology 185, Academic Press, San Diego, Calif.).

In some embodiments, a vector is a yeast expression vector. Examples of vectors for expression in yeast *Saccharomyces cerevisiae* include pYepSec1 (Baldari, et al.(1987) EMBO J. 6: 229-234), pMFa (Kuijan and Herskowitz (1982) Cell 30: 933-943), pJRY88 (Schultz et al. (1987) Gene 54: 113-123), pYES2 (Invitrogen Corporation, San Diego, Calif.), and picZ (Invitrogen Corp, San Diego, Calif.).

In some embodiments, a vector is capable of driving expression of one or more sequences in mammalian cells using a mammalian expression vector. Examples of mammalian expression vectors include pCDM8 (Seed (1987) Nature 329: 840) and pMT2PC (Kaufman et al. (1987) EMBO J. 6: 187-195). When used in mammalian cells, the expression vector's control functions are typically provided by one or more regulatory elements. For example, commonly used promoters are derived from polyoma, adenovirus 2, cytomegalovirus, simian virus 40, and others disclosed herein and known in the art. For other suitable expression systems for both prokaryotic and eukaryotic cells see, e.g., Chapters 16 and 17 of Sambrook et al. (1989) Molecular Cloning: A Laboratory Manual. 2nd ed., Cold Spring Harbor Laboratory, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y..

In some embodiments, the recombinant mammalian expression vector is capable of directing expression of the nucleic acid preferentially in a particular cell type (e.g.,

tissue-specific regulatory elements are used to express the nucleic acid). Tissue-specific regulatory elements are known in the art. Non-limiting examples of suitable tissue-specific promoters include the albumin promoter (liver-specific; Pinkert et al. (1987) *Genes Dev.* 1: 268-277), lymphoid-specific promoters (Calame and Eaton (1988) *Adv. Immunol.* 43: 235-275), in particular promoters of T cell receptors (Winoto and Baltimore (1989) *EMBO J.* 8: 729-733) and immunoglobulins (Baneiji et al. (1983) *Cell* 33: 729-740; Queen and Baltimore (1983) *Cell* 33: 741-748), neuron-specific promoters (e.g., the neurofilament promoter; Byrne and Ruddle (1989) *Proc. Natl. Acad. Sci. USA* 86: 5473-5477), pancreas-specific promoters (Edlund et al. (1985) *Science* 230: 912-916), and mammary gland-specific promoters (e.g., milk whey promoter; U.S. Pat. No. 4,873,316 and European Application Publication No. 264,166). Developmentally-regulated promoters are also encompassed, e.g., the murine hox promoters (Kessel and Gruss (1990) *Science* 249: 374-379) and the α -fetoprotein promoter (Campes and Tilghman (1989) *Genes Dev.* 3: 537-546).

In some aspects, the presently disclosed subject matter provides an expression vector comprising a cloning site for inserting at least one polynucleotide sequence encoding a polypeptide to be expressed, and at least one polynucleotide tag sequence comprising at least one AAG lysine codon that increases expression of the at least one polynucleotide sequence when the expression vector is introduced into a cell. In some embodiments, at least one polynucleotide tag sequence comprises at least one polylysine track comprising at least two consecutive AAG lysine codons. In some 5 embodiments, at least one polylysine track comprises at least two consecutive AAG lysine codons selected from the group consisting of (AAG)₂, (AAG)₃, (AAG)₆, and (AAG)₁₂. In some embodiments, at least one polylysine track comprises at least n consecutive AAG lysine codons (i.e., (AAG)_n), where n is a positive integer greater than or equal to 13.

In some aspects, the presently disclosed subject matter provides an expression vector comprising a cloning site for inserting at least one polynucleotide sequence encoding a polypeptide to be expressed, and at least one polynucleotide tag sequence comprising at least one AAA lysine codon that decreases expression of the at least one polynucleotide sequence when the expression vector is introduced into a cell. In some embodiments, at least one polynucleotide tag sequence comprises at least one polylysine track comprising at least two consecutive AAA lysine codons. wherein the and wherein the at least one polylysine track in b) comprises at least two consecutive AAA lysine codons selected from the group consisting of (AAA)₂, (AAA)₃, (AAA)₆, and (AAA)₁₂. In some embodiments, at least one polylysine track comprises at least n consecutive AAA lysine codons (i.e., (AAA)_n), where n is a positive integer greater than or equal to 13.

In some aspects, the presently disclosed subject matter provides an expression vector comprising at least one engineered polynucleotide sequence encoding a polypeptide to be expressed, the at least one engineered polynucleotide sequence comprising at least one engineered synonymous mutation of at least one AAA lysine codon to at least one AAG lysine codon in a coding sequence of the at least one polynucleotide sequence, wherein the synonymous mutation increases expression of the polypeptide to be expressed when the expression vector is introduced into a cell. In some 55 embodiments, at least one engineered polynucleotide sequence comprises at least one polylysine track comprising at least two consecutive lysine codons in the coding

sequence. In some embodiments, at least one polylysine track comprises at least two consecutive AAG lysine codons selected from the group consisting of (AAG)₂, (AAG)₃, (AAG)₆, and (AAG)₁₂. In some embodiments, at least one polylysine track comprises at least n consecutive AAG lysine codons (i.e., (AAG)_n where n is a positive integer greater than or equal to 13).

In some aspects, the presently disclosed subject matter provides an expression vector comprising at least one engineered polynucleotide sequence encoding a polypeptide to be expressed, the at least one engineered polynucleotide sequence comprising at least one engineered synonymous mutation of at least one AAG lysine codon to at least one AAA lysine codon in a coding sequence of the at least one polynucleotide sequence, wherein the synonymous mutation decreases expression of the polypeptide to be expressed when the expression vector is introduced into a cell.

In some embodiments, at least one engineered polynucleotide sequence comprises at least one polylysine track comprising at least two consecutive lysine codons in the coding sequence. In some embodiments, at least one polylysine track comprises at least two consecutive AAA lysine codons selected from the group consisting of (AAA)₂, (AAA)₃, (AAA)₆, and (AAA)₁₂. In some embodiments, at least one polylysine track comprises at least n consecutive AAA lysine codons (i.e., (AAA)_n where n is a positive integer greater than or equal to 13).

In some embodiments, at least one polylysine track comprises at least 11 consecutive A nucleotides in at least three consecutive lysine codons, prior to engineering the at least one engineered polynucleotide sequence to include the at least one engineered synonymous mutation.

Vectors can be designed for expression of CRISPR transcripts (e.g. nucleic acid transcripts, proteins, or enzymes) in prokaryotic or eukaryotic cells. For example, CRISPR transcripts can be expressed in bacterial cells such as *Escherichia coli*, insect cells (using baculovirus expression vectors), yeast cells, or mammalian cells. Suitable host cells are discussed further in Goeddel (1990) *Gene Expression Technology: Methods in Enzymology* 185, Academic Press, San Diego, Calif. Alternatively, the recombinant expression vector can be transcribed and translated in vitro, for example using T7 promoter regulatory sequences and T7 polymerase.

In aspects of the presently disclosed subject matter the terms “chimeric RNA”, “chimeric guide RNA”, “guide RNA”, “single guide RNA” and “synthetic guide RNA” are used interchangeably and refer to the polynucleotide sequence comprising the guide sequence. The term “guide sequence” refers to the about 20 bp sequence within the guide RNA that specifies a target sequence and may be used interchangeably with the terms “guide” or “spacer”.

A target sequence may comprise any polynucleotide, such as DNA or RNA polynucleotides. In some embodiments, a target sequence is located in the nucleus or cytoplasm of a cell. In some embodiments, the target sequence may be within an organelle of a eukaryotic cell, for example, mitochondrion or chloroplast. A sequence or template that may be used for recombination into the targeted locus comprising the target sequences is referred to as an “editing template” or “editing polynucleotide” or “editing sequence”. In aspects of the presently disclosed subject matter, an exogenous template polynucleotide may be referred to as an editing template. In an aspect of the presently disclosed subject matter the recombination is homologous recombination.

In some embodiments, a vector comprises one or more insertion sites, such as a restriction endonuclease recognition sequence (also referred to as a “cloning site”). In some

embodiments, one or more insertion sites (e.g. about or more than about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more insertion sites) are located upstream and/or downstream of one or more sequence elements of one or more vectors

In some embodiments, the degree of complementarity between a guide sequence and its corresponding target sequence, when optimally aligned using a suitable alignment algorithm, is about or more than about 50%, 60%, 75%, 80%, 85%, 90%, 95%, 97.5%, 99%, or more. Optimal alignment may be determined with the use of any suitable algorithm for aligning sequences, non-limiting example of which include the Smith-Waterman algorithm, the Needleman-Wunsch algorithm, algorithms based on the Burrows-Wheeler Transform (e.g. the Burrows Wheeler Aligner), ClustalW, Clustal X, BLAT, Novoalign (Novocraft Technologies, ELAND (Illumina, San Diego, Calif.), SOAP (available at soap.genomics.org.cn), and Maq (available at maq.sourceforge.net). In some embodiments, a guide sequence is about or more than about 5, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 35, 40, 45, 50, 75, or more nucleotides in length. In some embodiments, a guide sequence is less than about 75, 50, 45, 40, 35, 30, 25, 20, 15, 12, or fewer nucleotides in length.

Iii. Isolated Recombinant Cells

Aspects of the presently disclosed subject matter relate to an isolated recombinant cell comprising an expression vector of the presently disclosed subject matter. In some embodiments, host cells which contain the constructs and vectors of the presently disclosed subject matter are also encompassed, e.g. in vitro cells such as cultured cells, e.g., bacterial or eukaryotic cells which are used to store generate or manipulate the vectors, and the like. In some embodiments, the isolated recombinant cell or host cell is a mammalian cell. In some embodiments, the isolated recombinant cell or host cell is a human cell. In some embodiments, the cell is selected from the group consisting of a bacterial cell and a eukaryotic cell. In some embodiments, the eukaryotic cell is a mammalian cell. In some embodiments, the eukaryotic cell is a human cell.

Iv. Kits

Aspects of the presently disclosed subject matter relate to kits for modulating the expression levels of polypeptides in cells. In general, a presently disclosed kit contains some or all of the components, reagents, supplies, and the like to practice a method according to the presently disclosed subject matter.

In some embodiments, the term "kit" may refer to any intended article of manufacture (e.g., a package or a container) comprising at least one of the presently disclosed engineered nuclease, expression vector, isolated recombinant cell comprising the expression vector, and instructions for modulating expression of an endogenous polypeptide using an engineered nuclease or recombinant polypeptide of interest using the expression vector and/or the isolated recombinant cell. The kit can be packaged in a divided or undivided container, such as a carton, bottle, ampule, tube, etc. The presently disclosed compositions can be packaged in dried, lyophilized, or liquid form. Additional components provided can include vehicles for reconstitution of dried components. Preferably all such vehicles are sterile and apyrogenic so that they are suitable for injection into a patient without causing adverse reactions.

The polynucleotide sequences for use in the methods, expression vectors, and kits of the presently disclosed subject matter may encode one or more bioactive molecules functional in the treatment of a disease, disorder, or condition. The one or more bioactive molecules may be selected

from the group consisting of proteins, polypeptides, peptides, drugs, enzymes, and hormones.

In some embodiments, the kit can be used for high-throughput purification and quantification of a recombinant polypeptide of interest to be expressed. The kit may include a presently disclosed expression vector for expressing a recombinant polypeptide of interest in a host cell. In some embodiments, the kit includes an affinity chromatography resin, a proteolytic enzyme, an internal quantification standard, a matrix for MALDI-TOF mass spectrometry, and instructions for use. In some embodiments, the kit includes at least one buffer selected from the group consisting of a lysis buffer, a denaturing buffer, an affinity chromatography binding buffer, an affinity chromatography washing buffer, an affinity chromatography elution buffer, and a proteolytic digestion buffer. In some embodiments, the kit for high throughput purification and quantification includes at least one multi-well plate. In some embodiments, the kit for high throughput purification and quantification includes a partially or fully automated high throughput purification and quantification system.

V. Methods of Producing Polypeptides

Aspects of the presently disclosed subject matter relate to methods of producing a polypeptide of interest (e.g., a recombinant polypeptide) that involve culturing a recombinant cell of the presently disclosed subject matter in vitro under conditions suitable for the tunable expression of the polypeptide of interest in the cell. In some embodiments, the method optionally includes recovering the polypeptide of interest. In some embodiments, a kit of the presently disclosed subject matter can be used to produce a polypeptide of interest.

Vi. Recombinant Polypeptides

Aspects of the presently disclosed subject matter relate to recombinant polypeptides produced according to a method of the presently disclosed subject matter. In some aspects, a recombinant polypeptide of interest can be produced using an expression vector of the presently disclosed subject matter. In other aspects, a recombinant polypeptide of interest can be produced in an isolated recombinant cell of the presently disclosed subject matter. In certain aspects, a recombinant polypeptide of interest can be produced using a kit of the presently disclosed subject matter. As used herein, "polypeptide of interest" refers to any polypeptide for which the tunable regulation of its expression is desired in cells. In some embodiments, the polypeptide of interest comprises a therapeutic antibody, peptide, protein, or enzyme. In some embodiments, the polypeptide of interest comprises a naturally occurring polypeptide. In some embodiments, the polypeptide of interest comprises a variant of a naturally occurring polypeptide. In some embodiments, the polypeptide of interest comprises a fusion protein. In some embodiments, the polypeptide of interest comprises a label or tag. In some embodiments, the polypeptide of interest comprises a reporter. In some embodiments, the polypeptide of interest comprises at least one N-terminus HA tag and/or at least one C-terminus reporter, such as a fluorescent protein. In some embodiments, the polypeptide of interest is a naturally occurring mammalian protein or a variant thereof. In some embodiments, the polypeptide of interest comprises a human protein or a variant thereof. In some embodiments, the polypeptide of interest comprises a C-terminus polylysine track or tag (e.g., a polyA polynucleotide sequence tag, a polyAAG lysine polynucleotide sequence tag, etc.). In some embodiments, the polypeptide of interest comprises a N-terminus polylysine

track or tag (e.g., a polyA polynucleotide sequence tag, a polyAAG lysine polynucleotide sequence tag, etc.).

Vii. General Definitions

Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation. Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this presently described subject matter belongs.

Following long-standing patent law convention, the terms “a,” “an,” and “the” refer to “one or more” when used in this application, including the claims. Thus, for example, reference to “a subject” includes a plurality of subjects, unless the context clearly is to the contrary (e.g., a plurality of subjects), and so forth.

Throughout this specification and the claims, the terms “comprise,” “comprises,” and “comprising” are used in a non-exclusive sense, except where the context requires otherwise. Likewise, the term “include” and its grammatical variants are intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that can be substituted or added to the listed items.

For the purposes of this specification and appended claims, unless otherwise indicated, all numbers expressing amounts, sizes, dimensions, proportions, shapes, formulations, parameters, percentages, parameters, quantities, characteristics, and other numerical values used in the specification and claims, are to be understood as being modified in all instances by the term “about” even though the term “about” may not expressly appear with the value, amount or range. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are not and need not be exact, but may be approximate and/or larger or smaller as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art depending on the desired properties sought to be obtained by the presently disclosed subject matter. For example, the term “about,” when referring to a value can be meant to encompass variations of, in some embodiments, $\pm 100\%$ in some embodiments $\pm 50\%$, in some embodiments $\pm 20\%$, in some embodiments $\pm 10\%$, in some embodiments $\pm 5\%$, in some embodiments $\pm 1\%$, in some embodiments $\pm 0.5\%$, and in some embodiments $\pm 0.1\%$ from the specified amount, as such variations are appropriate to perform the disclosed methods or employ the disclosed compositions.

Further, the term “about” when used in connection with one or more numbers or numerical ranges, should be understood to refer to all such numbers, including all numbers in a range and modifies that range by extending the boundaries above and below the numerical values set forth. The recitation of numerical ranges by endpoints includes all numbers, e.g., whole integers, including fractions thereof, subsumed within that range (for example, the recitation of 1 to 5 includes 1, 2, 3, 4, and 5, as well as fractions thereof, e.g., 1.5, 2.25, 3.75, 4.1, and the like) and any range within that range.

EXAMPLES

The following examples are included to demonstrate various embodiments of the present disclosure. It should be appreciated by those of skill in the art that the techniques disclosed in the examples that follow represent techniques discovered by the inventors to function well in the practice of the invention, and thus can be considered to constitute preferred modes for its practice. However, those of skill in

the art should, in light of the present disclosure, appreciate that many changes can be made in the specific embodiments which are disclosed and still obtain a like or similar result without departing from the spirit and scope of the invention.

EXAMPLE 1: In Translational Regulation, Poly(A) Coding Sequences in Human Cells Unexpectedly Induce Ribosome Pausing Directly, without a Role for the Encoded Basic Peptide

Bioinformatic analysis can be used as an initial approach to ask whether there are evolutionary constraints that limit the abundance of polybasic amino acid residues. Runs of polybasic residues in coding sequences of genes from many eukaryotic organisms are under-represented when compared to runs of other amino acids (Karlin et al. (2002) Proc. Natl. Acad. Sci. U.S.A 99, 333-338). Interestingly, polyarginine runs have a similar abundance to polylysine runs at each segment length across multiple organisms (FIG. 1A, FIG. 1B). A series of mCherry reporters were developed to evaluate the effects of polybasic sequences on translation efficiency (output). The reporter construct consists of a double HA-tag, a run of control or polybasic sequences, followed by the mCherry reporter sequence (HAMCherry, FIG. 2A). As a control for DNA transfection and in vivo fluorescence measurements, a construct with green fluorescent protein (GFP) was also created. The reporters were used to ask whether the polybasic sequences influence translation of reporter sequences in neonatal human fibroblasts (HDFs) as well as in *Drosophila* S2 cells and Chinese hamster ovary cells (CHO) (FIG. 2B, FIG. 2C, FIG. 3A, FIG. 3B, FIG. 4A, FIG. 4B). Expression of the mCherry reporter was followed using fluorescence at 610 nm in vivo or western blot analyses of samples collected 48 hours after transfection (FIG. 2B, FIG. 2C). The stability of reporter mRNAs was determined using standard quantitative reverse transcription polymerase chain reaction assay (qRT-PCR, (Djuranovic et al. (2012) Science. 336, 237-240) (FIG. 2D). By careful primer design, this method allows estimation of the level of endonucleolytic cleavage on mRNAs with stalled ribosome complexes.

The results of DNA transfections indicate that strings of lysine codons specifically inhibit translation and decrease the stability of the mCherry reporter mRNA while up to 12 arginine codons (AGG and CGA) have much less, if any effect, on either translation or mRNA stability (FIG. 2B, FIG. 2C, FIG. 2D, FIG. 3A, FIG. 3B, FIG. 4A, FIG. 4B). The potency of translational repression by lysine codons is clearly seen with as few as six AAA-coded lysines (AAA6) and increases with the length of the homo-polymeric amino acid run. It is also noted that the levels of expressed mCherry reporters (FIG. 2B, FIG. 2C) correlate with the stability of their mRNAs (FIG. 2D), consistent with earlier published observations (Doma and Parker (2006) Nature 440, 561-564; Dimitrova et al. (2009) J. Biol. Chem. 284, 10343-10352; Tsuboi et al. (2012) Mol. Cell. 46, 518-529). To control for possible transcriptional artifacts due to the effects of homopolymeric sequence on transcription by RNA polymerase, mRNAs synthesized in vitro by T7 RNA polymerase were electroporated directly into HDF cells. Previous studies established that T7 RNA polymerase is able to transcribe such homopolymeric sequences with high fidelity (Koutmou et al. (2015) eLIFE 10.7554/eLife.05534; Djuranovic et al. (2012) Science. 336, 237-240). Results of the mRNA electroporation work reproduced DNA transfection experiments, consistent with models of translational repression triggered by lysine codons (FIG. 5A, FIG. 5B). To

assess whether the stability of polylysine reporter mRNAs is dependent on translation, the translation initiation inhibitor harringtonine (Fresno et al. (1977) Eur. J. Biochem. FEBS. 72, 323-330) was introduced into HDF cells prior to mRNA electroporation. In this case, a significant change in mRNA stability between wild type and polylysine-encoding mCherry constructs was not observed (FIG. 6A, FIG. 6B, FIG. 6C); these data indicate that accelerated decay of polylysine mCherry mRNAs is dependent on translation. Consistent with this observation, the insertion of 36As (sequence equivalent to twelve lysine AAA codons) after the stop codon, in the 3'UTR region, did not affect the protein expression level or mRNA stability of the assayed construct (FIG. 7A, FIG. 7B, FIG. 7C). Insertion of polylysine codons at different positions along the coding sequence drastically reduced reporter expression and mRNA levels independent of the relative position in the construct. As such, it follows that the observed changes in mRNA stability (FIG. 2D) result from translation-dependent processes.

The most striking observation from these data is that the production of polylysine constructs is codon dependent; runs of polylysine residues coded by AAA codons have a much larger effect on the protein output from reporter constructs than an equivalent run of lysine AAG codons (FIG. 2B, FIG. 2C, FIG. 2D, FIG. 3A, FIG. 3B, FIG. 4A, FIG. 4B, FIG. 5A, FIG. 5B, FIG. 6A, FIG. 6B, FIG. 6C, FIG. 7A, FIG. 7B, FIG. 7C, FIG. 8A, FIG. 8B, FIG. 8C). This effect is unlikely to be driven by the intron-less nature of the reporter since constructs containing human hemoglobin gene (delta chain, HBD) with two introns showed the same effect on protein output and RNA stability (FIG. 8A, FIG. 8B, FIG. 8C). It is also noted that this effect is unlikely to be driven simply due to tRNA^{Lys} abundance, since the relative protein expression and mRNA stability are comparable in cells from various species that do not share similar tRNA abundance profiles (gtrnadb.ucsc.edu/; FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 3A, FIG. 3B, FIG. 4A, FIG. 4B, FIG. 5A, FIG. 5B, FIG. 6A, FIG. 6B, FIG. 6C, FIG. 7A, FIG. 7B, FIG. 7C, FIG. 8A, FIG. 8B, FIG. 8C). Furthermore, the human genome encodes a comparable number of tRNA genes for AAA and AAG codons (gtrnadb.ucsc.edu/Hsapi19/) and general codon usage is similar (0.44 vs 0.56, AAA vs AAG). The generality of codon-dependent polylysine protein production was recently documented in *E. coli* cells where a single tRNA^{Lys}(UUU) decodes both AAA and AAG codons (Koutmou et al. (2015) eLIFE 10.7554/eLife.05534).

In light of these experimental observations, codon usage and the distribution of lysine codons in polylysine tracks in various species was systematically explored (FIG. 9). Remarkably, a strong under-representation of poly(A) nucleotide runs in regions coding for iterated lysines (even with as few as three lysines) in human genes is found (FIG. 9). When there are four iterated lysine residues, the difference between expected (from data for all lysine residues) and observed codon usage for four AAA codons in a row is over one order of magnitude (FIG. 10). Notably, similar

patterns of codon usage in lysine poly(A) tracks are observed in other vertebrates (FIG. 11).

Ribosome profiling data have the potential to reveal features of pausing on polybasic stretches throughout the genome (Ingolia (2014) Nat. Rev. Gen. 15, 205-213). A cumulative analysis of three ribosome profiling datasets from human cells for regions encoding four lysines in a row revealed that the occupancy pattern on four lysines encoded by three AAA and one AAG codon is different from the pattern for two, three and four AAG codons in four lysine-tracks (FIG. 12A). The latter three resemble the occupancy pattern for tracks of arginines (FIG. 13A, FIG. 13B), which is similar to the ribosome stalling on runs of basic amino acids observed by other researchers (Charneski and Hurst (2013) PLoS Biol. 11, e1001508). This suggests that the observed effect on protein output and mRNA stability is dependent on nucleotide, not simply the amino acid sequence. Additionally, the first example (with three AAA and one AAG codon) has a region of increased ribosome occupancy found additionally after the analyzed region (FIG. 12A). Together, these data suggest that attenuation of translation on poly(A) nucleotide tracks occurs via a different mechanism than just the interaction of positively charged residues with the negatively charged ribosomal exit tunnel.

In order to probe the potential impact of the observed disparities in codon distribution for runs of three and four consecutive lysine codons, runs of three lysine residues with various numbers of consecutive As (A9-A13) were inserted into the mCherry reporter construct (FIG. 12B). As in the previous experiments (FIG. 2B, FIG. 2C), the expression of the mCherry reporter as well as the stability of the mRNA was followed (FIG. 12C, FIG. 12D, FIG. 12E). It was found that the insertion of sequences with 12 or more consecutive As reduces mCherry reporter expression by more than 50% with comparable effects on mRNA stability. Importantly, in each construct, no more than three lysines are encoded so the increasing effect on protein output must result from consecutive As, not Ks.

Next, it was asked whether polylysine sequences from naturally occurring genes have the same general effect on expression of reporter protein. To take an unbiased approach, different lengths of homopolymeric lysine runs and various distributions of AAA and AAG codons were selected (FIG. 14A). Reporter constructs with lysine runs were electroporated into HDF cells and relative amounts of reporter expression and mRNA stability were evaluated (FIG. 14B, FIG. 14C). As with the designed sequences in FIG. 12B, the observed decreases in reporter protein expression and mRNA stability correlated with the number of consecutive A nucleotides and not with total number of lysine codons in the chosen sequences. The reporter experiments together (FIG. 2B, FIG. 2C, FIG. 2D, FIG. 3A, FIG. 3B, FIG. 4A, FIG. 4B, FIG. 5A, FIG. 5B, FIG. 6A, FIG. 6B, FIG. 6C, FIG. 7A, FIG. 7B, FIG. 7C, FIG. 8A, FIG. 8B, FIG. 8C, FIG. 12B, FIG. 12C, FIG. 12D, FIG. 12E, FIG. 14A, FIG. 14B, FIG. 14C) argue that the repressive effects of the polylysine sequence are caused by iterated poly(A) tracks rather than by runs of encoded lysine residues. Similar effects were recently documented in in vivo and in vitro experiments with *E. coli* cells or a purified translational system, respectively (Koutmou et al. (2015) eLIFE 10.7554/eLife.05534). The differences observed in expression of reporter sequences with poly(A) nucleotide tracks from human genes favor the possibility that such regions in natural genes play a "translational attenuator" role that can modulate overall protein expression.

TABLE 1

shows a table of overrepresentation of gene ontology terms for 456 genes containing polyA tracks in their coding regions up to P-value of 0.05					
Term	Background frequency	Sample frequency	Expected	+/-	P-value
nucleic acid binding (GO:0003676)	3799	149	7.614e+01	+	60651e-15
heterocyclic compound binding (GO:1901363)	5648	189	1.132e+02	+	5.339e-13
RNA binding (GO:0003723)	1500	79	3.006e+01	+	7.430e-13
organine cyclic compound binding (GO:0097159)	5717	190	1.146e+02	+	8.553e-13
poly(A) RNA binding (GO:0044822)	1122	62	2.249e+01	+	1.388e-10
binding (GO:0005488)	12444	316	2.494e+02	+	5.325e-09
DNA binding (GO:0003677)	2322	87	4.654e+01	+	1.491e-06
protein binding (GO:0005515)	8307	219	1.665e+02	+	3.725e-05
ion binding (GO:0043167)	5844	166	1.171e+02	+	3.815e-05
chromatin binding (GO:0003682)	409	26	8.197e+00	+	6.569e-05
zinc ion binding (GO:0008270)	1181	49	2.367e+01	+	2.713e-04
molecular function (GO:0003674)	15480	353	3.103e+02	+	3.165e-04
transition metal ion binding (GO:0046914)	1417	52	2.840e+01	+	3.844e-03
ATP binding (GO:0005524)	1430	52	2.886e+01	+	4.880e-03
nucleotide binding (GO:0000166)	2264	73	4.538e+01	+	6.077e-03
nucleoside phosphate binding (GO:1901265)	2265	73	4.540e+01	+	6.164e-03
adenyl ribonucleotide binding (GO:0032559)	1465	52	2.936e+01	+	9.081e-03
adenyl nucleotide binding (GO:0030554)	1483	52	2.972e+01	+	1.235e-02
protein serine/threonine kinase activity (GO:0004674)	434	22	8.698e+00	+	1.523e-02
nucleoside-triphosphate activity (GO:0017111)	707	30	1.417e+01	+	2.103e-02
purine ribonucleotide triphosphate binding (GO:0035639)	1760	58	3.527e+01	+	2.439e-02
purine ribonucleotide binding (GO:0032555)	1801	59	3.610e+01	+	2.486e-02
purine ribonucleoside binding (GO:0032550)	1769	58	3.545e+01	+	2.783e-02
purine nucleoside binding (GO:0001883)	1772	58	3.551e+01	+	2.908e-02
ribonucleoside binding (GO:0032549)	1773	58	3.553e+01	+	2.950e-02
ribonucleotide binding (GO:0032553)	1816	59	3.640e+01	+	3.087e-02
purine nucleotide binding (GO:0017676)	1821	59	3.650e+01	+	3.315e-02
nucleoside binding (GO:0001882)	1785	58	3.574e+01	+	3.409e-02
protein kinase activity (GO:0004672)	591	26	1.184e+01	+	3.421e-02
helicase activity (GO:0004386)	145	11	2.906e+00	+	3.556e-02
small molecule binding (GO:0036094)	2539	76	5.089e+01	+	4.353e-02

Based on the results with insertion of 12 consecutive A nucleotides (FIG. 12C) and endogenous A-rich sequences (FIG. 14B), it is proposed that a run of 11As in a stretch of 12 nucleotides (12A-1 pattern) will typically yield a measurable effect on protein expression. Since the A string did not require beginning in any particular codon frame, the sequence may not necessarily encode four consecutive lysines. As such, the 12A-1 pattern has been used to search the cDNA sequence database for multiple organisms (NCBI

RefSeq resource; Pruitt et al. (2014) Nucleic Acids Res. 42, D756-763). This query revealed over 1800 mRNA sequences from over 450 human genes; the proportion was similar in other vertebrates (Table 5). Gene ontology analyses revealed an over-representation of nucleic acid binding proteins, especially RNA binding and poly(A) RNA binding proteins (Table 1). The positions of poly(A) tracks are distributed uniformly along these identified sequences with no significant enrichment towards either end of the coding

region (FIG. 14D). The proteins encoded by these mRNAs are often conserved among eukaryotes; of the 7636 protein isoforms coded by mRNA with poly(A) tracks from human, mouse, rat, cow, frog, zebrafish and fruit fly, 3877 are classified as orthologous between at least two organisms. These orthologous proteins share very similar codon usage in the poly-lysine track, as seen in the example of the RASAL2 tumor suppressor protein (McLaughlin et al. (2013) Cancer Cell. 24, 365-378) (FIG. 15). These observations are consistent with the idea that poly(A) tracks may regulate specific sets of genes in these different organisms. Additional analyses of the ribosome profiling data for mRNAs from selected pools of genes (12A-1 pattern genes) showed an increased number of ribosome footprints (RPFs) in sequences following the poly(A) tracks (FIG. 13A, FIG. 13B). The observed pattern was similar, albeit more pronounced, to the pattern observed for four lysine tracks encoded by three AAA codons and one AAG (FIG. 2A), despite the fact that in many cases the selected pattern did not encode four lysines.

Given the strong sequence conservation and possible role in modulation of protein expression, the effects of mutations in poly(A) tracks were further explored. The reporter constructs containing poly(A) nucleotide tracks from endogenous genes (ZCRB1, MTDH and RASAL2) were used to evaluate effects of synonymous lysine mutations in these poly(A) tracks on protein expression (FIG. 16A, FIG. 16B, FIG. 16C, FIG. 17A, FIG. 17B, FIG. 17C, FIG. 18A, FIG. 18B, FIG. 18C). In each construct, mutations were made that changed selected AAG codons to AAA, increasing the length of consecutive As. Alternatively, AAA to AAG changes were introduced to create interruptions in poly(A) tracks. Reporter constructs with single AAG-to-AAA changes demonstrate consistent decreases in protein expression and mRNA stability. Conversely, AAA-to-AAG changes result in increases in protein expression and mRNA stability (FIG. 16B, FIG. 16C, FIG. 17A, FIG. 17B, FIG. 17C, FIG. 18A, FIG. 18B, FIG. 18C).

It was next asked whether the same synonymous mutations have similar effects when cloned in the full-length coding sequence of the ZCRB1 gene (FIG. 16D, FIG. 16E, FIG. 16F, FIG. 19A, FIG. 19B, FIG. 19C, FIG. 19D). Indeed, the effects on protein and mRNA levels observed with the mCherry reporter sequences are reproduced within the context of the complete coding sequence of the ZCRB1 gene (and mutated variant). Mutation of single AAG-to-AAA codons in the poly(A) track of the ZCRB1 gene (K137K; 411G>A) resulted in a significant decrease in both protein expression and mRNA stability (FIG. 16E, FIG. 16F, FIG. 19A, FIG. 19B, FIG. 19C, FIG. 19D); substitution of two AAA codons with synonymous AAG codons (K136K: 408 A>G; K139K:417A>G) resulted in increases in both recombinant ZCRB1 protein output and mRNA stability. Generally, mutations resulting in longer poly(A) tracks reduced protein expression and mRNA stability, while synonymous substitutions that result in shorter poly(A) nucleotide tracks increased both protein expression and mRNA stability. From these observations, it is suggested that synonymous mutations in poly(A) tracks could modulate protein production from these genes.

Poly(A) tracks resemble ribosome "slippery" sequences that have been associated with translational frame-shifts (Belfield et al. (2007) Nucleic Acids Res. 35, 1322-1332; Chen et al. (2014) Nature 512, 328-332). Recent studies suggest that polyA tracks can induce "sliding" of *E. coli* ribosomes resulting in frame-shifting (Koutmou et al. (2015) eLIFE 10.7554/eLife.05534; Yan et al. (2015) Cell 160(5),

870-81). Therefore, potential frame-shifted products of overexpressed ZCRB1 variants were looked for by immunoprecipitation using an engineered N-terminally located HA-tag. It was observed that the presence of a protein product of the expected size results from possible frame-shifting in the construct with increased length A tracts (ZCRB K137K (411G>A) mutant) (FIG. 21A). The presence of potential frame-shifted protein products was not observed in WT or control double synonymous mutations K136K(408 A>G); K139K(417A>G). Interestingly, it was noted that the K137K-synonymous change represents a recurrent cancer mutation found in the COSMIC database (COSMIC stands for Catalogue of Somatic Mutations in Cancer, cancer.sanger.ac.uk; Forbes et al. (2014) Nucleic Acids Res.) for ZCRB1 gene; cancersanger.ac.uk/cosmic/mutation/overview?id=109189). Similar results were obtained when immuno-precipitations were compared of overexpressed and HA-tagged wild type MTDH gene and a K451 K (1353 G>A) variant, yet another cancer-associated mutation (cancer.sanger.ac.uk/cosmic/mutation/overview?id=150510; FIG. 22A, FIG. 22B).

To further document the extent and direction of frame-shifting in the ZCRB1 transcript, polyA tracks were introduced from WT ZCRB1 and a K137K ZCRB1 mutant into a *Renilla* luciferase reporter gene. Single or double nucleotide(s) were introduced downstream in the reporter sequence following the A track, thus creating +1 and -1 frame-shift (FS) constructs, respectively (FIG. 20B). When compared to wild type ZCRB1 polyA track, the G>A mutant shows decreases in full length luciferase protein expression (approximately 40% reduction in "zero" frame); additionally, the G>A mutant exhibits an increase in expression of -1 FS frame construct (which is not observed in the wild type ZCRB1 poly(A) track -1 FS construct) (FIG. 20C). The total amount of luciferase protein activity from the -1 FS ZCRB1 G>A mutant construct is approximately 10% of that expressed from the "zero" frame mutant construct (FIG. 20C, FIG. 22A, FIG. 22B). No significant change in luciferase expression was detected in samples electroporated with +1 FS constructs where expression from these constructs resulted in background levels of luciferase activity (FIG. 21A, FIG. 21B).

Frame-shifting and recognition of out-of-frame premature stop codons can lead to nonsense-mediated mRNA decay (NMD) that results in targeted mRNA decay (Belew et al. (2011) Nucleic Acids Res. 39, 2799-2808; Belew et al. (2014) Nature. 512, 265-269). Previous efforts suggest that NMD may play a role in determining the stability of poly(A) track-containing mRNAs. Deletion of NMD factor Upf1p in yeast cells partially rescues mRNA levels from constructs with simple poly(A) tracks (Koutmou et al. (2015) eLIFE 10.7554/eLife.05534). The complete set of human poly(A) track-containing genes have been analyzed to see whether they would be likely targets for NMD as a result of frame-shifting on the poly(A) track (based on the usual rules for NMD; Lykke-Andersen et al. (2000) Cell. 103,1121-1131; Le Hir et al. (2001) EMBO J. 20, 4987-4997; Chang et al. (2007) Annu. Rev. Biochem. 76, 51-74; Popp and Maquat (2013) Annu. Rev. Genet. 47, 139-165). Based on the position of the poly(A) tracks, and their position relative to possible PTCs in the -1 and +1 frame, and the location of downstream exon-intron boundaries, a part of the genes of interest would likely be targeted by NMD as a result of frame-shifting during poly(A)-mediated stalling (these transcripts and position of PTCs are listed in Table 2). The considerable number of human poly(A) track genes may not

elicit NMD response since PTCs in both -1 and +1 frame following poly(A) tracks are less than 50 nucleotides away from established exon-intron boundaries. While the majority of frame-shift events seem to lead to proteins that would be truncated immediately after poly(A) tracks, in a few cases a novel peptide chain of substantial length may be produced

(Table 3). As such, the outcome of poly(A) track stalling and slipping may include a scenario in which a frame-shifted protein product is synthesized in addition to the full-length gene product (scheme shown in FIG. 20D). The possible role and presence of such fragments from poly(A) track genes and their variants is still to be elucidated.

TABLE 2

shows a table of mRNAs that have intron-exon boundary closer than 50 nucleotides downstream from a stop codon arising from frameshifting over polyA tracks. These genes would fall in the category of non "classical" NMD targets if frameshifting occurs on polyA track.								
mRNA GI number	end of polyA region	nearest stop codon	location of stop codon	location of downstream intron-exon boundary	mRNA GI number	end of polyA region	nearest stop codon	location of downstream intron-exon boundary
315360663	465	TAG	472	491	239787843	177	TAG	321
315360663	465	TGA	467	491	217035143	1437	TAG	1467
54792083	710	TAG	839	849	188035876	1539	TAA	1553
61743937	294	TAA	310	327	188035876	1539	TGA	1545
61743937	294	TAG	324	327	312283645	1473	TAA	1503
61743937	294	TGA	306	327	291084574	1802	TAG	2242
41350197	346	TGA	385	398	291084574	1802	TGA	1832
42542385	1274	TAA	1312	1342	312434029	1760	TAA	1790
115430109	97	TAA	139	171	21536424	476	TAA	508
115430109	97	TAG	321	340	21536424	476	TGA	515
115430109	97	TGA	162	171	312283632	1760	TAA	1790
38327633	431	TGA	452	470	393185915	487	TAG	498
148596997	396	TGA	411	416	393185915	487	TGA	523
146133847	192	TAA	473	499	187761329	1522	TGA	1553
146133847	192	TAG	200	246	557786140	263	TAA	265
189163500	2545	TAG	2788	2813	296278211	1102	TAA	1184
214010231	475	TAA	748	780	296278220	493	TAA	575
284795238	986	TAG	997	1014	257467646	2736	TGA	2785
284795238	986	TGA	1001	1014	385648269	3155	TAA	3253
332205944	3430	TAA	3688	3732	385648269	3155	TAG	3260
283549152	927	TGA	996	1015	27894332	928	TAG	1071
325197190	472	TAA	484	515	27894332	928	TGA	942
325197190	472	TGA	502	515	350606366	375	TAA	464
325197194	1713	TAA	1725	1756	350606366	375	TAG	434
325197194	1713	TGA	1743	1756	317008579	4447	TGA	4458
325197181	768	TAA	780	811	209915559	770	TAG	825
325197181	768	TGA	798	811	296278213	618	TAA	700
332205942	3451	TAA	3709	3753	350606370	621	TAA	710
378786661	207	TAA	304	352	350606370	621	TAG	680
378786661	207	TAG	351	352	350606369	438	TAG	497
378786661	207	TGA	364	408	385648267	3393	TAA	3491
367460089	3008	TAA	3256	3288	385648267	3393	TAG	3498
378786660	315	TAA	412	460	61744441	448	TGA	466
378786660	315	TAG	459	460	312283650	1601	TAA	1631
378786660	315	TGA	472	516	589811545	2995	TGA	3044
387528005	1266	TAG	1416	1437	387527983	928	TGA	942
57863258	885	TAG	938	947	197245388	940	TAA	1013
380420368	574	TAG	821	842	197245388	940	TGA	968
527317397	1917	TGA	1941	1958	262399360	1971	TAG	2015
543173130	2544	TAG	2847	2872	262399359	1962	TAG	2006
347954818	953	TGA	1010	1038	283806678	8679	TAG	8790
564473350	2511	TAA	2513	2521	189242611	609	TAG	944
585420399	1248	TAG	1381	1407	365812503	1512	TAG	1515
585420411	916	TAG	1049	1075	365812503	1512	TGA	1522
256542287	192	TAA	473	499	153792693	6916	TAG	6952
256542287	192	TAG	200	246	70980548	254	TAG	312
223634474	186	TAA	467	493	38683845	918	TAA	924
223634474	186	TAG	194	240	38683845	918	TAG	959
585866348	1248	TAG	1381	1407	38683845	918	TGA	981
110349755	889	TGA	926	951	148612837	2289	TAG	2586
110349753	889	TGA	926	951	47419908	2793	TAA	2807
544583528	866	TAA	980	1012	47419908	2793	TAG	2821
93277100	828	TAA	855	903	47419908	2793	TGA	2795
145699132	2771	TAA	2772	2777	21361597	1222	TGA	1225
93141224	645	TAG	649	667	291190786	1506	TAG	1514
527317380	612	TAA	693	696	67944632	259	TAG	391
32967604	1594	TAG	1683	1697	67944631	259	TAG	391
72377390	216	TAG	281	318	67944633	415	TAG	547
153251835	867	TAA	901	905	122114650	3206	TAA	3430
153251835	867	TGA	872	905	109715821	865	TAA	991
215272324	361	TGA	421	464	109715821	865	TAG	1082
46255020	219	TGA	226	247	109715821	865	TGA	917
116063563	3480	TAA	3534	3538	154354989	1512	TAG	1515

TABLE 2-continued

shows a table of mRNAs that have intron-exon boundary closer than 50 nucleotides downstream from a stop codon arising from frameshifting over polyA tracks. These genes would fall in the category of non "classical" NMD targets if frameshifting occurs on polyA track.								
mRNA GI number	end of polyA region	nearest stop codon	location of stop codon	location of downstream intron-exon boundary	mRNA GI number	end of polyA region	nearest stop codon	location of downstream intron-exon boundary
116063563	3480	TGA	3509	3538	154354989	1512	TGA	1522
194473999	931	TAA	998	1004	47419910	2895	TAA	2909
56550119	511	TAA	525	559	47419910	2895	TAG	2923
56550119	511	TAG	531	559	47419910	2895	TGA	2897
564473377	1659	TAA	1661	1669	216548486	1594	TGA	1632
151301227	2638	TAA	2698	2723	189491764	1859	TGA	1943
451172080	719	TAG	775	808	189242610	609	TAG	944
451172080	719	TGA	770	808	315360659	761	TAG	768
94538358	192	TAA	473	499	315360659	761	TGA	763
94538358	192	TAG	200	246	114155141	2355	TAA	2463
119637838	1257	TAG	1275	1304	114155141	2355	TAG	2450
119637838	1257	TGA	1288	1304	75677575	956	TAA	959
49640008	6092	TAA	6105	6154	75677575	956	TAG	971
49640008	6092	TGA	6114	6154	325197180	604	TAA	616
49640010	4801	TAA	4814	4863	325197180	604	TGA	634
49640010	4801	TGA	4823	4863	189409141	1081	TAA	1134
239787903	1353	TAA	1428	1472	189409141	1081	TAG	1156
239787903	1353	TAG	1469	1472	91208427	6031	TAA	6289
239787903	1353	TGA	1442	1472	115527086	2080	TGA	2087
55749880	458	TGA	519	549	125656164	2810	TAG	2926
225579091	2672	TAA	2683	2705	61635914	967	TGA	974
225579091	2672	TGA	2695	2705	183227692	581	TAA	618
125987600	747	TAA	858	889	183227692	581	TGA	625
187828563	1100	TAA	1112	1153	51873042	1122	TAG	1187
451172082	569	TGA	733	764	189163502	2545	TAG	2788
451172084	533	TAG	589	622	148746212	2045	TAG	2114
451172084	533	TGA	584	622	148746212	2045	TGA	2099
451172086	533	TGA	697	728	122937397	8679	TAG	8790
270483794	1195	TGA	1196	1203	284795235	1103	TAG	1114
209870068	878	TAA	940	985	284795235	1103	TGA	1118
72377376	305	TAG	370	407	392307008	2145	TAA	2393
63054847	785	TGA	789	820	124107605	853	TAA	861
573459699	766	TAA	781	801	124107605	853	TGA	864
573459699	766	TGA	893	915	385648266	3527	TAA	3625
116642876	1399	TAG	1450	1493	385648266	3527	TAG	3632
116642876	1399	TGA	1457	1493	325053711	956	TAA	959
188219548	1077	TAA	1084	1113	325053711	956	TAG	971
188219548	1077	TAG	1103	1113	325053712	956	TAA	959
544583488	906	TAA	1020	1052	325053712	956	TAG	971
543173126	2619	TAG	2922	2947	392307006	2628	TAA	2876
544346128	1063	TAG	1145	1189	459215048	2072	TAG	2147
544346128	1063	TGA	1232	1257	388490157	3286	TGA	3323
170650704	667	TAA	807	830	32483358	2382	TAA	2485
170650704	667	TAG	798	830	32483358	2382	TGA	2449
544583450	918	TAA	1032	1064	291290967	5965	TAG	6092
289547540	894	TAA	992	1033	291084578	1907	TAG	2347
367460086	2981	TAA	3229	3261	291084578	1907	TGA	1937
51243064	631	TGA	638	656	307746920	769	TAG	802
215599550	501	TAG	662	689	32483365	2349	TAA	2452
475505353	3017	TAA	3256	3290	32483365	2349	TGA	2416
122937226	703	TAA	731	738	531990837	1144	TAA	1153
122937226	703	TAG	855	869	531990837	1144	TGA	1184
122937395	987	TAA	1025	1028	32483360	2382	TAA	2485
122937395	987	TAG	1032	1072	32483360	2382	TGA	2449
122937395	987	TGA	1020	1028	22027649	707	TAG	787
57863256	971	TAG	1024	1033	22027649	707	TGA	782
451172081	569	TAG	625	658	350606365	491	TAA	580
451172081	569	TGA	620	658	350606365	491	TAG	550
87298936	1370	TAA	1453	1487	459215047	2072	TAG	2147
138175816	1579	TAA	1583	1610	291575138	897	TAG	1023
138175816	1579	TGA	1580	1610	531990838	858	TAA	868
150036261	475	TAA	575	602	531990838	858	TGA	899
150036261	475	TGA	476	493	353411933	747	TAG	751
315360661	566	TAG	573	592	313661425	3742	TGA	3757
315360661	566	TGA	568	592	257467647	3543	TGA	3592
38373672	3210	TAG	3474	3489	388490159	3786	TGA	3823
38373672	3210	TGA	3244	3290	531990840	822	TAA	832
544346132	1174	TAG	1256	1300	531990840	822	TGA	863
544346132	1174	TGA	1343	1368	354682004	903	TAG	1487
116805347	219	TGA	261	278	354682006	1473	TAA	1583
157502183	1552	TAA	1882	1906	354682006	1473	TAG	1484

TABLE 2-continued

shows a table of mRNAs that have intron-exon boundary closer than 50 nucleotides downstream from a stop codon arising from frameshifting over polyA tracks. These genes would fall in the category of non "classical" NMD targets if frameshifting occurs on polyA track.

mRNA GI number	end of polyA region	nearest stop codon	location of stop codon	location of downstream intron-exon boundary	mRNA GI number	end of polyA region	nearest stop codon	location of downstream intron-exon boundary
23111022	714	TAG	721	740	354682006	1473	TGA	1572
23111022	714	TGA	716	740	32483364	2349	TAA	2452
32967602	1594	TAG	1683	1697	32483364	2349	TGA	2416
573459727	352	TAA	367	387	305410828	910	TAG	942
573459727	352	TGA	479	501	467091961	1974	TAA	1981
573459714	617	TAA	632	652	467091961	1974	TAG	2146
573459714	617	TGA	744	766	467091961	1974	TGA	1990
574275032	622	TGA	629	633	401664559	8145	TAG	8151
574275429	622	TGA	629	633	163792200	1425	TAG	1542
574275427	622	TGA	629	633	119829186	1344	TAA	1374
574275776	622	TGA	629	633	119829186	1344	TGA	1363
574275778	557	TGA	564	568	305410830	1079	TAG	1111
169234948	317	TAG	360	370	284004924	591	TAA	593
169234948	317	TGA	336	370	224994180	967	TGA	974
557636701	3791	TAA	3834	3870	270483791	1363	TGA	1364
557636701	3791	TGA	3826	3870	305410832	747	TAG	780
344925844	1128	TGA	1241	1282	307746901	1016	TAG	1048
115298681	736	TGA	746	787	270132520	4540	TAA	4667
61743939	295	TAA	310	327	153791627	1121	TAG	1132
61743939	295	TAG	324	327	153791627	1121	TGA	1136
61743939	295	TGA	306	327	166235162	541	TAA	591
289547543	894	TAA	992	1033	193788631	841	TAG	883
7705934	473	TAG	718	750	193788631	841	TGA	857
209870072	451	TAA	514	559	194353965	6584	TAG	6589
209870078	917	TAA	979	1024	217330569	1221	TAG	1261
209870070	704	TAA	766	811	217330573	1408	TAG	1448
215599561	320	TAG	481	508	51873044	1026	TAG	1091
544583540	918	TAA	1032	1064	119395733	9174	TAG	9307
284172494	1574	TAA	1651	1685	217330567	1513	TAG	1553
239582713	995	TAG	1343	1371	51873046	1122	TAG	1187
225579094	2815	TAA	2826	2848	299782586	1712	TGA	1735
225579094	2815	TGA	2838	2848	289547522	1454	TAG	1494
110347419	1175	TAA	1181	1221	574269958	1379	TAA	1458
110347419	1175	TAG	1192	1221	574269958	1379	TAG	1385
110347419	1175	TGA	1187	1221	574272532	1583	TAA	1662
284795233	1121	TAG	1132	1149	574272532	1583	TAG	1589
284795233	1121	TGA	1136	1149	574272304	1192	TAA	1271
544583452	918	TAA	1032	1064	574272304	1192	TAG	1198
399498564	848	TAG	902	943	574271714	1478	TAA	1557
399498564	848	TGA	906	943	574271714	1478	TAG	1484
399498565	851	TAA	863	904	574271316	1379	TAA	1458
223005861	4124	TGA	4130	4134	574271316	1379	TAG	1385
155029543	670	TAG	810	848	574273241	1478	TAA	1557
155029541	801	TAG	941	979	574273241	1478	TAG	1484
157739935	1035	TAA	1036	1063	574269522	1427	TAA	1506
38202203	1157	TAA	1292	1305	574269522	1427	TAG	1433
388240807	1129	TAA	1135	1175	558472849	1026	TAG	1091
388240807	1129	TAG	1146	1175	156523967	855	TAA	867
388240807	1129	TGA	1141	1175	156523967	855	TAG	864
67782329	488	TAG	1321	1324	156523967	855	TGA	873
542133168	2773	TAA	2877	2887	112382209	1093	TAA	1108
542133168	2773	TAG	2838	2887	112382209	1093	TAG	1174
542133173	2334	TAA	2438	2448	112382209	1093	TGA	1097
542133173	2334	TAG	2399	2448	111120330	2052	TAA	2059
542133166	2690	TAA	2794	2804	111120330	2052	TAG	2224
542133166	2690	TAG	2755	2804	111120330	2052	TGA	2068
542133167	2827	TAA	2931	2941	88703042	692	TAA	713
542133167	2827	TAG	2892	2941	109255233	1177	TAG	1255
542133171	2646	TAA	2750	2760	88703040	587	TAA	608
542133171	2646	TAG	2711	2760	112789561	683	TGA	836
542133176	1321	TAA	1425	1435	126032349	10872	TAG	10941
542133176	1321	TAG	1386	1435	62244047	946	TAG	963
542133175	1388	TAA	1492	1502	226437633	186	TAA	452
542133175	1388	TAG	1453	1502	226437633	186	TAG	496
542133172	2571	TAA	2675	2685	226437633	186	TGA	209
542133172	2571	TAG	2636	2685	199559531	1216	TAA	1223
365192531	3074	TAA	3322	3354	189163519	4030	TAA	4062
239787905	1468	TAA	1543	1587	189163519	4030	TAG	4116
239787905	1468	TAG	1584	1587	189163519	4030	TGA	4051
239787905	1468	TGA	1557	1587	199559437	1154	TAA	1161
393185909	180	TAG	191	237	199559489	1301	TAA	1308

TABLE 2-continued

shows a table of mRNAs that have intron-exon boundary closer than 50 nucleotides downstream from a stop codon arising from frameshifting over polyA tracks. These genes would fall in the category of non "classical" NMD targets if frameshifting occurs on polyA track.								
mRNA GI number	end of polyA region	nearest stop codon	location of stop codon	location of downstream intron-exon boundary	mRNA GI number	end of polyA region	nearest stop codon	location of downstream intron-exon boundary
393185909	180	TGA	216	237	199558961	1216	TAA	1223
542133170	2639	TAA	2743	2753	205360986	300	TAA	372
542133170	2639	TAG	2704	2753	205360986	300	TAG	387
95147341	3226	TAA	3230	3264	205360986	300	TGA	304
95147341	3226	TAG	3238	3264	305410834	1016	TAG	1048
95147341	3226	TGA	3247	3264	32479524	1140	TAG	1659
34577113	1789	TAG	1857	1898	215982795	767	TAG	823
542133169	2639	TAA	2743	2753	349501056	1573	TAG	1613
542133169	2639	TAG	2704	2753	305410827	909	TAG	942
422398891	320	TAA	417	438	349501055	1573	TAG	1613
422398891	320	TAG	526	529	349501059	916	TAG	956
284795244	1103	TAG	1114	1131	349501060	1573	TAG	1613
284795244	1103	TGA	1118	1131	32479526	1373	TAG	1892
301500638	762	TAA	887	905	167860144	1425	TAA	1448
151301203	1982	TAA	2027	2050	167860144	1425	TGA	1473
151301203	1982	TGA	2030	2050	305410836	693	TAG	726
573459695	1552	TAA	1882	1906	386781570	946	TAG	963
422398886	320	TAA	417	438	167830435	2605	TGA	2714
296080784	530	TGA	663	703	167830432	658	TAG	857
300244517	499	TAA	535	547	167830432	658	TGA	660
300244517	499	TAG	517	547	594191052	1137	TAA	1416
300244517	499	TGA	502	547	594191052	1137	TAG	1236
296278216	1064	TAA	1146	1187	55956799	426	TAA	429
527317401	2486	TGA	2588	2594	113204614	8646	TAG	8716
313661424	3736	TGA	3751	3795	113204616	8646	TAG	8716
52145311	2119	TAA	2152	2183	296939603	2391	TAA	2823
52145311	2119	TGA	2139	2183	296939601	2391	TAA	2823
300797236	693	TAG	800	808	27765081	2101	TAA	2199
422398893	337	TAA	434	455	27765081	2101	TAG	2288
422398893	337	TAG	543	546	27765081	2101	TGA	2196
574274904	622	TGA	629	633	157266316	2444	TAA	2455
296278222	341	TAA	423	464	157266316	2444	TAG	2447
422398882	337	TAA	434	455	27765075	428	TAA	526
544346208	1454	TAG	1536	1580	27765075	428	TAG	615
544346208	1454	TGA	1623	1648	27765075	428	TGA	523
291084569	1795	TAG	2235	2249	207113159	4478	TGA	4515
291084569	1795	TGA	1825	1860	304376300	4364	TGA	4401
284795242	1103	TAG	1114	1131	207113158	4247	TGA	4284
284795242	1103	TGA	1118	1131	207113161	4367	TGA	4404
313661429	3451	TGA	3466	3510	207113163	4250	TGA	4287
324711032	1276	TAG	1294	1323	236463299	559	TAA	595
324711032	1276	TGA	1307	1323	236463299	559	TAG	574
313661427	3610	TGA	3625	3669	236463299	559	TGA	587
555943904	605	TAG	608	654	236463163	559	TAA	595
555943904	605	TGA	718	744	236463163	559	TAG	574
301500639	738	TAA	863	881	236463163	559	TGA	587
302191685	381	TAA	404	441	349501083	1110	TAG	1150
422398895	337	TAA	434	455	315360665	835	TAG	842
422398895	337	TAG	543	546	315360665	835	TGA	837
422398884	320	TAA	417	438	387527982	915	TAG	1058
422398884	320	TAG	544	547	387527982	915	TGA	929
300244513	239	TAA	275	287	402513678	237	TAA	246
300244513	239	TAG	257	287	402513678	237	TAG	257
300244513	239	TGA	242	287	305410841	927	TAG	959
313661433	3604	TGA	3619	3663	305410838	905	TAG	937
300244516	430	TAA	466	478	305410844	1074	TAG	1106
300244516	430	TAG	448	478	574957022	893	TAA	901
300244516	430	TGA	433	478	574957022	893	TAG	1028
527317400	2615	TGA	2717	2723	574957022	893	TGA	920
422398870	337	TAA	434	455	574957031	1965	TAA	1973
422398870	337	TAG	561	564	574957031	1965	TAG	2100
555943722	605	TAG	608	654	574957031	1965	TGA	1992
555943722	605	TGA	718	744	386268034	971	TAG	1012
284004933	616	TAA	618	625	386268035	971	TAG	1012
296278218	560	TAA	642	683	386268037	971	TAG	1012
574275184	577	TGA	584	588	223555916	1674	TAA	1693
574275536	666	TGA	673	677	223555916	1674	TGA	1687
386781586	575	TAG	592	638	396578113	1329	TAA	1399
300797255	693	TAG	800	808	396578113	1329	TGA	1452
386781549	946	TAG	963	1009	396578116	1117	TAA	1187
557947981	412	TAA	718	752	396578116	1117	TGA	1240

TABLE 2-continued

shows a table of mRNAs that have intron-exon boundary closer than 50 nucleotides downstream from a stop codon arising from frameshifting over polyA tracks. These genes would fall in the category of non "classical" NMD targets if frameshifting occurs on polyA track.

mRNA GI number	end of polyA region	nearest stop codon	location of stop codon	location of downstream intron-exon boundary	mRNA GI number	end of polyA region	nearest stop codon	location of downstream intron-exon boundary
557947981	412	TGA	430	435	95147334	3551	TAA	3660
110347424	1113	TAA	1119	1159	160707983	2206	TAG	2264
110347424	1113	TAG	1130	1159	154146192	1034	TAA	1086
110347424	1113	TGA	1125	1159	154146192	1034	TGA	1050
110347417	1104	TAA	1110	1150	145275203	1841	TAA	1907
110347417	1104	TAG	1121	1150	154146222	954	TAA	1006
110347417	1104	TGA	1116	1150	154146222	954	TGA	970
53729338	2711	TAA	2714	2749	120953299	4732	TAG	4829
53729336	2290	TAA	2293	2328	120953299	4732	TGA	4784
94538369	841	TAG	883	905	189409139	1021	TAA	1074
94538369	841	TGA	857	905	189409139	1021	TAG	1096
160707983	2206	TAA	2238	2272	239787837	233	TAG	377

TABLE 3

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.

Frameshift	Gene name	Sequence	ELMs
minus one	ALS2CR12	TKKFEMESGEE	ELME000117 ELME000085 ELME000064
minus one	APTX	LEFFQYRIL	ELME000355 ELME000370 ELME000120
minus one	ASTE1	EETEQQLF	ELME000370 ELME000236 ELME000120 ELME000352
minus one	BRCA1	QPNASQAQQKPTTHGR	ELME000202 ELME000353 ELME000239 ELME000070
minus one	C10orf90	RTKEKGDLTK	ELME000351
minus one	CASP5	DVLLYDTIFQIFNNRNCLS	ELME000020 ELME000336 ELME000370 ELME000335 ELME000120 ELME000352
minus one	CCDC146	SWNKKSKR	ELME000011 ELME000285 ELME000278
minus one	CCDC148	LGQEKTEVARNG	ELME000355
minus one	CCDC168	KKVSKLGSQGWRNQ	ELME000202 ELME000351 ELME000008 ELME000053
minus one	CDYL	RQSIWFGGKA	ELME000351 ELME000197
minus one	CENPQ	HLKDLSSSEQTKH	ELME000334 ELME000053
minus one	CEP290	HYQLQVQELTDL	ELME000086 ELME000163
minus one	CHEK1	YLNPWKKIDSAPLALL	ELME000182 ELME000085 ELME000084 ELME000355 ELME000081
minus one	CHRM5	EEKLYWQGNKLP	ELME000352 ELME000137
minus one	CNTLN	MLKMTRNGCCTFRNFLKDSFLLPHIY	ELME000146 ELME000336 ELME000079 ELME000337 ELME000355 ELME000370 ELME000369
minus one	CNTRL	AAQTRLSEL	ELME000086 ELME000285
minus one	CPNE3	TRIQVLSV	ELME000333 ELME000091 ELME000365
minus one	DHX36	TFGILKCSISEKSLRMECKRNW	ELME000336 ELME000162 ELME000368 ELME000063 ELME000103 ELME000360 ELME000370 ELME000108 ELME000064 ELME000100

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.					
Frameshift	Gene name	Sequence	ELMs		
minus one	DIAPH3	TDFVLWKASGTIQFNCK	ELME000368	ELME000085	ELME000370
			ELME000328		
minus one	DYNC1I2	TRRRKLLLLCKKNQILKKKGEKLK	ELME000146	ELME000149	ELME000106
		HCFKAWG	ELME000108	ELME000231	ELME000012
			ELME000233	ELME000102	
minus one	EIF2AK2	LLQKLLSKK	ELME000045	ELME000355	
minus one	FAM133A	SKDETEKEKD	ELME000285	ELME000220	ELME000064
minus one	FBXO38	DVYPSCSSTTASTVGNSSSHNTASQS	ELME000136	ELME000202	ELME000063
		PDF	ELME000159	ELME000197	ELME000239
			ELME000070	ELME000352	ELME000053
minus one	FILIP1	EILLAQNEPCPQSQLLHFPERRLQKV	ELME000202	ELME000173	ELME000335
		EEAHLQTGPHPLFR	ELME000108	ELME000012	ELME000352
			ELME000102		
minus one	GON4L	TKRKRDGRGQEGTLAYDLKDDMLDR	ELME000147	ELME000108	ELME000220
		TLEDGAKQHN	ELME000365	ELME000100	ELME000102
minus one	GOPC	KEAQLAEVKKLRKENEAL	ELME000232	ELME000351	ELME000335
			ELME000365	ELME000089	ELME000102
minus one	GPATCH4	KEAERGGRSYSI	ELME000091	ELME000351	
minus one	HMGXB4	RRTKREEREKSKRRTRCRPTRCS	ELME000271	ELME000202	ELME000101
			ELME000351	ELME000062	ELME000108
			ELME000220	ELME000276	ELME000100
			ELME000008	ELME000102	ELME000053
minus one	IFI16	PKKRLDPKG	ELME000106	ELME000108	ELME000100
minus one	IQCA1	EEKKGKTTQESQKTKERNKGEK	ELME000117	ELME000202	ELME000063
			ELME000220	ELME000064	ELME000352
			ELME000053		
minus one	JARID2	SHSQYHLSPPG	ELME000367	ELME000249	ELME000136
			ELME000159	ELME000285	
minus one	KNOP1	HQEGDALPGHKSPPSRSMESS	ELME000063	ELME000287	ELME000239
			ELME000053		
minus one	KRCC1	NKAKKGQRRKCFGTSLFLD	ELME000336	ELME000353	ELME000108
			ELME000102		
minus one	LPIN2	GERNTNRT	ELME000062		
minus one	MAST4	SGKVTKSLASALSLMIPGDMFAVSPL	ELME000155	ELME000182	ELME000367
		GSPMSPHLSDDPSSSRDSSPSRDSS	ELME000336	ELME000136	ELME000149
		AASASPHQPIVHSSGKNYGFTIRAIRV	ELME000147	ELME000337	ELME000085
		YVGDSDIYTVHHIVWNVEEGSPACQA	ELME000063	ELME000159	ELME000173
		GLKAGDLITHINGEPVHGLVHTEVIELLL	ELME000335	ELME000062	ELME000285
		KSG	ELME000313	ELME000153	ELME000365
			ELME000148	ELME000239	ELME000052
			ELME000321	ELME000053	
minus one	MED19	KRILNGKGRKRRRRKRIDIVQTTQVW	ELME000271	ELME000146	ELME000202
		AAPRPAAAAAY	ELME000101	ELME000103	ELME000093
			ELME000351	ELME000108	ELME000278
			ELME000012	ELME000233	ELME000369
			ELME000052	ELME000276	ELME000100
			ELME000270	ELME000102	
minus one	MGA	MGSDEFDISPRISKQEGSSASSVDLG	ELME000136	ELME000085	ELME000063
		QMF	ELME000159	ELME000062	ELME000365
			ELME000197	ELME000053	
minus one	MIS18BP1	SIPTYVKKRKTTHSSQMTVH	ELME000182	ELME000271	ELME000202
			ELME000063	ELME000062	ELME000285
			ELME000108	ELME000070	ELME000100

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.						
Frameshift	Gene name	Sequence	ELMs			
			ELME000008	ELME000270	ELME000102	
			ELME000053			
minus one	MKNK1	HQQPRACVY				
minus one	MORC4	IITEDSLPSLEAILNYSIFNRENDLLAQF DAIPGKKGTRVLIWNIRR	ELME000336	ELME000149	ELME000147	
			ELME000063	ELME000355	ELME000106	
			ELME000093	ELME000120	ELME000341	
			ELME000012	ELME000287	ELME000069	
			ELME000365	ELME000233	ELME000070	
			ELME000052	ELME000008	ELME000137	
minus one	MYH10	QAHIQDLEEQLDEEEGARQKLQLEKVT AEAKIKMEEIILLEDQNSKFIKEKKL MEDRIAECSSQLAEEEEKAKNLAKIR	ELME000342	ELME000146	ELME000117	
			ELME000149	ELME000202	ELME000106	
			ELME000333	ELME000335	ELME000353	
			ELME000365	ELME000233	ELME000239	
minus one	NEK3	QQNQDSFGK	ELME000353			
minus one	NGFRAP1	LIMANIHQENEEMEQQPMQNGEEDRPL GG	ELME000355			
minus one	NIPBL	ILTHRRLGVLQE	ELME000355	ELME000106	ELME000108	
			ELME000012	ELME000102		
minus one	NPIPB15	KTKQNPRSKNK	ELME000351			
minus one	NR3C1	FSRPLQESHKKP	ELME000117	ELME000336	ELME000355	
minus one	NUP85	RWCQVAPLSISS	ELME000351			
minus one	OSBPL1A	LSEALETLA	ELME000086	ELME000355		
minus one	PA2G4	GLQDCRECHQWGNIRRK	ELME000108	ELME000012	ELME000060	
			ELME000102			
minus one	PHLPP1	RRIHQYIHCHEKETWNCWAEAWWC RCPLSYQA	ELME000182	ELME000160	ELME000091	
			ELME000351	ELME000108	ELME000012	
			ELME000365	ELME000102		
minus one	PLXNC1	QILTSYIFGKQTAFLFASG	ELME000182	ELME000198	ELME000353	
			ELME000197	ELME000052		
minus one	PPP1R10	CHLRLPSQAP	ELME000354	ELME000367	ELME000202	
			ELME000334			
minus one	PRPF40A	QAKQLRKRNWEA	ELME000271	ELME000353	ELME000108	
			ELME000278	ELME000012	ELME000100	
			ELME000102			
minus one	PXK	FSSKEVKTICS	ELME000146	ELME000355		
minus one	RNF145	AAKEKLEAV	ELME000285	ELME000365	ELME000089	
minus one	SENPF7	YPRVSCYFQVITRKDTQSY	ELME000182	ELME000368	ELME000337	
			ELME000202	ELME000370	ELME000062	
			ELME000120	ELME000220	ELME000365	
			ELME000197	ELME000239	ELME000008	
			ELME000102	ELME000053		
minus one	SGOL1	VPQKKMHKSVSS	ELME000336			
minus one	SH3RF1	FVEVAFWRLH	ELME000368	ELME000355	ELME000370	
minus one	SLC46A3	SPIFAFQEEVQKKVSR	ELME000117	ELME000011	ELME000285	
minus one	SMAD5	CHGGTGESLEQSRTA	ELME000354	ELME000336	ELME000053	
minus one	SPATA16	ATSNC	ELME000085	ELME000285	ELME000070	
minus one	TAF1D	RYQPTGRPRGRPEGRNPIYS	ELME000103	ELME000093	ELME000351	
			ELME000122	ELME000108	ELME000097	
			ELME000012	ELME000095	ELME000102	

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.					
Frameshift	Gene name	Sequence	ELMs		
minus one	TAOK1	SGFQSRRIYSISKQKKK	ELME000011	ELME000285	ELME000108
			ELME000012	ELME000065	ELME000051
			ELME000061	ELME000102	
minus one	TDRD5	HNRRFARS	ELME000108	ELME000012	ELME000102
minus one	TFDP2	SGLACLPILLRNVRWR	ELME000285		
minus one	TMEM254	KKIEAKNGDPNDCSEFLRSVWVFWP QSIPIYNLGLPLGPFTQYLVDDHHTLLC NGYWLAWLIHVGESLYAIVLCK	ELME000020	ELME000155	ELME000182
			ELME000317	ELME000336	ELME000079
			ELME000368	ELME000160	ELME000202
			ELME000084	ELME000351	ELME000370
			ELME000335	ELME000120	ELME000081
			ELME000052		
minus one	U2SURP	IWNSSKN	ELME000355	ELME000070	
minus one	ULK4	RECWAVPLAAYTV	ELME000091	ELME000351	ELME000369
minus one	VEZF1	KLHLCALTA	ELME000091	ELME000351	
minus one	ZC3H13	WKSQERENGLIS	ELME000149	ELME000355	ELME000333
			ELME000231		
minus one	ZMYM5	IDAAEHLRYENEKNDGVLLLYT	ELME000149	ELME000084	ELME000355
			ELME000321		
minus one	ZRANB2	QRESSWSCY	ELME000080	ELME000199	ELME000368
			ELME000147	ELME000063	ELME000370
			ELME000062	ELME000353	
plus one	ABCC2	SLGPKMFQNPNG	ELME000146	ELME000285	
plus one	ALS2CR12	MTKKFEMESGEE	ELME000117	ELME000085	ELME000064
plus one	ANKHD1- EIF4EBP3	GTEKETGRR	ELME000093		
plus one	ANKHD1	GTEKETGRR	ELME000093		
plus one	ANKRD49	GKRPKQIASLGC	ELME000091	ELME000108	ELME000100
			ELME000270		
plus one	APAF1	ITNLSRLVVRPHTDAVYHACF	ELME000155	ELME000355	ELME000371
			ELME000070	ELME000052	
plus one	APTX	IGILSIQNTS	ELME000355	ELME000333	ELME000053
plus one	APTX	WNSFNTEYF	ELME000063	ELME000355	ELME000089
plus one	BBX	KKSKMDRHG	ELME000351		
plus one	BEND2	YQPCCIGICR	ELME000355		
plus one	BLM	VSSKSVSEGRDG	ELME000063	ELME000064	ELME000321
plus one	BRCA1	YNQMPVRHSRNLQLME	ELME000355	ELME000106	ELME000062
			ELME000120		
plus one	C10orf90	RTKEKGDLTK	ELME000351		
plus one	C16orf45	KLQKQREDE	ELME000351		
plus one	CAPN3	SPSSFRTEQTATRSWVWTRSQRRAK AKQA	ELME000146	ELME000011	ELME000147
			ELME000337	ELME000202	ELME000063
			ELME000285	ELME000108	ELME000365
			ELME000197	ELME000239	ELME000102
			ELME000053		
plus one	CASP5	DVLLYDTIFQIFNNRNCLS	ELME000020	ELME000336	ELME000370
			ELME000335	ELME000120	ELME000352

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.			
Frameshift	Gene name	Sequence	ELMs
plus one	CASP5	RMCCFMPSSRYSTTATASV	ELME000358 ELME000136 ELME000063 ELME000159 ELME000351 ELME000062 ELME000365 ELME000239 ELME000053
plus one	CCDC122	VLFNLKNELEHEKEIAAISAE	ELME000085 ELME000002 ELME000365
plus one	CCDC146	CLGTRSQN	ELME000354
plus one	CCDC148	WAKKKQKWQEME	ELME000271 ELME000201 ELME000355 ELME000278
plus one	CEP290	IIINFKCRSLQIF	ELME000355 ELME000106
plus one	CHD9	RRRYRREAI	ELME000101 ELME000103 ELME000351 ELME000108 ELME000012 ELME000089 ELME000102
plus one	CHRM5	WKRSGTGRGTASY	ELME000355 ELME000173 ELME000062 ELME000108 ELME000100 ELME000053
plus one	CNTRL	QHKLDYQNC	ELME000084 ELME000353
plus one	EIF2AK2	FYRNYSQRN	ELME000202 ELME000084 ELME000355 ELME000070
plus one	EIF5B	TEGQKTEF	ELME000086
plus one	ERC2	ATGPHRREGDTGR	ELME000285 ELME000108 ELME000102
plus one	ERO1LB	ARERLFSLLQG	ELME000045 ELME000368 ELME000149 ELME000370 ELME000285 ELME000231 ELME000061
plus one	EXOC1	NCFLCATVTTERPVQ	ELME000336 ELME000353
plus one	FAM133A	SQRMKQKQKRM	ELME000271 ELME000011 ELME000101 ELME000285 ELME000108 ELME000100 ELME000270 ELME000102
plus one	FAM227B	DSSFVSIYTHLWENVPRIFEALLIMESK	ELME000182 ELME000368 ELME000063 ELME000370 ELME000120 ELME000047 ELME000365 ELME000352
plus one	FAM81B	TEIVFQKYQIYKK	ELME000370
plus one	FILIP1	WRYYSWPRTSHVPSHNYIFQREDSR KWKRIRICRQAHIPYS	ELME000182 ELME000271 ELME000355 ELME000103 ELME000370 ELME000062 ELME000120 ELME000108 ELME000278 ELME000012 ELME000048 ELME000239 ELME000052 ELME000100 ELME000163 ELME000102 ELME000053
plus one	FOXP3	MRTPPHPVVIISAHTRKKFGLLEERGL RLPHRTAWFFFSV	ELME000358 ELME000155 ELME000367 ELME000085 ELME000063 ELME000106 ELME000333 ELME000091 ELME000351 ELME000370 ELME000365 ELME000102
plus one	GGNBP2	KKKSKILKCDEHIQKLGSCITDP	ELME000271 ELME000146 ELME000007 ELME000351 ELME000173 ELME000278 ELME000008
plus one	GYPC	GVETPPAKSAEKK	ELME000358 ELME000136 ELME000085 ELME000159 ELME000239
plus one	HMGXB4	REGQRERERRKAKKEHVGLPGVL	ELME000155 ELME000271 ELME000146 ELME000101 ELME000103 ELME000351 ELME000108 ELME000002 ELME000278 ELME000233 ELME000102
plus one	HYDIN	EIESDFLATNTTKAQEEQTSS	ELME000117 ELME000336 ELME000070 ELME000352 ELME000053

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.					
Frameshift	Gene name	Sequence	ELMs		
plus one	IGHMBP2	QRTSGHRSAHGGGL	ELME000085 ELME000053	ELME000062	ELME000353
plus one	IL1R2	DHSCDHFPQDHISFSGVKTDNPV	ELME000368 ELME000352	ELME000085 ELME000053	ELME000370
plus one	IQCA1	KKKKKEKQPKAKKQKKGTKEK	ELME000271 ELME000278	ELME000146 ELME000276	ELME000351 ELME000008
plus one	JARID2	GPTLSTISALL	ELME000085	ELME000063	ELME000365
plus one	KCNC1	KHIPRPQLGSPNY	ELME000155 ELME000159	ELME000199 ELME000351	ELME000136 ELME000005
plus one	KDM4D	HDCGGVSPFGKQ	ELME000136	ELME000159	
plus one	KNOP1	TRREMP SQATPSPGPWRAA	ELME000358 ELME000136 ELME000159	ELME000155 ELME000202 ELME000108	ELME000006 ELME000063 ELME000102
plus one	LARP7	DRVEASSLPEVRTGKRKRSSSEDAES LAPRSK	ELME000271 ELME000062 ELME000064 ELME000008 ELME000102	ELME000093 ELME000108 ELME000100 ELME000352	ELME000173 ELME000365 ELME000061 ELME000270
plus one	LOC101929870	QKSFPSEGGQRRSLFLDSNSRENLG WLAKLTREQNILPEAEKPHALSGGG	ELME000146 ELME000063 ELME000062 ELME000231 ELME000233	ELME000336 ELME000101 ELME000353 ELME000012 ELME000051	ELME000085 ELME000106 ELME000108 ELME000365 ELME000102
plus one	LPIN2	KEKEIQTGQ	ELME000351		
plus one	MPP3	PPMSPACEDTAAPFDEQQQEMAASAA FIDRHYGHLVDAVLVKEDLQGAYSQK VVLEKLSKDTHWVPVSWVR	ELME000146 ELME000368 ELME000085 ELME000333 ELME000326 ELME000197	ELME000336 ELME000147 ELME000063 ELME000370 ELME000313 ELME000233	ELME000136 ELME000202 ELME000159 ELME000120 ELME000365 ELME000052
plus one	NAA35	SPIEPRDHNEPSISEHVCWNV	ELME000147 ELME000064	ELME000285	ELME000365
plus one	NCOA7	SQTKCRDSLSCSYKDSYWEGR	ELME000336 ELME000285 ELME000053	ELME000162 ELME000064	ELME000062 ELME000321
plus one	NEK3	PSRIRIALG	ELME000146	ELME000012	ELME000365
plus one	NHLRC2	CTTLAGTGD	ELME000354		
plus one	NIPBL	KKRKAYEPK	ELME000271 ELME000278	ELME000351 ELME000100	ELME000108 ELME000102
plus one	NIPBL	RFLPTGGWGCYRR	ELME000106 ELME000012	ELME000351	ELME000370
plus one	NPIPBI5	KKQNKTHAPKTN	ELME000351	ELME000070	
plus one	NR3C1	NSAGHYRSLTRNL	ELME000146 ELME000120	ELME000085 ELME000334	ELME000353
plus one	OSBPL1A	CQKHWRWP	ELME000354 ELME000012	ELME000160 ELME000102	ELME000108
plus one	PDZD9	ERGVSNKVKTSVHNLSKTQTKLTV	ELME000336 ELME000365	ELME000202 ELME000070	ELME000091 ELME000352
plus one	PEG10	GVEEGARIQASIPTE	ELME000365	ELME000051	ELME000060
plus one	PPFIA2	RLGQLRGFMETEAQAQESLG	ELME000117 ELME000239	ELME000351	ELME000140

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.			
Frameshift	Gene name	Sequence	ELMs
plus one	PPP1R10	TVTYGCQAKPL	ELME000163
plus one	PRR14L	CEENVCRS	ELME000354 ELME000079
plus one	QTRTD1	LGKTGDHTMDIPGCLLYTKTGSAPHLT HHTL	ELME000182 ELME000336 ELME000147 ELME000085 ELME000355 ELME000173 ELME000052 ELME000053
plus one	RALGPS2	SSAPNAVAFTRRFNH	ELME000085 ELME000285 ELME000328 ELME000108 ELME000012 ELME000102
plus one	RBPJ	MERDGCSEQESQPCAFIG	ELME000117 ELME000202 ELME000064 ELME000352 ELME000053
plus one	RNF10	RNRSSCSAPQSSTP	ELME000085 ELME000063 ELME000351 ELME000062 ELME000239 ELME000070 ELME000053
plus one	RNF145	WLQRRNWRQC	ELME000355 ELME000108 ELME000102
plus one	RYR1	RKISQSAQT	ELME000202 ELME000085 ELME000351 ELME000008
plus one	SEN7	HIRGCPVTSKSSPERQLKVMLTNVLWT DLGRKFRKTLPRND	ELME000146 ELME000336 ELME000136 ELME000063 ELME000106 ELME000159 ELME000093 ELME000062 ELME000153 ELME000278 ELME000365 ELME000064 ELME000239 ELME000052 ELME000102 ELME000053
plus one	SEN7	YPRVSCYFQVITRKGLTTK	ELME000182 ELME000146 ELME000368 ELME000337 ELME000370 ELME000062 ELME000120 ELME000365 ELME000197 ELME000239 ELME000102
plus one	SGOL1	FPKKKCTNLSVP	ELME000271 ELME000367 ELME000365 ELME000233 ELME000070 ELME000008
plus one	SLC26A8	KKGPERAPFLVSFVQ	ELME000336 ELME000149 ELME000351 ELME000328
plus one	SLC46A3	AQFLHSRRKFRKKCHV	ELME000271 ELME000336 ELME000011 ELME000285 ELME000108 ELME000278 ELME000102
plus one	SLC4A7	KEEAERMLQDDDDTVHLPFEGGSLLQI PVKA	ELME000155 ELME000367 ELME000149 ELME000147 ELME000091 ELME000351 ELME000335 ELME000240 ELME000052
plus one	SLC05A1	SVDAVSDDDLKEKSNNSEQADKKVS SMGFGKDVRDLPRAAVRI	ELME000063 ELME000106 ELME000091 ELME000198 ELME000285 ELME000365 ELME000197 ELME000233 ELME000070 ELME000008
plus one	SLC05A1	SVDAVSDDDLKEKSNNSEQADKKVS SMGFGKDVRGVIIIPVPSAGVGIVLGGYI	ELME000085 ELME000063 ELME000198 ELME000285 ELME000365 ELME000197 ELME000233 ELME000070 ELME000008
plus one	SPG11	IFLKKRKEL	ELME000011 ELME000355 ELME000108 ELME000100 ELME000102
plus one	SYCP1	QENTNIFIGNT	ELME000353
plus one	TCF25	EKQEKQHGRSIGKRTRRYRSHRPRED	ELME000101 ELME000103 ELME000093 ELME000108 ELME000278 ELME000012 ELME000048 ELME000100 ELME000352 ELME000102

TABLE 3-continued

shows a table showing peptides arising from possible frame-shifting on polyA tracks. Direction of frame-shift, gene name and peptide sequences following polyA track are shown in table. Additional analyses of possible eukaryotic linear motifs (ELMs) found in these peptides is included.			
Frameshift	Gene name	Sequence	ELMs
plus one	TDRD5	ATTEDLQEA	ELME000285 ELME000052
plus one	TERF1	SRRATESRIPVSKSQP	ELME000146 ELME000062 ELME000285 ELME000108 ELME000239 ELME000008 ELME000102
plus one	TFDP2	QVDWPAYQFCSGMSESG	ELME000085 ELME000063 ELME000370 ELME000353
plus one	THOC2	KERCTALQDKLLEEEKKOMEHVORVL QRLKLEKDNWL	ELME000351 ELME000062 ELME000002
plus one	TMEM254	KENRSQEWPRK	ELME000202 ELME000351 ELME000070
plus one	TNRC6B	ATQKVTEQKTKVPE	ELME000011 ELME000285 ELME000053
plus one	TRAPPC10	QHPSPNLYCGQ	ELME000182 ELME000136 ELME000159 ELME000353 ELME000163
plus one	TRDN	CRSRTTGKKTKGKERKTCGTSKVTKE RTLRLN	ELME000354 ELME000147 ELME000202 ELME000063 ELME000093 ELME000173 ELME000062 ELME000220 ELME000064 ELME000052 ELME000102 ELME000053
plus one	TRDN	RNKDTGERTEES	ELME000351 ELME000053
plus one	TRPC1	IASGIPGFVLIYIDVWPVQL	ELME000020 ELME000182 ELME000085 ELME000355 ELME000333 ELME000091 ELME000120 ELME000083 ELME000047 ELME000365 ELME000081
plus one	ULK4	LESAGLFPWLHTQC	ELME000086 ELME000085 ELME000355
plus one	VEZF1	QNFICVHLLQ	ELME000353
plus one	WNK1	QEESSLKQQVEQSSASQTGIKQLPSAS TGIPTASTTSASVSTQVE	ELME000147 ELME000202 ELME000085 ELME000063 ELME000106 ELME000353 ELME000365 ELME000064 ELME000239 ELME000053
plus one	ZCRB1	LLNQKKKLRLK	ELME000271 ELME000011 ELME000355 ELME000278 ELME000102
plus one	ZDHHC3	DEHESRFWPPLLSRLGQPLCHARPRE GRPVPVCGLEKDPDRHGHSSTSPHHST TVPSVLMNV	ELME000136 ELME000368 ELME000063 ELME000159 ELME000370 ELME000173 ELME000365 ELME000233 ELME000239 ELME000369 ELME000352 ELME000053
plus one	ZFHX3	STPFSFHS	ELME000285
plus one	ZMYM5	LLMLQNTDYMKMRKMMVCCCT	ELME000355 ELME000122 ELME000120 ELME000095 ELME000102

In conclusion, the presently disclosed subject matter demonstrates that lysine coding poly(A) nucleotide tracks in human genes act as translational attenuators. It is shown that the effect is dependent on nucleotide, not amino acid sequence, and the attenuation occurs in a distinct manner from previously described polybasic amino acid runs. These “poly(A) translational attenuators” are highly conserved across vertebrates, implying that they might play an important role in balancing gene dosage. Presence of such a regulatory function is further supported by negative selection against single nucleotide variants in human poly(A) segments both in dbSNP and COSMIC databases (FIG. 23; Table 4; Based on dbSNP data, it was found that variations in polyA region are less common than in randomly chosen section of the same length in genes that do not contain polyA

segment (1,8k segments vs 71k segments, one random all transcripts, mean of 0.44 vs 0.49 variations per segment, p-value 0.008 with permutation test and 0.009 with Welch t-test). Almost 300 genes from the original set of 456 had no variation within polyA segment reported in dbSNP). However, it is not yet clear what the effects stemming from synonymous mutation in poly(A) tracks are. Results point to either alterations in protein-levels (altered gene dosage) or to the production of frame-shifted products in the cell. As such, these translational attenuation mechanisms may supplement the already large number of mechanisms through which synonymous mutations can exert biological effects (reviewed in Hunt et al. (2014) Trends Genet. TIG. 30, 308-321).

TABLE 4

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
AASDH	c.342delA	p.K114fs*14	Deletion - Frameshift
ABCA5	c.733G>A	p.E245K	Substitution - Missense
ABCA5	c.742_743insA	p.I248fs*12	Insertion - Frameshift
ABCA5	c.742A>T	p.I248L	Substitution - Missense
ABCA5	c.742delA	p.I248fs*1	Deletion - Frameshift
ABCC2	c.882G>C	p.K294N	Substitution - Missense
ACBD3	c.568delA	p.R190fs*43	Deletion - Frameshift
ADAL	c.644delA	p.E218fs*33	Deletion - Frameshift
ADAL_ENST00000428046	c.563delA	p.E191fs*33	Deletion - Frameshift
AHI1	c.910_911insA	p.T304fs*6	Insertion - Frameshift
AHI1	c.910delA	p.T304fs*23	Deletion - Frameshift
AHI1	c.911C>A	p.T304K	Substitution - Missense
AIM2	c.1027A>C	p.T343P	Substitution - Missense
AIM2	c.1027delA	p.T343fs?	Deletion - Frameshift
AKD1_ENST00000424296	c.2091A>G	p.K697K	Substitution - coding silent
AKD1_ENST00000424296	c.2098_2099delGA	p.E700fs*33	Deletion - Frameshift
AKD1_ENST00000424296	c.2098G>A	p.E700K	Substitution - Missense
AL118506.1	c.48G>A	p.K16K	Substitution - coding silent
ALS2CR12	c.777A>C	p.K259N	Substitution - Missense
ALS2CR12	c.778A>G	p.K260E	Substitution - Missense
ANKHD1	c.4385A>G	p.K1462R	Substitution - Missense
ANKHD1	c.4386G>A	p.K1462K	Substitution - coding silent
ANKHD1-EIF4EBP3	c.4385A>G	p.K1462R	Substitution - Missense
ANKHD1-EIF4EBP3	c.4386G>A	p.K1462K	Substitution - coding silent
ANKRD12	c.2806_2809delAAAC	p.K936fs*22	Deletion - Frameshift
ANKRD1	c.215_216insA	p.K73fs*10	Insertion - Frameshift
ANKRD1	c.216G>T	p.K72N	Substitution - Missense
ANKRD1	c.223C>T	p.L75L	Substitution - coding silent
ANKRD26	c.1340_1341insA	p.N447fs*5	Insertion - Frameshift
ANKRD32_ENST00000265140	c.987A>C	p.E329D	Substitution - Missense
ANKRD36C	c.2517_2518insA	p.Q840fs*11	Insertion - Frameshift
ANKRD36C	c.2713A>G	p.K905E	Substitution - Missense
ANKRD36C	c.2716T>C	p.C906R	Substitution - Missense
ANKRD36C_ENST00000420871	c.2517_2518insA	p.Q840fs*11	Insertion - Frameshift
ANKRD36C_ENST00000420871	c.2713A>G	p.K905E	Substitution - Missense
ANKRD36C_ENST00000420871	c.2716T>C	p.C906R	Substitution - Missense
ANKRD49	c.200delA	p.M70fs*32	Deletion - Frameshift
APAF1	c.1798_1799delAA	p.N602fs*23	Deletion - Frameshift
APAF1	c.1798delA	p.N602fs*8	Deletion - Frameshift
APAF1	c.1799A>G	p.K600R	Substitution - Missense
ARHGAP18	c.1418T>G	p.M473R	Substitution - Missense
ARHGAP18	c.492delA	p.K164fs*54	Deletion - Frameshift
ASH1L	c.2134delA	p.R712fs*23	Deletion - Frameshift
ASTE1	c.1884_1885delAA	p.R632fs*10	Deletion - Frameshift
ASTE1	c.1892A>G	p.K631R	Substitution - Missense
ASTE1	c.1894_1895insA	p.R632fs*11	Insertion - Frameshift
ASTE1	c.1894delA	p.R632fs*33	Deletion - Frameshift
ATAD2	c.354_355insA	p.E119fs*18	Insertion - Frameshift
ATAD2	c.354delA	p.E119fs*8	Deletion - Frameshift
ATL1	c.1665G>T	p.K555N	Substitution - Missense
ATR	c.2320_2321insA	p.I774fs*3	Insertion - Frameshift
ATR	c.2320delA	p.I774fs*5	Deletion - Frameshift
BARD1	c.623_624insA	p.K209fs*5	Insertion - Frameshift
BARD1	c.623delA	p.K208fs*4	Deletion - Frameshift
BARD1_ENST00000260947	c.623_624insA	p.K209fs*5	Insertion - Frameshift
BARD1_ENST00000260947	c.623delA	p.K208fs*4	Deletion - Frameshift
BAT2D1	c.464delA	p.E158fs*66	Deletion - Frameshift
BAT2D1_ENST00000392078	c.464delA	p.E158fs*66	Deletion - Frameshift
BEND5	c.545A>G	p.K182R	Substitution - Missense
BEND5	c.545delA	p.K182fs*15	Deletion - Frameshift
BEND5	c.546G>A	p.K182K	Substitution - coding silent
BEND5_ENST00000371833	c.1052A>G	p.K351R	Substitution - Missense
BEND5_ENST00000371833	c.1052delA	p.K351fs*15	Deletion - Frameshift
BEND5_ENST00000371833	c.1053G>A	p.K351K	Substitution - coding silent
BPTF	c.2874delA	p.I961fs*1	Deletion - Frameshift
BPTF_ENST00000335221	c.3252delA	p.I1087fs*1	Deletion - Frameshift
BRCA1	c.1960_1961insA	p.Y655fs*18	Insertion - Frameshift
BRCA1	c.1961_1961delA	p.K654fs*47	Deletion - Frameshift
BRCA1	c.1961_1962insA	p.Y655fs*18	Insertion - Frameshift
BRCA1	c.1961delA	p.K654fs*47	Deletion - Frameshift
BRCA1_ENST00000471181	c.1961delA	p.K654fs*47	Deletion - Frameshift
BRCA2	c.8941G>A	p.E2981K	Substitution - Missense
C10orf68	c.555G>A	p.K185K	Substitution - coding silent
C10orf6	c.1002A>G	p.E334E	Substitution - coding silent
C10orf90	c.1991A>T	p.K664M	Substitution - Missense
C10orf90	c.1992_1993GA>TT	p.K664_K665>N*	Complex - compound substitution

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
C10orf90	c.1998delA	p.E667fs*7	Deletion - Frameshift
C10orf96	c.605A>C	p.E202A	Substitution - Missense
C12orf45	c.544delA	p.K184fs*>2	Deletion - Frameshift
C13orf40	c.2236_2237insA	p.I746fs*40	Insertion - Frameshift
C13orf40	c.2236delA	p.I746fs*1	Deletion - Frameshift
C14orf102	c.261G>T	p.K87N	Substitution - Missense
C14orf102	c.268_269insA	p.R90fs*7	Insertion - Frameshift
C14orf102	c.268delA	p.R90fs*69	Deletion - Frameshift
C14orf23	c.342_343insA	p.T117fs*20	Insertion - Frameshift
C14orf23	c.342delA	p.T117fs*8	Deletion - Frameshift
C14orf23	c.345A>C	p.K115N	Substitution - Missense
C14orf23	c.346_347insAAC	p.K116_T117insQ	Insertion - In frame
C14orf38	c.1890A>G	p.K630K	Substitution - coding silent
C14orf38	c.1894_1895insA	p.N632fs*6	Insertion - Frameshift
C14orf38	c.1895_1896insA	p.N632fs*6	Insertion - Frameshift
C16orf45	c.310G>T	p.E104*	Substitution - Nonsense
C16orf45	c.317C>A	p.T106N	Substitution - Missense
C16orf88	c.647A>C	p.K216T	Substitution - Missense
C16orf88	c.652delA	p.I218fs*41	Deletion - Frameshift
C18orf34	c.874delA	p.M292fs*3	Deletion - Frameshift
C18orf34_ENST00000383096	c.874delA	p.M292fs*3	Deletion - Frameshift
C1orf131	c.416G>A	p.R139K	Substitution - Missense
C1orf9	c.850G>T	p.E284*	Substitution - Nonsense
C1orf9_ENST00000367723	c.1327G>T	p.E443*	Substitution - Nonsense
C2orf77	c.407A>G	p.K136R	Substitution - Missense
C2orf77_ENST00000447353	c.407A>G	p.K136R	Substitution - Missense
C3orf77	c.1885delA	p.A632fs*5	Deletion - Frameshift
C3orf77_ENST00000309765	c.1885delA	p.A632fs*5	Deletion - Frameshift
C6orf103_ENST00000367493	c.1813delA	p.K607fs*>6	Deletion - Frameshift
C6orf10	c.1543G>A	p.E515K	Substitution - Missense
CAMKK2_ENST00000392474	c.1601C>A	p.T534K	Substitution - Missense
CAMSAP1L1	c.3748_3749insA	p.Q1253fs*12	Insertion - Frameshift
CAMSAP1L1	c.3749delA	p.K1252fs*19	Deletion - Frameshift
CAPN3_ENST00000397163	c.1788_1789insA	p.T599fs*33	Insertion - Frameshift
CASP5	c.153_154delAA	p.K51fs*3	Deletion - Frameshift
CASP5	c.154delA	p.T52fs*26	Deletion - Frameshift
CASP5_ENST00000393141	c.240_241delAA	p.K80fs*3	Deletion - Frameshift
CASP5_ENST00000393141	c.241delA	p.T81fs*26	Deletion - Frameshift
CCBL1_ENST00000427720	c.375_376delAA	p.K125fs*>34	Deletion - Frameshift
CCDC108_ENST00000295729	c.438delA	p.K146fs*3	Deletion - Frameshift
CCDC148	c.1260delA	p.K420fs*15	Deletion - Frameshift
CCDC150	c.838_839insA	p.E284fs*13	Insertion - Frameshift
CCDC150	c.839delA	p.E284fs*14	Deletion - Frameshift
CCDC150	c.847A>G	p.K283E	Substitution - Missense
CCDC150	c.850G>A	p.E284K	Substitution - Missense
CCDC175	c.1890A>G	p.K630K	Substitution - coding silent
CCDC34_ENST00000328697	c.720A>G	p.L240L	Substitution - coding silent
CCDC34_ENST00000328697	c.731_732insA	p.N244fs*3	Insertion - Frameshift
CCDC34_ENST00000328697	c.731delA	p.N244fs*28	Deletion - Frameshift
CCT8L1	c.1642_1643insA	p.I552fs*6	Insertion - Frameshift
CCT8L2	c.1654_1655insA	p.I552fs*6	Insertion - Frameshift
CCT8L2	c.1654delA	p.I552fs*4	Deletion - Frameshift
CD46	c.509A>G	p.N170S	Substitution - Missense
CDHR3_ENST00000542731	c.2259 - 2delA	p.?	Unknown
CDKL2	c.222_223insA	p.R75fs*7	Insertion - Frameshift
CDKL2	c.222delA	p.K74fs*5	Deletion - Frameshift
CDYL	c.216_217insA	p.G75fs*13	Insertion - Frameshift
CDYL	c.217delA	p.K76fs*25	Deletion - Frameshift
CDYL	c.219A>G	p.K73K	Substitution - coding silent
CEP164	c.336_337insA	p.E117fs*88	Insertion - Frameshift
CEP164	c.337delA	p.K116fs*22	Deletion - Frameshift
CEP164	c.347A>G	p.K116R	Substitution - Missense
CEP290	c.828delA	p.E277fs*16	Deletion - Frameshift
CHD2	c.3724_3725insA	p.Y1246fs*13	Insertion - Frameshift
CHD2	c.3725delA	p.K1245fs*4	Deletion - Frameshift
CHD2_ENST00000394196	c.3725delA	p.K1245fs*4	Deletion - Frameshift
CHD7	c.1922A>T	p.K641I	Substitution - Missense
CHD7_ENST00000423902	c.1922A>T	p.K641I	Substitution - Missense
CHEK1	c.668A>C	p.E223A	Substitution - Missense
CHEK1	c.668delA	p.T226fs*14	Deletion - Frameshift
CHEK1	c.675A>G	p.K225K	Substitution - coding silent
CIR1	c.865delA	p.I289fs*51	Deletion - Frameshift
CNTRL_ENST00000373855	c.1328delA	p.I446fs*1	Deletion - Frameshift
COL17A1	c.1170delA	p.E391fs*12	Deletion - Frameshift
COL17A1	c.1171G>C	p.E391Q	Substitution - Missense
CPNE3	c.771G>A	p.K257K	Substitution - coding silent

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
CPNE3	c.771G>T	p.K257N	Substitution - Missense
CWC27	c.995delA	p.V335fs*1	Deletion - Frameshift
CWF19L2_ENST00000282251	c.279G>T	p.K93N	Substitution - Missense
CWF19L2_ENST00000282251	c.287delA	p.K96fs*41	Deletion - Frameshift
DCLRE1C	c.1706A>C	p.K569T	Substitution - Missense
DCLRE1C	c.1708delA	p.R570fs*6	Deletion - Frameshift
DCLRE1C_ENST00000378278	c.2051A>C	p.K684T	Substitution - Missense
DCLRE1C_ENST00000378278	c.2053delA	p.R685fs*6	Deletion - Frameshift
DDX18	c.327G>T	p.K109N	Substitution - Missense
DDX18	c.333G>T	p.K111N	Substitution - Missense
DDX59	c.1294A>G	p.K432E	Substitution - Missense
DDX59_ENST00000331314	c.1294A>G	p.K432E	Substitution - Missense
DENR	c.317delA	p.K108fs*10	Deletion - Frameshift
DHX36	c.2560delA	p.R854fs*4	Deletion - Frameshift
DHX36	c.2564A>T	p.K855I	Substitution - Missense
DHX36	c.460_461insA	p.M154fs*3	Insertion - Frameshift
DHX36	c.460delA	p.M154fs*27	Deletion - Frameshift
DHX36	c.461_462insA	p.M154fs*3	Insertion - Frameshift
DHX36	c.578delA	p.N193fs*25	Deletion - Frameshift
DIAPH2	c.208G>T	p.E70*	Substitution - Nonsense
DIAPH3	c.952A>C	p.K318Q	Substitution - Missense
DIAPH3	c.958delA	p.I320fs*20	Deletion - Frameshift
DNAH6	c.6440G>T	p.R2147I	Substitution - Missense
DNAJC1	c.578_579insA	p.T194fs*18	Insertion - Frameshift
DNAJC2	c.590_591insA	p.N197fs*4	Insertion - Frameshift
DNAJC2	c.590delA	p.N197fs*8	Deletion - Frameshift
DSEL	c.2905_2906insA	p.R969fs*4	Insertion - Frameshift
DSEL	c.2905delA	p.R969fs*16	Deletion - Frameshift
DSEL	c.2910A>C	p.K970N	Substitution - Missense
DSEL	c.2910A>T	p.K970N	Substitution - Missense
DYNC1I2	c.97delA	p.E36fs*34	Deletion - Frameshift
DYNC2H1_ENST00000398093	c.832A>C	p.N278H	Substitution - Missense
EEA1	c.2428A>C	p.K810Q	Substitution - Missense
EFCAB7	c.1138_1139insA	p.I380fs*7	Insertion - Frameshift
EHBP1	c.1026delA	p.N344fs*2	Deletion - Frameshift
EIF2AK2	c.1531G>T	p.E511*	Substitution - Nonsense
EIF3J	c.222G>C	p.K74N	Substitution - Missense
EIF3J	c.223delA	p.I77fs*1	Deletion - Frameshift
EIF3J	c.229delA	p.I77fs*1	Deletion - Frameshift
EML6	c.4063delA	p.K1357fs*9	Deletion - Frameshift
EML6	c.4071G>T	p.K1357N	Substitution - Missense
ENSG00000121031	c.10811delA	p.N3604fs*48	Deletion - Frameshift
ENSG00000121031	c.496_497insA	p.I166fs*11	Insertion - Frameshift
ENSG00000121031	c.496delA	p.I166fs*6	Deletion - Frameshift
ENSG00000174501	c.4764_4765insA	p.Q1589fs*11	Insertion - Frameshift
ENSG00000174501	c.4960A>G	p.K1654E	Substitution - Missense
ENSG00000174501	c.4963T>C	p.C1655R	Substitution - Missense
ENSG00000188423	c.651A>G	p.K217K	Substitution - coding silent
ENSG00000188423	c.658_659delGA	p.E220fs*33	Deletion - Frameshift
ENSG00000188423	c.658G>A	p.E220K	Substitution - Missense
ENSG00000225516	c.313A>C	p.K105Q	Substitution - Missense
ENSG00000268852	c.52A>T	p.K18*	Substitution - Nonsense
ERC2	c.1528delA	p.T510fs*21	Deletion - Frameshift
ERC2_ENST00000288221	c.1528delA	p.T510fs*21	Deletion - Frameshift
ERCC4	c.1461G>C	p.K487N	Substitution - Missense
ERICH1	c.487delA	p.R163fs*3	Deletion - Frameshift
ERO1LB	c.188A>C	p.K63T	Substitution - Missense
ESCO2	c.1117G>C	p.D373H	Substitution - Missense
F5	c.3096G>C	p.K1032N	Substitution - Missense
F5	c.3102A>C	p.K1034N	Substitution - Missense
F8	c.3632A>C	p.K1211T	Substitution - Missense
F8	c.3637delA	p.I1213fs*5	Deletion - Frameshift
F8	c.3638T>G	p.I1213S	Substitution - Missense
F8_ENST00000360256	c.3632A>C	p.K1211T	Substitution - Missense
F8_ENST00000360256	c.3637delA	p.I1213fs*5	Deletion - Frameshift
F8_ENST00000360256	c.3638T>G	p.I1213S	Substitution - Missense
FAM133A	c.150T>G	p.N50K	Substitution - Missense
FAM178A	c.1002A>G	p.E334E	Substitution - coding silent
FAM186A_ENST00000327337	c.2261delA	p.K754fs*2	Deletion - Frameshift
FAM200B	c.170_173delAAGT	p.S59fs*9	Deletion - Frameshift
FAM200B_ENST00000422728	c.170_173delAAGT	p.S59fs*9	Deletion - Frameshift
FAM83A_ENST00000536633	c.1065delA	p.K357fs*8	Deletion - Frameshift
FAM9A	c.477_478insA	p.Q160fs*8	Insertion - Frameshift
FAM9A	c.477delA	p.K159fs*3	Deletion - Frameshift
FAM9A	c.480A>G	p.Q160Q	Substitution - coding silent
FASTKD1	c.2213G>T	p.R738I	Substitution - Missense

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
FASTKD1	c.2216A>G	p.K739R	Substitution - Missense
FASTKD1	c.2220delA	p.K740fs*10	Deletion - Frameshift
FAT4	c.12401delA	p.K4136fs*17	Deletion - Frameshift
FBXO38	c.2083delA	p.N697fs*32	Deletion - Frameshift
FBXO38	c.2085A>G	p.K695K	Substitution - coding silent
FERMT2	c.452A>C	p.K151T	Substitution - Missense
FERMT2	c.455delA	p.K152fs*4	Deletion - Frameshift
FERMT2	c.456G>A	p.K152K	Substitution - coding silent
FERMT2	c.456G>AG	p.K153fs*5	Complex - frameshift
FEZ2	c.837G>T	p.K279N	Substitution - Missense
FEZ2	c.838A>G	p.K280E	Substitution - Missense
FLG	c.476_477insA	p.E160fs*10	Insertion - Frameshift
FLG	c.477_478insA	p.E160fs*10	Insertion - Frameshift
FLI45831	c.312G>T	p.K104N	Substitution - Missense
FRA10AC1	c.694_695insA	p.R232fs*4	Insertion - Frameshift
FRA10AC1	c.694delA	p.R232fs*>84	Deletion - Frameshift
FRA10AC1_ENST00000371426	c.694_695insA	p.R232fs*4	Insertion - Frameshift
FRA10AC1_ENST00000371426	c.694delA	p.R232fs*67	Deletion - Frameshift
GIMAP7	c.776delA	p.I261fs*1	Deletion - Frameshift
GOLGA4	c.4091delA	p.V1367fs*10	Deletion - Frameshift
GPR110	c.95_96insA	p.E33fs*6	Insertion - Frameshift
GPR110	c.95_96insT	p.K32fs*7	Insertion - Frameshift
GPR110_ENST00000371243	c.95_96insA	p.E33fs*6	Insertion - Frameshift
GRK4_ENST00000398052	c.656delA	p.R222fs*2	Deletion - Frameshift
GRK4_ENST00000398052	c.666_669delAATA	p.I223fs*10	Deletion - Frameshift
GRK4_ENST00000398052	c.668T>A	p.I223K	Substitution - Missense
GRLF1	c.570A>T	p.K190N	Substitution - Missense
GRLF1_ENST00000317082	c.570A>T	p.K190N	Substitution - Missense
GRLF1_ENST00000317082	c.578A>C	p.K193T	Substitution - Missense
HELLS	c.454delA	p.N154fs*29	Deletion - Frameshift
HERC5	c.409_410insA	p.I140fs*19	Insertion - Frameshift
HERC5	c.410delA	p.I140fs*1	Deletion - Frameshift
HERC5	c.416A>T	p.K139I	Substitution - Missense
HMGXB4	c.1163delA	p.K391fs*33	Deletion - Frameshift
HMGXB4	c.1173G>A	p.K391K	Substitution - coding silent
HMMR	c.1990_1991delAA	p.K666fs*3	Deletion - Frameshift
HMMR	c.1990delA	p.K666fs*11	Deletion - Frameshift
IQGAP2	c.4364G>T	p.R1455I	Substitution - Missense
ITIH5_ENST00000397146	c.2020C>A	p.Q674K	Substitution - Missense
ITPR2	c.3123 - 1G>A	p.?	Unknown
ITPR2	c.3127G>T	p.E1043*	Substitution - Nonsense
JMJD1C	c.5357A>G	p.E1786G	Substitution - Missense
JMJD1C	c.5364delA	p.E1789fs*45	Deletion - Frameshift
JMJD1C_ENST00000399262	c.6068A>G	p.E2023G	Substitution - Missense
JMJD1C_ENST00000399262	c.6075delA	p.E2026fs*45	Deletion - Frameshift
KCNC1	c.1362_1363insA	p.K458fs*16	Insertion - Frameshift
KCNC1	c.1363delA	p.K457fs*20	Deletion - Frameshift
KCNC1_ENST00000265969	c.1362_1363insA	p.K458fs*16	Insertion - Frameshift
KCNC1_ENST00000265969	c.1363delA	p.K457fs*20	Deletion - Frameshift
KCNQ1	c.1257G>T	p.K419N	Substitution - Missense
KCNQ1	c.1258delA	p.K422fs*10	Deletion - Frameshift
KDM2B_ENST00000377071	c.75A>C	p.K25N	Substitution - Missense
KDM2B_ENST00000377071	c.77delA	p.K26fs*81	Deletion - Frameshift
KDM2B_ENST00000377071	c.82_83delAC	p.T28fs*8	Deletion - Frameshift
KDM4D	c.271delA	p.K93fs*4	Deletion - Frameshift
KIAA1279	c.1509delA	p.I506fs*1	Deletion - Frameshift
KIAA1731	c.1649G>A	p.R550K	Substitution - Missense
KIAA1731_ENST00000325212	c.1649G>A	p.R550K	Substitution - Missense
KIAA2018	c.3045A>C	p.K1015N	Substitution - Missense
KIAA2018	c.3046_3047delAA	p.N1016fs*8	Deletion - Frameshift
KIAA2018	c.3046_3047insA	p.N1016fs*9	Insertion - Frameshift
KIAA2018	c.3046A>C	p.N1016H	Substitution - Missense
KIAA2018	c.3047delA	p.N1016fs*23	Deletion - Frameshift
KIAA2026_ENST00000399933	c.2069delA	p.K690fs*3	Deletion - Frameshift
KIAA2026_ENST00000399933	c.2074T>C	p.L692L	Substitution - coding silent
KIF5B	c.1045G>T	p.E349*	Substitution - Nonsense
KIF6	c.1458G>C	p.K486N	Substitution - Missense
KIF6_ENST00000287152	c.1458G>C	p.K486N	Substitution - Missense
KRCC1	c.715A>G	p.K239E	Substitution - Missense
LAMB4	c.4803A>T	p.K1601N	Substitution - Missense
LMOD2_ENST00000458573	c.1432A>C	p.K478Q	Substitution - Missense
LMOD2_ENST00000458573	c.1437G>T	p.K479N	Substitution - Missense
LRRIC17	c.597A>C	p.K199N	Substitution - Missense
LRRIC17	c.818A>G	p.E273G	Substitution - Missense
LRRIC17_ENST00000393217	c.4615_4616insA	p.I1542fs*8	Insertion - Frameshift
LRRIC17_ENST00000393217	c.4616delA	p.I1542fs*17	Deletion - Frameshift

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
LRR1Q1_ENST00000393217	c.4623A>G	p.K1541K	Substitution - coding silent
LRR1Q1_ENST00000393217	c.4624A>G	p.I1542V	Substitution - Missense
LRR1Q1_ENST00000393217	c.4625T>A	p.I1542N	Substitution - Missense
LRR1Q1_ENST00000393217	c.5095_5096insA	p.N1702fs*13	Insertion - Frameshift
LRR1Q1_ENST00000393217	c.5096delA	p.N1702fs*20	Deletion - Frameshift
LTN1	c.1597_1598delAA	p.N536fs*2	Deletion - Frameshift
LTN1	c.1607delA	p.N536fs*33	Deletion - Frameshift
LTN1	c.1608T>G	p.N536K	Substitution - Missense
MAP7D3	c.2567A>C	p.K856T	Substitution - Missense
MAP9	c.1733A>C	p.K578T	Substitution - Missense
MARCKS	c.454delA	p.K155fs*12	Deletion - Frameshift
MCF2	c.773T>A	p.I258K	Substitution - Missense
MCF2	c.774A>T	p.I258I	Substitution - coding silent
MCF2	c.780_781insA	p.L261fs*6	Insertion - Frameshift
MCF2_ENST00000370573	c.773T>A	p.I258K	Substitution - Missense
MCF2_ENST00000370573	c.774A>T	p.I258I	Substitution - coding silent
MCF2_ENST00000370573	c.780_781insA	p.L261fs*6	Insertion - Frameshift
MCF2_ENST00000370578	c.1208T>A	p.I403K	Substitution - Missense
MCF2_ENST00000519895	c.953T>A	p.I318K	Substitution - Missense
MCF2_ENST00000519895	c.954A>T	p.I318I	Substitution - coding silent
MCF2_ENST00000519895	c.960_961insA	p.L321fs*6	Insertion - Frameshift
MIS18BP1	c.471delA	p.K157fs*24	Deletion - Frameshift
MLH3	c.1755_1756insA	p.E586fs*3	Insertion - Frameshift
MLH3	c.1755delA	p.E586fs*24	Deletion - Frameshift
MLH3	c.1756G>T	p.E586*	Substitution - Nonsense
MLH3_ENST00000355774	c.1755_1756insA	p.E586fs*3	Insertion - Frameshift
MLH3_ENST00000355774	c.1755delA	p.E586fs*24	Deletion - Frameshift
MLH3_ENST00000355774	c.1756G>T	p.E586*	Substitution - Nonsense
MORC1	c.2634A>C	p.E878D	Substitution - Missense
MORC1	c.2641_2642insA	p.I881fs*11	Insertion - Frameshift
MORC1	c.2641delA	p.I881fs*1	Deletion - Frameshift
MPP3	c.1538C>T	p.T513M	Substitution - Missense
MPP6	c.910delA	p.K306fs*4	Deletion - Frameshift
MTDH	c.1341G>A	p.K447K	Substitution - coding silent
MTDH	c.1342A>T	p.K448*	Substitution - Nonsense
MTIF2	c.1975T>A	p.F659I	Substitution - Missense
MYCBP2	c.1124delA	p.K375fs*4	Deletion - Frameshift
MYCBP2_ENST00000357337	c.1124delA	p.K375fs*4	Deletion - Frameshift
MYCBP2_ENST00000407578	c.1238delA	p.K413fs*4	Deletion - Frameshift
MYT1L	c.182G>T	p.R61I	Substitution - Missense
MYT1L	c.187A>G	p.T63A	Substitution - Missense
NAA16	c.1883G>C	p.R628T	Substitution - Missense
NAA35	c.1692G>A	p.K564K	Substitution - coding silent
NAA35	c.1693delA	p.K567fs*6	Deletion - Frameshift
NEK1	c.3156A>G	p.K1052K	Substitution - coding silent
NEK1_ENST00000507142	c.3156A>G	p.K1052K	Substitution - coding silent
NHLRC2	c.1517_1518insA	p.N508fs*13	Insertion - Frameshift
NHLRC2	c.1522A>C	p.N508H	Substitution - Missense
NIPBL	c.1507delA	p.R505fs*35	Deletion - Frameshift
NIPBL_ENST00000448238	c.1507delA	p.R505fs*35	Deletion - Frameshift
NKRF	c.700A>C	p.K234Q	Substitution - Missense
NKRF_ENST00000542113	c.745A>C	p.K249Q	Substitution - Missense
NKTR	c.1297_1298insA	p.V436fs*2	Insertion - Frameshift
NOL7	c.707delA	p.K238fs*16	Deletion - Frameshift
NOL7	c.716delA	p.N240fs*14	Deletion - Frameshift
NOP58	c.1564delA	p.K524fs*>6	Deletion - Frameshift
NUFIP1	c.494delA	p.K165fs*38	Deletion - Frameshift
NUP85	c.127 + 2T>C	p.?	Unknown
OR6C76	c.921_922insA	p.H312fs*>2	Insertion - Frameshift
OR6C76	c.921C>A	p.H307Q	Substitution - Missense
OR6C76	c.922_923delAA	p.K311fs*>2	Deletion - Frameshift
OR6C76	c.922delA	p.K311fs*>2	Deletion - Frameshift
OR6C76	c.932A>C	p.K311T	Substitution - Missense
PA2G4	c.1108delA	p.K372fs*16	Deletion - Frameshift
PA2G4	c.1116G>T	p.K372N	Substitution - Missense
PARP14	c.4200G>T	p.K1400N	Substitution - Missense
PARP14_ENST00000474629	c.4689G>T	p.K1563N	Substitution - Missense
PCDH7	c.2685A>G	p.K895K	Substitution - coding silent
PCDH7_ENST00000361762	c.2826A>G	p.K942K	Substitution - coding silent
PCDHA12	c.552delA	p.D187fs*8	Deletion - Frameshift
PCDHA12	c.559G>T	p.D187Y	Substitution - Missense
PDCL2	c.253A>G	p.K85E	Substitution - Missense
PDCL2	c.262A>C	p.K88Q	Substitution - Missense
PKD2L2	c.904delA	p.I304fs*1	Deletion - Frameshift
PKD2L2	c.916G>T	p.E306*	Substitution - Nonsense
PKD2L2_ENST00000508883	c.916G>T	p.E306*	Substitution - Nonsense

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
PLXNC1	c.4192delA	p.I1400fs*21	Deletion - Frameshift
PLXNC1	c.4196A>C	p.K1399T	Substitution - Missense
PNISR	c.806A>G	p.K269R	Substitution - Missense
PNISR	c.807_808insA	p.A270fs*6	Insertion - Frameshift
PPFIA2	c.2527A>G	p.K843E	Substitution - Missense
PPFIA2	c.2529A>G	p.K843K	Substitution - coding silent
PPP1R10	c.924A>G	p.K308K	Substitution - coding silent
PPP1R10	c.930delA	p.V311fs*79	Deletion - Frameshift
PPP2R3C	c.67_68insA	p.S23fs*2	Insertion - Frameshift
PPP2R3C	c.67delA	p.S23fs*5	Deletion - Frameshift
PRKDC	c.10814delA	p.N3605fs*48	Deletion - Frameshift
PRKDC	c.496_497insA	p.I166fs*11	Insertion - Frameshift
PRKDC	c.496delA	p.I166fs*6	Deletion - Frameshift
PRPF40A	c.1167 + 3A>T	p.?	Unknown
PRPF40A_ENST00000359961	c.1560 + 3A>T	p.?	Unknown
PRPF40A_ENST00000410080	c.1479 + 3A>T	p.?	Unknown
PRR11	c.58delA	p.E23fs*9	Deletion - Frameshift
PTHLH_ENST00000354417	c.557delA	p.K186fs*12	Deletion - Frameshift
PTPLAD1	c.1074G>T	p.K358N	Substitution - Missense
PTPLAD1	c.1077A>C	p.K359N	Substitution - Missense
PTPRC	c.2404G>T	p.E802*	Substitution - Nonsense
PTPRZ1	c.5636delA	p.K1879fs*34	Deletion - Frameshift
PTPRZ1_ENST00000393386	c.5636delA	p.K1879fs*34	Deletion - Frameshift
PXK	c.1367_1368insA	p.R459fs*15	Insertion - Frameshift
PXK_ENST00000356151	c.1367_1368insA	p.R459fs*15	Insertion - Frameshift
PYHIN1	c.416_417insA	p.P142fs*3	Insertion - Frameshift
PYHIN1	c.423_424insA	p.P142fs*3	Insertion - Frameshift
PYHIN1	c.424C>A	p.P142T	Substitution - Missense
PYHIN1_ENST00000368135	c.424C>A	p.P142T	Substitution - Missense
PYHIN1_ENST00000392254	c.424C>A	p.P142T	Substitution - Missense
Q99543-2	c.590_591insA	p.N197fs*4	Insertion - Frameshift
Q99543-2	c.590delA	p.N197fs*8	Deletion - Frameshift
RAD23B_ENST00000457811	c.312_313delAA	p.K108fs*38	Deletion - Frameshift
RALGAPA1	c.5582delA	p.N1861fs*6	Deletion - Frameshift
RALGAPA1_ENST00000307138	c.5582delA	p.N1861fs*6	Deletion - Frameshift
RASAL2	c.1097A>G	p.E366G	Substitution - Missense
RASAL2	c.1104_1105insA	p.D372fs*4	Insertion - Frameshift
RASAL2	c.1104G>C	p.K368N	Substitution - Missense
RASAL2	c.1105_1106insA	p.D372fs*4	Insertion - Frameshift
RASAL2	c.1105delA	p.K371fs*7	Deletion - Frameshift
RASAL2	c.1111_1112insA	p.D372fs*4	Insertion - Frameshift
RASAL2	c.1112_1113insA	p.D372fs*4	Insertion - Frameshift
RASAL2_ENST00000367649	c.1151A>G	p.E384G	Substitution - Missense
RASAL2_ENST00000367649	c.1158_1159insA	p.D390fs*4	Insertion - Frameshift
RASAL2_ENST00000367649	c.1158G>C	p.K386N	Substitution - Missense
RASAL2_ENST00000367649	c.1159_1160insA	p.D390fs*4	Insertion - Frameshift
RASAL2_ENST00000367649	c.1159delA	p.K389fs*7	Deletion - Frameshift
RASAL2_ENST00000367649	c.1165_1166insA	p.D390fs*4	Insertion - Frameshift
RASAL2_ENST00000367649	c.1166_1167insA	p.D390fs*4	Insertion - Frameshift
RASAL2_ENST00000462775	c.707A>G	p.E236G	Substitution - Missense
RASAL2_ENST00000462775	c.714_715insA	p.D242fs*4	Insertion - Frameshift
RASAL2_ENST00000462775	c.714G>C	p.K238N	Substitution - Missense
RASAL2_ENST00000462775	c.715_716insA	p.D242fs*4	Insertion - Frameshift
RASAL2_ENST00000462775	c.715delA	p.K241fs*7	Deletion - Frameshift
RBM43	c.205G>T	p.E69*	Substitution - Nonsense
RBM43	c.213_214insA	p.V72fs*18	Insertion - Frameshift
RBM43	c.213delA	p.V72fs*13	Deletion - Frameshift
RBMX2	c.491delA	p.K166fs*29	Deletion - Frameshift
RBMX2	c.498_499insA	p.K170fs*30	Insertion - Frameshift
RBMX2	c.503A>C	p.K168T	Substitution - Missense
RBMX2	c.505A>G	p.K169E	Substitution - Missense
RBMX2	c.506A>G	p.K169R	Substitution - Missense
RBMX2	c.511_514delGAAA	p.E171fs*23	Deletion - Frameshift
RBPJ	c.202delA	p.E71fs*21	Deletion - Frameshift
RBPJ	c.204A>G	p.K68K	Substitution - coding silent
RBPJ_ENST00000348160	c.205delA	p.E72fs*21	Deletion - Frameshift
RBPJ_ENST00000348160	c.207A>G	p.K69K	Substitution - coding silent
RDX	c.628A>C	p.N210H	Substitution - Missense
REV3L	c.4543G>A	p.E1515K	Substitution - Missense
REV3L	c.4543G>T	p.E1515*	Substitution - Nonsense
REV3L	c.4550T>C	p.I1517T	Substitution - Missense
REV3L_ENST00000358835	c.4777G>T	p.E1593*	Substitution - Nonsense
REV3L_ENST00000358835	c.4784T>C	p.I1595T	Substitution - Missense
RG9MTD1	c.384delA	p.K131fs*3	Deletion - Frameshift
RIF1	c.4507delA	p.K1505fs*18	Deletion - Frameshift
RNASEH2B	c.917A>C	p.K306T	Substitution - Missense

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
RNASEH2B	c.926T>A	p.I309N	Substitution - Missense
RNASEH2B	c.926T>C	p.I309T	Substitution - Missense
RNF145	c.68A>G	p.K23R	Substitution - Missense
RNF145	c.68delA	p.K23fs*17	Deletion - Frameshift
RNF145	c.69G>A	p.K23K	Substitution - coding silent
RNF145	c.70A>G	p.K24E	Substitution - Missense
RNF145	c.71A>G	p.K24R	Substitution - Missense
RNF145	c.79_80delAA	p.N27fs*43	Deletion - Frameshift
RNF145	c.80delA	p.N27fs*13	Deletion - Frameshift
RNF145	c.81C>A	p.N27K	Substitution - Missense
RNPC3	c.346_347insA	p.R120fs*3	Insertion - Frameshift
RNPC3	c.347delA	p.R120fs*18	Deletion - Frameshift
ROCK1_ENST00000399799	c.149A>C	p.K50T	Substitution - Missense
RPL9	c.150G>A	p.K50K	Substitution - coding silent
RPL9	c.158A>G	p.K53R	Substitution - Missense
RPL9	c.159G>T	p.K53N	Substitution - Missense
RSPO3	c.659_660insA	p.P223fs*2	Insertion - Frameshift
RYR1	c.8508G>T	p.K2836N	Substitution - Missense
RYR1	c.8509delA	p.T2839fs*89	Deletion - Frameshift
SAT1_ENST00000379251	c.439delA	p.N150fs*13	Deletion - Frameshift
SCAF11	c.2995_2999delGAAAA	p.E999fs*2	Deletion - Frameshift
SCAF11	c.3002_3003insA	p.N1001fs*2	Insertion - Frameshift
SCAF11	c.3002delA	p.N1001fs*5	Deletion - Frameshift
SCAPER	c.2603delA	p.N868fs*8	Deletion - Frameshift
SCAPER	c.2605A>T	p.K869*	Substitution - Nonsense
SCAPER	c.2613delA	p.A872fs*4	Deletion - Frameshift
SCAPER_ENST00000538941	c.1867A>T	p.K623*	Substitution - Nonsense
SCAPER_ENST00000538941	c.1875delA	p.A626fs*4	Deletion - Frameshift
SEC63	c.1586delA	p.K529fs*4	Deletion - Frameshift
SEC63	c.1587G>T	p.K529N	Substitution - Missense
SENP7	c.230A>G	p.K77R	Substitution - Missense
SENP7_ENST00000394095	c.230A>G	p.K77R	Substitution - Missense
SEPT7_ENST00000469679	c.683C>A	p.A228E	Substitution - Missense
SH3RF1	c.2151G>A	p.K717K	Substitution - coding silent
SHPRH	c.495_496insA	p.E166fs*7	Insertion - Frameshift
SHPRH	c.495delA	p.E166fs*3	Deletion - Frameshift
SLC16A12_ENST00000371790	c.83G>C	p.R28T	Substitution - Missense
SLC22A9	c.1005A>C	p.K335N	Substitution - Missense
SLC22A9	c.995delA	p.K335fs*67	Deletion - Frameshift
SLC45A2	c.865A>C	p.K289Q	Substitution - Missense
SLC46A3	c.154A>C	p.K52Q	Substitution - Missense
SLC46A3_ENST00000380814	c.154A>C	p.K52Q	Substitution - Missense
SLC4A7	c.3450G>C	p.K1150N	Substitution - Missense
SLCO5A1	c.1148A>G	p.K383R	Substitution - Missense
SLCO5A1	c.1150T>G	p.F384V	Substitution - Missense
SLTM	c.1539delA	p.E514fs*8	Deletion - Frameshift
SLTM	c.1540G>T	p.E514*	Substitution - Nonsense
SMAD5	c.137A>G	p.K46R	Substitution - Missense
SMC1B_ENST00000357450	c.2444C>G	p.T815S	Substitution - Missense
SMC2	c.814delA	p.I274fs*1	Deletion - Frameshift
SMC2	c.822A>G	p.I274M	Substitution - Missense
SMC2_ENST00000303219	c.822A>G	p.I274M	Substitution - Missense
SMC2L1	c.814delA	p.I274fs*1	Deletion - Frameshift
SMC2L1	c.822A>G	p.I274M	Substitution - Missense
SMNDC1	c.567C>A	p.N189K	Substitution - Missense
SNX6	c.538A>T	p.K180*	Substitution - Nonsense
SP100_ENST00000341950	c.1349_1350insA	p.K455fs*>20	Insertion - Frameshift
SPATA1	c.1345delA	p.I452fs*1	Deletion - Frameshift
SPEF2	c.711G>A	p.K237K	Substitution - coding silent
SPEF2	c.712A>G	p.K238E	Substitution - Missense
SPEF2_ENST00000356031	c.2649_2650delAA	p.K886fs*2	Deletion - Frameshift
SPEF2_ENST00000356031	c.2649delA	p.K886fs*8	Deletion - Frameshift
SPEF2_ENST00000356031	c.711G>A	p.K237K	Substitution - coding silent
SPEF2_ENST00000356031	c.712A>G	p.K238E	Substitution - Missense
SPINK5	c.2459delA	p.K823fs*101	Deletion - Frameshift
SREK1IP1	c.363A>C	p.K121N	Substitution - Missense
SYCP1	c.2891_2892insA	p.L968fs*5	Insertion - Frameshift
SYCP1	c.2892delA	p.K967fs*2	Deletion - Frameshift
SYCP2	c.3062_3063insT	p.N1024fs*3	Insertion - Frameshift
SYCP2	c.3071delA	p.N1024fs*26	Deletion - Frameshift
TAF1B	c.187_188delAA	p.N66fs*3	Deletion - Frameshift
TAF1B	c.187_189delAAA	p.K65delK	Deletion - In frame
TAF1B	c.187delA	p.N66fs*26	Deletion - Frameshift
TAF1B	c.198C>A	p.N66K	Substitution - Missense
TAF1D	c.281_282insA	p.K95fs*28	Insertion - Frameshift
TAF1D	c.281delA	p.K94fs*26	Deletion - Frameshift

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
TAF7L	c.1331A>C	p.Q444P	Substitution - Missense
TAF7L	c.1333A>C	p.K445Q	Substitution - Missense
TAOK1_ENST00000261716	c.1738G>A	p.E580K	Substitution - Missense
TCF25	c.384_385insA	p.Q132fs*27	Insertion - Frameshift
TCF25	c.385delA	p.K131fs*17	Deletion - Frameshift
TCF25	c.393_394insA	p.Q132fs*27	Insertion - Frameshift
TCOF1	c.4127delA	p.E1379fs*>33	Deletion - Frameshift
TCOF1_ENST00000504761	c.4358delA	p.E1456fs*>33	Deletion - Frameshift
TCP1	c.730A>C	p.T244P	Substitution - Missense
TDRD5	c.1243G>A	p.E415K	Substitution - Missense
TET1_ENST00000373644	c.57C>A	p.N19K	Substitution - Missense
TET1_ENST00000373644	c.58delA	p.K22fs*23	Deletion - Frameshift
TEX10	c.11_13delAAA	p.K4delK	Deletion - In frame
TEX10	c.13delA	p.R5fs*10	Deletion - Frameshift
TEX15	c.5122delA	p.R1708fs*3	Deletion - Frameshift
TFAM	c.432delA	p.E148fs*2	Deletion - Frameshift
TFAM	c.441 + 2T>G	p.?	Unknown
TFAM_ENST00000395377	c.376delA	p.T126fs*6	Deletion - Frameshift
THOC2	c.2531A>G	p.K844R	Substitution - Missense
THOC2_ENST00000245838	c.2768A>G	p.K923R	Substitution - Missense
TIF1	c.2574G>A	p.K858K	Substitution - coding silent
TMEM97	c.518_519insA	p.*177fs?	Insertion - Frameshift
TMEM97	c.519delA	p.K176fs?	Deletion - Frameshift
TNRC6A	c.104A>C	p.K35T	Substitution - Missense
TNRC6B_ENST00000301923	c.179A>G	p.K60R	Substitution - Missense
TNRC6B_ENST00000301923	c.180G>A	p.K60K	Substitution - coding silent
TNRC6B_ENST00000301923	c.182A>C	p.K61T	Substitution - Missense
TNRC6B_ENST00000454349	c.113A>G	p.K38R	Substitution - Missense
TNRC6B_ENST00000454349	c.114G>A	p.K38K	Substitution - coding silent
TNRC6B_ENST00000454349	c.116A>C	p.K39T	Substitution - Missense
TPR	c.2055A>C	p.E685D	Substitution - Missense
TPR	c.2056A>T	p.K686*	Substitution - Nonsense
TRDN	c.1145A>G	p.K382R	Substitution - Missense
TRIM24	c.2676G>A	p.K892K	Substitution - coding silent
TRIM59	c.594G>T	p.Q198H	Substitution - Missense
TRIM59	c.604delA	p.S202fs*3	Deletion - Frameshift
TRMT6	c.458delA	p.K153fs*9	Deletion - Frameshift
TRPC1	c.509A>G	p.K170R	Substitution - Missense
TTF1	c.1007_1008insA	p.K337fs*9	Insertion - Frameshift
TTF1	c.1007delA	p.K336fs*87	Deletion - Frameshift
TWISTNB	c.932_933insA	p.K312fs*6	Insertion - Frameshift
TWISTNB	c.932delA	p.K311fs*15	Deletion - Frameshift
TWISTNB	c.933G>T	p.K311N	Substitution - Missense
TWISTNB	c.935_936insG	p.R313fs*5	Insertion - Frameshift
TWISTNB	c.938G>A	p.R313K	Substitution - Missense
ULK4_ENST00000301831	c.1778delA	p.K593fs*17	Deletion - Frameshift
USP36	c.2874_2879delGAAAAA	p.K959_K960delKK	Deletion - In frame
USP36_ENST00000312010	c.2874_2879delGAAAAA	p.K959_K960delKK	Deletion - In frame
USP36_ENST00000312010	c.2874G>A	p.K958K	Substitution - coding silent
USP40	c.3468G>T	p.K1156N	Substitution - Missense
USP40	c.3477_3478insA	p.Q1160fs*20	Insertion - Frameshift
USP40	c.3477delA	p.K1159fs*12	Deletion - Frameshift
USP40_ENST00000450966	c.3468G>T	p.K1156N	Substitution - Missense
USP40_ENST00000450966	c.3477_3478insA	p.Q1160fs*20	Insertion - Frameshift
USP40_ENST00000450966	c.3477delA	p.K1159fs*12	Deletion - Frameshift
VAX1	c.473A>C	p.K158T	Substitution - Missense
VAX1	c.477A>G	p.K159K	Substitution - coding silent
VAX1	c.477delA	p.K159fs*>28	Deletion - Frameshift
VAX1	c.478C>G	p.Q160E	Substitution - Missense
VAX1	c.480A>C	p.Q160H	Substitution - Missense
VAX1	c.483G>T	p.K161N	Substitution - Missense
WDHD1	c.1827G>A	p.K609K	Substitution - coding silent
WNK1	c.1738_1749ins?	p.?	Unknown
WNK1	c.1739delA	p.K583fs*11	Deletion - Frameshift
WNK1	c.1740A>G	p.E580E	Substitution - coding silent
WNK1	c.1749G>T	p.K583N	Substitution - Missense
WNK1_ENST00000537687	c.1739delA	p.K583fs*11	Deletion - Frameshift
WNK1_ENST00000537687	c.1740A>G	p.E580E	Substitution - coding silent
WNK1_ENST00000537687	c.1749G>T	p.K583N	Substitution - Missense
ZC3H13	c.3719G>C	p.S1240T	Substitution - Missense
ZC3H13	c.3719G>T	p.S1240I	Substitution - Missense
ZCCHC9	c.189G>T	p.K63N	Substitution - Missense
ZCCHC9	c.196G>T	p.E66*	Substitution - Nonsense
ZCRB1	c.411G>A	p.K137K	Substitution - coding silent
ZCRB1	c.419_420insA	p.K141fs*4	Insertion - Frameshift
ZCRB1	c.419delA	p.K140fs*13	Deletion - Frameshift

TABLE 4-continued

Shows a table of genes with mutations within the polyA region reported in the COSMIC database.			
Gene name	mutation (nucleotide)	mutation (protein)	Type of mutation
ZFH3	c.1031A>T	p.N344I	Substitution - Missense
ZFR	c.1074_1075insA	p.E359fs*27	Insertion - Frameshift
ZFR	c.1074delA	p.E359fs*4	Deletion - Frameshift
ZMAT1	c.1276A>T	p.K426*	Substitution - Nonsense
ZMAT1	c.1277A>T	p.K426I	Substitution - Missense
ZMAT1	c.1278A>G	p.K426K	Substitution - coding silent
ZMAT1_ENST00000372782	c.1789A>T	p.K597*	Substitution - Nonsense
ZMAT1_ENST00000372782	c.1790A>T	p.K597I	Substitution - Missense
ZMAT1_ENST00000372782	c.1791A>G	p.K597K	Substitution - coding silent
ZMYM5_ENST00000337963	c.1934delA	p.N645fs*>25	Deletion - Frameshift
ZNF236	c.1373delA	p.M461fs*1	Deletion - Frameshift
ZNF236_ENST00000543926	c.1373delA	p.M461fs*1	Deletion - Frameshift
ZNF34	c.664delA	p.T222fs*15	Deletion - Frameshift
ZNF518A	c.2777delA	p.T929fs*2	Deletion - Frameshift
ZNF518A_ENST00000371192	c.2777delA	p.T929fs*2	Deletion - Frameshift
ZNF600	c.194_195insA	p.L66fs*4	Insertion - Frameshift
ZNF644	c.871delA	p.R291fs*7	Deletion - Frameshift
ZNF644	c.872G>T	p.R291I	Substitution - Missense

Materials and Methods for Example 1

Cell Culture

HDF cells were cultured in Dulbecco's modified Eagle's medium (Gibco) and supplemented with 10% fetal bovine serum, 5% MEM non-essential amino acids (100×, Gibco), 5% penicillin and streptomycin (Gibco), and L-glutamine (Gibco). T-Rex-CHO cells were grown in Ham's F12K medium (ATCC) with the same supplements. *Drosophila* S2 cells were cultured in Express Five SFM Medium (Invitrogen) supplemented with 100 units per milliliter penicillin, 100 units per milliliter streptomycin (Gibco) and 45 ml of 200 mM L-glutamine (Gibco) per 500 ml of medium.

Plasmids and mRNA were introduced to the cells by the Neon® Transfection System (Invitrogen) with 100 µl tips according to cell specific protocols (www.lifetechnologies.com/us/en/home/life-science/cell-culture/transfection/transfection-selection-misc/neon-transfection-system/neon-protocols-cell-line-data.html). Cells electroporated with DNA plasmids were harvested after 48 hours if not indicated differently. Cells electroporated with mRNA were harvested after 4 hours, if not indicated differently. All transfections in S2 cells were performed using Effectene reagent (Qiagen). DNA Constructs

mCherry reporter constructs were generated by PCR amplification of an mCherry template with forward primers containing the test sequence at the 5' end and homology to mCherry at the 3' end. The test sequence for each construct is listed in Table 5. The PCR product was purified by NucleoSpin Gel and PCR Clean-up kit (Macherey-Nagel) and integrated into the pcDNA-DEST40, pcDNA-DEST53 or pMT-DEST49 expression vector by the Gateway cloning system (Invitrogen). Luciferase constructs were generated by the same method.

Whole gene constructs were generated by PCR amplification from gene library database constructs from Thermo (MTDH CloneId:5298467) or Life Technologies GeneArt Strings DNA Fragments (ZCRB1) and cloned in pcDNA-DEST40 vector for expression. Synonymous mutations in the natural gene homopolymeric lysine runs were made by site directed mutagenesis. Human beta-globin gene (HBD, delta chain) was amplified from genomic DNA isolated from HDF cells. Insertions of poly(A)-track, AAG-codons or pre-mature stop codon in HBD constructs were made by site directed mutagenesis. Sequences of inserts are in the Table 5.

TABLE 5

Sequences of mCherry inserts		SEQ ID NO:
Construct	Sequence inserted between 2HA tag and mCherry	
WT	No sequence inserted	
GAA ₁₂	GAA GAA GAA GAA GAA GAA GAA GAA GAA GAA GAA GAA	1
AAG ₆	AAG AAG AAG AAG AAG AAG	2
AAG ₉	AAG AAG AAG AAG AAG AAG AAG AAG AAG	3
AAG ₁₂	AAG AAG AAG AAG AAG AAG AAG AAG AAG AAG AAG AAG	4
STOP	TAA	
AAA ₆	AAA AAA AAA AAA AAA AAA	5
AAA ₉	AAA AAA AAA AAA AAA AAA AAA AAA AAA	6
AAA ₁₂	AAA AAA AAA AAA AAA AAA AAA AAA AAA AAA AAA AAA	7
CGA ₁₂	CGA CGA CGA CGA CGA CGA CGA CGA CGA CGA CGA CGA	8
AGG ₁₂	AGG AGG AGG AGG AGG AGG AGG AGG AGG AGG AGG AGG	9
A ₉	GCA GCG AAA AAA AAA TCC GTG	10
A ₁₀	GCA AAA AAA AAA GTG	11
A ₁₁	GCA GCA AAA AAA AAA ACC GTG	12
A ₁₂	GCA GAA AAA AAA AAA ACC GTG	13
A ₁₃	GCA GAA AAA AAA AAA AAC GTG	14
SLU7	GAG AAG AAG AAG AAG AAA AAG AAG AAG AAG AAG CAT	15
MTDH	TCC AAA AAG AAA AAA AAG AAA AAG AAG AAG CAA GGT	16
Nop58	GAG AAA AAG AAG AAA AAG AAA AAA AAG AAG AAG AAG AAG	17

TABLE 5-continued

Sequences of mCherry inserts			
Construct	Sequence inserted between 2HA tag and mCherry	SEQ ID NO.	
ZCRB1	CCA AAG AAG AAA GAA AAA AAG AAA AAA AAG AAA GCT	18	
RASAL2	GTG GAA AAA AAG AAA AAA AAG GAC AAG AAT AAT TAT	19	
ZCRB1	CCA AAG AAG AAA GAA AAA AAG AAA AAA AAG AAA GCT	20	
ZCRB1 G>A	CCA AAG AAG AAA GAA AAA AAA AAA AAA AAG AAA GCT	21	
ZCRB1 A>G	CCA AAG AAG AAA GAA AAG AAG AAA AAG AAG AAA GCT	22	
RASAL2	GTG GAA AAA AAG AAA AAA AAG GAC AAG AAT AAT TAT	23	
RASAL2 G>A	GTG GAA AAA AAA AAA AAA AAG GAC AAG AAT AAT TAT	24	
RASAL2 A>G	GTG GAA AAA AAG AAA AAG AAG GAC AAG AAT AAT TAT	25	
RASAL2 A>G(3)	GTG GAA AAG AAG AAG AAG AAG GAC AAG AAT AAT TAT	26	

In Vitro mRNA Synthesis

Capped and polyadenylated mRNA was synthesized in vitro using m Message mMachine T7 Transcription Kit (LifeTechnologies) following manufacturers procedures. The quality of mRNA was checked by electrophoresis and sequencing of RT-PCR products.

RNA Extraction and Qrt-PCR

Total RNA was extracted from cells using the RiboZol RNA extraction reagent (Amresco) according to the manufacturer's instructions. 400 µl of RiboZol reagent was used per well of 6 or 12 well plates for RNA extraction. Precipitated nucleic acids were treated by Turbo DNase (Ambion) and total RNA was dissolved in RNase-free water and stored at -20° C. RNA concentration was measured by Nanodrop (0D260/280). iScript Reverse Transcription Supermix (Biorad) was used with 1 µg of total RNA following the manufacturer's protocol. iQ SYBR Green Supermix (Biorad) protocol was used for qRT-PCR on the CFX96 Real-Time system with Bio-Rad CFX Manager 3.0 software. Cycle threshold (Ct) values were normalized to the neomycin resistance gene expressed from the same plasmid.

Western Blot Analysis

Total cell lysates were prepared with passive lysis buffer (Promega). Blots were blocked with 5% milk in 1×TBS 0.1% Tween-20 (TBST) for 1 hour. HRP-conjugated or primary antibodies were diluted by manufacturer recommendations and incubated overnight with membranes. Membranes were washed 4 times for 5 minutes in TBST and prepared for imaging or secondary antibody was added for additional one hour incubation. Images were generated by Bio-Rad Molecular Imager ChemiDoc XRS System with Image Lab software by chemiluminescence detection or by the LI-COR Odyssey Infrared Imaging System. Blots imaged by the LI-COR system were first incubated for 1 hr with Pierce DyLight secondary antibodies.

Immunoprecipitation

Total cell lysates were prepared with passive lysis buffer (Promega) and incubated with Pierce anti-HA magnetic beads overnight at 4° C. Proteins were eluted by boiling the beads with 1× SDS sample buffer for 7 minutes. Loading of protein samples was normalized to total protein amounts.

Cell Imaging

HDF cells were electroporated with the same amount of DNA plasmids and plated in 6 well plates with the optically clear bottom. Prior to imaging, cells were washed with a fresh DMEM media without Phenol-Red and incubated 20 minutes with DMEM media containing 0.025% Hoechst 33342 dye for DNA staining. Cells were washed with DMEM media and imaged in Phenol-Red free media using an EVOS-FL microscope (40× objective). Images were analyzed using EVOS-FL software.

Sequence data and variation databases: Sequence data were derived from NCBI RefSeq resource (Pruitt et al. (2014) RefSeq: an update on mammalian reference sequences. *Nucleic Acids Res.* 42, D756-763), on February 2014. Two variations databases were used: dbSNP (Sherry et al. (2001) *Nucleic Acids Res.* 29, 308-311), build 139 and COSMIC, build v70 (Forbes et al. (2014) *Nucleic Acids Res.*).

mRNA Mapping

As some inconsistencies between transcripts and proteins were observed in some of the sequence databases, before starting the analyses protein sequences were mapped to mRNA sequences using exonerate tool (Slater & Birney (2005) *BMC Bioinformatics* 6, 31), using protein2genome model and requiring a single best match. In case of multiple best matches (when several transcripts had given identical results), a first one was chosen, as the choice of corresponding isoform (as this was the most common reason for multiple matches) did not influence downstream analyses.

Ribosome Profiling Data

Three independent studies of ribosome profiling data from human cells were analyzed. These were: GSE51424 prepared by Gonzales and coworkers (Gonzalez et al. (2014) *J. Neurosci. Off. J. Soc. Neurosci.* 34, 10924-10936) from which samples: SRR1562539, SRR1562540 and SRR1562541 were used; GSE48933 prepared by Rooijers and coworkers (Rooijers et al. (2013) *Nat. Commun.* 4) from which samples: SRR935448, SRR935449, SRR935452, SRR935453, SRR935454 and SRR935455 were used; GSE42509 prepared by Loayza-Puch and coworkers (Loayza-Puch et al. (2013) *Genome Biol.* 14, R32) from which samples SRR627620-SRR627627 were used. The data were analyzed similarly to the original protocol created by Ingolia and coworkers (Ingolia et al. (2012) *Nat. Protoc.* 7, 1534-1550), with modifications reflecting the fact that reads were mapped to RNA data, instead of genome.

Raw data were downloaded and adapters specific for each experiments were trimmed. Then the reads were mapped to human noncoding RNAs with bowtie 1.0.1 (Langmead et al. (2009) *Genome Biol.* 10, R25) (bowtie-p 12-t-un) and unaligned reads were mapped to human RNAs (bowtie-p 12-v 0-a-m 25-best-strata-suppress 1,6,7,8). The analysis of occupancy was originally done in a similar way to Charneski and Hurst ((2013) *PLoS Biol.* 11, e100150817), however, given that genes with polyA were not highly expressed and the data were sparse (several positions with no occupancy), instead of mean of 30 codons prior to polyA position, it was decided to normalize only against occupancy of codon at the position 0 multiplied by the average occupancy along the gene. Occupancy data were visualized with R and Ggplot2 library using geom_boxplot aesthetics. On all occupancy graphs, the upper and lower "hinges" correspond to the first

and third quartiles (the 25th and 75th percentiles). The upper and lower whiskers extend from hinges at 1.5*IQR of the respective hinge.

Variation Analysis

To assess the differences in SNPs in polyA regions vs random region of the same length in other genes, the same distribution of lengths in both cases needed to be used. The distribution of lengths for polyA regions identified as mentioned above (12 As allowing for one mismatch) up to length 19 (longer are rare) is presented in FIG. 27. Using the same distribution of lengths, one random region of length drawn from the distribution randomly placed along each gene from all human protein coding RNAs was selected. Distributions of number of SNPs per segment for all polyA segments and for one random segment for each mRNA were compared using Welch Two Sample t-test, Wilcoxon rank sum test with continuity correction and two sample permutation test with 100000 permutations.

Abundance of Polytracks in Protein Sequences

Abundance was expressed by a following equation:

$$\text{Abundance} = \frac{1}{-\log_{10} \frac{N_P}{N_R}}$$

where NP is number of proteins with K+ polytrack (at least 2, at least 3, etc.) and NR is the total number of occurrences of a particular amino acid. It is to normalize against variable amino acid presence in different organisms. All isoforms of proteins were taken into account.

Other Analyses

List of human essential genes was obtained from the work of Georgi and coworkers (PLoS Genet 9, e1003484 (2013)). Gene Ontology analyses were done using Term Enrichment Service at amigo.geneontology.org/rte. Most of graphs were prepared using the R and GGPLOT2 library. For FIG. 14A, the values of the Y-axis were computed by 1D gaussian kernel density estimates implemented in R software. Custom Perl scripts were used to analyze and merge the data.

EXAMPLE 2: PolyA Tracks can be Used to Regulate Gene Expression in *Escherichia Coli*

We have recently identified polyA tracks as a regulator of gene expression. This mechanism is used endogenously in most eukaryotic genomes and regulates approximately 2% of human genes (Arthur, et al., 2015; Habich, et al., 2016). The polyA track causes ribosomal stalling and frameshifting during translation elongation, leading to mRNA instability and degradation of nascent protein products (Arthur, et al., 2015; Koutmou, et al., 2015). The translation elongation cycle is an ideal target for a universal method of gene regulation because it is the most highly conserved step in protein biosynthesis between prokaryotes and eukaryotes (Melnikov, et al., 2012). We have thus reasoned that polyA tracks, due to their versatility in lengths and sequence composition, can be used as a system to create programmable hypomorphic mutants and regulate gene expression in wide variety of model organisms (FIG. 29).

We have generated a fluorescent reporter gene that has an insertion of defined polyA tracks in order to control the amount of expression (FIG. 29A). The reporter consists of either a constitutive or inducible promoter driving expression of the mCherry fluorescence and other reporter proteins. A double HA-tag was added at the beginning of the coding

sequence for detection on western blot analysis. The polyA track is inserted directly after the HA-tag. The length of the polyA track varies from 9 to 36 consecutive adenine nucleotides, adding 3 to 12 lysine residues to the protein sequence (FIG. 29A). To control for the effects of polybasic peptide arising from sequential lysine residues (Kuroha, et al., 2010; Brandman, et al., 2012), we generated control reporters with consecutive lysine AAG codons. We hypothesized that as the length of the polyA track is increased, expression of the reporter gene products will decrease (FIGS. 29B and 29C). These reporters can be transiently transfected, recombined or inserted into the genome of cell cultures or whole organisms. Likewise, endogenous genes can be edited to include a polyA tracks in their open reading frames (ORF's) using genome editing methodology.

We first tested whether polyA tracks can be used in single cell model organisms to attenuate gene expression from a defined reporter gene. To show that polyA tracks can be used to control gene expression in *E. coli* cells we created a set of reporters with increasing length of polyA tracks under the arabinose-inducible promoter pBAD (FIG. 31). We transformed the chemically competent *E. coli* cells (Top10 strain) with plasmids expressing HA-mCherry, HA-(AAG)_n-mCherry or HA-polyA-mCherry. All *E. coli* cell cultures were induced at the same optical density and monitored for both cell growth and fluorescence of the mCherry constructs during induction. While *E. coli* containing wild type and LysAAG controls (6×, 9× and 12× lysine AAG codons) show no significant differences in the amount of mCherry fluorescence, cell cultures containing constructs with polyA tracks show progressively less fluorescence with increasing length of the polyA track (FIG. 26A). Addition of 9 and 12A's in a row (3 or 4 LysAAA codons) consistently reduced fluorescence of mCherry reporter by 15-35%. Further additions in the length of the polyA track resulted in constant decrease of mCherry reporter fluorescence where 36 consecutive adenine nucleotides resulted in a barely visible expression of the reporter (<5% of wild type). Western blot analyses of equal amounts of *E. coli* cell lysates expressing different polyA track and control reporters confirmed mCherry fluorescence data and indicated again that protein abundances of polyA track reporters strongly depend on the length of polyA track (FIG. 26B). Reporters with 9 and 12As in the row (3 and 4LysAAA codons, respectively) show reduction in protein abundances in the range of 20-40% of the wild type mCherry and constructs with more than 27As in the row (9 and more LysAAA codons) were hardly detectable by western blot analyses.

EXAMPLE 3: PolyA Tracks can be Used to Regulate Gene Expression in Protozoan *Tetrahymena thermophila*

We previously showed that polyA tracks can influence expression of the reporter genes in *S. cerevisiae* (Koutmou, et al., 2015) cells. To test whether polyA tracks can regulate gene expression in another model single cell eukaryotic organism, we monitored the effect of various length tracks on YFP expression in the protozoan *T. thermophila*. The genome of *T. thermophila* has extremely high AT content (>75%) and has been extensively used as microbial animal model [Collins, K. and M. A. Gorovsky (2005). "*Tetrahymena thermophila*." Curr Biol 15(9): R317-318.] (Eisen, et al., 2006). Our *T. thermophila* reporter contained the coding sequence of a Macronucleus-Localized Protein of unknown function (MLP1, TTHERM_00384860) fused to eYFP protein (FIG. 45A-D). The fusion with MLP1 directed YFP to

Tetrahymena macronuclei to allow easier quantification of YFP levels (FIG. 26C). These two proteins were fused, separated by linkers containing an HA-tag (MLP1-HA-YFP (WT)) and polyA tracks of 18, 27, or 36As, (AAA)6, (AAA)9, or (AAA)12, respectively, or 12 LysAAG (AAG) 12 codons inserted as a control. All constructs were expressed upon cadmium-induction of the upstream MTT1 promoter. Just as in our *E. coli* experiments with mCherry reporter, the YFP gene containing increasing lengths of polyA tracks exhibited a progressive decrease in total protein accumulation, measured by fluorescence, relative to the HA-linker fusion or LysAAG insertion controls (FIG. 26D). The construct with 18As in a row (6LysAAA) showed approximately 50% reduction in protein fluorescence, while constructs with 27 and 36As required 20 times longer exposure for detection of YFP by microscopy. The construct with 12LysAAG codons showed fluorescence that was readily 4-5 fold lower than WT construct. This effect could be attributed to polybasic peptide stalling that was observed earlier in *S. cerevisiae* cells (Koutmou, et al., 2015; Kuroha, et al., 2010; Brandman, et al., 2012). We further confirmed our fluorescence results by Western blot analyses (FIG. 26E). The MLP1-HA-YFP-fusion (WT) was readily visible whereas the polyA track-YFP fusions showed attenuation of expression. Insertion of 18A's (6LysAAA) showed expression levels that were approximately 35% of the HA-YFP control, while 27 and 36A's (9 and 12LysAAA) constructs were at the limit of detection (FIG. 26E). Insertion of 12LysAAG codons in the fusion protein, showed 6- to 8-fold reduction in expression of fusion protein. Since the different fusion proteins were all transcribed from the MTT1 promoter under identical induction conditions, we reasoned that amount of mRNA produced for each construct should be equivalent. We used qRT-PCR to quantify the steady state level of YFP mRNA of each fusion (FIG. 26F). The steady-state mRNA levels of polyA track-YFP constructs strongly reflected the decreasing YFP protein accumulation relative to the WT. Insertion of 18As (6LysAAA codons) reduced mRNA levels to approximately 30-35% of WT levels, while insertion of 27 and 36As (9 and 12LysAAA codons, respectively) reduced mRNA levels to less than 5% of the HA-YFP construct (FIG. 26D). Interestingly, while the attenuation at the protein level was stronger for the insertion of 12LysAAG codons than for 6LysAAA codons the trend was lost at mRNA levels. This can be due to the different pathways that resolve polybasic peptide stalling and polyA track-induced stalling and frameshifting in eukaryotic cells (Arthur, et al., 2015; Koutmou, et al., 2015). Nonetheless, we were able to control expression of reporter genes using polyA tracks in a similar manner in *T. thermophila*, a single cell AT-rich protozoan, as we did previously in *S. cerevisiae* and above mentioned *E. coli* cells (Koutmou, et al., 2015).

EXAMPLE 4: PolyA Tracks can be Used for Gene Expression Regulation in Plant Tissues

To test whether polyA tracks can attenuate gene expression in plants, we transiently co-expressed HA-polyA-mCherry with an YFP construct as internal control in the model plant *Nicotiana benthamiana* (FIG. 33). The expression of mCherry and YFP was assessed by fluorescence imaging (FIG. 27A). Like single cell cultures, *N. benthamiana* epidermal cells showed attenuated mCherry fluorescence proportional to the length of the polyA tracks (6,9 and 12 LysAAA codons) compared to the HA-mCherry and 12 LysAAG control constructs (FIG. 27A). The fluorescence data for each construct revealed the same trend of gene

expression regulation as in *T. thermophila* cells (FIG. 26D). As fluorescence in this assay was not quantifiable, protein abundance was determined by semi-quantitative Western blot analysis of *N. benthamiana* leaves infiltrated with the HA-polyA-mCherry. The levels of HA-mCherry proteins were normalized to levels of the cis-linked selectable marker phosphinotricin acetyl transferase (BAR) in the same sample (FIG. 27B and FIG. 27C). The addition of a polyA track with 18As (6 LysAAA) decreased protein accumulation to approximately 70% of HA-mCherry levels. Further reduction of mCherry protein accumulation, to 30% and below detection limit was observed in 9 LysAAA and 12 LysAAA constructs, respectively (FIG. 27C). Parallel analyses of steady state mRNA levels of transcripts with increasing lengths of polyA tracks showed progressively reduced levels of polyA track mRNAs when compared to transcript levels of the HA-mCherry and AAG-containing control constructs (FIG. 27D). mRNA levels were reduced to approximately 50-55% of WT expression for 6LysAAA transcripts, while 9 and 12LysAAA constructs had reduced mRNA levels to approximately 30 and 20% of controls, respectively (FIG. 27D). These results indicate that polyA tracks affect both mRNA and protein levels and can be used to regulate the gene expression in plants.

EXAMPLE 5: PolyA Tracks can be Used to Regulate Gene Expression in Human Tissue Cultures

To further assess the universality of polyA tracks on protein expression, we tested our reporter series in human tissue cultures using HeLa cells. Plasmids with HA-mCherry, HA-12LysAAG-mCherry and HA-polyA-mCherry reporters, driven by the constitutively active CMV promoter, were electroporated into HeLa cells for transient expression. Protein abundances were assessed by Western blot analyses 24 hours after electroporation (FIG. 27E). As in our previous study on expression of endogenous and synthetic polyA tracks in various human tissue cultures (Arthur, et al., 2015), constructs with increasing length of polyA tracks (6, 9 and 12 LysAAA) were expressed less than control constructs and reduction in protein expression was proportional with the length of polyA track. Construct with 18As (6LysAAA) displayed approximately 3-fold reduction in expression compared to WT construct. Insertion of 27 and 36As (9 and 12LysAAA, respectively), exhibited 6 and 25-fold reduction of HA-mCherry expression compared to WT (FIG. 27F). Our control construct with 12LysAAG codons did not show any reduction in protein levels compared to WT construct (FIG. 27E and FIG. 27F). This again indicates differences between translational stalling induced by polybasic peptides (Arthur, et al., 2015; Koutmou, et al., 2015; Kuroha, et al., 2010; Brandman, et al., 2012), which seems to be cell or organism specific and unpredictable, and polyA track-induced ribosomal stalling and frameshifting (Arthur, et al., 2015; Koutmou, et al., 2015) which is clearly dependent on the length of polyA tracks and conserved between multiple organisms and tissue cultures. mRNA stability of the reporters corresponded to protein expression as it was seen in our previous report (data not shown) (Arthur, et al., 2015). Together with our previous study (Arthur, et al., 2015), our results indicate that polyA tracks can easily be used to regulate expression of reporters or genes transiently transfected in diverse eukaryotic tissues and cultured cell systems, such as *N. benthamiana* and

human cell cultures, as well as other mammalian or insect tissue culture systems (Arthur, et al., 2015).

EXAMPLE 6: PolyA Tracks can be Used for Gene Expression Regulation in all Tissues of Model Organism

We next sought to test whether polyA tracks can be used to regulate reporter gene expression in complex, multicellular organisms. We chose fruit fly, *D. melanogaster*, due to the well-developed tools in the manipulation of endogenous genetic loci, as well as for the easier assessment of our mCherry reporter screen. Using the PhiC31-integrase approach (Groth, et al., 2004), we generated single transgene insertions of the HA-mCherry and HA-12LysAAG-mCherry controls, and HA-polyA-mCherry (6, 9 and 12LysAAA) constructs in the identical genomic location in the third chromosome (FIG. 34). All constructs contained Upstream Activation Sequence (UAS) followed by HSP70 promoter which actively transcribes mCherry reporter mRNAs in response to expression of GAL4 protein (Duffy, 2002). To drive expression of mCherry in all tissues, each transgenic line was crossed to a line that carried the Tub-GAL4 driver line that expresses GAL4 protein in all tissues and a UAS-linked GFP transgene, which allowed us to use GFP expression for normalization of the mCherry reporter genes (FIG. 34).

Expression of mCherry was assessed by fluorescence imaging of formaldehyde fixed salivary glands (SG), central nervous system (CNS), and proventriculus (PV) dissected from otherwise wild-type third instar larvae (FIG. 32A). Wild type HA-mCherry expressed well in all imaged tissues. Addition of a polyA track with 18As (6LysAAA) reduced mCherry expression to approximately 30% of the wild type construct in all three tissues (FIG. 28B, FIG. 28C, FIG. 28D, and FIG. 35). Constructs with 27As and 36As (9 and 12LysAAA codons, respectively) reduced expression of mCherry in all assayed tissues to approximately 20% and 10% of wild type levels, respectively (FIG. 28B, FIG. 28C, FIG. 28D, and FIG. 35). Western blot analysis on cell lysates produced from five fruit fly larvae for each independent construct confirmed our quantification of fluorescence imaging data (FIG. 40). As in the previous experiments with *T. Thermophila* and tissue cultures (FIG. 30 and FIG. 31), mRNA stability of polyA track constructs in fruit fly larvae showed inverse correlation with the length of polyA track (FIG. 36) and concordance with protein abundances measured by Western blot analyses. Insertion of 12LysAAG codons had moderate effect on levels of mCherry mRNA and protein and was in the range of 18As (6LysAAA) insertion construct (FIG. 28B, FIG. 28C, FIG. 28D, FIG. 36, and FIG. 36). Our data indicate that individual tissues of a complex multicellular organism, such as fruit fly, are equally sensitive to gene expression attenuation mediated by polyA tracks. Therefore, one can use polyA track constructs to create hypomorphic alleles and allelic series, in complex multicellular organisms with similar relative gene expression attenuation efficiency in the different tissues.

EXAMPLE 7: PolyA Tracks Control Gene Expression Independently of the Promoter Strength

Our data from fruit fly experiment indicated that the ratio between reporters with polyA track insertion and control is maintained in all tissues (FIG. 27B, FIG. 27C, FIG. 27D, FIG. 35, FIG. 36, FIG. 37). This suggests that inserted polyA tracks maintain their capacity of gene regulation indepen-

dently of the strength of mRNA transcription, which is known to have a large dynamic range across genes and cell types (Li, et al., 2014).

To systematically evaluate how differences in the strength of transcription would affect gene regulation and hypomorphic expression of reporters with polyA track insertion, we used human Flp-In™ T-Rex™ 293 cell lines. Using a protocol for generation of stable and inducible expression cell lines, we have generated cells with a single insertion of our mCherry control and 36A polyA track construct in the defined chromosomal locus (FIG. 38). The strength of transcription in these cell lines was varied by use of increasing amounts of doxycycline (0.001 to 0.1 mg/ml of Dox) present in the growth media and levels of transcription was assayed in relation to constitutively expressed hygromycin B phosphotransferase. Dose-dependence response of doxycycline-inducible CMV promoter for both polyA track and control mCherry transcript ranged over two orders of magnitude. At the same time, relative expression of polyA track construct was kept constant at 5-8% of expression of control construct based on the Western blot analysis (FIG. 29A and FIG. 29B). Moreover, relative mRNA levels of control and polyA track constructs measured by the normalized ratios did not change under different transcriptional strengths (FIG. 29C). The steady state amount of 36As polyA track construct was constantly in the range between 1-3% of the normalized control construct. The same results are obtained using stable cell lines that express HA-tagged human hemoglobin (delta chain, WT-HBD) and an 18As HBD construct (HBD-6LysAAA) with polyA track inserted in the second exon of the HBD coding sequence (FIG. 39). Expression of the HBD-6LysAAA protein was 3-fold reduced compared to WT-HBD construct, based on Western blot analysis (FIG. 40), and mRNA levels were approximately 20% of the HBD-WT mRNA levels, measured by qRT-PCR (FIG. 41). The relative ratios of WT-HBD and HBD-6LysAAA protein and mRNA levels were constant for different doxycycline induction levels. Together with previous data, showing regulated expression of mCherry reporter in different tissues of the transgenic fruit fly, these data demonstrate that polyA tracks can control gene expression independently of the promoter strength associated with assayed gene, keeping relative ratio of the protein levels between wild type and polyA track-attenuated product constant.

EXAMPLE 8: PolyA-Tracks can be Used to Create Sets of Hypomorphic Mutants in Functional Genes

The polyA tracks are mainly composed of lysine residues, AAA or AAG codons, which can be problematic when expressed as tags due to their charge and specific modifications (ubiquitination, acetylation, SUMOylation and hydroxylation). These features of poly-lys chains can further influence protein function as well as cell homeostasis (Brandman, et al., 2012; Dimitrova, et al., 2009; Choe, et al., 2016; Yonashiro, et al., 2016).

We tested our ability to regulate gene expression of functional proteins in both prokaryotic and eukaryotic cell system. In *E. coli*, the chloramphenicol acetyltransferase (CAT) gene confers resistance to the broad spectrum antibiotic chloramphenicol (CAM) in a dose-dependent manner (Shaw, et al., 1991). To show that we can regulate expression of CAT protein by insertion of polyA tracks we assessed *E. coli* survival under increasing concentrations of CAM in comparison with wild type CAT gene. To control for the influence of additional lysine residues in the N-terminus of CAT protein we also inserted 10LysAAG codons in the

N-terminus of CAT gene. Expression of WT-CAT, AAG10-CAT and polyA-CAT constructs was driven by the inducible arabinose promoter (pBAD, FIG. 42). All *E. coli* cultures were pulse induced, with addition of 0.1% arabinose, and growth was monitored on LB plates using different amounts of CAM in the media. WT-CAT and AAG10-CAT control constructs were able to survive CAM selection to the same level of CAM concentration in the media (75 mg/ml CAM, FIG. 30A). Therefore, the function of CAT protein is not affected by the addition of 10 consecutive Lys residues. PolyA tracks in constructs led to increased CAM sensitivity, of *E. coli* cells, which correlated with the length of the polyA tracks inserted in CAT gene (FIG. 30A). While majority of constructs could grow on minimal addition of CAM in the media (15 mg/ml), constructs with 24, 27 or 30As (8, 9 and 10LysAAA) were unable to grow on LB-plates with CAM concentration of 30 mg/ml. Furthermore, survivability of *E. coli* cells with CAT constructs having 15, 18 and 21As (5, 6 and 7LysAAA) on one hand, and 9 and 12As (3 and 4LysAAA) on the other hand, was impaired when cells were grown on LB-plates with final CAM concentrations of 50 mg/ml or 75 mg/ml, respectively (FIG. 30A). The survivability of *E. coli* cultures with different CAT constructs was in concordance with expression levels of CAT protein assayed by Western blot analyses (FIG. 43). The insertion of 10LysAAG codons in CAT gene did not affect *E. coli* cell growth on CAM selective media or levels of CAT protein expression arguing that insertion of multiple lysine residues in the N-terminus is not detrimental for the function and stability of CAT protein. These data demonstrate that polyA tracks can regulate levels of certain enzyme expression (CAT) in *E. coli* cells proportionally with their length.

To test ability of polyA tracks to regulate expression and function of protein in a eukaryotic cell we monitored how polyA tracts affect expression of N-succinyl-5-aminoimidazole-4-carboxamide ribotide synthetase (Ade1) in *Saccharomyces cerevisiae* (FIG. 30B). Disruption of the ADE1 gene results in the storage of a red pigment due to the buildup of a metabolic byproduct of the adenine biosynthesis pathway. Yeast cells that are *ade1Δ* are a dark red color; reintroduction of functional Ade1 protein restores the wild type white coloration in a dose-dependent manner. Differences in colony color and ability to grown on adenine dropout media (SD-Ade) have been utilized to differentiate strains of yeast prions (Liebman, et al., 2012), assess mitotic stability (Hieter, et al., 1985), and monitor gene expression (Mano, et al., 2013).

To survey how polyA tracts affect expression of Ade1, we transformed *ade1Δ* strains of *S. cerevisiae* with single copy plasmids (p416) encoding polyA-ADE1-FLAG, with the polyA tracks containing 18, 27 or 36As (6, 9, or 12 LysAAA, FIG. 44). Control plasmids contained no insertions (WT) or 12 LysAAG codons. Transformants were spotted onto plates to monitor color phenotype and growth on media lacking adenine (SD-Ade, FIG. 30B). The empty vector control exhibited a dark red coloration and inability to grow on SD-Ade, consistent with disruption of the ADE1 locus, while the wild type Ade1-FLAG restored both the white phenotype and growth on SD-Ade. Yeasts containing constructs with polyA track length of 18, 27 and 36A showed progressively pinker coloration and poorer growth on SD-Ade; however, the control 12LysAAG construct conferred a nearly-WT white color and strong growth on SD-Ade (FIG. 30B). Dot blot analysis of Ade1 protein expression, normalized to total protein, was in accordance with our phenotypic results and revealed visibly reduced amounts of expression for constructs with insertion of 9 and 12LysAAA codons

(FIG. 44). Expression of Ade1 protein with 12 lysine residues at the N-terminus, as in the case of 12LysAAG, did not impair function of the assayed protein (Ade1) and show similar results as insertion of 6LysAAA codons. Therefore, addition of polyA tracks to functional genes in both *E. coli* and *S. cerevisiae* preserved protein function but regulated protein abundance and as such polyA tracks could potentially be used in creation of hypomorphic gene mutants with fixed levels of protein expression.

Methods for Examples 2-8

E. Coli Experiments

mCherry reporter constructs used for expression in *E. coli* cells were subcloned using LR clonase recombination (Thermo Fisher Scientific) from pENTR/D-Topo constructs used in this study or in previous studies (Arthur, et al., 2015; Koutmou, et al., 2015). The resulting pBAD-DEST49 vector constructs express Thioredoxin (Thrx) fusion protein as Thrx-HA tag-insert-mCherry. For assaying expression of mCherry reporter all constructs were expressed in 2 ml *E. coli* Top10 strain grown in LB-Carbencillin (LB-Carb; final concentration 100 ug/ml). The cells were grown to optical density at 600 nm (OD600) of 0.4 at 37° C. and induced with addition of arabinose (0.5% w/v). Fluorescence of mCherry reporter for each construct was measured in triplicates 2 to 4 hours after induction using Biotek Synergy H4 plate reader (Excitation 475±9, Emission 620±9). The amount of fluorescence was normalized to number of cells measured by OD600. To additionally check for expression of fusion proteins, 200 ul of the cells was harvested 2 hr post-induction, resuspended in 100 ul of 2×SDS sample buffer and analyzed by SDS-PAGE followed by western blot analysis using HA-tag specific probe. Images of western blot analyses were generated by Bio-Rad Molecular Imager ChemiDoc XRS System with Image Lab software for chemiluminescence detection.

The chloramphenicol acetyltransferase (CAT) constructs used for functional protein studies were created by amplification of the CAT gene from pENTR/D-Topo vector (Thermo Fisher Scientific) using primers listed in Table 6. Constructs were subcloned into pBAD-DEST49 vector for use in functional assays. *E. coli* Top10 cells freshly transformed with pBAD-DEST49 plasmids expressing CAT reporters with different polyA tracks as well as CAT control reporters were grown in liquid LB-Carb media (100 ug/ml). For the chloramphenicol (CAM) survivability assay *E. coli* cells were grown to OD600=0.4 and non-induced (NI) fractions were spotted on LB-Carb plates (Carb 100 ug/ml) without chloramphenicol or to LB-Carb/CAM plates with raising amount of chloramphenicol in the media (CAM 15-100 ug/ml). The residual amount of the cells was induced for 1 hour with arabinose (final concentration 0.1% (w/v)). Cells were washed twice in M9 minimal media, resuspended in the starting volume of LB-Carb media and 5 ul of cells was spotted as induced (I) fraction on LB-Carb and LB-Carb-CAM plates. Plates were incubated overnight at 37° C. and imaged 24 hours post induction using Bio-Rad Molecular Imager ChemiDoc XRS System.

TABLE 6

Oligos used for generation of <i>E. coli</i> expressing CAT constructs			
SEQ ID NO:	Construct/ Oligo Name	Primer Sequence	
27	CAT WT For	CACCATGCACCATCACCATCACCATGAAA AAAAATCACTGGATATACCAACCGTTGATA TATCCC	

91

TABLE 6-continued

Oligos used for generation of <i>E. coli</i> expressing CAT constructs		
SEQ ID NO:	Construct/Oligo Name	Primer Sequence
28	CAT 10xAG	CACCATGCACCATCACCATCACCATGAG AAGAAGAAGAAGAAGAAGAAGAAGA AGATCACTGGATATACCACCGTTGATATA TCCC
29	CAT 3xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAATCACTGGATATACCACCGTT GATATATCCC
30	CAT 4xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATATACCAC GTTGATATATCCC
31	CAT 5xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATATACC ACCGTTGATATATCCC
32	CAT 6xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATAT ACCACCGTTGATATATCCC
33	CAT 7xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATATACC TATACCACCGTTGATATATCCC
34	CAT 8xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATATACC GGATATACCACCGTTGATATATCCC
35	CAT 9xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATATACC CTGGATATACCACCGTTGATATATCCC
36	CAT 10xAAA	CACCATGCACCATCACCATCACCATGAAA AAAAAAAAAAAAATCACTGGATATACC TCATGGATATACCACCGTTGATATATCCC
37	CAT Rev	CATTACAGATCTTCTTCAGAAATAAGTTTT TGTTCCGCCCGCCCTGCCACTCATCGC AG

Saccharomyces cerevisiae Experiments

In order to conduct functional studies with polyA track hypomorphic attenuation in *S. cerevisiae* cells the ADE1 locus was deleted from 74D-964 yeast strain via homologous recombination. Resultant *ade1Δ* strains were transformed with an empty vector or a plasmid-based ADE1 containing variable length of polyA tracks as well as WT and 12AAG insertion constructs. Constructs were generated by performing PCR on ADE1 isolated from the yeast genomic tiling library (Open Biosystems) with primers listed in Table 6. PCR products were digested and ligated into a p416 vector backbone containing the ADE1 endogenous promoter. Clones were verified via sequencing and correct constructs were transformed into *ade1Δ* deletions strains via the PEG-LiOAc method (Table 6). To generate dilution spottings, three colonies were picked from each transformation plate and grown overnight in selective media. In the morning, cultures were normalized to OD₆₀₀=1.0 and 10 ul of cells spotted onto rich media, SD-Ura, and SD-Ade for phenotypic analysis.

Relative protein abundance was determined via Western dot blotting. Briefly, yeast transformants were picked from selection plates to inoculate 10 mL of SD-Ura and grown overnight to ~OD=0.6. In the morning, cells were harvested and lysed in buffer (25 mM Tris-HCl pH 7.5, 50 mM KCl, 1 mM EDTA, Roche protease inhibitor cocktail) via

92

mechanical disruption with acid-washed glass beads (Sigma). Total protein was normalized to 1 mg/ml via Bradford assay, and 20 μg of total protein was spotted onto a nitrocellulose membrane. Western blotting was performed by overnight incubation with anti-Flag (Sigma M2, 1:1000 in 5% milk) and goat anti-rabbit (Sigma, 1:10,000 in 5% milk) antibodies followed by detection with chemiluminescence (Amersham ECL).

TABLE 7

Primers used for generation of ADE1 constructs		
SEQ ID No:	Name	Sequence
38	FwdAde1-Spe1WT	5'GGactagtATGTCAATTACGAAGACTGAACTGG
39	FwdAde1-Spe16AAA	5'GGactagtATGAAAAAAAAAAAAAAAAAAAA TCAATTACGAAGACTGAACCTGG
40	FwdAde1-Spe19AAA	5'GGactagtATGAAAAAAAAAAAAAAAAAAAA AAAAAAAAATCAATTACGAAGACTGAACCTGG
41	FwdAde1-Spe112AAA	5'GGactagtATGAAAAAAAAAAAAAAAAAAAA AAAAAAAAATCAATTACGAAGACTGAACCTGG
42	FwdAde1-Spe112AAG	5'GGactagtATGAAGAAGAAGAAGTCAATTACG AAGACTGAACCTGG
43	RevAde1-Cla1FLAG	5'GGatcgatTTACTTGTCTCATCGTCCTT GTAGTCGTGAGACCATTAGACCC
44	FwdAde1-PromSac1	5'GGgagctcACACGATAGCAAAGCAG
45	RevAde1-PromXba1	5'GGtctagaTATCGTTAATATTTCG

TABLE 8

<i>S. Cerevisiae</i> strains used in this study		
Name	Strain Background	Genotype
HT 971	74D-694	Mat A, <i>ade1::KANMX4</i> , <i>trp1-289(UAG)</i> , <i>his3Δ-200</i> , <i>ura3-52</i> , <i>leu2-3</i> , 112
HT 972	74D-694	Mat A, <i>ade1::KANMX4</i> , <i>trp1-289(UAG)</i> , <i>his3Δ-200</i> , <i>ura3-52</i> , <i>leu2-3</i> , 112

Tetrahymena thermophila Experiments

T. thermophila strain B2086 (II) was used for all experiments reported. Similar results were obtained with strain CU428 [(VII) *mpr1-1/mpr1-1*]. To assess the effect of polyA-tracks on protein accumulation, we modified a fluorescent protein tagging vector, pBSICY-gtw (Mott, et al., 2011) so as to fuse YFP to the carboxyl-terminus of a macronucleus-localized protein of unknown function (MLP, THERM_00384860), separated by a Gateway recombination cassette (Invitrogen/Life Technologies, Inc), and expressed from the cadmium inducible MTT1 promoter (Shang, et al., 2002). The MLP gene coding region was amplified with oligonucleotides 5' ALM Bsi' 5'-CAC CCG TAC GAA TAA AAT GAG CAT TAA TAA AGA AGA AGT-3' (SEQ ID. No: 46) and 3' ALM RV 5'-GAT ATC TTC AAT TTT AAT TTT TCT TCG AAG TTG C 3' (SEQ ID NO: 47) and cloned into pENTR-D in a topoisomerase mediated reaction prior to digesting with BsiWI and EcoRV and

inserting into BsiWI/PmeI digested pBSICY-gtw. Subsequently, LR Clonase II was used to insert a linker containing the sequence coding for an HA epitope tag alone (WT) or the tag plus different length of polyA tracks or AAG insertions in place of the Gateway cassette.

The expression cassette is located within the 5' flanking region of a cycloheximide resistant allele of the rpl29 gene to direct its integration into this genomic locus. These constructs were linearized with PvuI and SacI in the region flanking the *Tetrahymena* rpl29 sequences and introduced into starved *Tetrahymena* cells by biolistic transformation (Cassidy-Hanley, et al., 1997; Bruns, et al., 2000). Transformants were selected in 1× SPP medium containing 12.5 µg/ml cycloheximide. To control for copy number, PCR assays with primers MTT2386 5'-TCTTAGCTACGTGATTCACG-3' (SEQ ID NO: 48) and Chx-117, 5'-ATGTGTTATTAATCGATTGAT-3' (SEQ ID NO: 49) and Chx85r, 5'-TCTCTTTTCATGCATGCTAGC-3' (SEQ ID NO: 50) verified that all rpl29 loci contained the integrated expression construct.

Transgene expression was induced by addition of 0.4 µg/ml CdCl₂ and cells were grown 12-16 hours before monitoring protein accumulation. YFP accumulation was visualized by epifluorescence microscopy as previously described (Matsuda, et al., 2010). Whole cells extracts were generated by boiling concentrated cell pellets in 1× Laemmli sample buffer, followed by were fractionation on 10% SDS polyacrylamide gels and transfer to nitrocellulose. YFP accumulation was monitored with mouse anti-GFP antisera (G28R anti-GFP (OAEA00007) antibody, Aviva Systems Biology) and normalized to acetylated alpha Tubulin (6-11B-1 monoclonal Anti-Acetylated Tubulin antibody (T7451) Sigma-Aldrich). qPCR analysis was done using 5'-AGGCCTACAAGACCAAGGGT-3' (SEQ ID NO: 51) and 5'-AGAGCGGTTTTGACGTTGGA-3' (SEQ ID NO: 52) primers for *T. Thermophila* ribosomal protein L21 (rpl21) which was used for normalization. Primers 5'-CCCGTATGACGTACCGGATTATG-3' (SEQ ID NO: 53) and 5'-ACTTCAGGGTCAGCTTGCC-3' (SEQ ID NO: 54) were used for detection and estimation of fusion protein transcript levels using SybrGreen master mix and CFX96 Touch™ Real time PCR Detection System (BioRad). Normalized ΔCt values were used to calculate fold ratio between WT, 12LysAAG and polyA track constructs.

Nicotiano *Benthamiana* Experiments

Constructs for expression of HA-tagged mCherry reporters that were already cloned in pEntryD-TOPO vector were sub-cloned to pEarleyGate 100 (ABRC stock number CD3-724) through LR reaction using LR clonase (Invitrogen™). The mCherry reporter constructs, pEARLY100 and pBIN61 plasmids were individually electroporated into *Agrobacterium tumefaciens* strains GV3101 (Koncz, et al., 1986). The strain carrying pBIN61 construct expressing p19 protein from tomato bushy stunt virus was co-infiltrated with the reporter constructs to suppress post-transcriptional gene silencing (Voinnet, et al., 2003). The *Agrobacterium* suspensions carrying the reporter constructs were infiltrated into the leaves of 5- to 6-week-old *N. benthamiana* plants as described in Joensuu, J. J. et al. Briefly, saturated over-night cultures were spun-down and resuspended in the infiltration solution (3.2 g/L Gamborg's B5 plus vitamins, 20 g/L sucrose, 10 mM MES pH 5.6, 200 µM 4'-Hydroxy-3',5'-dimethoxyacetophenone) to a final OD₆₀₀=1.0; *Agrobacterium* suspensions carrying the reporter constructs were individually mixed with suspensions carrying the pBIN61 construct in 1:1 ratio prior to infiltrations. These suspensions were infiltrated into separate segments of two young leaves

on each of eight different *N. benthamiana* plants, which served as biological replicates. For control, 1:1 suspension of *A. tumefaciens* carrying pEARLY 100 with no insert along with pBIN61 was used. The infiltrated plants were maintained in a controlled growth chamber conditions at 22° C., with a 16h photoperiod.

Samples of the abaxial epidermis of *N. benthamiana* leaves infiltrated with different mCherry reporter constructs were collected 6 days post-infiltration. Infiltration was performed as described in the previous section, with the addition of an YFP-expressing construct pEARLY104 (ABRC stock number CD3-686), which served as infiltration control. The samples were visualized for fluorescence by confocal laser-scanning microscopy using a Leica TCS SP2 confocal microscope. Samples for RNA and total soluble protein (TSP) extraction were separately collected from the infiltrated plants 6 days post-infiltration using a cork borer (7.1 mm in diameter); each sample contained equal amounts of leaf tissue (2 leaf discs) collected from each of the segments on the two leaves infiltrated with the same construct.

Analysis of mCherry protein accumulation was carried out by Western blot as described in Gutiérrez et al., 2013 and Conley et al., 2009. Briefly, phosphate-buffered saline (PBS: 8 g/L NaCl, 1.16 g/L Na₂HPO₄, 0.2 g/L KH₂PO₄, 0.2 g/L KCl, pH 7.4), supplemented with 1 mM EDTA, 1 mM phenylmethanesulfonylfluoride (PMSF), 1 µg/ml leupeptin 0.1% Tween-20 and 100 mM sodium L-ascorbate was used for total soluble protein (TSP) extractions. Bradford assay (Biorad) was used to quantify TSP in the extracts using a standard curve (r²=0.99) of known concentrations of Bovine Serum Albumin (BSA). Sample extracts (25 µg TSP for mCherry and 5 µg TSP for phosphinotricin acetyl transferase [BAR] protein detection) were separated by SDS-PAGE, blotted onto nitrocellulose membrane and probed with a primary anti-HA tag antibody (Genscript) for mCherry, or anti-Phosphinotricin acetyl transferase antibody (Abcam) for BAR, both at 1:2000 dilution, followed by HRP-conjugated secondary antibody (Biorad) at 1:5000 dilution. The blots were washed (3 times×10 min) in 1×Tris-buffered Saline (TBS, 50 mM Tris, 150 mM NaCl, pH 7.5) containing 0.1% Tween (Sigma) and images were obtained after 1 min incubation with the enhanced chemiluminescence (ECL) detection system (GE Healthcare). Numerical values for protein accumulation were derived from the detected band intensities on the analyzed images using TotalLab TL 100 software (Nonlinear Dynamics, Durham, USA). The mCherry accumulation values were normalized for Basta accumulation detected in the same sample. Normalized values of the mCherry protein accumulation for each reporter construct were presented as the mean of eight biological replicates±SE; Tukey's honest significance test (JMP software, SAS Institute Inc.) was used to identify significantly different means (α=0.05).

For quantitative RT-PCR (qPCR), total RNA was extracted using an RNeasy plant mini kit coupled with DNase treatment (Qiagen). The purified RNA (500 ng) was reverse-transcribed using the Maxima first-strand cDNA synthesis kit (Thermo Fisher Scientific). The resulting cDNA (2 ng/µl) was quantified by qPCR using the Maxima SYBR Green/ROX qPCR master mix (Thermo Fisher Scientific) and CFX384 Touch™ Real-Time PCR Detection System (Biorad). Cycle threshold (Ct) values were normalized to phosphinotricin N-acetyltransferase (BAR) gene expressed in the same plasmid used for transient expression. Primer sequences used: For mCherry-mCherryFWD: 5'-GGCTACCCATACGATGTTCC-3' (SEQ ID NO: 55);

mCherryREV: 5'-CCTCCATGTGCACCTTGAAG-3' (SEQ ID NO: 56); for BASTA-BAR-F3: 5'-TCAAGAGCGTGGTCGCTG-3' (SEQ ID NO: 57) and BAR-R3: 5'-CAAATCTCGGTGACGGGCAG-3' (SEQ ID NO: 58).

Drosophila melanogaster Experiments

Reporter gene expression was achieved with the GAL4/UAS system. The UAS-mCherry transgene plasmids were constructed from the phiC31 integrase plasmid, pJFRC28-10XUAS-IVS-GFP-p10 (Addgene plasmid #36431) (Pfeiffer, et al., 2012). GFP was removed by digestion with KpnI and XbaI and replaced with HA-mCherry and HA-polyA-mCherry. Transgenic fly lines were obtained by injecting P{CaryP}attP2 embryos with each pJFRC28 mCherry construct to achieve site-specific, single insertion on the third chromosome at the attP2 landing site (Rainbow Transgenic Flies, Inc.). Injected GO adult flies were backcrossed to w1118 flies. Red-eyed progeny indicated successful germline integration of the UAS-mCherry expression cassette. Male red-eye progeny were crossed to female w;TM3 Sb/TM6 Tb flies followed by sib-crosses of the F1 progeny to generate homozygous UAS-mCherry transgenic lines. Insertion was confirmed by Sanger sequencing of PCR amplified mCherry from genomic DNA of individual flies. Each mCherry transgenic fly line was crossed to a TubGal4 UAS-GFP driver line (derived from BSC42734) to achieve mCherry expression in all tissues. GFP expression was used for normalization. All flies were maintained at 25° C.

Third instar larvae from each cross were fixed in formaldehyde and dissected to recover the salivary glands (SG), intact central nervous system (CNS), and proventriculus (PV). The tissues were mounted on glass cover slips and confocal images were taken on a Zeiss Imager 2 upright microscope using identical parameters for all images of each tissue type. Fluorescence intensity of the mCherry and GFP were quantified with Zen 9 software.

Total RNA was extracted from each cross by pooling 5 third instar larvae in 1.5 ml RNase-free Eppendorf tubes which were then frozen in dry ice. Frozen samples were homogenized using 1.5 ml pestles (Fisherbrand, RNase- and DNase-free). After homogenization, 1 ml RiboZol reagent (Amresco) was added and extraction was completed according to manufacturer's instructions. Total RNA samples were treated with TURBO DNA-free kit (Ambion) to remove potential genomic DNA. cDNA synthesis was performed with iScript Reverse Transcription Supermix (Bio-Rad) with 1 µg of total RNA in a 20 µl reaction. RT-qPCR was performed in the Bio-Rad CFX96 Real-Time System with iQ SYBR Green Supermix (Bio-Rad). The mCherry transcript was detected with the following primers: 5'-TGACGTACCGGATTATGCAA-3' (SEQ ID NO: 59) and 5'-ATATGAAGTACAGGGGACAGG-3' (SEQ ID NO: 60). Cycle threshold (Ct) values were normalized to EF1 with the following primers: 5'-GCGTGGGTTTGTGATCAGTT-3' (SEQ ID NO: 61) and 5'-GATCTTCTCCTTGCCCATCC-3' (SEQ ID NO: 62)) (Ponton, et al., 2011).

For western blot analysis, five third instar larvae from each cross were collected and frozen in dry ice. Frozen samples were homogenized using 1.5 ml pestles (Fisherbrand, RNase- and DNase-free). After homogenization, SDS sample buffer was added and the samples were boiled for 10 minutes. Anti-HA was used to detect mCherry expression. Samples were normalized with anti-GFP.

H. sapiens Cell Culture Experiments

mCherry reporter constructs used for transient expression in human cells were subcloned using LR clonase recombi-

nation (Thermo Fisher Scientific) from pEntryD-Topo constructs used in other experiments or in previous studies (Arthur, et al., 2015). DNA fragments for constructs used for creation of inducible and stable cell lines were PCR amplified, purified and ligated into pcDNA 5/FRT/TO vector (Thermo Fisher Scientific).

HeLa cells were cultured in Dulbecco's modified Eagle's medium (DMEM) (Gibco) and supplemented with 10% fetal bovine serum, 5% minimum essential medium nonessential amino acids (100×, Gibco), 5% penicillin and streptomycin (Gibco), and L-glutamine (Gibco). FLP-In T-REx Hek-293 cells were grown in the same media with addition of 5 µg/ml of blastocidin and 100 µg/ml of Zeocin for non-recombined cells, or 5 µg/ml of blastocidin and 100 µg/ml of hygromycin for growth of stable cell lines expressing mCherry or HBD constructs.

Plasmids were introduced to the tissue culture cells by the Neon Transfection System (Thermo Fisher Scientific) using 100-µl tips according to cell-specific protocols (www.lifetechnologies.com/us/en/home/life-science/cell-culture/transfection/transfection-selection-misc/neon-transfection-system/neon-protocols-cell-line-data.html). HeLa cells, used for transient expression, were electroporated with 1.5 µg of DNA plasmids and were harvested 24 hours after the electroporation. FLP-In T-REx Hek-293 cells were electroporated with plasmids, selected for positive clones as described by protocol (https://tools.thermofisher.com/content/sfs/manuals/flipinsystem_man.pdf). Expression of polyA track and control constructs was induced by addition of various amounts of doxycycline from a common stock (1 µg/ml) and harvested 24 or 48 hours after induction, if not indicated differently.

Total RNA was extracted from cells using the RiboZol RNA extraction reagent (Amresco) according to the manufacturer's instructions or using GenElute™ Direct. RiboZol reagent (500 µl) was used in each well of 6- or 12-well plates for RNA extraction. Precipitated nucleic acids were treated with Turbo deoxyribonuclease (Ambion), and total RNA was dissolved in ribonuclease-free water and stored at -20° C. RNA concentration was measured by NanoDrop (OD260/280). iScript Reverse Transcription Supermix (Bio-Rad) was used with 1 µg of total RNA following the manufacturer's protocol. RT-qPCR was performed in the Bio-Rad CFX96 Real-Time System with iQ SYBR Green Supermix (Bio-Rad). For both transient expression samples and stable cell line samples, the mCherry transcript was detected with the following primers: 5'-TGACGTACCGGATTATGCAA-3' (SEQ ID NO: 63) and 5'-ATATGAAGTACAGGGGACAGG-3' (SEQ ID NO: 64). Cycle threshold (Ct) values were normalized to the neomycin resistance gene expressed from the same plasmid for transient expression (5'-CTGAATGAAGTACAGGACGA-3' (SEQ ID NO: 65) and 5'-ATACTTTCTCGGCAGGAGCA-3' (SEQ ID NO: 66)) or hygromycin (5'-GATGTAGGAGGGCGTGGATA-3' (SEQ ID NO: 67) and 5'-ATAGGTCAGGCTCTCGCTGA-3' (SEQ ID NO: 68) or actin gene for stable cell lines (5'-AGAAAATCTGGCACCACACC-3' (SEQ ID NO: 69) and 5'-AGAGGCGTACAGGGATAGCA-3' (SEQ ID NO: 70)).

Total cell lysates were prepared with passive lysis buffer (Promega). Blots were blocked with 5% milk in 1× tris-buffered saline-0.1% Tween 20 (TBST) for 1 hour. Horseradish peroxidase-conjugated or primary antibodies were diluted according to the manufacturer's recommendations and incubated overnight with membranes. The membranes were washed four times for 5 min in TBST and prepared for imaging, or secondary antibody was added for additional 1 hour of incubation. Images were generated by Bio-Rad

Molecular Imager ChemiDoc XRS System with Image Lab software by chemiluminescence detection or by the LI-COR Odyssey Infrared Imaging System. Blots imaged by the LI-COR system were first incubated for 1 hour with Pierce DyLight secondary antibodies.

Discussion for EXAMPLES 2-8

We have presented a rapid method of generating hypomorphic mutations in a reporter or gene of interest. Insertion of a polyA track into a coding sequence shows predictable and robust attenuation of gene expression in all tested cell culture and model organism systems. The length of the polyA track can be manipulated to achieve full-range of expression levels, allowing for the generation of an allelic series from complete knockout to wild-type expression for the study of gene function. This method can also be used in synthetic biology applications that require precise gene control and modeling of metabolic and signaling networks (Chuang, et al., 2010).

The use of polyA tracks overcomes many of the challenges present in current methods of generating hypomorphic mutations and controllable gene expression. For instance, a recent approach to attenuate gene expression in *E. coli* is mutagenesis of the Shine-Dalgarno sequence in the gene of interest. The expression levels from all possible six-mer Shine-Dalgarno sequences were experimentally determined and the information is available in the EMOPEC database (Bonde, et al., 2016). However, this valuable resource would have to be generated anew to use this approach in other prokaryotes and it could not be used in eukaryotic systems. Additionally, many orthogonal translation systems rely on modified Shine-Dalgarno sequences (Hui, et al., 1987; Hui, et al., Methods Enzymol., 1987; Lee, et al., 1996; Rackham, et al., 2005). Use of an orthogonal translation system would prohibit use of the Shine-Dalgarno sequence for expression regulation. The polyA track system of gene regulation and creation of hypomorphic mutations overcomes this issue due to its dependency on regulation of translation elongation cycle which is conserved between prokaryotes and eukaryotes (Melnikov, et al., 2012).

Hypomorphic mutations have been generated in eukaryotic cell systems by insertion of an antibiotic resistance gene into introns (Meyers, et al., 1998) or the 3'-untranslated region of genes (Breslow, et al., 2008). Insertion of the neomycin resistance gene (neo) into an intron introduces a cryptic splice site that causes aberrant splicing of transcripts, effectively reducing gene expression (Meyers, et al., 1998). The reliance on stochastic cryptic splicing events leads to unpredictable changes in transcript expression and is rather gene dependent. Insertion of neo in various genes have resulted in expression of a functionally null allele (Nagy, et al., 1998), hypomorphic expression (Meyers, et al., 1998; Hirotsune, et al., 1998), or no change in expression (Wolpowitz, et al., 2000). Our system of polyA tracks gives predictable gene expression attenuation in variety of different eukaryotic systems and, furthermore, shows relative gene expression attenuation efficiency in the different tissues of the same organism.

We have primarily introduced polyA tracks at the N-terminal regions of reporter genes due to the uniformity of the construct design and to reduce potential frameshifting effects (Arthur, et al., 2015; Koutmou, et al., 2015). We do not anticipate this to be a major limitation of this method. Our *Tetrahymena* reporters place the polyA tracks at the N-terminus of YFP, but at the C-terminus of the linked *Tetrahymena* gene (FIG. 32). Furthermore, insertion of polyA track in the second exon of the human beta globin gene (HBD) gene, an unstructured loop of the protein, argue

that polyA tracks can be introduced at various positions in the gene (FIG. 39, FIG. 40, and FIG. 41). Additionally, we have shown previously that naturally occurring polyA track sequences exist in the human genome and that potential frameshifted products are efficiently degraded by non-sense mediated decay mechanisms (Arthur, et al., 2015).

PolyA tracks that are used endogenously in eukaryotic genomes are typically interrupted by other nucleotides at various positions within the A-rich sequence. We have observed that the position of the interrupting nucleotide, in combination with the length of the A-rich sequence, modulate gene expression (Arthur, et al., 2015). This characteristic indicates that polyA mediated regulation can be further developed for even more precise control of gene expression. Lastly, approximately 2% of human genes are endogenously regulated by polyA tracks, including many well-studied, disease-associated genes, such as BRCA1, TCOF1 and MTDH among others (Arthur, et al., 2015; Habich, et al., 2016). As we showed in our previous study, synonymous mutations of the internal polyA track of such genes can allow investigators to dramatically change expression levels of these genes without manipulation of protein sequence or the gene regulatory elements such as promoters and enhancers (Arthur, et al., 2015).

The addition of a polyA track to the target gene will result in additional lysine residues in the protein product. Like any protein tag, it is important to consider the effects of the additional residues when studying the functionality of the protein. We have shown that the function and stability of two structurally diverse proteins, CAT and Ade1, are not affected by up to 12 additional lysine residues. To control for possible effects of the poly-lysine tracks, investigators can create an allele with the same number of lysine residues encoded by AAG codons. The AAG codons will have minimal effect on expression levels while encoding a synonymous protein. Furthermore, the flexibility in polyA track placement within the coding sequence allows investigators to choose the most suitable insertion site for the protein of interest.

The conservation of the polyA track sequences in the multiple genes across vertebrates as well as our analysis of mutation rates of polyA tracks (36As) inserted in the defined locus of *D. melanogaster* genome argue that polyA tracks can be used to create stable hypomorphic gene alleles. Our results are in the range of already described hypermutability (approximately 8%) of the short tandem repeats (STRs) and BAT-40 microsatellite (40As) located in the second intron of the 3-beta-hydroxysteroid dehydrogenase gene. The distinction is that our data show general mutation rates for the whole fruitfly after more than 30 generations while in the case of the mentioned study the mutation rate is dependent on the cell type. An additional study found that the mutation rate in polyA region, 10As in this case, is in the range of 10⁻⁴ per cell per generation. As such, the authors argue that approximately 1% of cells will be affected by a polyA region mutation in 100 generations. Similar rates were observed in the other studies with approximately 10⁻⁶-10⁻² mutation rate for the different lengths of homopolymeric regions or STRs. PolyA tracks used in our study tend to operate on the shorter side of the length distribution of STRs and as such should have similar if not even lower rates of mutations.

PolyA tracks that are used endogenously in eukaryotic genomes are typically interrupted by other nucleotides at various positions within the A-rich sequence, which further reduces potential hypermutability effects. We have observed that the position of the interrupting nucleotide or codon, in combination with the length of the A-rich sequence, modu-

lates gene expression (FIG. 45A-D). These observations suggest that polyA-mediated regulation can be further developed for even more precise control of gene expression. Lastly, approximately 2% of human genes are endogenously regulated by polyA tracks, including many well-studied, disease-associated genes, such as BRCA1, TCOF1 and MTDH among others. As we showed in our previous study, synonymous mutations of the internal polyA track of such genes can allow investigators to dramatically change expression levels of these genes without manipulation of protein sequence or gene regulatory elements such as promoters and enhancers. The use of our method is not restricted only to these genes, and we feel that the synthetic biology field will benefit from this application. Control of biosynthetic pathways for production of useful secondary metabolites, antibiotics, or recombinant antibodies, as well as introduction of controllable retrosynthetic and fully engineered pathways or ultimate control of metabolic pathways in the modeling of diseases are just a few among the multiple possible applications of this method in the future.

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SEQUENCE LISTING

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Sequence total quantity: 116
SEQ ID NO: 1          moltype = DNA  length = 36
FEATURE              Location/Qualifiers
misc_feature          1..36
                      note = SYNTHESIZED
source                1..36
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 1
gaagaagaag aagaagaaga agaagaagaa gaagaa          36

SEQ ID NO: 2          moltype = DNA  length = 18
FEATURE              Location/Qualifiers
misc_feature          1..18
                      note = SYNTHESIZED
source                1..18
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 2
aagaagaaga agaagaag          18

SEQ ID NO: 3          moltype = DNA  length = 27
FEATURE              Location/Qualifiers
misc_feature          1..27
                      note = SYNTHESIZED
source                1..27
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 3
aagaagaaga agaagaagaa gaagaag          27

SEQ ID NO: 4          moltype = DNA  length = 36
FEATURE              Location/Qualifiers
misc_feature          1..36
                      note = SYNTHESIZED
source                1..36
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 4
aagaagaaga agaagaagaa gaagaagaag aagaag          36

SEQ ID NO: 5          moltype = DNA  length = 18
FEATURE              Location/Qualifiers
misc_feature          1..18
                      note = SYNTHESIZED
source                1..18
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 5

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aaaaaaaaa aaaaaaaaa	18
SEQ ID NO: 6	moltype = DNA length = 27
FEATURE	Location/Qualifiers
misc_feature	1..27
	note = SYNTHESIZED
source	1..27
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 6	
aaaaaaaaa aaaaaaaaaa aaaaaaa	27
SEQ ID NO: 7	moltype = DNA length = 36
FEATURE	Location/Qualifiers
misc_feature	1..36
	note = SYNTHESIZED
source	1..36
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 7	
aaaaaaaaa aaaaaaaaaa aaaaaaa aaaaaa	36
SEQ ID NO: 8	moltype = DNA length = 36
FEATURE	Location/Qualifiers
misc_feature	1..36
	note = SYNTHESIZED
source	1..36
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 8	
cgacgacgac gacgacgacg acgacgacga cgacga	36
SEQ ID NO: 9	moltype = DNA length = 36
FEATURE	Location/Qualifiers
misc_feature	1..36
	note = SYNTHESIZED
source	1..36
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 9	
aggaggagga ggaggaggag gaggaggagg aggagg	36
SEQ ID NO: 10	moltype = DNA length = 21
FEATURE	Location/Qualifiers
misc_feature	1..21
	note = SYNTHESIZED
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 10	
gcagcgaaaa aaaaatccgt g	21
SEQ ID NO: 11	moltype = DNA length = 15
FEATURE	Location/Qualifiers
misc_feature	1..15
	note = SYNTHESIZED
source	1..15
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 11	
gcaaaaaaaaa aagtg	15
SEQ ID NO: 12	moltype = DNA length = 21
FEATURE	Location/Qualifiers
misc_feature	1..21
	note = SYNTHESIZED
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 12	
gcagcaaaaa aaaaaaccgt g	21
SEQ ID NO: 13	moltype = DNA length = 21
FEATURE	Location/Qualifiers
misc_feature	1..21
	note = SYNTHESIZED
source	1..21
	mol_type = other DNA
	organism = synthetic construct

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SEQUENCE: 13
gcagaaaaaa aaaaaaccgt g                                21

SEQ ID NO: 14      moltype = DNA  length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                   note = SYNTHESIZED
source            1..21
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 14
gcagaaaaaa aaaaaaacgt g                                21

SEQ ID NO: 15      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 15
gagaagaaga agaagaaaaa gaagaagaag aagcat                36

SEQ ID NO: 16      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 16
tccaaaaaga aaaaaaagaa aaagaagaag caaggt                36

SEQ ID NO: 17      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 17
gagaaaaaga agaaaaagaa aaaaaagaga gagaga                36

SEQ ID NO: 18      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 18
ccaagaaga aagaaaaaaa gaaaaaaaag aaagct                36

SEQ ID NO: 19      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 19
gtggaaaaaa agaaaaaaa ggacaagaat aattat                36

SEQ ID NO: 20      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 20
ccaagaaga aagaaaaaaa gaaaaaaaag aaagct                36

SEQ ID NO: 21      moltype = DNA  length = 36
FEATURE           Location/Qualifiers
misc_feature       1..36
                   note = SYNTHESIZED
source            1..36
                   mol_type = other DNA

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                                organism = synthetic construct
SEQUENCE: 21
ccaaagaaga aagaaaaaaa aaaaaaaag aaagct 36

SEQ ID NO: 22      moltype = DNA length = 36
FEATURE           Location/Qualifiers
misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 22
ccaaagaaga aagaaaagaa gaaaaagaag aaagct 36

SEQ ID NO: 23      moltype = DNA length = 36
FEATURE           Location/Qualifiers
misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 23
gtggaaaaaa agaaaaaaa ggacaagaat aattat 36

SEQ ID NO: 24      moltype = DNA length = 36
FEATURE           Location/Qualifiers
misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 24
gtggaaaaaa aaaaaaaa ggacaagaat aattat 36

SEQ ID NO: 25      moltype = DNA length = 36
FEATURE           Location/Qualifiers
misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 25
gtggaaaaaa agaaaaagaa ggacaagaat aattat 36

SEQ ID NO: 26      moltype = DNA length = 36
FEATURE           Location/Qualifiers
misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 26
gtggaaaaga agaagaagaa ggacaagaat aattat 36

SEQ ID NO: 27      moltype = DNA length = 65
FEATURE           Location/Qualifiers
misc_feature      1..65
                  note = SYNTHESIZED
source            1..65
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 27
caccatgcac catcaccatc accatgaaaa aaaaatcact ggatatacca ccgttgatat 60
atccc 65

SEQ ID NO: 28      moltype = DNA length = 89
FEATURE           Location/Qualifiers
misc_feature      1..89
                  note = SYNTHESIZED
source            1..89
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 28
caccatgcac catcaccatc accatgagaa gaagaagaag aagaagaaga agaagaagat 60
cactggatat accaccgttg atatatccc 89

SEQ ID NO: 29      moltype = DNA length = 68
FEATURE           Location/Qualifiers
misc_feature      1..68

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source          note = SYNTHESIZED
                1..68
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 29
caccatgcac catcaccatc accatgaaaa aaaaaaaatc actggatata ccaccgttga 60
tatatccc                                     68

SEQ ID NO: 30      moltype = DNA length = 71
FEATURE           Location/Qualifiers
misc_feature       1..71
source            note = SYNTHESIZED
                1..71
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 30
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa atcactggat ataccaccgt 60
tgatatatcc c                                     71

SEQ ID NO: 31      moltype = DNA length = 74
FEATURE           Location/Qualifiers
misc_feature       1..74
source            note = SYNTHESIZED
                1..74
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 31
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa aaaatcactg gatataccac 60
cgttgatata tccc                                     74

SEQ ID NO: 32      moltype = DNA length = 77
FEATURE           Location/Qualifiers
misc_feature       1..77
source            note = SYNTHESIZED
                1..77
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 32
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa aaaaaaatca ctggatatac 60
caccgttgat atatccc                                     77

SEQ ID NO: 33      moltype = DNA length = 80
FEATURE           Location/Qualifiers
misc_feature       1..80
source            note = SYNTHESIZED
                1..80
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 33
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa aaaaaaaaaa tcactggata 60
taccaccgtt gatatatccc                                     80

SEQ ID NO: 34      moltype = DNA length = 83
FEATURE           Location/Qualifiers
misc_feature       1..83
source            note = SYNTHESIZED
                1..83
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 34
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa aaaaaaaaaa aaatcactgg 60
atataccacc gttgatatat ccc                                     83

SEQ ID NO: 35      moltype = DNA length = 86
FEATURE           Location/Qualifiers
misc_feature       1..86
source            note = SYNTHESIZED
                1..86
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 35
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa aaaaaaaaaa aaaaaatcac 60
tgatataacc accgttgata tatccc                                     86

SEQ ID NO: 36      moltype = DNA length = 89
FEATURE           Location/Qualifiers
misc_feature       1..89
source            note = SYNTHESIZED
                1..89

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                                mol_type = other DNA
                                organism = synthetic construct
SEQUENCE: 36
caccatgcac catcaccatc accatgaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaat 60
cactggatat accaccgttg atatatccc 89

SEQ ID NO: 37      moltype = DNA length = 60
FEATURE           Location/Qualifiers
misc_feature      1..60
                  note = SYNTHESIZED
source            1..60
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 37
cattacagat cttcttcaga aataagtttt tgttcgccc cgccctgcc ctcacgcag 60

SEQ ID NO: 38      moltype = DNA length = 33
FEATURE           Location/Qualifiers
misc_feature      1..33
                  note = SYNTHESIZED
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 38
ggactagtat gtcaattacg aagactgaac tgg 33

SEQ ID NO: 39      moltype = DNA length = 51
FEATURE           Location/Qualifiers
misc_feature      1..51
                  note = SYNTHESIZED
source            1..51
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 39
ggactagtat gaaaaaaaaa aaaaaaaaaa caattacgaa gactgaactg g 51

SEQ ID NO: 40      moltype = DNA length = 60
FEATURE           Location/Qualifiers
misc_feature      1..60
                  note = SYNTHESIZED
source            1..60
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 40
ggactagtat gaaaaaaaaa aaaaaaaaaa aaaaaaaatc aattacgaag actgaactgg 60

SEQ ID NO: 41      moltype = DNA length = 69
FEATURE           Location/Qualifiers
misc_feature      1..69
                  note = SYNTHESIZED
source            1..69
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 41
ggactagtat gaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaatca attacgaaga 60
ctgaactgg 69

SEQ ID NO: 42      moltype = DNA length = 69
FEATURE           Location/Qualifiers
misc_feature      1..69
                  note = SYNTHESIZED
source            1..69
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 42
ggactagtat gaagaagaag aagaagaaga agaagaagaa gaagaagtca attacgaaga 60
ctgaactgg 69

SEQ ID NO: 43      moltype = DNA length = 53
FEATURE           Location/Qualifiers
misc_feature      1..53
                  note = SYNTHESIZED
source            1..53
                  mol_type = other DNA
                  organism = synthetic construct
SEQUENCE: 43
ggatcgattt acttgctgct atcgctcctg tagtcgtgag accatttaga ccc 53

SEQ ID NO: 44      moltype = DNA length = 25

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FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = SYNTHESIZED	
source	1..25	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 44		
gggagctcac acgatagcaa agcag		25
SEQ ID NO: 45	moltype = DNA length = 24	
FEATURE	Location/Qualifiers	
misc_feature	1..24	
	note = SYNTHESIZED	
source	1..24	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 45		
ggtctagata tcgttaatat ttgc		24
SEQ ID NO: 46	moltype = DNA length = 39	
FEATURE	Location/Qualifiers	
misc_feature	1..39	
	note = SYNTHESIZED	
source	1..39	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 46		
caccggtacg aataaaatga gcattaataa agaagaagt		39
SEQ ID NO: 47	moltype = DNA length = 34	
FEATURE	Location/Qualifiers	
misc_feature	1..34	
	note = SYNTHESIZED	
source	1..34	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 47		
gatatcttca attttaattt ttcttcgaag ttgc		34
SEQ ID NO: 48	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 48		
tccttagctac gtgattcacg		20
SEQ ID NO: 49	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = SYNTHESIZED	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 49		
atgtgttatt aatcgattga t		21
SEQ ID NO: 50	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 50		
tctctttcat gcatgctagc		20
SEQ ID NO: 51	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 51		
aggcctacaa gaccaagggt		20

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SEQ ID NO: 52	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 52		
agagcggttt tgacgttgga		20
SEQ ID NO: 53	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = SYNTHESIZED	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 53		
cccgatgac gtaccggatt atg		23
SEQ ID NO: 54	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
misc_feature	1..19	
	note = SYNTHESIZED	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 54		
acttcagggt cagcttgcc		19
SEQ ID NO: 55	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 55		
ggctacccat acgatgttc		20
SEQ ID NO: 56	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 56		
cctccatgtg caccttgaag		20
SEQ ID NO: 57	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = SYNTHESIZED	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 57		
tcaagagcgt ggctcgctg		18
SEQ ID NO: 58	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 58		
caaatctcgg tgacgggcag		20
SEQ ID NO: 59	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 59		
tgacgtaccg gattatgcaa		20

-continued

SEQ ID NO: 60	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 60		
atatgaactg aggggacagg		20
SEQ ID NO: 61	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 61		
gcgtgggttt gtgatcagtt		20
SEQ ID NO: 62	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 62		
gatcttctcc ttgcccatcc		20
SEQ ID NO: 63	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 63		
tgacgtaccg gattatgcaa		20
SEQ ID NO: 64	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 64		
atatgaactg aggggacagg		20
SEQ ID NO: 65	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 65		
ctgaatgaac tgcaggacga		20
SEQ ID NO: 66	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 66		
atactttctc ggcaggagca		20
SEQ ID NO: 67	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = SYNTHESIZED	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 67		

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gatgtaggag ggcgtggata	20
SEQ ID NO: 68	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = SYNTHESIZED
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 68	
ataggtcagg ctctcgctga	20
SEQ ID NO: 69	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = SYNTHESIZED
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 69	
agaaaatctg gcaccacacc	20
SEQ ID NO: 70	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = SYNTHESIZED
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 70	
agaggcgtac agggatagca	20
SEQ ID NO: 71	moltype = DNA length = 12
FEATURE	Location/Qualifiers
misc_feature	1..12
	note = SYNTHESIZED
source	1..12
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 71	
gcagcgggtga gc	12
SEQ ID NO: 72	moltype = AA length = 4
FEATURE	Location/Qualifiers
REGION	1..4
	note = SYNTHESIZED
source	1..4
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 72	
AAVS	4
SEQ ID NO: 73	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = SYNTHESIZED
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 73	
AAKKKVS	7
SEQ ID NO: 74	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = SYNTHESIZED
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 74	
AAKKKTV	7
SEQ ID NO: 75	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = SYNTHESIZED
source	1..7
	mol_type = protein
	organism = synthetic construct

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SEQUENCE: 75		
AAKKKNV		7
SEQ ID NO: 76	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = SYNTHESIZED	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 76		
AEKKKNV		7
SEQ ID NO: 77	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = SYNTHESIZED	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 77		
PKKKEKKKKK KA		12
SEQ ID NO: 78	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 78		
tccaaaaaga aaaaaaagaa aaagaagaag caa		33
SEQ ID NO: 79	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 79		
tccaagaaga agaagaagaa gaagaagaag caa		33
SEQ ID NO: 80	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 80		
tccaaaaaaa aaaaaaagaa aaagaagaag caa		33
SEQ ID NO: 81	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 81		
tccaaaaaaaaaaaaaaaaaaaaa caa		33
SEQ ID NO: 82	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = SYNTHESIZED	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 82		
SKKKKKKKKK Q		11
SEQ ID NO: 83	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = SYNTHESIZED	
source	1..10	
	mol_type = protein	

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SEQUENCE: 83	organism = synthetic construct	
EKKKKKDKNN		10
SEQ ID NO: 84	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 84		
aagaagaaag aaaaaaagaa aaaaaagaaa gct		33
SEQ ID NO: 85	moltype = DNA length = 37	
FEATURE	Location/Qualifiers	
misc_feature	1..37	
	note = SYNTHESIZED	
source	1..37	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 85		
aagaagaaag aaaaaaagaa aaaaaagaaa gctctaa		37
SEQ ID NO: 86	moltype = DNA length = 38	
FEATURE	Location/Qualifiers	
misc_feature	1..38	
	note = SYNTHESIZED	
source	1..38	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 86		
aagaagaaag aaaaaaagaa aaaaaagaaa gctcctaa		38
SEQ ID NO: 87	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 87		
aagaagaaag aaaaaaaaaa aaaaaagaaa gct		33
SEQ ID NO: 88	moltype = DNA length = 37	
FEATURE	Location/Qualifiers	
misc_feature	1..37	
	note = SYNTHESIZED	
source	1..37	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 88		
aagaagaaag aaaaaaaaaa aaaaaagaaa gctctaa		37
SEQ ID NO: 89	moltype = DNA length = 38	
FEATURE	Location/Qualifiers	
misc_feature	1..38	
	note = SYNTHESIZED	
source	1..38	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 89		
aagaagaaag aaaaaaaaaa aaaaaagaaa gctcctaa		38
SEQ ID NO: 90	moltype = DNA length = 33	
FEATURE	Location/Qualifiers	
misc_feature	1..33	
	note = SYNTHESIZED	
source	1..33	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 90		
tccaaaaaga aaaaaaaaaa aaagaagaag caa		33
SEQ ID NO: 91	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
misc_feature	1..15	
	note = SYNTHESIZED	
source	1..15	

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SEQUENCE: 91	mol_type = other DNA	
gcaaaaaaaaa atgtg	organism = synthetic construct	15
SEQ ID NO: 92	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = SYNTHESIZED	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 92		5
AKKNV		
SEQ ID NO: 93	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
misc_feature	1..15	
	note = SYNTHESIZED	
source	1..15	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 93		15
gcgaaaaaaaa aatcc		
SEQ ID NO: 94	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = SYNTHESIZED	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 94		5
AKKKS		
SEQ ID NO: 95	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
misc_feature	1..15	
	note = SYNTHESIZED	
source	1..15	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 95		15
caaaaaaaaa atgtg		
SEQ ID NO: 96	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = SYNTHESIZED	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 96		5
QKKNV		
SEQ ID NO: 97	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
misc_feature	1..15	
	note = SYNTHESIZED	
source	1..15	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 97		15
gaaaaaaaaa aggtg		
SEQ ID NO: 98	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = SYNTHESIZED	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 98		5
EKKKV		
SEQ ID NO: 99	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
misc_feature	1..15	
	note = SYNTHESIZED	

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source	1..15 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 99 caaaaaaaaaa aggtg		15
SEQ ID NO: 100 FEATURE REGION	moltype = AA length = 5 Location/Qualifiers 1..5 note = SYNTHESIZED	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 100 QKKKV		5
SEQ ID NO: 101 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = SYNTHESIZED	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 101 gcgaaaaaaaa aaaatgtg		18
SEQ ID NO: 102 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = SYNTHESIZED	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 102 AKKKNV		6
SEQ ID NO: 103 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = SYNTHESIZED	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 103 gcgaaaaaaaa aaaagggtg		18
SEQ ID NO: 104 FEATURE REGION	moltype = AA length = 6 Location/Qualifiers 1..6 note = SYNTHESIZED	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 104 AKKKKV		6
SEQ ID NO: 105 FEATURE misc_feature	moltype = DNA length = 58 Location/Qualifiers 1..58 note = SYNTHESIZED	
source	1..58 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 105 gtaccggatt atgcgaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaagt gagcaagg		58
SEQ ID NO: 106 FEATURE misc_feature	moltype = DNA length = 58 Location/Qualifiers 1..58 note = SYNTHESIZED	
source	1..58 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 106 gtaccggatt atgcgaaaaa aaaaaaaaaa ctgaaaaaaa aaaaaaaagt gagcaagg		58
SEQ ID NO: 107 FEATURE misc_feature	moltype = DNA length = 58 Location/Qualifiers 1..58	

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source	note = SYNTHESIZED 1..58 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 107		
gtaccggatt atgcgaaaaa aaaaaaaaaa tacaaaaaaa aaaaaaaagt gagcaagg		58
SEQ ID NO: 108	moltype = DNA length = 58	
FEATURE	Location/Qualifiers	
misc_feature	1..58	
	note = SYNTHESIZED	
source	1..58 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 108		
gtaccggatt atgcgaaaaa aaaaaaaaaa cccaaaaaaa aaaaaaaagt gagcaagg		58
SEQ ID NO: 109	moltype = DNA length = 55	
FEATURE	Location/Qualifiers	
misc_feature	1..55	
	note = SYNTHESIZED	
source	1..55 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 109		
gtaccggatt atgcgaaaaa aaaaaaaaaa aaaaaaaagt gag caagg		55
SEQ ID NO: 110	moltype = DNA length = 42	
FEATURE	Location/Qualifiers	
misc_feature	1..42	
	note = SYNTHESIZED	
source	1..42 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 110		
tgcacccaaa aaaattttaca aaaaaaccgt gagcaagggc ga		42
SEQ ID NO: 111	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = SYNTHESIZED	
source	1..12 mol_type = protein organism = synthetic construct	
SEQUENCE: 111		
DVEKKKKKDK NN		12
SEQ ID NO: 112	moltype = DNA length = 36	
FEATURE	Location/Qualifiers	
misc_feature	1..36	
	note = SYNTHESIZED	
source	1..36 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 112		
gatgtggaaa aaaagaaaaa aaaggacaag aataat		36
SEQ ID NO: 113	moltype = DNA length = 36	
FEATURE	Location/Qualifiers	
misc_feature	1..36	
	note = SYNTHESIZED	
source	1..36 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 113		
gacgtggaaa aaaagaaaaa aaaggacaag aataat		36
SEQ ID NO: 114	moltype = DNA length = 36	
FEATURE	Location/Qualifiers	
misc_feature	1..36	
	note = SYNTHESIZED	
source	1..36 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 114		
gatgtggaaa agaagaaaaa aaaggacaaa aataat		36
SEQ ID NO: 115	moltype = DNA length = 36	
FEATURE	Location/Qualifiers	

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misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 115
gatgtggaaa agaagaaaaa aaaggacaaa aacaac          36

SEQ ID NO: 116    moltype = DNA length = 36
FEATURE
misc_feature      1..36
                  note = SYNTHESIZED
source            1..36
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 116
gatgtggaaa agaagaaaaa aaaggataaa aacaac          36

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What is claimed is:

1. A method for modulating the level of expression of a polypeptide in a cell, the method comprising: introducing into the cell an expression vector comprising a polynucleotide sequence encoding a polypeptide to be expressed, comprising:

- (a) introducing into an open reading frame of the polynucleotide sequence a polynucleotide tag sequence comprising at least three consecutive AAA lysine codons, wherein the AAA lysine codons decrease expression of the polynucleotide sequence when the expression vector is introduced into a cell relative to a reference vector without the at least three consecutive AAA lysine codons; or
- (b) introducing into an open reading frame of the polynucleotide sequence: a polynucleotide tag sequence comprising at least one AAG lysine codon, wherein the at least one AAG lysine codon increases expression of the at least one polynucleotide sequence when the expression vector is introduced into a cell relative to a reference vector without the at least one AAG lysine codon, thereby modulating the level of expression of the polypeptide in the cell; and

wherein the expression vector comprises a cloning site having at least two endonuclease recognition sequences for inserting the at least one polynucleotide sequence and at least one polynucleotide tag sequence in the open reading frame of the polynucleotide sequence encoding a polypeptide to be expressed between a start codon and the cloning site.

2. The method of claim 1, wherein modulating the level of expression comprises: increasing the amount of consecutive A nucleotides by inserting a polynucleotide tag comprising at least three AAA lysine codons into the open reading frame of the polynucleotide sequence encoding the polypeptide.

3. The method of claim 2, wherein the polynucleotide tag sequence comprising at least three consecutive AAA lysine

codons is inserted into an open reading frame at a position selected from the group consisting of in a 5' untranslated region (UTR), in an exon, at an exon/intron boundary, and in a 3' UTR.

4. The method of claim 1, wherein modulating the level of expression comprises: decreasing the amount of consecutive A nucleotides by inserting a polynucleotide tag comprising at least one AAG lysine codon into the open reading frame of the polynucleotide sequence encoding the polypeptide in the cell.

5. The method of claim 4, wherein the at least one AAG lysine codon is inserted into polynucleotide sequence for expression at the N-terminus or C-terminus of the polypeptide in the cell.

6. The method of claim 1, wherein the polynucleotide tag sequence in (b) comprises at least one polylysine track comprising at least two consecutive AAG lysine codons.

7. A method for modulating the level of expression of a polypeptide in a cell, the method comprising: introducing into the cell an expression vector comprising a polynucleotide sequence encoding a polypeptide to be expressed comprising:

introducing into an open reading frame of the polynucleotide sequence at least one synonymous G to A nucleotide mutation into at least one AAG lysine codon, wherein the at least synonymous mutation decreases expression of the polynucleotide sequence when the expression vector is introduced into a cell relative to a reference vector without the a synonymous mutation in lysine codon.

8. The method of claim 7, wherein at least one polylysine track comprises the at least one lysine codon, and wherein the at least one polylysine track comprises a combination of AAG and AAA lysine codons or a plurality of AAG lysine codons.

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